

**Dubai Metro, UAE
Green Line – Al-Qusais Depot**

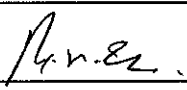
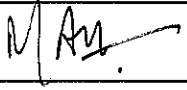
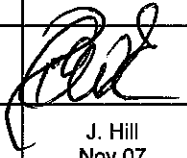
**Diesel Fuel Tank
(Building 21)**

**Structural Design Calculations
(DCP4 Submission)**

Nov 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001-D-GAW-WDT-DR-DCC-545014	Controlled Copy No.:
Document Title: Green Line – Al-Qusais Depot Diesel Fuel Tank (Building 21) Structural Design Calculations (DCP4 Submission)	
Covering Letter / Transmittal Ref. No.:	Date of Issue: 20 Nov 2007

Revision, Review and Approval/Authorization Records

				
A2	DCP4 Submission	D. Shah Nov 07	M. Lake Nov 07	J. Hill Nov 07
A1	DCP3 Submission	D. Shah Aug 07	P. Scott Aug 07	J. Hill Aug 07
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

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Rev A2	Page 2

Table

1	Introduction.....	04
2	Building Form and Structural System	04
3	Peer Review of Red Line building and comparison.....	04
4	Drawings.....	05

Appendix 'A'

	Red Line Rashidiya Depot, Diesel Fuel Station Building-21, Design Calculations (Enhanced Design)	06
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Appendix 'B'

	Response to Engineer Comments.....	
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1 Introduction

The Dubai Metro Project is under construction by the Dubai Rapid Link Consortium (DURL). The Civil Works are contracted to the Japan Turkey Metro Joint Venture (JTMJV), and the Rail Systems Works are contracted to Mitsubishi Heavy Industries (MHI). The rolling stock will be provided by Kinki Sharyo.

The diesel fuel tank is located adjacent to the wheel monitoring equipment. The structure will provide cover to the filling station spanning across the operational track.

This submission for Diesel fuel tank (Building 21) in Al-Qusais Depot is made by the Designers for the JTMJV, and as such concentrates on the scope of the Civil Works Contract, which includes the Architecture, Structure and M&E Design.

2 Building Form and Structural System

The diesel fuel tank and filling station is located adjacent to the wheel monitoring equipment. The structure will provide cover to the filling station spanning across the operational track. This building is identical as that of Red Line Rashidiya Main Depot, Building 21 Diesel Fuel Station.

A steel portal frame canopy with shell type roof is proposed in the harmony of other identical buildings. The building is to be clad with insulated standing seam aluminium sheeting, fixed to cold formed purlins which span onto structural frame.

A 300 mm thick Ground level slab monolithic with Reinforced Concrete Beam is proposed to house the facilities. The RC Pedestal raised from RC pad footing will support structural steel columns with holding down bolts fixed in position at formation level.

3 Peer review of Red Line building and comparison

Essentially this building is identical as that of Red Line Rashidiya Main Depot, Building 21 Diesel Fuel Tank. So the form and structural system has been replicated from the Rashidiya building.

The structural report DM001-E-MAV-CDP-DR-DCC-371953 Rev.A1 (Ref.Sect.5) for the Rashidiya building 21-Diesel Fuel Station and the Engineer's Comments and their responses have been peer reviewed for loadings, analysis, strength & deflection checks for all the members and foundation design. The drawings have also been peer reviewed for the design intent translation on to the details, which have then been replicated onto the Green Line Depot building-21-Diesel Fuel Tank as listed in sect.4.

The basis of this report is to utilise Red Line Main Depot Diesel Fuel Station (Bldg-21) to Green Line Al-Qusais Depot Diesel Fuel Tank (Bldg-21) with sufficient review to allow submission of Green Line DCP4 Phase.

4 Drawings

To complement the purpose of this submission, the following Structural drawings are submitted along with this report.

	<u>Drawing Title</u>	<u>Drawing No.</u>
Structural Drawings		
1	Al-Qusais-Depot - General Notes And Typical Details Sheet 1 of 3	DM001-D-GDP-CDP-DD-DCC-551990
2	Al-Qusais-Depot - General Notes And Typical Details Sheet 2 of 3	DM001-D-GDP-CDP-DD-DCC-551991
3	Al-Qusais-Depot - General Notes And Typical Details Sheet 3 of 3	DM001-D-GDP-CDP-DD-DCC-551992
4	Al-Qusais Depot Building 21, Diesel Fuel Tank - Ground Slab Layout Plan	DM001-D-GAW-WDT-DD-DCC-551725
5	Al-Qusais Depot Building 21, Diesel Fuel Tank - Roof Layout Plan	DM001-D-GAW-WDT-DD-DCC-551726
6	Al-Qusais Depot Building 21, Diesel Fuel Tank - Section A-A & B-B	DM001-D-GAW-WDT-DD-DCC-551727
7	Al-Qusais Depot Building 21, Diesel Fuel Tank - Ground Slab Details	DM001-D-GAW-WDT-DD-DCC-551728
8	Al-Qusais Depot Building 21, Diesel Fuel Tank - Fuel Tank Details - Sheet 1	DM001-D-GAW-WDT-DD-DCC-551729
9	Al-Qusais Depot Building 21, Diesel Fuel Tank - Roof Setting Out - Plan & Section	DM001-D-GAW-WDT-DD-DCC-551730
10	Al-Qusais Depot Building 21, Diesel Fuel Tank - Fuel Tank Details - Sheet 2	DM001-D-GAW-WDT-DD-DCC-551731

Appendix 'A'
Red Line Rashidiya Depot
Diesel Fuel Station Building-21
Design Calculations (Enhanced Design)

**Dubai Metro, UAE
Rashidiya Depot**

**Structural Design Calculation
(Enhanced Design)
Building 21 – Diesel Fuel Station**

May 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001-E-MAV-CDP-DR-DCC-371953	Controlled Copy No.:
Document Title: Main Depot- Rashidiya Structural Design Calculation (Enhanced Design) Building 21 – Fuel Station	
Covering Letter / Transmittal Ref. No.: To be advised	Date of Issue: 29 May 2007

Revision, Review and Approval/Authorization Records

A1	Issued for Approval (ENHANCED DESIGN)	CK Lam May 07	P. Shepherd-Smith May 07	B. Maney May 07
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

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4	Meinhardt Infrastructure (CK Lam)

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Table of Contents

1	TYPICAL PORTAL FRAME 3D ANALYSIS	001
2	FOUNDATION DESIGN	033
3	SECONDARY STEEL BEAM & CONNECTION DESIGN.....	036
4	UNDERGROUND FUEL TANK DESIGN.....	043

1 TYPICAL PORTAL FRAME 3D ANALYSIS

Document No.: **DM001-E-MAV-CDP-DR-DCC-371953**

Rev A1

Date: 29 May 2007

Page



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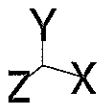
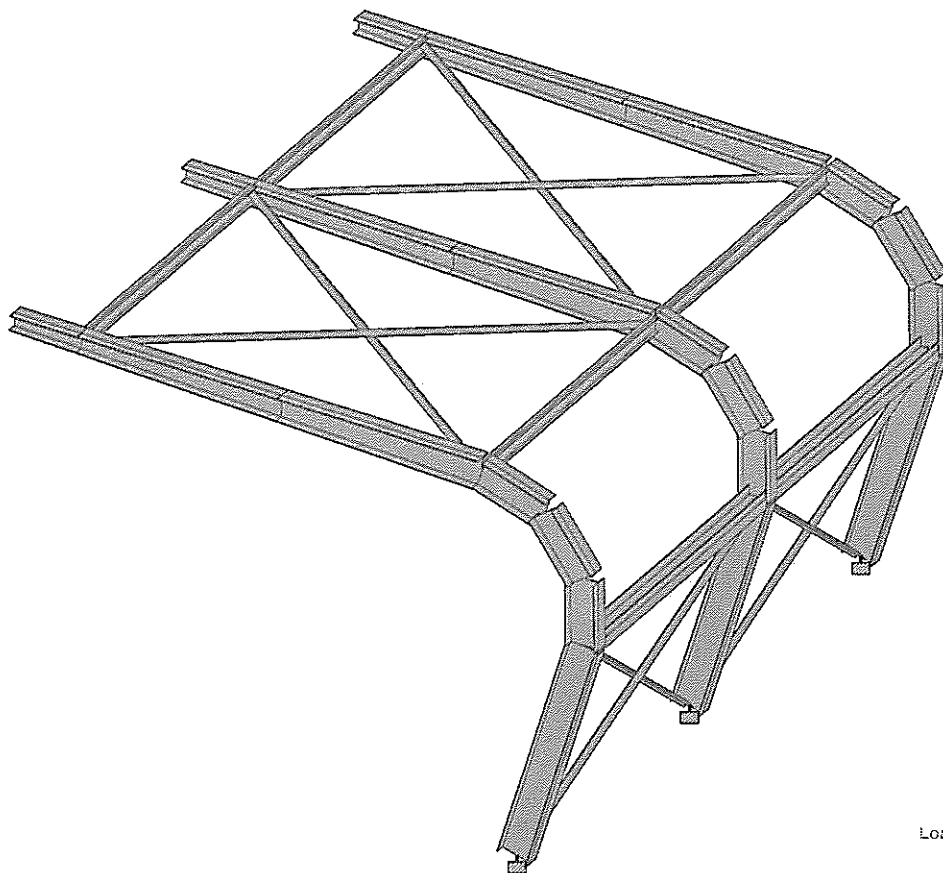
Date May 2007

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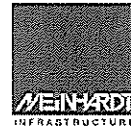
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File Fuel Station-R1.STD

Date/Time 16-May-2007 19:52



Load 1



Site:	Dubai Metro Project - Rashidiya Fuel Station Loadings	Date:	16-May-07	Job No.:	68010.01
		Designed by:	CJA	Sheet No.:	

Column spacing = 4 m

Roof Loadings :

DL

W_{DL} :

Sheeting	0.1 kPa		
Purlins	0.1 kPa		
Insulation	0.3 kPa		
	<u>0.5 kPa</u>	x 4 =	2 kN/m

LL

0.6 kPa x 4 = 2.4 kN/m

Notional horizontal load, P_h

(Load Case of 1.4DL + 1.6LL)

Tributary load area for the roof

= 6×4
= 24 m²

Total Dead Load
12 kN

Total Live Load
14 kN
(Including A/C equipment)

Tributary load area for the wall

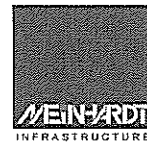
= 4×4
= 16 m²

Total Dead Load
8 kN
(excluding the M&E Finishes)

Total Live Load

Therefore, total ULS load, W_u = 34 kN

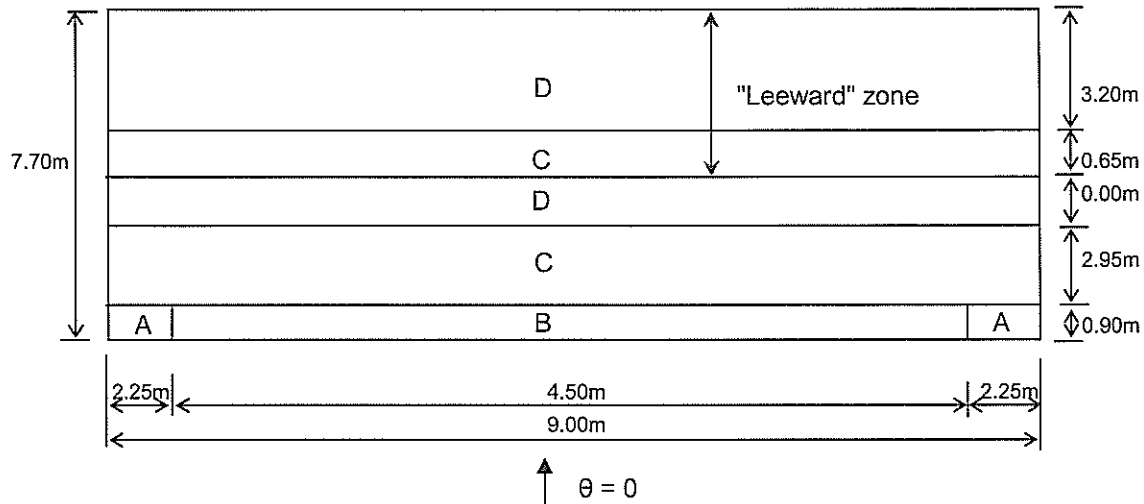
P_h = 1.5 % of W_u
= 0.5 kN



Site:	Dubai Metro Project - Rashidiya Fuel Station	Date:	16-May-07	Job No.:	068010.01
	Wind Load Computation	Designed by:	CJA	Sheet No.:	

Average Cpe for roof (Wind direction normal to eave)

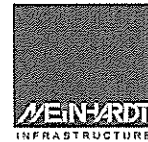
PLAN



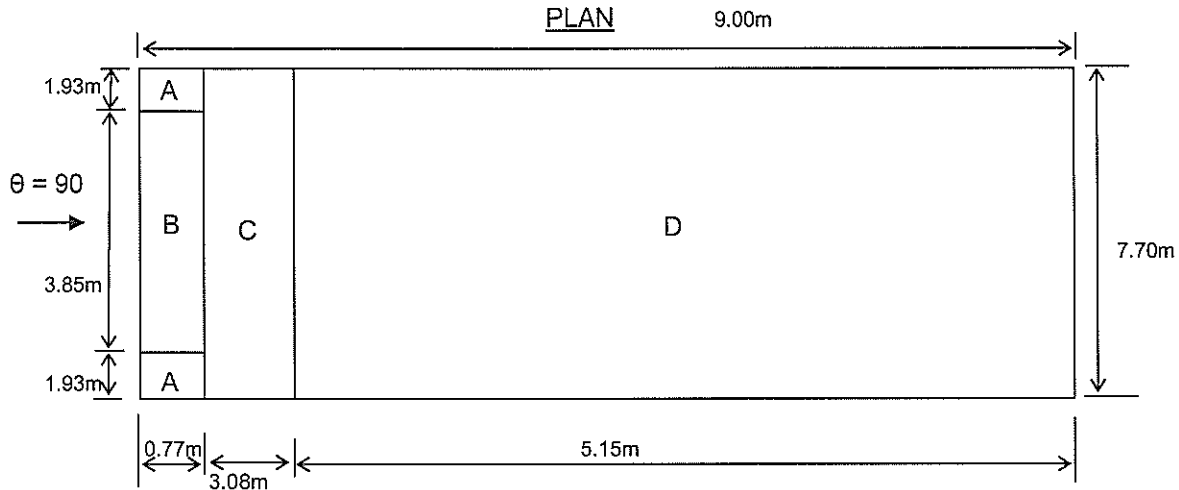
Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 9 \text{ m} \\
 H &= 5.2 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 9 \\
 &= 9 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	4.1	-2			
B	4.1	-1.4			
C	26.6	-0.7	5.9	-0.7	-0.7
D	0.0	-0.2	28.8	-0.2	0.2
Total	34.7		34.7		
Average		-0.93		-0.28	0.05



Site:	Dubai Metro Project - Rashidiya Fuel Station Wind Load Computation	Date:	16-May-07	Job No.:	068010.01
		Designed by:	CJA	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 7.7 \text{ m} \\
 H &= 5.2 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 7.7 \\
 &= 7.7 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

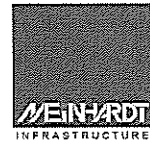
For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

Wind Load Computation

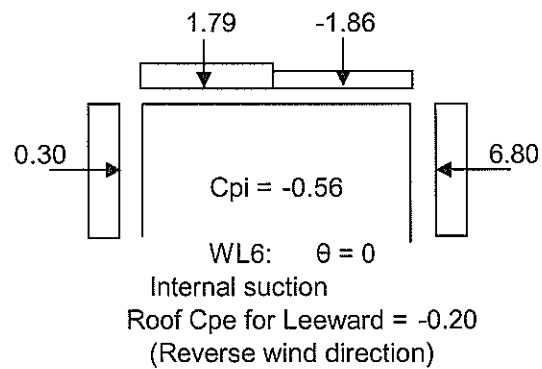
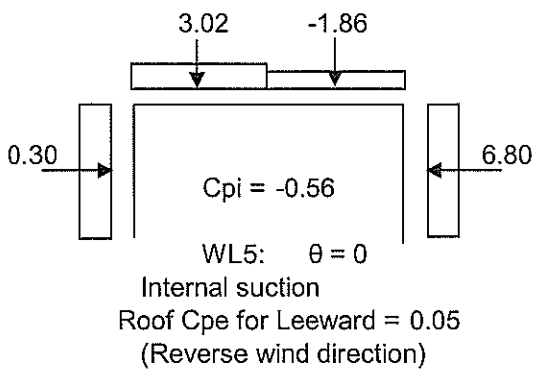
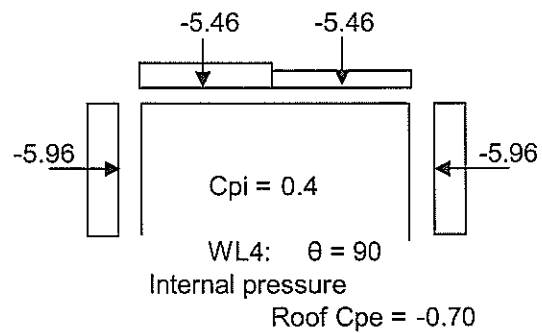
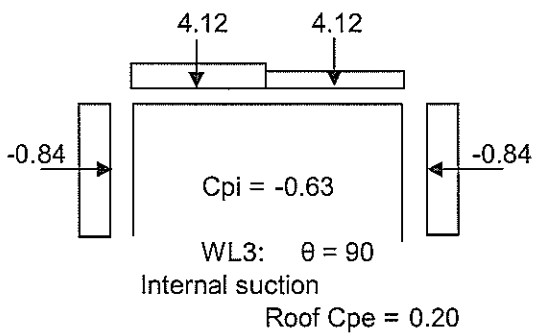
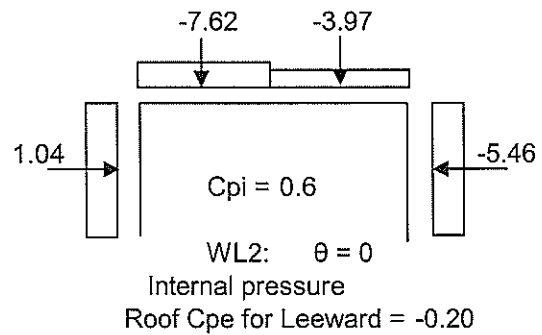
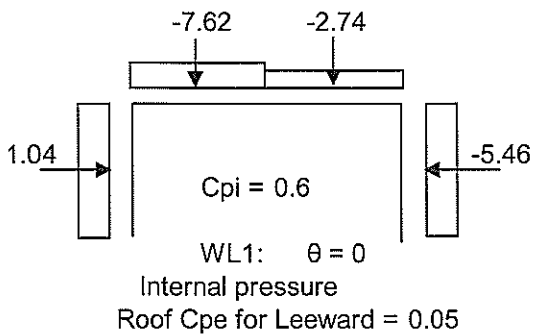
$$\begin{aligned}
 D &= 7.70 \text{ m} \\
 H &= 5.2 \text{ m} \\
 D/H &= 1.48
 \end{aligned}$$

Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.93	-0.28 0.05	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.81	-0.5 -0.5	-0.8 (Zone B)	-0.8 (Zone B)



Site:	Dubai Metro Project - Rashidiya Fuel Station Wind Load Computation	Date:	16-May-07	Job No.:	068010.01
		Designed by:	CJA	Sheet No.:	

$$\text{Wind Load} = 4 \times q \times (C_{pe} - C_{pi}) \quad \text{kN/m} \quad (\text{For unit width})$$



Site: Dubai Metro Project - Rashidiya
Fuel Station
Earthquake Loading

Date: 16-May-07

Job No.: 68010.01

Designed by: CJA

Sheet No.:

Estimation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
Height of the Column h_n = Given below

Design Information:

- 1) Seismic Zone = 2A (from Design Brief)
2) Occupancy Category = 1 (Tanle- 16-K, Conservatively adopt 1)
3) Lateral Force procedure = **Static Procedure** (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

- 4) **Base Shear** $V_{base-S} = (C_v \times I \times W) / (R \times T)$ (Section 1630.2.1)
Seismic Coefficient $C_v = 0.32$ (Table 16-R assuming soil profile type S_D)
Importance Factor $I = 1.00$ (Table 16-K)
Numeric. Coef: Strength & Ductility $R = 4.50$ (Table 16-N for OMRT steel only)
Elastic Fundamental Period of Vibration $T = C_t (h_n)^{3/4}$ Sec.
Numerical Coefficient $C_t = 0.0853$ (Steel frame t.b.c)

- 5) **Limiting Values of Base shear** $V_{base-S-Max} = (2.5 \times C_a \times I \times W) / R = 4.20$
 $V_{base-S-Min} = 0.11 \times C_a \times I \times W = 0.83$

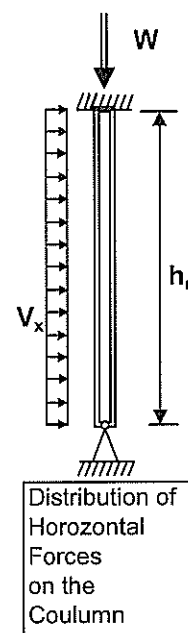
Design Earthquake Loads

(Section 1630.6)

- $E_x = \rho \times \text{Min. } (E_h \text{ or } V_{base-S-Max}) + E_v$
 $E_{x-max} = \Omega \times \text{Min. } (E_h \text{ or } V_{base-S-Max})$
 $V_{base-S} = E_h =$ Earthquake load in the element due to the base shear, V , as set out above (for non-structural components refer to section 1632)
Vert. component of EQ ground motion $E_v = 0.5 C_a I D$ (Table 16-K)
Seismic Coefficient $C_a = 0.22$ (Table 16-Q, for soil type S_D)
Seismic Force Amplification Factor $\Omega = 2.80$ (Table 16-N)
Dead Load on the element $W = D =$ Given below
 $\rho = 2 - 6.1 / (r_{max} \sqrt{A_B}) = 1.25$ (conservative) ($1.0 \leq \rho \leq 1.5$)
 $r_{max} =$ The maximum element-storey shear ratio
 $A_B =$ Ground floor area of the structure

Distribution of Horizontal Forces:Distribution of Horizontal Force on the Column $V_x = E_x / h_n$

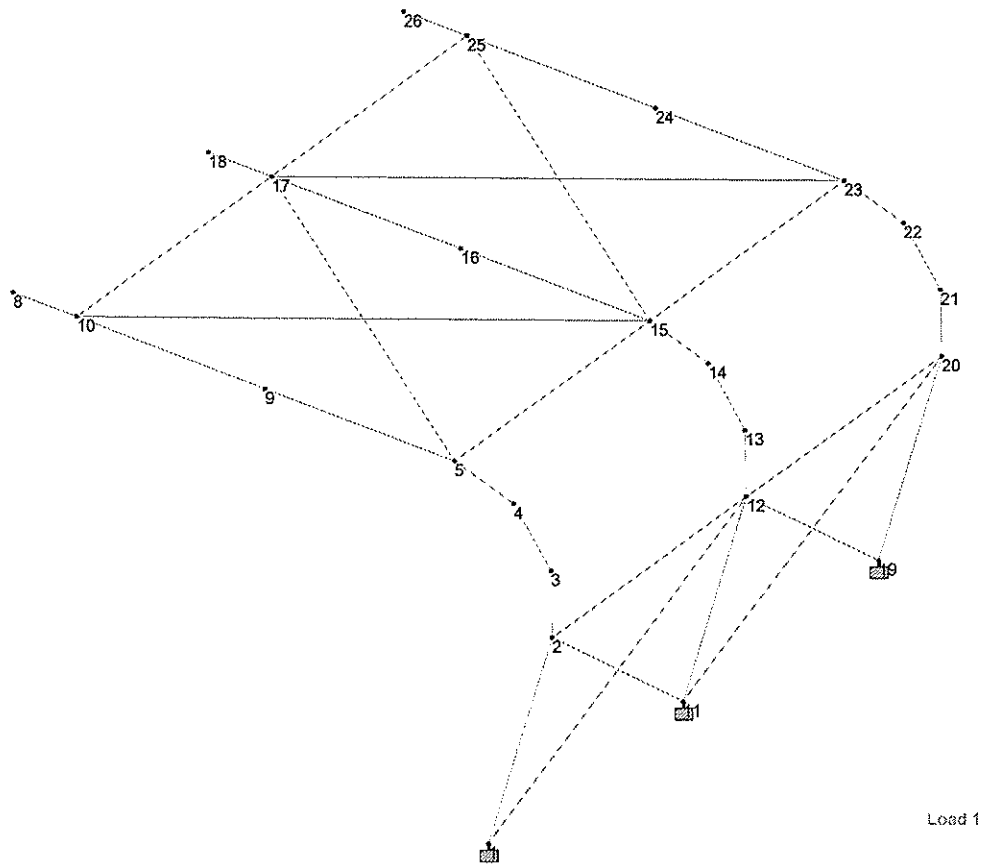
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	34	4.00	0.24	10.1	4.20	3.784	9.0	11.8	2.3	2.9





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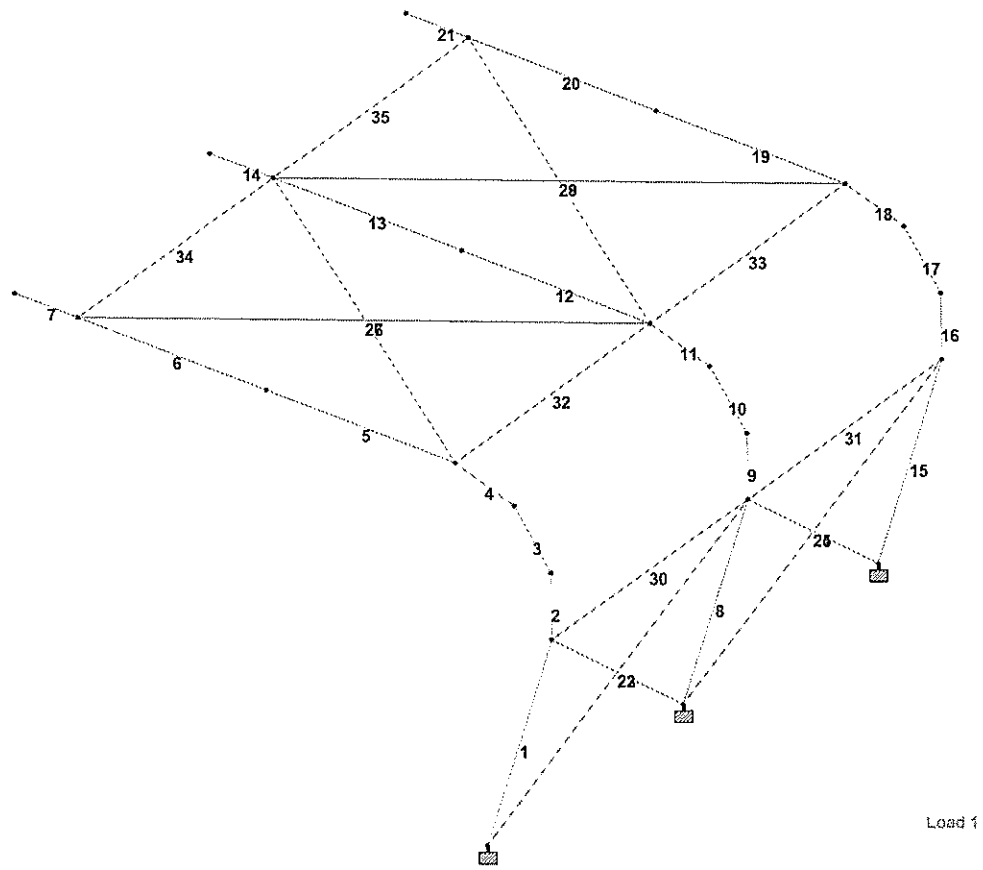
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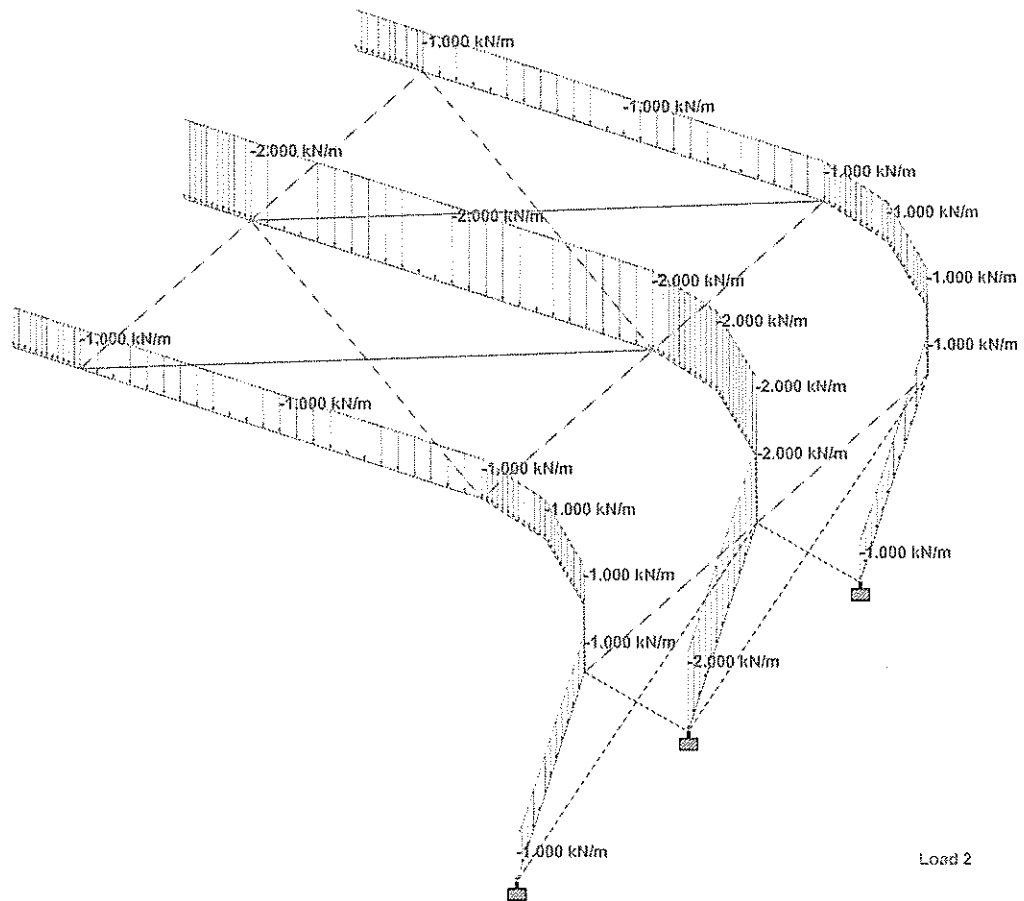
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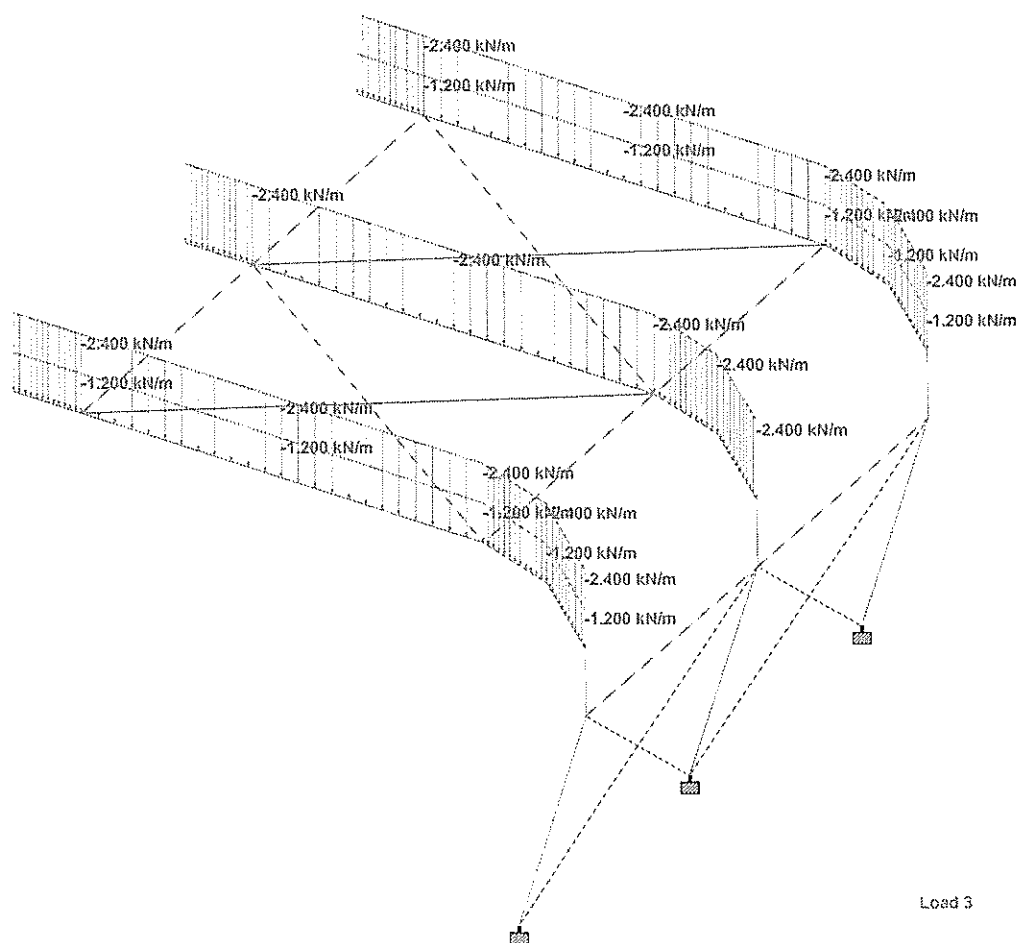
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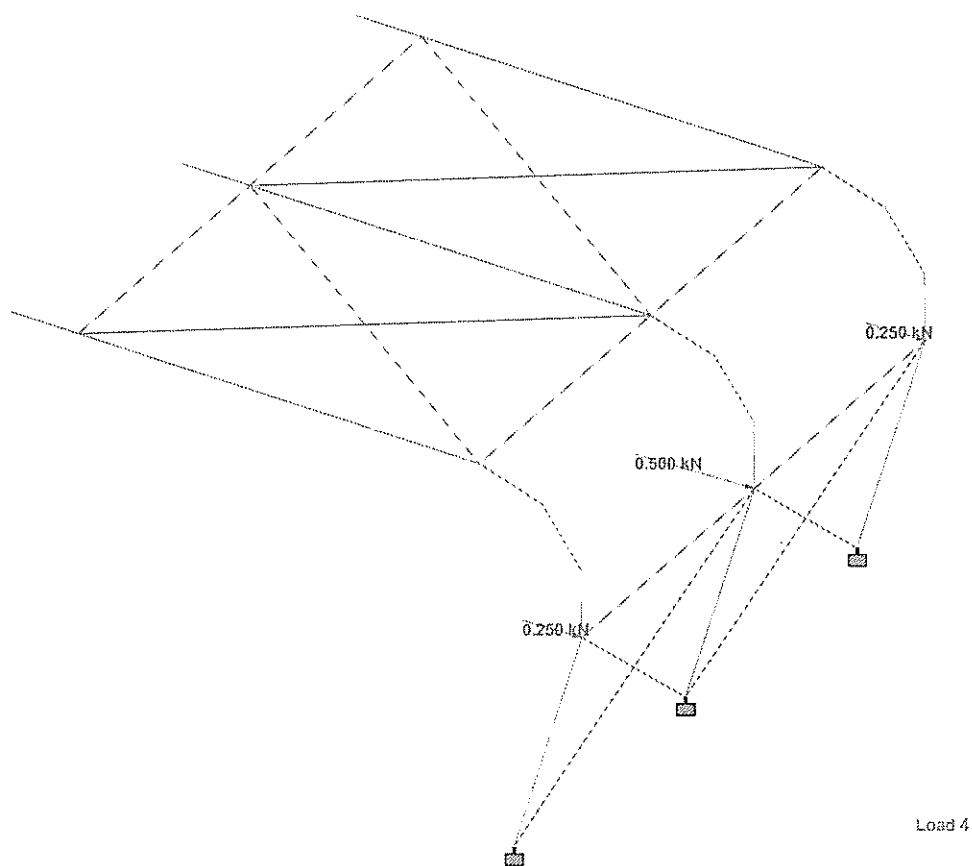
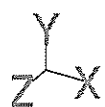
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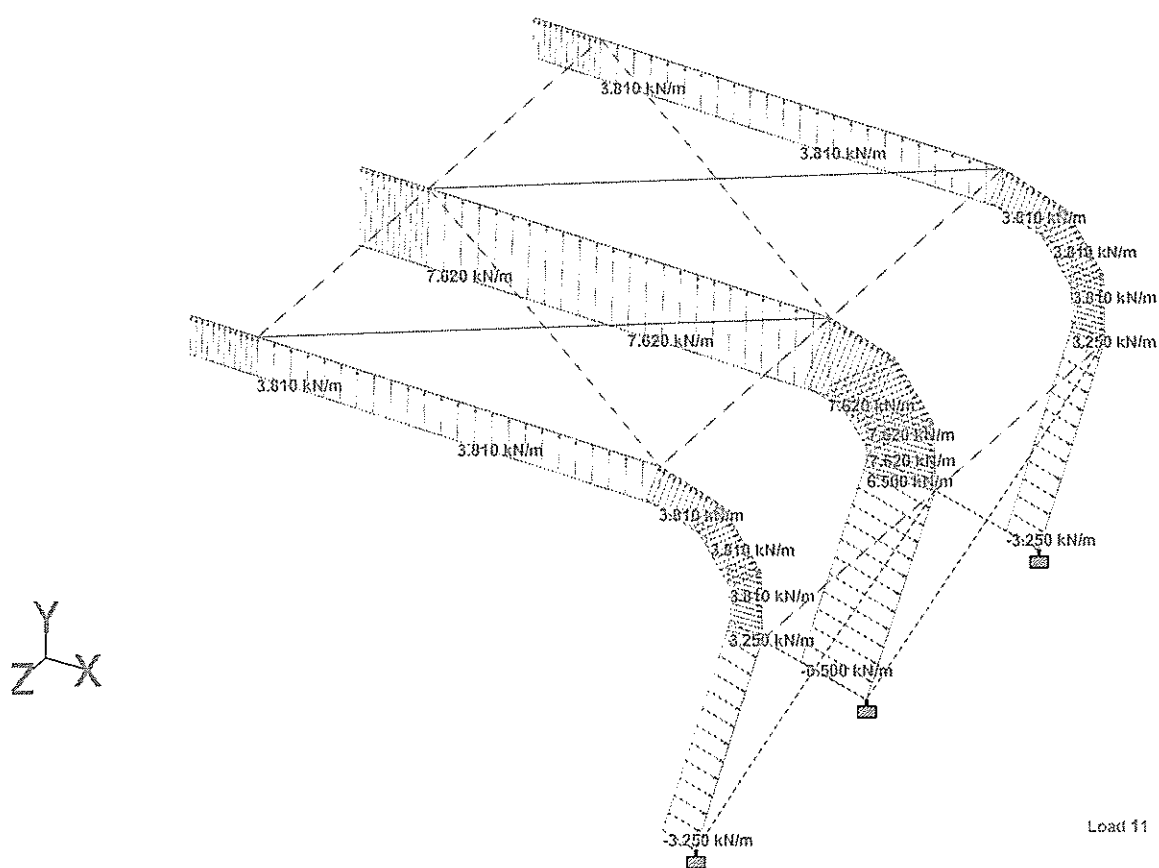
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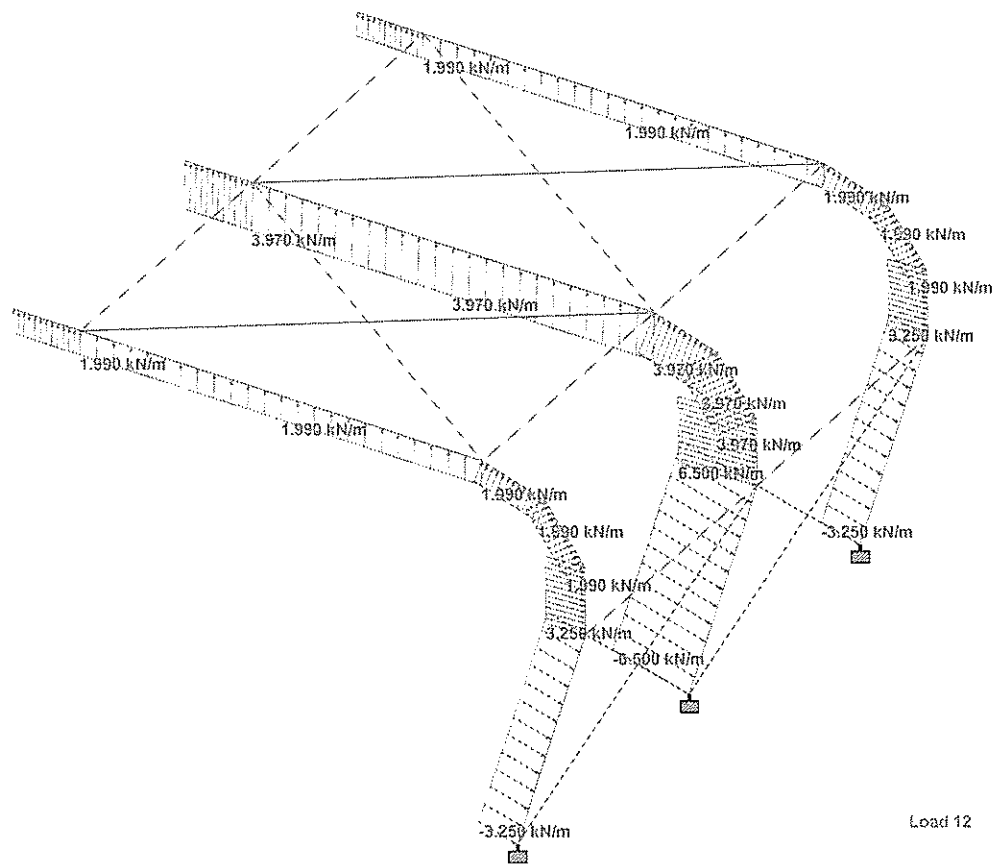
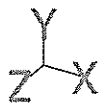
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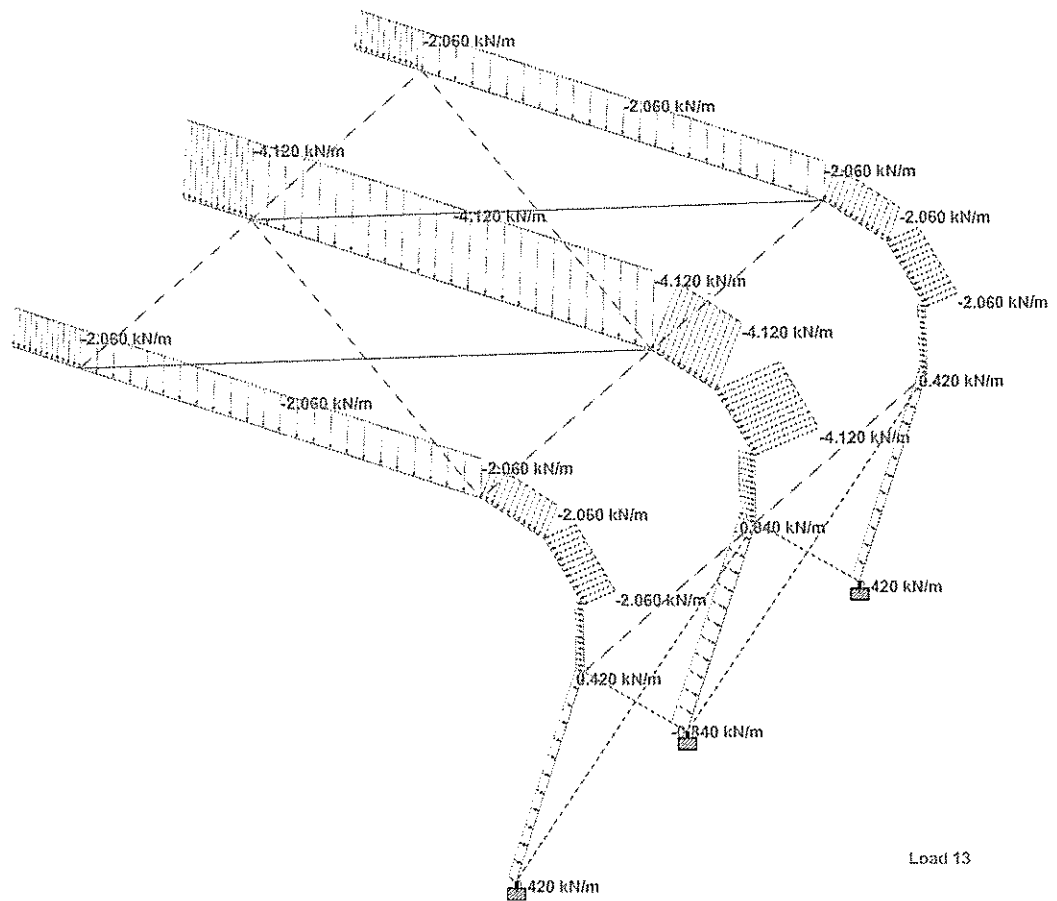


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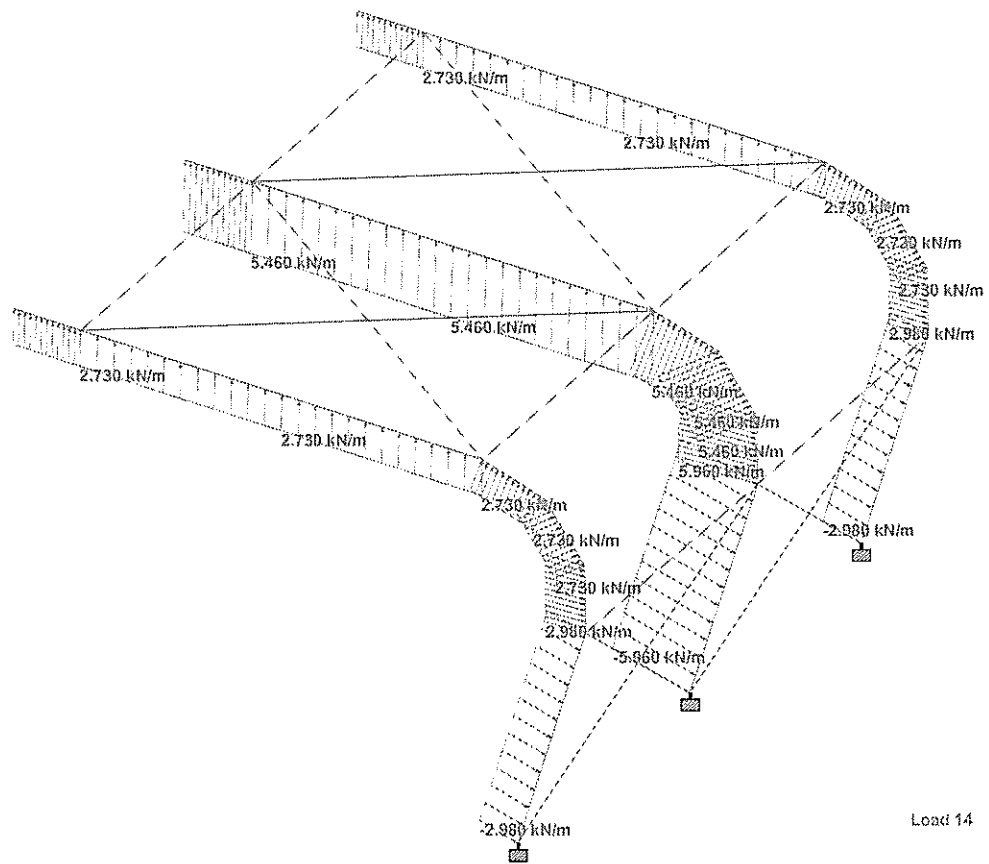
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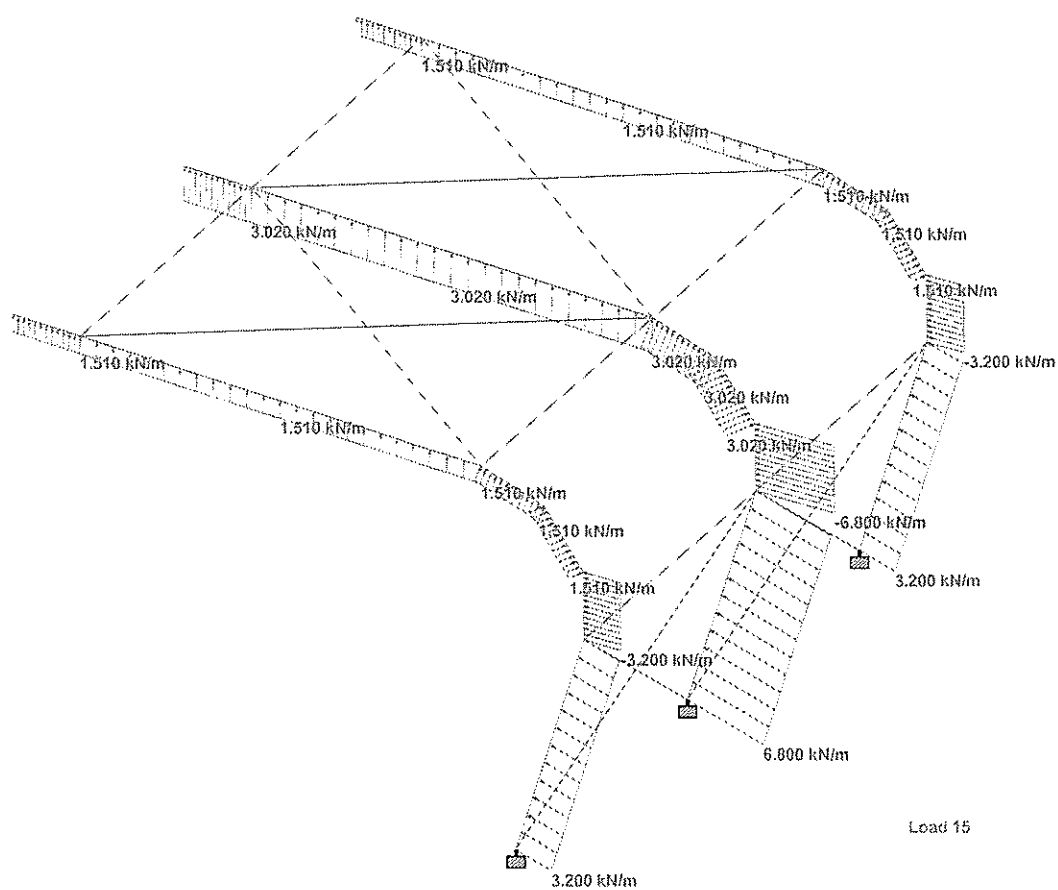
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Client

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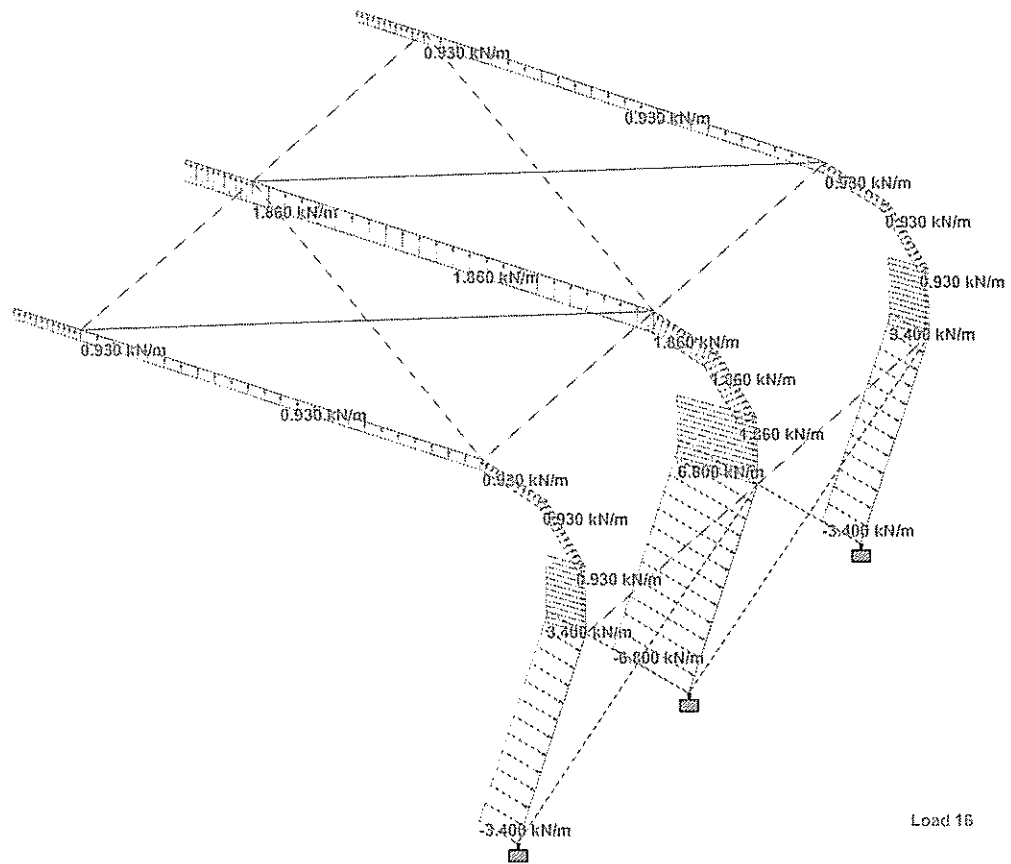
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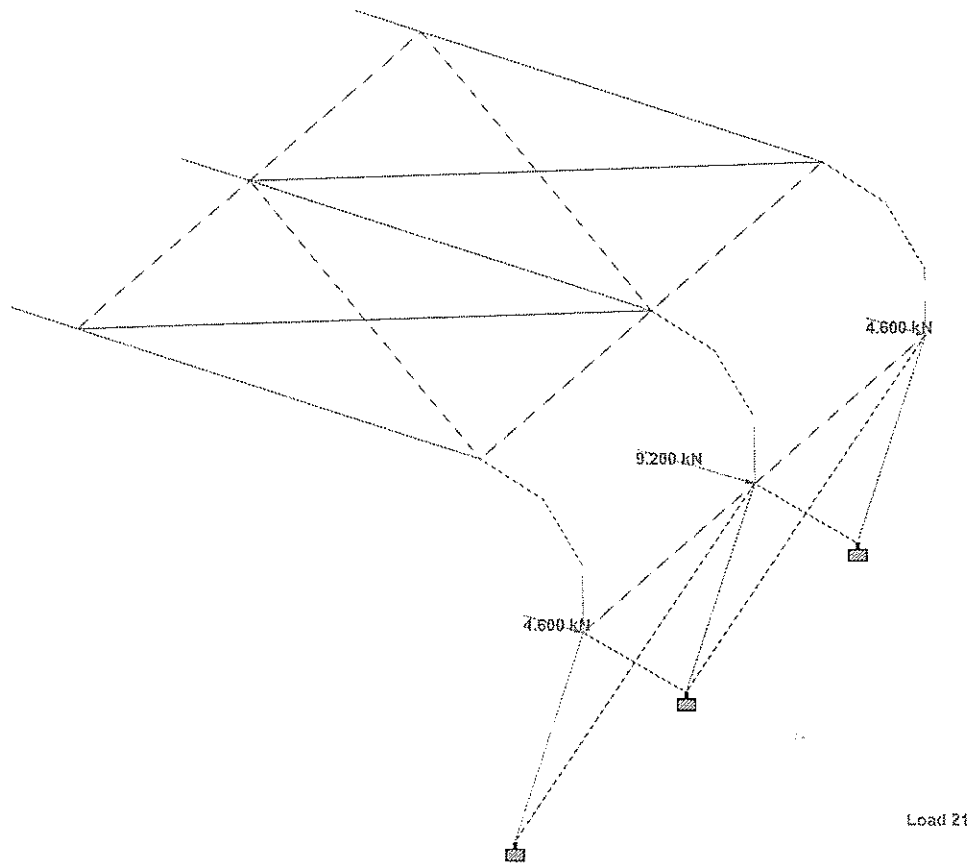
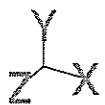
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File Fuel Station-R1.STD

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* Research Engineers, Intl.
* Date= Nov 10, 2007
* Time= 13:53:15
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* USER ID: MPL

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4. ENGINEER DATE MAY 2007
5. END JOB INFORMATION
6. UNIT METER K
7. UNIT METER K
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10. 8 -6.759 5.213 4; 9 -3.16135 5.08729 4; 10 -5.85971 5.18158 4; 11 0 0 0
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15. 1 1 2; 2 2 3; 3 4; 4 4 5; 5 9; 6 9 10; 7 10 8; 8 11 12; 9 12 13; 10 13 14
16. 11 14 15; 12 15 16; 13 16 17; 14 17 18; 15 19 20; 16 20 21; 17 21 22; 18 22 23
17. 19 23 24; 20 24 25; 21 25 26; 22 26 27; 23 27 28; 24 28 29; 25 29 30; 26 30 31
18. 31 32; 32 33; 33 34; 34 35; 35 36; 36 37; 37 38; 38 39; 39 40; 40 41; 41 42; 42 43
19. 43 44; 44 45; 45 46; 46 47; 47 48; 48 49; 49 50; 50 51; 51 52; 52 53; 53 54; 54 55
20. 55 56; 56 57; 57 58; 58 59; 59 60; 60 61; 61 62; 62 63; 63 64; 64 65
21. ISOTROPIC STEEL
22. E 2.05E+008
23. POISSON 0.3
24. DENSITY 76.8195
25. ALPHA 1.2E-005
26. DAMP 0.03
27. END DEFINE MATERIAL
28. MEMBER PROPERTIES BRITISH
29. 1 TO 4 TO 11 TO 18 TAPERED 0.4 0.008 0.4 0.25 0.012 0.25 0.012
30. 5 TO 12 TAPERED 0.4 0.008 0.314 0.25 0.012 0.25 0.012
31. 13 TO 20 TAPERED 0.238 0.008 0.28 0.25 0.012 0.25 0.012
32. 21 TO 28 TAPERED 0.238 0.008 0.28 0.25 0.012 0.25 0.012
33. 29 TO 36 TAPERED 0.238 0.008 0.28 0.25 0.012 0.25 0.012
34. 37 TO 44 TAPERED 0.238 0.008 0.28 0.25 0.012 0.25 0.012
35. 45 TO 52 TAPERED 0.238 0.008 0.28 0.25 0.012 0.25 0.012
36. CONSTANTS
37. BETA 90 MEMB 30 31
38. MATERIAL STEEL ALL
39. SUPPORTS
40. 1 11 19 FIXED
41. STAAD SPACE
42. MEMBER TENSION
43. 22 TO 33
44. 34 TO 45
45. SELFWEIGHT Y -1
46. LOAD 2 SLL
47. 1 TO 7 15 TO 21 UNI GY -1.
48. 8 TO 14 UNI GY -2
49. LOAD 3 LL
50. MEMBER LOAD
51. 3 TO 7 17 TO 21 UNI GY -1.2
52. 3 TO 7 30 TO 14 17 TO 21 UNI GY -2.4
53. LOAD 4 PH (NOTIONAL HORIZONTAL LOAD)
54. JOINT LOAD
55. 2 20 FX 0.25
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Fuel Station-R1

150. LOAD COMB 2011 DL + LL + WL1
 151. 1 1.0 2 1.0 3 1.0 11 1.0
 152. LOAD COMB 2012 DL + LL + WL2
 0 STAAD SPACE
 153. 1 1.0 2 1.0 3 1.0 12 1.0
 154. LOAD COMB 2013 DL + LL + WL3
 155. 1 1.0 2 1.0 3 1.0 13 1.0
 156. LOAD COMB 2014 DL + LL + WL4
 157. 1 1.0 2 1.0 3 1.0 14 1.0
 158. PERFORM ANALYSIS
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-- PAGE NO. 4

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER-ELEMENTS/SUPPORTS = 24/ 35/
 ORIGINAL/FINAL BAND-WIDTH= 10/ 6/
 TOTAL DEGREES OF FREEDOM = 126
 SIZE OF STIFFNESS MATRIX = 11, 6 DOUBLE KILLO-WORDS
 REQ'D/AVAIL. DISK SPACE = 12.1/ 402172.2 MB

0 **START ITERATION NO. 2
 **START ITERATION NO. 3
 **NOTE-Tension/Compression converged after 3 iterations, Case= 1
 0 **START ITERATION NO. 2
 **START ITERATION NO. 3
 **NOTE-Tension/Compression converged after 3 iterations, Case= 2
 0 **START ITERATION NO. 2
 **START ITERATION NO. 3
 **NOTE-Tension/Compression converged after 3 iterations, Case= 3
 0 **START ITERATION NO. 2
 **START ITERATION NO. 3
 **NOTE-Tension/Compression converged after 2 iterations, Case= 4
 0 **START ITERATION NO. 2
 **START ITERATION NO. 3
 **NOTE-Tension/Compression converged after 3 iterations, Case= 11
 0 **START ITERATION NO. 2
 **START ITERATION NO. 3
 **NOTE-Tension/Compression converged after 3 iterations, Case= 12
 0 STAAD SPACE
 0 **START ITERATION NO. 2
 **START ITERATION NO. 3
 **NOTE-Tension/Compression converged after 3 iterations, Case= 13
 0 **START ITERATION NO. 2
 **START ITERATION NO. 3
 **NOTE-Tension/Compression converged after 3 iterations, Case= 14
 0 **START ITERATION NO. 2
 **START ITERATION NO. 3
 **NOTE-Tension/Compression converged after 3 iterations, Case= 15

Page 3

Fuel Station-R1

**START ITERATION NO. 2
 **NOTE-Tension/Compression converged after 2 iterations, Case= 16
 0 STAAD SPACE
 0 **START ITERATION NO. 2
 **NOTE-Tension/Compression converged after 2 iterations, Case= 21

159. LOAD LIST 1001 1002 1011 TO 1014 1021 1111 TO 1114 1121 --
 160. 1211 TO 1214 1221 1321 2001 2002 2011 TO 2014
 161. PRINT SUPPORT REACTION LIST 1 2 11 19
 0 SUPPORT REACTION LIST 1
 0 STAAD SPACE
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-- PAGE NO. 6

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

 JOINT LOAD FORCE-X FORCE-Y FORCE-Z MOM-X MOM-Y MOM-Z
 0 1 1001 -2.01 79.67 0.10 0.49 -0.12 -167.00
 1002 -2.31 79.67 0.11 0.49 -0.12 -166.04
 1011 -31.65 27.35 2.22 0.52 -0.13 200.04
 1012 -34.86 -17.56 2.52 0.44 -0.11 217.78
 1013 -25.34 46.34 1.22 0.47 -0.12 217.78
 1017 14.41 1.41 1.41 0.51 -0.13 98.83
 1112 -27.67 4.59 1.39 0.38 -0.10 115.53
 1121 0.04 85.28 0.28 1.35 -0.33 -213.42
 1113 1.98 59.84 0.31 1.54 -0.38 -154.00
 1213 1.92 50.02 0.29 1.41 -0.35 -137.31
 1014 -26.50 38.31 1.62 0.49 -0.13 35.75
 1114 -28.98 5.04 1.88 0.53 -0.13 136.70
 1214 -29.04 -4.78 1.86 0.41 -0.10 133.40
 1021 -5.89 43.61 0.16 0.40 -0.10 -66.39
 1121 4.86 43.60 0.00 0.39 -0.10 -101.86
 1221 -3.25 22.09 0.14 0.58 -0.07 -59.74
 1321 3.21 51.88 0.07 0.55 -0.09 -103.56
 2001 -0.59 38.70 0.07 0.33 -0.08 -75.67
 2002 -26.37 22.79 1.85 0.44 -0.11 75.79
 2011 -21.11 38.61 1.02 0.39 -0.10 2.75
 2012 0.03 71.06 0.23 1.13 -0.28 -177.85
 2013 -22.08 31.92 1.35 0.41 -0.10 29.80
 2014 4.03 85.21 0.00 0.00 -0.00 -184.70
 11 1001 3.61 85.21 0.00 0.00 -0.00 -183.55
 1011 -21.39 15.54 0.00 0.00 0.00 54.07
 1111 -28.83 -11.42 0.00 0.00 0.00 171.49
 1211 -28.72 0.00 0.00 0.00 0.00 186.86
 1012 -23.08 24.53 0.00 0.00 0.00 -86.54
 1112 -30.60 27.84 0.00 0.00 0.00 186.50
 1312 -30.60 27.84 0.00 0.00 0.00 108.90
 1013 4.37 103.40 0.00 0.00 0.00 -243.08
 1113 1.15 91.08 0.00 0.00 0.00 -175.19
 1213 1.26 76.39 0.00 0.00 0.00 -157.82
 1014 -20.06 32.20 0.00 0.00 0.00 7.72
 1114 -27.29 8.01 0.00 0.00 0.00 117.41
 1214 -27.17 -6.69 0.00 0.00 0.00 134.79
 1021 -6.61 54.63 0.00 0.00 0.00 -69.76
 1121 8.68 54.66 0.00 0.00 0.00 -111.90
 1221 -7.91 33.06 0.00 0.00 0.00 -18.02
 1321 7.38 33.08 0.00 0.00 0.00 -60.16
 2001 2.48 57.85 0.00 0.00 0.00 -50.10
 2002 1.30 47.80 0.00 0.00 0.00 -45.10
 2011 -19.72 37.21 0.00 0.00 0.00 -14.20
 2012 -19.72 37.21 0.00 0.00 0.00 -202.57
 2013 3.59 86.17 0.00 0.00 0.00 6.43
 2014 -16.72 26.83 0.00 0.00 0.00 --
 0 STAAD SPACE
 0

7

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

 JOINT LOAD FORCE-X FORCE-Y FORCE-Z MOM-X MOM-Y MOM-Z
 0 STAAD SPACE
 0

Page 4

MAX/MIN FORCE VALUES FOR MEMB						2 - AMONGST ALL SECT LOCATIONS								
	FZ	DIST	LD	FY	DIST LD	MZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	4.89	0.90	1113			286.05	0.00	1013						
	-0.06	0.00	1221			0.57	0.00	1113				76.09	c	0.00 1013
MIN.	-20.98	0.00	1211			-152.61	0.00	1211						
	-0.31	0.90	1113			0.06	0.90	1221				28.40	T	0.90 1211
3	0.00	MAX	44.93	1013		283.60	1013					-0.06	1221	0.19 1113
		MIN	-30.03	1211		-135.65	1211					-0.31	1113	0.03 1211
	0.18	MAX	43.82	1013		275.62	1013					-0.06	1221	0.13 1113
		MIN	-32.25	1211		-130.32	1211					-0.31	1113	0.01 1211
	0.36	MAX	42.71	1013		267.83	1013					-0.06	1221	0.07 1211
		MIN	-28.47	1211		-125.13	1211					-0.31	1113	0.00 1211
	0.54	MAX	41.60	1013		259.23	1013					-0.36	1113	0.02 1113
		MIN	-26.90	1211		-120.23	1211					-0.36	1113	0.00 1211
	0.72	MAX	40.49	1013		252.83	1013					-0.06	1221	-0.01 1221
		MIN	-26.90	1211		-115.17	1211					-0.31	1113	-0.04 1011
	0.90	MAX	39.38	1013		245.68	1013					-0.06	1221	-0.02 1211
		MIN	-26.12	1211		-110.40	1211					-0.31	1113	-0.10 1113

	MAX/MIN FORCE VALUES FOR MEMB						3. AMONGST ALL SECT LOCATIONS					
	F _y /F _z	DIST	LD	MZ	DIST	LD	FY	DIST	LD	FX	DIST	LD
MAX.	44.93	0.00	1013		283.60	0.00	1013					
	-0.06	0.00	1221		0.19	0.00	1113			59.20	c	0.00 1013
MIN.	-30.03	0.00	1211		-135.65	0.00	1211					
	-30.31	0.90	1113		-0.10	0.90	1113			14.90		0.90 1211

[illegible]

MAX/MIN FORCE VALUES FOR MEMB				1, AMONGST ALL SECT LOCATIONS			
	FV/	LD		NZ/	LD		
	DIST	LD		DIST	LD	FX	DIST
MAX.	63.16	0.00	1013	245.58	0.00	1013	
	0.00	0.00	1221	-0.04	0.00	1221	
MIN.	-29.95	0.00	1211	-110.40	0.00	1211	
	-0.31	0.90	1113	-0.50	0.90	1113	
						22.20	c
						2.72	0.90 1211

5	0.00	MAX	58.59	1013	191.92	1013	0.05	1113	-0.10	1221
		MIN	-25.38	1211	-84.95	1211	0.01	1321	-0.52	1113
0.54	MAX		52.63	1013	161.57	1013	0.05	1113	-0.09	1221
	MIN		-23.41	1211	-71.77	1211	0.01	1321	-0.50	1113
1.08	MAX		49.08	1013	133.79	1013	0.05	1113	-0.09	1221
	MIN		-21.43	1211	-59.66	1211	0.01	1321	-0.46	1113
1.62	MAX		44.32	1013	108.37	1013	0.05	1113	-0.04	1221
	MIN		-19.49	1211	-48.99	1211	0.01	1321	-0.47	1113
2.16	MAX		39.19	1013	88.91	1013	0.02	1113	-0.07	1221
	MIN		-17.66	1211	-39.68	1211	0.01	1321	-0.47	1113
2.70	MAX		34.86	1013	65.80	1211	0.05	1113	-0.07	1221
	MIN		-15.47	1211	-29.75	1211	0.01	1321	-0.39	1113

[illegible]

Page 6

19	1001	-2.01	79.67	-0.10	-0.49	0.12	-167.00
	1002	-2.31	79.67	-0.11	-0.49	0.12	-166.04
	1011	-1.65	27.35	-0.52	-0.52	0.13	90.94
	1111	-5.08	17.35	-2.52	-0.37	0.14	201.98
	1012	-2.53	46.26	-0.52	-0.47	0.11	21.76
	1013	-2.53	14.41	-0.51	-0.51	0.13	98.83
	1112	-2.61	4.59	-1.19	-0.38	0.10	115.53
	1212	-2.61	85.28	-0.28	-1.35	0.33	-213.42
	1015	0.04	85.28	-0.15	-1.34	0.38	-134.00
	1113	1.98	59.84	-0.11	-0.49	0.13	137.31
	1213	1.92	58.32	-0.29	-0.49	0.33	153.75
	1014	-5.05	30.31	-1.92	-0.41	0.10	153.40
	1114	-5.05	4.78	-0.43	-0.43	0.10	153.40
	1214	-5.04	43.61	-0.16	-0.40	0.10	-166.39
	1021	-5.89	43.61	-0.16	-0.39	0.10	-101.66
	1121	4.86	43.60	0.00	-0.39	0.07	-19.94
	1221	-5.25	22.09	-0.14	-0.29	0.07	-55.21
	1002	-5.51	52.86	0.02	-0.28	0.07	-159.59
	1003	-0.53	28.70	-0.26	-0.35	0.06	-72.97
	1017	-0.53	38.63	-0.26	-0.39	0.10	72.75
	1018	-1.11	71.91	-0.23	-1.13	0.28	-177.85
	1019	-1.11	31.92	-0.23	-1.41	0.10	-29.80
	2013	-0.03	31.92	-0.23	-1.41	0.10	-29.80
	2014	-2.08	31.92	-0.23	-1.41	0.10	-29.80

END OF LATEST ANALYSIS RESULT

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162. PRINT FORCE ENVELOPE NSECTION 5 ALL
1) FORCE ENVELOPE NSECTION 5 0
0) STAAD SPACE

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MEMBER FORCE ENVELOPE

ALL UNITS ARE KN METE

DISTANCE
EY L

1		MAX	38.11	1031	217.78	1211	0.31	1113	-0.29	1321
0.00	MIN		1.03	1321	-213.42	1013	0.06	1321	-1.58	1113
0.64	MAX		35.24	1011	200.63	1211	0.31	1113	-0.75	1321
	MIN		3.29	1321	-228.84	1013	0.06	1321	-1.36	1113
1.28	MAX		32.39	1321	165.33	1211	0.06	1321	-1.36	1113
	MIN		3.29	1011	-172.51	1211	0.36	1113	-0.18	1321
1.92	MAX		29.51	1011	172.51	1211	0.36	1113	-0.18	1321
	MIN		20.19	1321	-258.39	1013	0.06	1321	-0.98	1113
2.56	MAX		26.64	1011	161.52	1211	0.31	1113	-0.14	1321
	MIN		-0.09	1321	-272.42	1013	0.06	1321	-0.78	1113
3.20	MAX		23.77	1011	186.05	1211	0.31	1113	-0.11	1321
	MIN		-0.37	1321	-286.05	1013	0.06	1321	-0.58	1113

MAX/MIN FORCE VALUES FOR MEMB	1, AMONGST ALL SECT LOCATIONS
1	1

	MAX.	FE	DIST	LD	MY	DIST	LD	FX	DIST	LD
2	0.00	MAX	4.38	1113	286.05	1013	-0.06	1221	0.57	1113
		MIN	-20.98	1211	-152.61	1211	-0.31	1113	0.11	1221
	0.18	MAX	4.48	1113	285.59	1013	-0.06	1221	0.52	1113
		MIN	-20.06	1211	-148.92	1211	-0.31	1113	0.10	1221
	0.36	MAX	4.58	1113	285.12	1013	-0.06	1221	0.46	1113
		MIN	-19.24	1211	-145.38	1211	-0.31	1113	0.09	1221
	0.54	MAX	4.69	1113	284.63	1013	-0.06	1221	0.41	1113
		MIN	-18.43	1211	-141.99	1211	-0.31	1113	0.08	1221
	0.72	MAX	4.79	1113	284.12	1013	-0.06	1221	0.35	1113
		MIN	-17.59	1211	-138.63	1211	-0.31	1113	0.29	1113
	0.90	MAX	4.89	1113	283.60	1013	-0.06	1221	0.26	1113
		MIN	-16.80	1211	-135.65	1211	-0.31	1113	0.06	1221
		STAND SPACE						PAGE NO.	9	

Page 5

Fuel Station-RL																				
0.00			3.20 2014			0.00			3.20 2014			41.93 T			3.20 1211					

9																				
MAX.		0.00	MAX	8.47	1013	314.74	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014			
		MIN	-3.61	1212	-175.85	1211	-175.85	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014		
0.18		MAX	8.64	1013	313.20	1013	313.20	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014		
		MIN	-1.97	1212	-175.53	1211	-175.53	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014		
0.36		MAX	8.82	1013	311.63	1013	311.63	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014		
		MIN	-1.52	1211	-175.31	1211	-175.31	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014		
0.54		MAX	9.52	1321	170.97	1211	170.97	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014		
		MIN	-1.52	1321	308.40	1013	308.40	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014		
0.72		MAX	9.17	1013	176.33	1211	176.33	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014		
		MIN	-1.53	1321	306.73	1013	306.73	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014		
0.90		MAX	9.34	1013	177.19	1211	177.19	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014		
		MIN	-1.54	1321					87.63	C	0.00	1013	52.93	T	0.90	1211				

9. AMONGST ALL SECT LOCATIONS																				
MAX/MIN FORCE VALUES FOR MEMB		FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD	FX	DIST	LD	FX	DIST	LD	FX	DIST	LD	
		FZ	DIST	LD	NY	DIST	LD													
MAX.		9.34	0.90	1013	314.74	0.00	1013													
		MIN.	-3.61	0.00	1001	0.00	0.00	1001												
		0.00	0.90	1212	-177.19	0.90	1211													

10																				
MAX.		0.00	MAX	56.17	1013	306.73	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	
		MIN	-25.75	1211	-177.19	1211	-177.19	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
0.18		MAX	54.64	1013	296.76	1013	296.76	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	-24.11	1211	-172.70	1211	-172.70	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
0.36		MAX	55.13	1013	182.97	1211	182.97	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	-21.58	1211	-179.55	1211	-179.55	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
0.54		MAX	51.58	1013	164.82	1211	164.82	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	-20.84	1211	-161.01	1211	-161.01	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
0.72		MAX	50.05	1013	268.51	1013	268.51	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	-19.20	1211	-157.71	1211	-157.71	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
0.90		MAX	48.52	1013	259.65	1013	259.65	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	-17.57	1211					--	PAGE NO. 13										
STAAD SPACE																				

10. AMONGST ALL SECT LOCATIONS																				
MAX/MIN FORCE VALUES FOR MEMB		FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD	FX	DIST	LD	FX	DIST	LD	FX	DIST	LD	
		FZ	DIST	LD	NY	DIST	LD													
MAX.		56.17	0.00	1013	306.73	0.00	1013													
		MIN.	-25.75	0.00	1001	0.00	0.00	1001												
		0.00	0.90	1214	-177.19	0.00	1211													

11																				
MAX.		0.00	MAX	73.92	1013	259.65	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	
		MIN	-42.19	1211	-157.71	1211	-157.71	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
0.18		MAX	47.97	1013	156.31	1211	156.31	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	-40.70	1211	-139.71	1211	-139.71	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
0.36		MAX	70.07	1013	143.04	1211	143.04	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	-39.26	1211	-136.10	1211	-136.10	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
0.54		MAX	68.14	1013	221.27	1013	221.27	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	-37.79	1211	-129.43	1211	-129.43	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
0.72		MAX	66.21	1013	209.17	1013	209.17	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	-36.33	1211	-129.43	1211	-129.43	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
0.90		MAX	64.28	1013	197.42	1013	197.42	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	-34.87	1211	-123.02	1211	-123.02	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014

11. AMONGST ALL SECT LOCATIONS																				
MAX/MIN FORCE VALUES FOR MEMB		FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD	FX	DIST	LD	FX	DIST	LD	FX	DIST	LD	
		FZ	DIST	LD	NY	DIST	LD													
MAX.		73.92	0.00	1013	259.65	0.00	1013													
		MIN.	-42.19	0.00	1001	-157.71	0.00	1001												
		0.00	0.90	1214	-157.71	0.00	1211													

12																				
MAX.		0.00	MAX	65.47	1013	197.42	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	
		MIN	-45.81	1211	-123.02	1211	-123.02	1211	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014

12. AMONGST ALL SECT LOCATIONS																				
MAX/MIN FORCE VALUES FOR MEMB		FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD	FX	DIST	LD	FX	DIST	LD	FX	DIST	LD	
		FZ	DIST	LD	NY	DIST	LD													
MAX.		35.82	0.00	1112	188.86	0.00	1211													
		MIN.	-11.72	3.20	1211	-314.74	3.20	1013												

8																				
MAX.		58.59	0.00	1013	191.92	0.00	1013	0.05	1113	0.00	1011	11.42	C	0.00	1011	0.00	2014	0.00	2014	
		MIN.	-25.38	0.00	1211	-84.95	0.00	1211	-0.52	0.00	1113	0.56	T	2.70	1321	0.00	1011	0.00	2014	
		0.01	2.70	1321																

6																				
MAX.		34.86	1013	65.81	1013	0.05	1113	0.01	1321	-0.07	1221									
		MIN	-35.47	1211	-28.76	1211	-0.39	1113	0.01	1321	-0.39	1113								
0.54		MAX	33.17	1013	-28.06	1211	-0.36	1211	0.01	1321	-0.36	1211								
		MIN	-35.42	1013	-31.36	1013	-0.05	1211	0.01	1321	-0.05	1211								
1.08		MAX	-11.47	1211	-13.71	1211	-0.33	1113	0.01	1321	-0.33	1113								
		MIN	-11.47	1211	-13.71	1211	-0.33	1113	0.01	1321	-0.33	1113								
1.62		MAX	-20.71	1013	-20.80	1013	0.05	1113	0.01	1321	-0.30	1113								
		MIN	-9.46	1211	-9.56	1211	0.01	1321	0.01	1321	-0.30	1113								
2.16		MAX	-16.00	1013	-10.89	1013	0.05	1113	0.01	1321	-0.28	1113								
		MIN	-7.44	1211	-5.00	1211	0.01	1321	0.01	1321	-0.28	1113								
2.70		MAX	-11.31	1013	-3.52	1013	0.05	1113	0.01	1321	-0.25	1113								
		MIN	-5.42	1211	-1.52	1211	0.01	1321	0.01	1321	-0.25	1113								
STAAD SPACE																				

6. AMONGST ALL SECT LOCATIONS																				
MAX/MIN FORCE VALUES FOR MEMB		FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD	FX	DIST	LD	FX	DIST	LD	FX	DIST	LD	
		FZ	DIST	LD	NY	DIST	LD													
MAX.		34.86	0.00	1013	65.81	0.00	1013	10.83	C	0.00	1011									
		MIN.	-15.47	0.00	1113	-29.76	0.00	1211	0.70	T	2.70	1321								
		0.01	2.70	1321																

7																				
MAX.		7.81	1013	3.51	1013	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	0.00	2014	
		MIN	-3.28	1211	-1.52	1211	0.00	2014	0.00	2014	0.00	2014	0.00	201						

Fuel Station-11									
MAX/MIN	FORCE VALUES FOR MEMB	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FX	DIST LD	FX	DIST LD
0.54	MAX	59.50	1013	163.68	1013	0.00	2014	0.00	2014
1.08	MIN	-39.50	1211	-100.52	1211	0.00	2014	0.00	2014
0.54	MAX	53.54	1013	133.16	1013	0.00	2014	0.00	2014
1.62	MIN	-33.18	1211	-80.36	1211	0.00	2014	0.00	2014
2.16	MAX	47.59	1013	105.85	1013	0.00	2014	0.00	2014
2.70	MIN	-30.89	1211	-82.73	1211	0.00	2014	0.00	2014
0.54	MAX	40.89	1013	97.09	1013	0.00	2014	0.00	2014
2.70	MIN	-26.13	1211	-47.09	1211	0.00	2014	0.00	2014
0.54	MAX	35.71	1013	60.87	1013	0.00	2014	0.00	2014
2.70	MIN	-22.20	1211	-33.87	1211	0.00	2014	0.00	2014

STAAD SPACE

12, AMONGST ALL SECT LOCATIONS									
MAX/MIN	FORCE VALUES FOR MEMB	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FX	DIST LD	FX	DIST LD
MAX.	65.47	0.00	1013	197.42	0.00	1013	0.00	1013	0.00
MIN.	-43.81	0.00	1211	-123.02	0.00	1211	0.00	1211	0.00
0.00	2.70	2014	0.00	2.70	2014	0.53	2.70	2014	0.53
13	0.00	MAX	35.71	1013	60.87	1013	0.00	2014	0.00
0.54	MIN	-22.20	1211	-33.87	1211	0.00	2014	0.00	2014
1.08	MAX	29.78	1013	43.18	1013	0.00	2014	0.00	2014
1.62	MIN	-17.86	1211	-23.06	1211	0.00	2014	0.00	2014
2.16	MAX	23.86	1013	28.70	1013	0.00	2014	0.00	2014
2.70	MIN	-13.51	1211	-14.59	1211	0.00	2014	0.00	2014
0.54	MAX	17.86	1013	18.41	1013	0.00	2014	0.00	2014
2.16	MIN	-13.51	1211	-14.59	1211	0.00	2014	0.00	2014
2.70	MAX	14.81	1013	9.39	1013	0.00	2014	0.00	2014
2.70	MIN	-8.41	1211	-4.69	1211	0.00	2014	0.00	2014
0.54	MAX	8.41	1013	4.41	1013	0.00	2014	0.00	2014
2.70	MIN	-0.45	1211	-3.27	1211	0.00	2014	0.00	2014

13, AMONGST ALL SECT LOCATIONS									
MAX/MIN	FORCE VALUES FOR MEMB	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FX	DIST LD	FX	DIST LD
MAX.	35.71	0.00	1013	60.87	0.00	1013	0.00	1013	0.00
MIN.	-22.20	0.00	1211	-33.87	0.00	1211	0.00	1211	0.00
0.00	2.70	2014	0.00	2.70	2014	0.30	2.70	2014	0.30
14	0.00	MAX	9.82	1013	4.42	1013	0.00	2014	0.00
0.18	MIN	-7.28	1211	-3.28	1211	0.00	2014	0.00	2014
0.36	MAX	5.89	1013	1.59	1013	0.00	2014	0.00	2014
0.54	MIN	-4.37	1211	-1.18	1211	0.00	2014	0.00	2014
0.72	MAX	3.93	1013	0.71	1013	0.00	2014	0.00	2014
0.90	MIN	-2.58	1211	-0.78	1211	0.00	2014	0.00	2014
0.54	MAX	1.26	1013	0.13	1013	0.00	2014	0.00	2014
2.70	MIN	-0.45	1211	-0.13	1211	0.00	2014	0.00	2014

STAAD SPACE

14, AMONGST ALL SECT LOCATIONS									
MAX/MIN	FORCE VALUES FOR MEMB	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FX	DIST LD	FX	DIST LD
MAX.	9.82	0.00	1013	4.42	0.00	1013	0.00	1001	0.00
MIN.	-7.28	0.00	1211	-3.28	0.00	1211	0.00	1211	0.00
0.00	0.90	2014	0.00	0.90	2014	0.23	0.90	2014	0.23
15	0.00	MAX	38.11	1011	217.78	1211	-0.06	1321	1.58
0.64	MIN	-1.03	1321	-213.42	1013	-0.31	1113	0.29	1321
1.28	MAX	35.24	1011	200.63	1211	-0.06	1321	1.38	1113
2.70	MIN	-0.75	1321	-228.84	1013	-0.31	1113	0.25	1321
0.64	MAX	32.38	1011	185.53	1211	-0.06	1321	1.18	1113
2.70	MIN	-0.47	1321	-243.61	1013	-0.31	1113	0.22	1321

Page 9

Fuel Station-11									
MAX/MIN	FORCE VALUES FOR MEMB	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FX	DIST LD	FX	DIST LD
1.92	MAX	29.51	1011	172.50	1211	-0.06	1321	0.98	1113
2.56	MIN	-0.19	1321	-258.33	1013	-0.31	1113	0.18	1321
3.20	MAX	26.64	1011	161.52	1211	-0.06	1321	0.78	1113
0.54	MIN	-0.09	1321	-272.42	1013	-0.31	1113	0.14	1321
0.54	MAX	23.77	1011	152.61	1211	-0.06	1321	0.58	1113
2.70	MIN	-0.37	1321	-286.05	1013	-0.31	1113	0.11	1321

15, AMONGST ALL SECT LOCATIONS

MAX/MIN	FORCE VALUES FOR MEMB	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FX	DIST LD	FX	DIST LD
MAX.	38.11	0.00	1011	217.78	0.00	1211	81.43	C	0.00
MIN.	-0.06	0.00	1321	-286.05	3.20	1013	30.33	T	3.20
0.31	3.20	1113	0.11	3.20	1221				
16	0.00	MAX	4.38	1113	286.05	1013	0.31	1113	-0.11
0.18	MIN	-20.88	1211	-152.61	1211	0.06	1221	-0.57	1113
0.36	MAX	4.48	1113	285.59	1013	0.31	1113	-0.10	1221
0.54	MIN	-20.06	1211	-148.92	1211	0.06	1221	-0.52	1113
0.72	MAX	4.58	1113	285.12	1013	0.31	1113	-0.09	1221
0.90	MIN	-19.24	1211	-145.38	1211	0.06	1221	-0.46	1113
0.54	MAX	4.69	1113	284.63	1013	0.31	1113	-0.08	1221
0.72	MIN	-18.43	1211	-141.99	1211	0.06	1221	-0.41	1113
0.90	MAX	4.79	1113	284.12	1013	0.31	1113	-0.07	1221
0.90	MIN	-17.61	1211	-138.75	1211	0.06	1221	-0.35	1113
0.90	MAX	4.89	1113	283.60	1013	0.31	1113	-0.06	1221
0.90	MIN	-16.80	1211	-135.65	1211	0.06	1221	-0.29	1113

STAAD SPACE

16, AMONGST ALL SECT LOCATIONS									
MAX/MIN	FORCE VALUES FOR MEMB	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FX	DIST LD	FX	DIST LD
MAX.	4.89	0.90	1113	286.05	0.00	1013	0.00	1013	0.00
MIN.	-20.88	0.00	1211	-152.61	0.00	1211	76.09	C	0.00
0.06	0.90	1221	-0.57	0.00	1113	28.40	T	0.90	1211
17	0.00	MAX	44.93	1013	283.60	1013	0.31	1113	-0.03
0.18	MIN	-30.03	1211	-135.65	1211	0.06	1221	-0.19	1113
0.36	MAX	43.82	1013	275.62	1013	0.31	1113	-0.01	1211
0.54	MIN	-29.25	1211	-130.32	1211	0.06	1221	-0.13	1113
0.72	MAX	42.71	1013	267.83	1013	0.31	1113	0.00	1211
0.90	MIN	-28.47	1211	-125.13	1211	0.06	1221	-0.07	1113
0.90	MAX	41.60	1013	260.25	1013	0.31	1113	0.02	1011
0.90	MIN	-27.69	1211	-120.08	1211	0.06	1221	-0.02	1113
0.90	MAX	40.49	1013	252.87	1013	0.31	1113	0.04	1211
0.90	MIN	-26.90	1211	-118.28	1013	0.31	1113	0.00	1211
0.90	MAX	28.42	1211	110.40	1211	0.06	1221	0.02	1221

17, AMONGST ALL SECT LOCATIONS									
MAX/MIN	FORCE VALUES FOR MEMB	FY/ FZ	DIST LD	MZ/ MY	DIST LD	FX	DIST LD	FX	DIST LD
MAX.	44.93	0.00	1013	283.60	0.00	1013	0.00	1013	0.00
MIN.	-30.03	0.00	1211	-135.65	0.00	1211	59.20	C	0.00
0.06	0.90	1221	-0.19	0.00	1113	14.90	T	0.90	1211
18	0.00	MAX	63.51	1013	245.68	1013	0.31	1113	0.22
0.18	MIN	-29.95	1211	-110.40	1211	0.06	1221	0.04	1221
0.36	MAX	61.99	1013	234.38	1013	0.31	1113	0.28	1113
0.54	MIN	-29.28	1211	-105.06	1211	0.06	1221	0.05	1221
0.72	MAX	60.47	1013	223.35	1013	0.31	1113	0.33	1113
0.90	MIN	-28.60	1211	-99.85	1211	0.06	1221	0.06	1221
0.90	MAX	58.94	1013	212.60	1013	0.31	1113	0.39	1113
0.90	MIN	-27.93	1211	-94.76	1211	0.06	1221	0.07	1211
0.90	MAX	57.42	1013	200.72	1013	0.31	1113	0.08	1221
0.90	MIN	-26.58	1211	-84.95	1211	0.06	1221	0.09	1221

Page 10

MAX/MIN FORCE VALUES FOR MEMB 18, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	FZ/	DIST	LD	FX	DIST	LD
MAX.	63.51	0.00	1013	245.68	0.00	1013	22.20	C	0.00 1013
	0.31	0.00	1113	0.50	0.90	1113			
MIN.	-29.95	0.00	1211	-110.40	0.00	1211	2.22	C	0.90 1211
	0.06	0.90	1221	0.04	0.00	1221			

19	0.00	MAX	58.59	1013	191.92	1013	-0.01	1321	0.52	1113
		MIN	-25.38	1211	-84.95	1211	-0.05	1113	0.10	1221
	0.54	MAX	53.83	1013	161.57	1013	-0.01	1321	0.50	1113
		MIN	-23.41	1211	-71.77	1211	-0.05	1113	0.09	1221
	1.08	MAX	49.08	1013	133.79	1013	-0.01	1321	0.47	1113
		MIN	-21.43	1211	-59.66	1211	-0.05	1113	0.09	1221
	1.62	MAX	41.45	1013	148.67	1013	-0.01	1321	0.44	1113
		MIN	-19.35	1211	-68.67	1211	-0.05	1113	0.08	1221
	2.16	MAX	39.59	1013	85.91	1013	-0.01	1321	0.41	1113
		MIN	-17.46	1211	-38.66	1211	-0.05	1113	0.07	1221
	2.70	MAX	34.86	1013	65.81	1013	-0.01	1321	0.39	1113
		MIN	-15.47	1211	-29.76	1211	-0.05	1113	0.07	1221

MAX/MIN FORCE VALUES FOR MEMB 19, AMONGST ALL SECT LOCATIONS										
	FY/	DIST	LD	FZ/	DIST	LD	FX	DIST	LD	
MAX.	58.59	0.00	1013	191.92	0.00	1013	11.42	C	0.00	1011
	-0.01	0.00	1321	0.52	0.00	1113				
	-25.38	0.00	1211	-84.95	0.00	1211				
MIN.	-0.05	2.70	1113	0.07	2.70	1221	0.56	T	2.70	1321

20	0.00	MAX	34.86	1013	65.81	1013	-0.01	1321	0.39	1113
		MIN	-15.47	1211	-29.76	1211	-0.05	1113	0.07	1221
	0.54	MAX	30.14	1013	48.26	1013	-0.01	1321	0.36	1113
		MIN	-13.47	1211	-21.95	1211	-0.05	1113	0.06	1221
	1.08	MAX	25.42	1013	33.26	1013	-0.01	1321	0.33	1113
		MIN	-11.47	1211	-15.21	1211	-0.05	1113	0.05	1221
	1.62	MAX	20.16	1013	29.80	1013	-0.01	1321	0.18	1113
		MIN	-9.76	1211	-48.67	1211	-0.05	1113	0.04	1221
	2.16	MAX	18.00	1013	10.89	1013	-0.01	1321	0.58	1113
		MIN	-7.44	1211	-5.00	1211	-0.05	1113	0.02	1221
	2.70	MAX	11.31	1013	3.52	1013	-0.01	1321	0.25	1113
		MIN	-5.42	1211	-1.52	1211	-0.05	1113	0.00	1211

MAX/MIN FORCE VALUES FOR MEMB 20, AMONGST ALL SECT LOCATIONS									
	FY/ FZ	DIST	LD	FY/ FZ	DIST	LD	FX	DIST	LD
MAX.	34.86	0.00	1013	65.81	0.00	1013	10.83	C	0.00 1011
	-0.01	0.00	1321	0.39	0.00	1113			
MIN.	-15.47	0.00	1211	-29.76	0.00	1211	0.70	T	2.70 1321
	-0.05	2.70	1113	0.00	2.70	1211			

21	0.00	MAX	7.81	1013	3.51	1013	0.00	2014	0.00	2014
		MIN	-3.38	1211	-1.52	1211	0.00	2014	0.00	2014
	0.18	MAX	6.25	1013	2.25	1013	0.00	2014	0.00	2014
		MIN	-2.71	1211	-0.97	1211	0.00	2014	0.00	2014
	0.36	MAX	4.68	1013	1.26	1013	0.00	2014	0.00	2014
		MIN	-2.92	1211	-0.53	1211	0.00	2014	0.00	2014
	0.54	MAX	3.92	1013	0.83	1013	0.00	2014	0.00	2014
		MIN	-1.35	1211	-0.24	1211	0.00	2014	0.00	2014
	0.72	MAX	1.56	1013	0.14	1013	0.00	2014	0.00	2014
		MIN	-0.68	1211	-0.06	1211	0.00	2014	0.00	2014
	0.90	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014				

MAX/MIN FORCE VALUES FOR MEMB 21, AMONGST ALL SECT LOCATIONS										
	FY/	DIST	LD	FZ/	DIST	LD	FX	DIST	LD	
MAX.	7.81	0.00	1013	3.51	0.00	1013	0.00	2014	0.00	2014
	-3.38	0.00	1211	-1.52	0.00	1211				
MIN.	-3.38	0.00	1211	-1.52	0.00	1211	0.00	2014	0.00	2014
	-0.05	2.70	1113	0.00	2.70	1211				

Fuel Station-RI									
	FY/ FZ	DIST DIST	LD LD	MZ/ MY	DIST DIST	LD LD	FX	DIST	LD
MAX.	7.81	0.00	1013	3.51	0.00	1013	0.25	C	0.00 1001
	0.00	0.00	1001	0.00	0.00	1001			
MIN.	-3.38	0.00	1211	-1.52	0.00	1211	0.00		0.90 2014
	0.00	0.90	2014	0.00	0.90	2014			

22	0.00	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	1.02	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	2.05	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	3.07	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	4.10	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	5.12	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	STAAD SPACE									

MAX/MIN FORCE VALUES FOR MEMB 22, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	FZ/	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1001	0.00	0.00	1001	0.10	C	0.00 1121
	0.00	0.00	1001	0.00	0.00	1001			
MIN.	0.00	5.12	2014	0.00	5.12	2014	3.14	T	5.12 1211
	0.00	5.12	2014	0.00	5.12	2014			

23	0.00	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	1.02	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	2.05	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	3.07	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	4.10	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	5.12	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	STAAD SPACE									

MAX/MIN FORCE VALUES FOR MEMB 23, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	FZ/	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1001	0.00	0.00	1001	0.00	C	0.00 1121
	0.00	0.00	1001	0.00	0.00	1001			
MIN.	0.00	5.12	2014	0.00	5.12	2014	0.00	T	5.12 1221
	0.00	5.12	2014	0.00	5.12	2014			

24	0.00	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	1.02	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	2.05	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	3.07	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	4.10	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	5.12	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	STAAD SPACE									

MAX/MIN FORCE VALUES FOR MEMB 24, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	FZ/	DIST	LD	FX	DIST	LD
	FZ	DIST	LD		NY	DIST			
MAX..	0.00	0.00	1001		0.00	0.00	1001		
								Page 12	

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			Fuel Station-rl		
4.00	MAX	3.76	1113	12.09	1011 0.21 1113
	MIN	-3.53	1011	-17.40	1113 0.01 1211
	STAAD SPACE				--- PAGE NO. 25

MAX/MIN FORCE VALUES FOR MEMB						34. AMONGST ALL SECT LOCATIONS					
	FY	DIST	LD	FZ	DIST LD	MZ	DIST	LD	FX	DIST	LD
MAX.	4.93	0.00	1113	12.09	4.00	1011					
	0.21	0.00	1113	0.60	4.00	1113			7.54	C	0.00 1011
MIN.	-3.53	4.00	1011	-17.40	4.00	1113					
	0.01	4.00	1211	-0.25	0.00	1113			0.55	T	4.00 1321

	0.00	0.80	1.60	2.40	3.20	4.00
MAX	3.53	2.99	3.13	2.93	2.73	2.53
MIN	-2.99	-4.00	-4.23	-4.46	-4.70	-4.93
MTN	17.49	16.30	16.76	17.53	18.87	20.00
MAX	12.09	12.90	13.11	13.33	13.67	14.00
MIN	-12.90	-13.11	-13.33	-13.67	-14.00	-14.21
MTN	-0.21	-0.21	-0.21	-0.21	-0.21	-0.21
MAX	0.60	0.43	0.26	0.09	-0.08	-0.25
MIN	-0.43	-0.60	-0.76	-0.91	-1.06	-1.21
MTN	0.03	0.03	0.03	0.03	0.03	0.03
MAX	0.26	0.26	0.26	0.26	0.26	0.26
MIN	-0.21	-0.21	-0.21	-0.21	-0.21	-0.21
MTN	0.09	0.09	0.09	0.09	0.09	0.09
MAX	0.21	0.21	0.21	0.21	0.21	0.21
MIN	-0.21	-0.21	-0.21	-0.21	-0.21	-0.21
MTN	0.12	0.12	0.12	0.12	0.12	0.12
MAX	0.12	0.12	0.12	0.12	0.12	0.12
MIN	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12
MTN	0.08	0.08	0.08	0.08	0.08	0.08
MAX	0.21	0.21	0.21	0.21	0.21	0.21
MIN	-0.21	-0.21	-0.21	-0.21	-0.21	-0.21
MTN	0.11	0.11	0.11	0.11	0.11	0.11
MAX	0.11	0.11	0.11	0.11	0.11	0.11
MIN	-0.11	-0.11	-0.11	-0.11	-0.11	-0.11
MTN	0.00	0.00	0.00	0.00	0.00	0.00
MAX	0.00	0.00	0.00	0.00	0.00	0.00
MIN	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00
MTN	0.00	0.00	0.00	0.00	0.00	0.00

	MAX/MIN FORCE VALUES FOR MEMB				35° AMONGST ALL SECT LOCATIONS				FX	DYST	LD	
	FZ	FY	LD	DYST	MZ	MY	LD	DYST				
MAX.	3.53	0.00	1011	12.09	0.00	1011			7.54	C	0.00	1011
	-0.01	0.00	1211	0.60	0.00	1113						
	-4.93	4.00	1113	-17.40	4.00	1113						
MIN.	-0.21	4.00	1113	-0.25	4.00	1113			0.55		4.00	1321

END OF FORCE ENVELOPE FROM INTERNAL STORAGE *****

163. PRINT JOINT DISPLACEMENTS ALL
PRINT DISPLACE ALL

--- PAGE NO. 26

JOINT DISPLACEMENT (CM)		RADIANS)			STRUCTURE TYPE = SPACE		
JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
1	1001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1011	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1111	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1211	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1012	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1112	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1212	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1013	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1113	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1213	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1014	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1114	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1214	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1021	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1121	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1221	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1321	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2012	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2013	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2014	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	1001	-1.7307	0.5095	-0.0316	-0.0010	-0.0026	-0.0123
	1002	-1.7307	0.5076	-0.0315	-0.0010	-0.0026	-0.0123

	Fug ¹ Station-Fl		Fug ² Station-Fl		Fug ³ Station-Fl	
1.60	MAX	0.00	0.00	0.00	0.00	0.00
	MIN	0.00	0.00	0.00	0.00	0.00
2.40	MAX	0.00	0.00	0.00	0.00	0.00
	MIN	0.00	0.00	0.00	0.00	0.00
3.20	MAX	0.00	0.00	0.00	0.00	0.00
	MIN	0.00	0.00	0.00	0.00	0.00
4.00	MAX	0.00	0.00	0.00	0.00	0.00
	MIN	0.00	0.00	0.00	0.00	0.00

MAX/MIN FORCE VALUES FOR MEMB				31, AMONGST ALL SECT LOCATIONS			
FX	FZ	LD	DIST	FX	FZ	LD	DIST
MAX.	0.00	0.00	1001	0.00	0.00	1001	0.00
	0.00	0.00	1001	0.00	0.00	1001	0.00
	0.00	4.00	2014	0.00	4.00	2014	0.00
MIN.	0.00	4.00	2014	0.00	4.00	2014	4.00

[illegible]

	MAX/MIN FORCE VALUES FOR MEMB				32. AMONGST ALL SECT LOCATIONS					
	FZ/	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	0.00	1001	0.00	0.00	1001			
	0.00	0.00	0.00	1001	0.00	0.00	1001			
	0.00	4.00	4.00	2014	0.00	4.00	2014	0.00	C	0.00 1321
MIN.	0.00	0.00	4.00	2014	0.00	4.00	2014			
	0.00	0.00	4.00	2014	0.00	4.00	2014	0.06	T	4.00 1011

	0.00	MAX	0.00	2014	0.00	2014	0.00	2014
33	0.00	MAX	0.00	2014	0.00	2014	0.00	2014
	MIN		0.00	2014	0.00	2014	0.00	2014
	0.80	MAX	0.00	2014	0.00	2014	0.00	2014
	MIN		0.00	2014	0.00	2014	0.00	2014
	1.60	MAX	0.00	2014	0.00	2014	0.00	2014
	MIN		0.00	2014	0.00	2014	0.00	2014
	2.40	MAX	0.00	2014	0.00	2014	0.00	2014
	MIN		0.00	2014	0.00	2014	0.00	2014
	3.20	MAX	0.00	2014	0.00	2014	0.00	2014
	MIN		0.00	2014	0.00	2014	0.00	2014
	4.00	MAX	0.00	2014	0.00	2014	0.00	2014
	MIN		0.00	2014	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB				33, AMONGST ALL SECT LOCATIONS						
	PZ/	FZ/	LD	DIST	LD	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	0.00	1001	0.00	0.00	1001			
	0.00	0.00	0.00	1001	0.00	0.00	1001			
	0.00	0.00	0.00	2014	0.00	4.00	2014	0.00	C	0.00 1321
MIN.	0.00	0.00	4.00	2014	0.00	4.00	2014			
	0.00	0.00	4.00	2014	0.00	4.00	2014	0.06	T	4.00 1011

	MAX	4.93	1113	0.00	1221	0.21	1113	0.00	1211
0.00	MAX	-2.73	1011	-0.04	1011	0.01	1211	-0.21	1113
0.80	MAX	-2.73	1113	3.97	1113	0.01	1211	-0.21	1113
1.60	MAX	4.46	1113	4.33	1011	0.21	1113	0.09	1113
2.40	MAX	-2.93	1011	-7.53	1113	0.01	1211	0.01	1011
3.20	MAX	4.23	1113	6.76	1011	0.21	1113	0.26	1113
4.00	MAX	-3.13	1011	-11.01	1113	0.01	1211	0.02	1211
4.80	MAX	4.00	1113	9.34	1011	0.01	1211	0.43	1113
5.60	MAX	-3.33	1011	-14.30	1113	0.01	1211	0.03	1211
6.40	MAX	-3.33	1011	-14.30	1113	0.01	1211	0.03	1211

FUEL STATION-R1									
STRUCTURE TYPE = SPACE									
JOINT DISPLACEMENT (CM) RADIANS									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN		

0	5	1001	-5.0310	-2.3486	-0.3323	-0.0035	0.0007	0.0233	
		1002	-5.0186	-2.3461	-0.3323	-0.0035	0.0007	0.0233	
		1011	0.9011	0.0436	-0.2848	-0.0026	0.0006	-0.0011	
		1111	3.8992	1.3817	-0.3323	-0.0032	0.0007	-0.0146	
		1211	4.4067	1.6182	-0.2373	-0.0021	0.0005	-0.0169	
		1112	1.2095	-0.8207	-0.2848	-0.0028	0.0006	-0.0076	
		1112	1.4369	0.3734	-0.3323	-0.0034	0.0007	-0.0043	
		1212	1.9443	0.6058	-0.2373	-0.0024	0.0005	-0.0066	
		1013	-6.1284	-2.7888	-0.9625	-0.0105	0.0020	0.0279	
		1113	-4.3019	-1.9277	-1.1229	-0.0124	0.0023	0.0169	
		1213	-3.7944	-1.6863	-1.0280	-0.0113	0.0021	0.0169	
		1014	-0.3872	-0.4759	-0.2848	-0.0027	0.0006	0.0042	
		1114	2.3962	0.7757	-0.3273	-0.0033	0.0007	-0.0084	
		1214	2.9037	1.0121	-0.2373	-0.0023	0.0005	-0.0107	
		1021	-2.3110	-1.1396	-0.2848	-0.0031	0.0006	0.0112	
		1121	-2.7682	-1.2298	-0.2848	-0.0031	0.0006	0.0123	
		1221	-2.9132	-0.4869	-0.2136	-0.0024	0.0004	0.0047	
		1321	-1.3704	-0.5771	-0.2136	-0.0024	0.0004	0.0053	
		2001	-3.3030	-1.5418	-0.2373	-0.0025	0.0005	-0.0153	
		2002	-2.2858	-1.0665	-0.2373	-0.0025	0.0005	-0.0153	
		2011	0.7509	-0.0839	-0.2373	-0.0022	0.0005	-0.0010	
		2012	-1.0079	-0.6839	-0.2373	-0.0023	0.0005	-0.0064	
		2013	-5.1070	-2.3240	-0.8021	-0.0087	0.0017	0.0232	
		2014	-0.3227	-0.3966	-0.2373	-0.0023	0.0005	0.0035	
	8	1001	-5.6838	-21.0614	0.0577	-0.0006	0.0007	0.0326	
		1002	-5.6707	-21.0396	0.0576	-0.0006	0.0007	0.0325	
		1011	0.9147	0.3350	0.0438	0.0062	0.0003	-0.0002	
		1111	4.2827	12.2645	0.0524	0.0046	0.0005	-0.0185	
		1211	4.8567	14.4050	0.0355	0.0056	0.0003	-0.0219	
		1012	-1.4366	-7.3895	0.0460	0.0027	0.0005	0.0116	
		1112	1.5394	3.2526	0.0550	0.0005	0.0006	-0.0047	
		1212	2.1134	5.3930	0.0381	0.0015	0.0004	-0.0080	
		1013	-6.9073	-25.1005	0.1703	0.0062	0.0021	0.0389	
		1113	-4.8430	-17.4103	0.1999	0.0029	0.0026	0.0271	
		1213	-4.6590	-15.2698	0.1831	0.0023	0.0023	0.0269	
		1014	-2.6590	-6.8473	0.0452	0.0042	0.0004	0.0069	
		1114	3.1848	8.9877	0.0371	0.0032	0.0003	-0.0136	
		1214	3.6291	-10.2602	0.0496	-0.0023	0.0006	0.0160	
		1021	-3.1117	-11.0637	0.0508	-0.0021	0.0006	0.0171	
		1121	-1.0502	-4.4143	0.0374	-0.0025	0.0005	0.0069	
		1221	-1.5328	-5.2177	0.0386	-0.0023	0.0005	0.0081	
		2001	-3.7318	-13.8323	0.0413	-0.0007	0.0005	0.0214	
		2002	-2.5834	-9.5917	0.0418	-0.0017	0.0005	0.0149	
		2011	0.7623	0.2792	0.0365	0.0052	0.0003	-0.0002	
		2012	-1.1972	-6.1579	0.0384	0.0023	0.0004	0.0097	
		2013	-5.7561	-20.9171	0.1419	0.0054	0.0018	0.0324	
		2014	-0.4320	-3.5903	0.0377	0.0035	0.0018	0.0058	
	0	STAND SPACE							

FUEL STATION-R1									
STRUCTURE TYPE = SPACE									
JOINT DISPLACEMENT (CM) RADIANS									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN		

0	9	1001	-5.2844	-9.6167	-0.1733	-0.0021	0.0007	0.0299	
		1002	-5.2717	-9.6059	-0.1733	-0.0021	0.0007	0.0299	
		1011	0.9096	0.2328	-0.1109	-0.0017	0.0005	-0.0049	
		1111	4.0524	5.7206	-0.1253	0.0006	0.0006	-0.0173	
		1211	4.5855	6.6909	-0.1260	0.0017	0.0004	-0.0204	
		1012	-1.2954	-3.3139	-0.1498	-0.0001	0.0005	0.0105	
		1112	1.4799	1.5769	-0.1740	-0.0015	0.0007	-0.0046	
		1212	2.0130	2.5472	-0.1248	-0.0005	0.0005	-0.0076	
		1013	-6.4306	-11.4482	-0.5007	-0.0084	0.0021	0.0356	
		1113	-4.5111	-7.9131	-0.5834	-0.0112	0.0025	0.0247	
		1213	-3.9780	-6.9428	-0.5341	-0.0101	0.0023	0.0217	
		1014	-0.4356	-1.8995	-0.1502	0.0006	0.0005	-0.0062	
		1114	2.4830	3.2270	-0.1755	-0.0006	0.0004	-0.0072	
		1214	3.0131	4.5545	-0.1485	-0.0007	0.0004	-0.0127	
		2001	-2.9017	-5.0541	-0.1480	-0.0027	0.0006	0.0157	
		2002	-0.9854	-1.9838	-0.1110	-0.0024	0.0005	0.0063	
		2011	-1.4336	-2.3835	-0.1108	-0.0023	0.0005	0.0074	
		2012	-3.4694	-6.3137	-0.1235	-0.0017	0.0005	0.0196	
		2013	-2.4011	-4.3697	-0.1235	-0.0022	0.0005	0.0136	
		2014	0.7580	0.1982	-0.1257	-0.0014	0.0004	-0.0003	
	0	STAND SPACE							

FUEL STATION-R1									
STRUCTURE TYPE = SPACE									
JOINT DISPLACEMENT (CM) RADIANS									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN		

0	3	1001	-3.0242	0.4904	-0.1229	-0.0010	-0.0017	0.0163	
		1002	-3.0150	0.4885	-0.1229	-0.0010	-0.0017	0.0164	
		1011	0.7298	-0.1610	-0.1166	-0.0009	-0.0013	-0.0020	
		1111	2.5625	-0.4659	-0.1315	-0.0010	-0.0016	-0.0011	
		1211	2.8678	-0.5153	-0.0979	-0.0008	-0.0014	-0.0132	
		1012	-2.8678	-0.5153	-0.0979	-0.0008	-0.0014	-0.0132	
		1112	1.0126	-0.1989	-0.1247	-0.0010	-0.0016	-0.0039	
		1212	1.3179	-0.2483	-0.0911	-0.0007	-0.0012	-0.0055	
		1013	-3.7194	0.6116	-0.3486	-0.0029	-0.0049	-0.0197	
		1113	-2.6282	0.4355	-0.4022	-0.0033	-0.0057	0.0138	
		1213	-2.3229	0.3862	-0.3686	-0.0030	-0.0053	0.0138	
		1014	-0.0853	-0.0197	-0.1132	-0.0009	-0.0014	-0.0020	
		1114	1.6116	-0.3011	-0.1275	-0.0010	-0.0016	-0.0069	
		1214	1.9169	-0.3504	-0.0939	-0.0007	-0.0011	-0.0020	
		1021	-1.3583	0.2129	-0.1027	-0.0008	-0.0015	-0.0077	
		1121	-1.6959	0.2817	-0.1024	-0.0008	-0.0015	-0.0088	
		1221	-0.5182	0.0766	-0.0756	-0.0006	-0.0011	-0.0031	
		1321	-0.8558	0.1453	-0.0873	-0.0007	-0.0012	-0.0043	
		2001	-1.9855	0.3219	-0.0873	-0.0007	-0.0012	-0.0037	
		2002	-1.3744	0.2226	-0.0857	-0.0007	-0.0012	-0.0074	
		2011	0.6082	-0.1342	-0.0971	-0.0008	-0.0011	-0.0016	
		2012	-0.4989	-0.0365	-0.0923	-0.0007	-0.0011	-0.0038	
		2013	-0.0995	-0.3077	-0.2305	-0.0024	-0.0041	-0.0167	
		2014	-0.0711	-0.0164	-0.0943	-0.0027	-0.0041	-0.0167	
	4	1001	-4.3712	-0.4783	-0.2835	-0.0024	0.0003	0.0202	
		1002	-4.3597	-0.4786	-0.2835	-0.0024	0.0003	0.0201	
		1011	0.8608	-0.0682	-0.2530	-0.0021	0.0004	-0.0016	
		1111	3.4783	0.1911	-0.2912	-0.0024	0.0004	-0.0132	
		1211	3.9194	0.2394	-0.2115	-0.0018	0.0003	-0.0153	
		1012	-0.9982	-0.2204	-0.2478	-0.0021	0.0003	-0.0062	
		1112	1.3094	0.0136	-0.2851	-0.0024	0.0003	-0.0041	
		1212	1.7505	0.0619	-0.2054	-0.0017	0.0002	-0.0062	
		1013	-5.5331	-0.5331	-0.8147	-0.0068	0.0007	0.0242	
		1113	-3.7555	-0.3755	-0.9465	-0.0080	0.0007	0.0168	
		1213	-3.3144	-0.1576	-0.8668	-0.0073	0.0007	0.0148	
		1014	-0.7256	-0.2876	-0.7256	-0.0021	0.0003	0.0032	
		1114	2.1352	-0.0868	-0.7256	-0.0021	0.0003	0.0032	
		1214	2.1352	-0.0868	-0.7256	-0.0021	0.0003	0.0032	
		1021	-1.9856	-0.2151	-0.2405	-0.0020	0.0003	-0.0006	
		1121	-2.4180	-0.2375	-0.2405	-0.0020	0.0003	-0.0006	
		1221	-0.7812	-0.1127	-0.1793	-0.0015	0.0002	-0.0040	
		1321	-1.2036	-0.1046	-0.1794	-0.0015	0.0002	-0.0051	
		2001	-2.8698	-0.3140	-0.2021	-0.0017	0.0002	-0.0132	
		2002	-1.9863	-0.2174	-0.2007	-0.0017	0.0002	-0.0092	
		2011	0.7174	-0.0569	-0.2108	-0.0018	0.0003	-0.0013	
		2012	-0.8318	-0.1836	-0.2065	-0.0017	0.0002	-0.0052	
		2013	-4.4496	-0.4609	-0.6790	-0.0057	0.0005	-0.0202	
		2014	-0.2296	-0.1314	-0.2083	-0.0018	0.0003	-0.0026	
	0	ST							

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE

JOINT LOAD X-TRANS Y-TRANS Z-TRANS X-ROTAN Y-ROTAN Z-ROTAN

13	1001	-3.2042	0.5260	0.0000	0.0000	0.0000	0.0169
	1002	-3.1945	0.5240	0.0000	0.0000	0.0000	0.0000
	1003	-0.5042	-0.0982	0.0000	0.0000	0.0000	-0.0021
	1004	2.3389	-0.4098	0.0000	0.0000	0.0000	-0.0114
	1005	2.6827	-0.4637	0.0000	0.0000	0.0000	-0.0133
	1006	-0.7779	0.1101	0.0000	0.0000	0.0000	0.0048
	1007	0.8431	-0.1667	0.0000	0.0000	0.0000	-0.0033
	1008	1.1869	-0.2206	0.0000	0.0000	0.0000	-0.0032
	1009	1.0848	0.0000	0.0000	0.0000	0.0000	0.0157
	1010	3.1242	0.5085	0.0000	0.0000	0.0000	0.0147
	1011	-2.7804	0.4546	0.0000	0.0000	0.0000	0.0021
	1012	1.4179	-0.0317	0.0000	0.0000	0.0000	-0.0065
	1013	1.7618	-0.3121	0.0000	0.0000	0.0000	-0.0084
	1014	-1.4782	0.2290	0.0000	0.0000	0.0000	0.0084
	1015	-1.8354	0.3053	0.0000	0.0000	0.0000	0.0095
	1016	-0.5951	0.0831	0.0000	0.0000	0.0000	0.0038
	1017	-0.9523	0.1595	0.0000	0.0000	0.0000	0.0048
	1018	-2.1101	0.3456	0.0000	0.0000	0.0000	0.0112
	1019	-1.4849	0.2402	0.0000	0.0000	0.0000	0.0048
	1020	0.4202	-0.0519	0.0000	0.0000	0.0000	-0.0048
	1021	-0.6827	0.0918	0.0000	0.0000	0.0000	0.0040
	1022	-3.4830	0.5741	0.0000	0.0000	0.0000	0.0183
	1023	-0.2376	0.0264	0.0000	0.0000	0.0000	0.0017
14	1001	-4.5932	-0.4724	0.0000	0.0000	0.0000	0.0207
	1002	-4.5834	-0.4730	0.0000	0.0000	0.0000	0.0207
	1003	0.6714	0.0226	0.0000	0.0000	0.0000	-0.0025
	1004	3.2650	0.2565	0.0000	0.0000	0.0000	-0.0138
	1005	3.7670	0.3164	0.0000	0.0000	0.0000	-0.0161
	1006	-1.1822	-0.1802	0.0000	0.0000	0.0000	0.0061
	1007	1.1024	0.0199	0.0000	0.0000	0.0000	-0.0061
	1008	1.6044	0.0797	0.0000	0.0000	0.0000	-0.0061
	1009	-5.9759	-0.6033	0.0000	0.0000	0.0000	0.0267
	1010	-3.9603	-0.4738	0.0000	0.0000	0.0000	0.0180
	1011	-0.4637	-0.4139	0.0000	0.0000	0.0000	0.0180
	1012	1.9419	-0.0954	0.0000	0.0000	0.0000	-0.0026
	1013	2.4438	0.1188	0.0000	0.0000	0.0000	-0.0101
	1014	-2.1756	-0.2725	0.0000	0.0000	0.0000	0.0105
	1015	-2.6090	-0.2509	0.0000	0.0000	0.0000	0.0115
	1016	-0.9128	-0.1455	0.0000	0.0000	0.0000	0.0048
	1017	-1.3462	-0.1239	0.0000	0.0000	0.0000	0.0059
	1018	-3.0276	-0.3140	0.0000	0.0000	0.0000	0.0137
	1019	-2.1413	-0.2318	0.0000	0.0000	0.0000	0.0098
	1020	0.5595	0.0189	0.0000	0.0000	0.0000	-0.0021
	1021	-0.9852	-0.1501	0.0000	0.0000	0.0000	0.0059
	1022	-4.9799	-0.5028	0.0000	0.0000	0.0000	0.0043
	1023	-0.3856	-0.0795	0.0000	0.0000	0.0000	0.0022
	0	STAAD SPACE					

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE

JOINT LOAD X-TRANS Y-TRANS Z-TRANS X-ROTAN Y-ROTAN Z-ROTAN

		STAAD SPACE		JOINT DISPLACEMENT (CM)		RADIAN(S)		STRUCTURE TYPE = SPACE				PAGE NO	
		LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN					
11	1001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1004	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1008	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1009	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1010	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1011	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1012	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1013	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1014	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1015	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1016	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1017	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1018	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1019	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
	1020	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
1021	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1022	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1023	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1024	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1025	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1026	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1027	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1028	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1029	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1030	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1031	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1032	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1033	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1034	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1035	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1036	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1037	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1038	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1039	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1040	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1041	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1042	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1043	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1044	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1045	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1046	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1047	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1048	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1049	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1050	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1051	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1052	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1053	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1054	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1055	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1056	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1057	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1058	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1059	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1060	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1061	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1062	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1063	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1064	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1065	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1066	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1067	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1068	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1069	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1070	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1071	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1072	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1073	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1074	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1075	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1076	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1077	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1078	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1079	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1080	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1081	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1082	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1083	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1084	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1085	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1086	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1087	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1088	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1089	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1090	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1091	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
1092	0.0000	0.00											

	Fuel] Station-R1	
1021	-2.4335	0.1482
1121	-2.9017	0.1480
1221	-1.9534	0.0036
1321	-1.3882	0.0035
2001	-3.4694	0.0055
2002	-6.3117	0.0137
2003	-4.4694	0.0017
2002	-4.3657	0.0022
2011	-0.1401	0.0005
2012	0.1982	0.0036
2013	0.1982	-0.0034
2014	-1.0785	-0.0014
2015	-2.1616	0.0001
2016	-1.1748	0.0004
2017	-1.5829	0.0004
2018	-1.5829	-0.0005
2019	-0.3630	0.1152
2020	-0.3630	-0.0005
2021	-0.3630	0.0005
2022	-0.3630	0.0005
2023	-0.3630	0.0005
2024	-0.3630	0.0005
2025	-0.3630	0.0005
2026	-0.3630	0.0005
2027	-0.3630	0.0005
2028	-0.3630	0.0005
2029	-0.3630	0.0005
2030	-0.3630	0.0005
2031	-0.3630	0.0005
2032	-0.3630	0.0005
2033	-0.3630	0.0005
2034	-0.3630	0.0005
2035	-0.3630	0.0005
2036	-0.3630	0.0005
2037	-0.3630	0.0005
2038	-0.3630	0.0005
2039	-0.3630	0.0005
2040	-0.3630	0.0005
2041	-0.3630	0.0005
2042	-0.3630	0.0005
2043	-0.3630	0.0005
2044	-0.3630	0.0005
2045	-0.3630	0.0005
2046	-0.3630	0.0005
2047	-0.3630	0.0005
2048	-0.3630	0.0005
2049	-0.3630	0.0005
2050	-0.3630	0.0005
2051	-0.3630	0.0005
2052	-0.3630	0.0005
2053	-0.3630	0.0005
2054	-0.3630	0.0005
2055	-0.3630	0.0005
2056	-0.3630	0.0005
2057	-0.3630	0.0005
2058	-0.3630	0.0005
2059	-0.3630	0.0005
2060	-0.3630	0.0005
2061	-0.3630	0.0005
2062	-0.3630	0.0005
2063	-0.3630	0.0005
2064	-0.3630	0.0005
2065	-0.3630	0.0005
2066	-0.3630	0.0005
2067	-0.3630	0.0005
2068	-0.3630	0.0005
2069	-0.3630	0.0005
2070	-0.3630	0.0005
2071	-0.3630	0.0005
2072	-0.3630	0.0005
2073	-0.3630	0.0005
2074	-0.3630	0.0005
2075	-0.3630	0.0005
2076	-0.3630	0.0005
2077	-0.3630	0.0005
2078	-0.3630	0.0005
2079	-0.3630	0.0005
2080	-0.3630	0.0005
2081	-0.3630	0.0005
2082	-0.3630	0.0005
2083	-0.3630	0.0005
2084	-0.3630	0.0005
2085	-0.3630	0.0005
2086	-0.3630	0.0005
2087	-0.3630	0.0005
2088	-0.3630	0.0005
2089	-0.3630	0.0005
2090	-0.3630	0.0005
2091	-0.3630	0.0005
2092	-0.3630	0.0005
2093	-0.3630	0.0005
2094	-0.3630	0.0005
2095	-0.3630	0.0005
2096	-0.3630	0.0005
2097	-0.3630	0.0005
2098	-0.3630	0.0005
2099	-0.3630	0.0005
2100	-0.3630	0.0005
2101	-0.3630	0.0005</

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
25	1001	-5.5816	-18.1347	0.0012	0.0006	-0.0007	0.0325
	1002	-5.5686	-18.1156	0.0012	0.0006	-0.0007	0.0325
	1011	-5.0140	0.3155	0.0054	-0.0062	-0.0003	-0.0002
	1111	4.2245	10.5996	0.0053	-0.0046	-0.0005	-0.0185
	1211	4.7880	12.4400	0.0053	-0.0056	-0.0003	-0.0218
	1012	-1.4001	-6.3431	0.0036	-0.0027	-0.0005	0.0116
	1112	1.5247	2.8312	0.0032	-0.0005	-0.0006	-0.0047
	1212	2.0882	4.6716	0.0032	-0.0004	-0.0004	-0.0088
	1013	-0.7628	-2.4672	0.0017	-0.0006	-0.0004	0.0070
	1113	-0.7582	-2.4972	0.0016	-0.0006	-0.0004	0.0070
	1213	-4.1845	-13.1368	0.0000	0.0089	-0.0021	0.0237
	1014	-0.4967	-3.6879	0.0004	0.0042	-0.0004	0.0069
	1114	2.5787	5.9289	0.0040	-0.0022	-0.0006	-0.0033
	1214	3.1422	7.7684	0.0040	-0.0032	-0.0003	-0.0135
26	1001	-2.5789	-8.8242	0.0008	0.0023	-0.0006	0.0159
	1021	-3.0580	-9.5267	0.0008	0.0021	-0.0006	0.0171
	1121	-1.0284	-3.7898	0.0004	0.0025	-0.0005	0.0069
	1221	-1.5074	-4.4923	-0.0004	0.0023	-0.0005	0.0081
	2001	-3.6646	-11.9993	0.0007	0.0007	-0.0005	0.0214
	2002	-2.3567	-8.2552	0.0004	0.0017	-0.0005	0.0144
	2011	-0.1667	-0.5859	0.0003	0.0007	-0.0004	0.0097
	2012	-1.5544	-5.7859	0.0003	-0.0023	-0.0004	0.0037
	2013	-1.5544	-5.7859	0.0003	-0.0023	-0.0004	0.0037
	2014	-0.4139	-3.0732	0.0006	0.0035	-0.0003	0.0037
	2101	-5.6838	-21.0614	0.0036	-0.0006	-0.0007	0.0326
	2102	0.5707	-21.0396	0.0577	0.0006	-0.0007	0.0325
	2111	0.9147	0.3350	0.0438	-0.0062	-0.0003	-0.0002
	2112	4.2827	12.2645	0.0242	-0.0046	-0.0005	-0.0185
2121	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
STAAD SPACE	1012	-1.4366	-7.3895	0.0460	-0.0027	-0.0005	0.0116
	1112	1.5394	3.2526	0.0550	-0.0005	-0.0006	-0.0047
	1212	2.1134	5.3930	0.0581	-0.0015	-0.0004	-0.0088
	1013	-0.8473	-2.4603	0.0017	-0.0006	-0.0004	0.0070
	1113	-0.8473	-2.4603	0.0016	-0.0006	-0.0004	0.0070
	1213	-4.7690	-15.2698	0.0000	0.0099	-0.0021	0.0237
	1014	-0.5184	-4.3083	0.0004	0.0089	-0.0021	0.0237
	1114	2.6108	4.3083	0.0452	-0.0042	-0.0004	0.0069
	1214	3.1848	8.9877	0.0540	-0.0032	-0.0006	-0.0102
	1021	-2.6291	-8.8242	0.0371	-0.0032	-0.0003	-0.0136
	1121	-3.1117	-11.0637	0.0496	0.0023	-0.0006	0.0160
	1221	-1.0502	-4.4543	0.0508	0.0021	-0.0006	0.0171
	1321	-1.5328	-5.2177	0.0374	0.0025	-0.0005	0.0069
	2001	-3.7318	-13.8323	0.0413	0.0007	-0.0005	0.0214
2002	-2.3834	-8.5917	0.0418	0.0017	-0.0005	0.0145	
2011	-0.1623	-0.5859	0.0003	0.0007	-0.0004	0.0097	
2012	-1.5561	-5.7859	0.0003	-0.0023	-0.0004	0.0037	
2013	-1.5561	-5.7859	0.0003	-0.0023	-0.0004	0.0037	
2014	-0.4320	-3.5903	0.0006	-0.0035	-0.0003	0.0038	
2101	-5.6838	-21.0614	0.0036	-0.0006	-0.0007	0.0326	
2102	0.5707	-21.0396	0.0577	0.0006	-0.0007	0.0325	
2111	0.9147	0.3350	0.0438	-0.0062	-0.0003	-0.0002	
2112	4.2827	12.2645	0.0242	-0.0046	-0.0005	-0.0185	
2121	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2122	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2123	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2124	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2125	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2126	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2127	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2128	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2129	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2130	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2131	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2132	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2133	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2134	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2135	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2136	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2137	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2138	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2139	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2140	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2141	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2142	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2143	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2144	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2145	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2146	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2147	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2148	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2149	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2150	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2151	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2152	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2153	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2154	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2155	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2156	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2157	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2158	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2159	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2160	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2161	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2162	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2163	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2164	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2165	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2166	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2167	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2168	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2169	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2170	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
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2172	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2173	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2174	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2175	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2176	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2177	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2178	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2179	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2180	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2181	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2182	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2183	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2184	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2185	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2186	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2187	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2188	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2189	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2190	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2191	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2192	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
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2194	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2195	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2196	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2197	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2198	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2199	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	
2200	4.8567	14.4050	0.0355	-0.0056	-0.0003	-0.0219	

END OF LATEST ANALYSIS RESULT

```

164. PARAMETER 1
165. CODE BS5950
166. PY 345000 MEMB 1 TO 21 30 TO 35
167. PY 900000 MEMB 22 TO 29
168. SBLT 1 ALL
169. CHECK CODE ALL
STEEL DESIGN 0

```

STAAD.Pro CODE CHECKING - (BSI)
Page 24

STAAD SPACE	Fuel	Station-#1	0.0020
1014	-0.0853	-0.0197	0.1132
1114	1.6116	-0.3504	-0.0016
1214	1.9169	-0.3041	-0.0096
1021	-1.3583	0.0939	-0.0011
1121	-1.6959	0.2817	0.0015
1221	-0.5182	0.0766	0.0088
1011	-0.8526	0.1453	0.0031
1111	-1.3945	0.0852	0.0043
1211	-1.3925	0.2226	0.0174
1001	-0.6082	0.0971	-0.0016
2011	-0.6082	0.0971	-0.0016
2012	-0.4989	0.0565	0.0011
2013	-0.4989	0.0565	0.0011
2014	-0.0711	0.0164	0.0041
2101	-4.3712	-0.4783	0.0011
2102	-4.3197	-0.4786	0.0024
2103	-0.8608	-0.0682	0.0201
2104	3.4783	0.1911	-0.0004
2111	3.9194	0.2394	-0.0004
2112	-0.9082	-0.2294	-0.0018
2113	1.7204	0.0139	-0.0003
2114	-1.7204	0.0139	-0.0003
2105	-1.3192	-0.5531	-0.0017
2115	3.7555	-0.3746	0.0068
2116	-3.7555	-0.3746	0.0068
2117	-0.2756	-0.1576	0.0080
2118	0.2756	-0.1576	0.0080
2119	-0.3144	-0.3263	-0.0007
2120	0.3144	-0.3263	-0.0007
2121	2.1525	0.0868	0.0071
2122	2.5936	0.1331	0.0021
2123	-1.9956	-0.2435	-0.0003
2124	-2.4180	-0.2375	-0.0003
2125	-0.7812	-0.1127	-0.0002
2126	-1.2036	-0.0046	0.0020
2127	1.2036	-0.0046	0.0020
2128	-1.9863	-0.2174	-0.0001
2129	-1.9863	-0.2174	-0.0001
2130	0.7174	-0.0569	0.0017
2131	-0.7174	-0.0569	0.0017
2132	-4.4318	-0.1836	0.0018
2133	-4.4318	-0.1836	0.0018
2134	-0.8995	-0.4699	0.0057
2135	-0.8995	-0.4699	0.0057
2136	-0.2296	-0.1314	0.0002
2137	-0.2296	-0.1314	0.0002
2138	-0.0018	0.0018	-0.0003
2139	-0.0018	0.0018	-0.0003
2140	-0.0018	0.0018	-0.0003
2141	-0.0018	0.0018	-0.0003
2142	-0.0018	0.0018	-0.0003
2143	-0.0018	0.0018	-0.0003
2144	-0.0018	0.0018	-0.0003
2145	-0.0018	0.0018	-0.0003
2146	-0.0018	0.0018	-0.0003
2147	-0.0018	0.0018	-0.0003
2148	-0.0018	0.0018	-0.0003
2149	-0.0018	0.0018	-0.0003
2150	-0.0018	0.0018	-0.0003
2151	-0.0018	0.0018	-0.0003
2152	-0.0018	0.0018	-0.0003
2153	-0.0018	0.0018	-0.0003
2154	-0.0018	0.0018	-0.0003
2155	-0.0018	0.0018	-0.0003
2156	-0.0018	0.0018	-0.0003
2157	-0.0018	0.0018	-0.0003
2158	-0.0018	0.0018	-0.0003
2159	-0.0018	0.0018	-0.0003
2160	-0.0018	0.0018	-0.0003
2161	-0.0018	0.0018	-0.0003
2162	-0.0018	0.0018	-0.0003
2163	-0.0018	0.0018	-0.0003
2164	-0.0018	0.0018	-0.0003
2165	-0.0018	0.0018	-0.0003
2166	-0.0018	0.0018	-0.0003
2167	-0.0018	0.0018	-0.0003
2168	-0.0018	0.0018	-0.0003
2169	-0.0018	0.0018	-0.0003
2170	-0.0018	0.0018	-0.0003
2171	-0.0018	0.0018	-0.0003
2172	-0.0018	0.0018	-0.0003
2173	-0.0018	0.0018	-0.0003
2174	-0.0018	0.0018	-0.0003
2175	-0.0018	0.0018	-0.0003
2176	-0.0018	0.0018	-0.0003
2177	-0.0018	0.0018	-0.000

0	ZU14	-0.2296	-0.1314	0.2083	0.0018	-0.0003	0.0026
0	STAAD SPACE					---	PAGE NO. 36

	JOINT DISPLACEMENT (CM)	RADIANS	STRUCTURE TYPE = SPACE
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2	0.0000	0.0000	0.0000
3	0.0000	0.0000	0.0000
4	0.0000	0.0000	0.0000
5	0.0000	0.0000	0.0000
6	0.0000	0.0000	0.0000
7	0.0000	0.0000	0.0000
8	0.0000	0.0000	0.0000
9	0.0000	0.0000	0.0000
10	0.0000	0.0000	0.0000
11	0.0000	0.0000	0.0000
12	0.0000	0.0000	0.0000
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70	0.0000	0.0000	0.0000
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74	0.0000	0.0000	0.0000
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84	0.0000	0.0000	0.0000
85	0.0000	0.0000	0.0000
86	0.0000	0.0000	0.0000
87	0.0000	0.0000	0.0000
88	0.0000	0.0000	0.0000

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
23	1001	-5.5310	-2.3486	0.3323	0.0035	-0.0007	0.0233
	1002	-5.0186	-2.3461	0.3323	0.0035	-0.0007	0.0233
	1011	0.3011	0.0436	0.2848	0.0026	-0.0006	-0.0011
	1111	3.8992	1.3817	0.3323	0.0032	-0.0007	-0.0146
	1211	4.4067	1.6182	0.2373	0.0021	-0.0005	-0.0169
	1012	-1.2095	-0.8207	0.2848	0.0007	-0.0006	-0.0076
	1112	1.4369	0.3734	0.3323	0.0034	-0.0007	-0.0043
	1212	1.9443	0.6998	0.2373	0.0024	-0.0005	-0.0066
	1013	-6.1284	-2.7889	0.3625	0.0105	-0.0020	0.0279
	1113	-4.7419	-2.1862	0.3625	0.0105	-0.0020	0.0279
	1213	-4.7419	-2.1862	1.3758	0.0117	-0.0021	0.0165
	1014	-0.3877	-0.4959	0.2848	0.0023	-0.0006	0.0042
	1114	2.3962	0.7757	0.3323	0.0033	-0.0007	-0.0084
	1214	2.9037	1.0121	0.2373	0.0023	-0.0005	-0.0107
24	1001	-2.3110	-1.1296	0.2848	0.0031	-0.0006	0.0122
	1101	-2.7682	-1.2798	0.2848	0.0031	-0.0006	0.0122
	1201	-0.9132	-0.4869	0.2136	0.0024	-0.0004	0.0047
	1321	-1.3704	-0.5771	0.2136	0.0024	-0.0004	0.0059
	2001	-3.3030	-1.5418	0.2373	0.0025	-0.0005	0.0153
	2002	-2.2858	-1.0665	0.2373	0.0026	-0.0005	0.0106
	2011	-0.7509	0.0363	0.3373	0.0027	-0.0005	-0.0010
	2012	-1.0707	0.2849	0.3373	0.0027	-0.0005	-0.0013
	2013	-1.0707	0.2849	0.3373	0.0027	-0.0005	-0.0013
	2014	-0.3227	-0.3666	0.2373	0.0021	-0.0005	0.0035
	2015	-5.2844	-0.6167	0.1733	0.0021	-0.0007	0.0239
	1002	-5.2717	0.6059	0.1733	0.0021	-0.0007	0.0239
	1011	0.8096	0.2378	0.1509	-0.0017	-0.0005	-0.0004
	1111	4.5855	5.7206	0.1263	-0.0017	-0.0006	-0.0173
	1211	-1.2954	6.6809	0.3373	-0.0017	-0.0004	-0.0204
25	1012	1.4799	1.5769	0.1498	0.0015	-0.0005	-0.0105
	1112	1.4799	1.5769	0.1740	0.0015	-0.0007	-0.0046
	1212	2.0130	2.3472	0.1248	0.0005	-0.0005	-0.0078
	1013	-6.4306	-2.9482	0.3807	0.0094	-0.0021	0.0349
	1113	-3.9788	-2.9482	0.3807	0.0094	-0.0021	0.0349
	1213	-3.9788	-6.9428	1.5341	0.0101	-0.0023	0.0117
	1014	-0.4356	-1.8995	0.1502	-0.0006	-0.0003	0.0062
	1114	-2.4830	3.2270	0.1745	-0.0006	-0.0006	-0.0097
	1214	3.1611	4.1973	0.1252	-0.0004	-0.0004	-0.0127

Page 23

Page 26

2 FOUNDATION DESIGN

Document No.: **DM001-E-MAV-CDP-DR-DCC-371953**

Rev A1

Date: 29 May 2007

Page

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

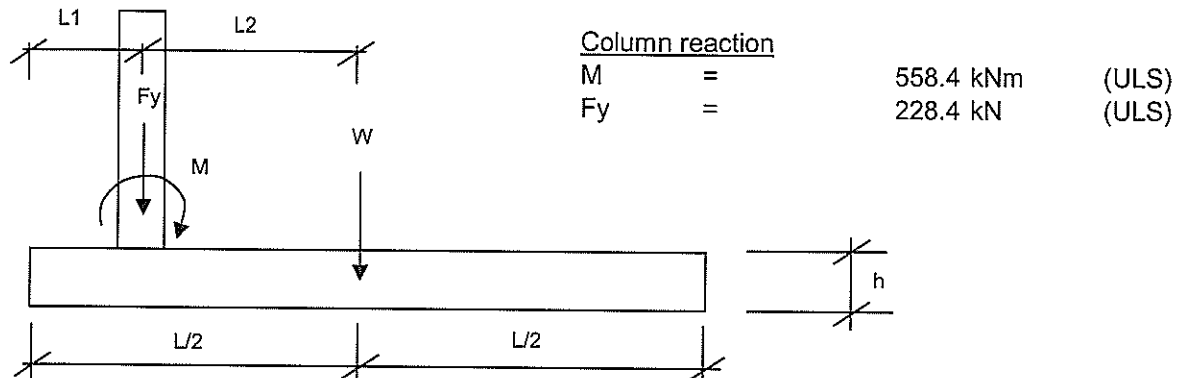


Site Dubai Depot
Fuel Station (Rashidiya)

Date May '07 Job No. 068010

Designed by Rudy Sheet No.

Footing Design



Assume footing dimension

L	=	2.5 m	Depth of soil above the footing, Hs	=	1.0 m
B	=	9.0 m	Density of the backfill soil, γ_{soil}	=	20.0 kN/m ³
h	=	0.5 m	W (soil)	=	450.0 kN
L1	=	0.4 m			
L2	=	0.9 m	Safe bearing capacity	=	150.0 kN/m ²
W (selfwt)	=	243.0 kN			

Equivalent bending moment at the center of the footing, M_c

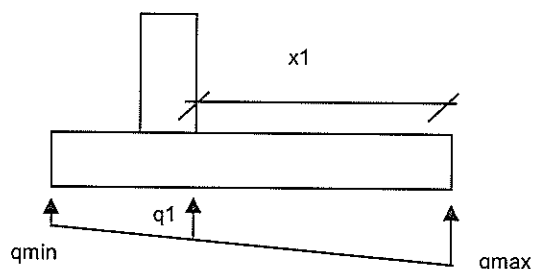
$$= (M - F_y \cdot L_2) / \gamma_L = 1.4$$

$$= 256.1 \text{ kNm}$$

Total vertical load at the center of the footing, F_t

$$= F_y / \gamma_L + W(\text{soil}) + W(\text{selfwt})$$

$$= 856.1 \text{ kN}$$



Eccentricity due to the resultant force, e

$$= M_c / F_t$$

$$= 0.299 \text{ m} \geq L/6 = 0.417 \text{ m}$$

The whole base is under compression

Effective stress length, L'	=	2.5 m	
Base pressure, q_{max}	=	13.65 kN/m ²	Ok
q_{min}	=	9.11 kN/m ²	

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai Depot Fuel Station (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Design of footing reinforcement

Distance between the footing edge to the column edge, x_1	=	2.125 m
Stress at the face of the column, q_1	=	9.79 kN/m ²
cover to the main reinforcement	=	83 mm
effective depth of the footing, d	=	367 mm
d'	=	83 mm
f_{cu}	=	40 N/mm ²
f_y	=	460 N/mm ²

Design bending reinforcement

Maximum ULS moment at the face of column, M_1		
=	$B*[0.5*q_1*(x_1^2/3)+0.5*q_{max}*(2*x_1^2/3)]*\gamma_L$	= 351.8 kNm

K'	=	0.156
$K = M_1/f_{cu}Bd^2$	=	0.00726
z	=	242.3 mm
x	=	277.1 mm

A_s' req	=	0.0 mm ²
A_s req	=	3628.0 mm ²
A_s req (min)	=	585.0 mm ² / m

Provide	60	T12		
A_s provided	=	6785.8 mm ² (spacing =	149.7 mm)	Ok
	=	755.35 mm ² / m		

Design shear reinforcement

Maximum ULS shear at the face of column, V_1		
=	$B*0.5*[q_1+q_{max}]*x_1*\gamma_L$	= 313.9 kN

V_1/bd	=	0.10 N/mm ²	$100A_s/Bd$	=	0.205445
			$400/d$	=	1.089918

v_c	=	0.41 N/mm ²		
$v_c + 0.4$	=	0.81 N/mm ²		
$0.8f_{cu}^{0.5}$	=	5.0 N/mm ²	<=	5 N/mm ²

A_{sv}/sv req	=	0	mm ² / mm	
Provide	T	@300 c/c	(2 legs)	
A_{sv}/sv prov	=	0	mm ² / mm	

3 SECONDARY STEEL BEAM & CONNECTION DESIGN

Document No.: **DM001-E-MAV-CDP-DR-DCC-371953**

Rev A1

Date: 29 May 2007

Page

Project: Dubai Depot - Fuel Station
Subject: Design of Secondary beams
Description:

Job No.: 068010.021

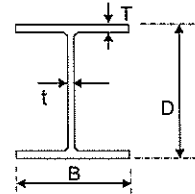
Designed by: *Rudra* Sheet No.**Secondary Beam Design**

load, $w = (1.4 \times 0.8 + 1.6 \times 0.6) \times 4 = 8.32 \text{ kN/m}$
 span length, $L = 4 \text{ m}$
 Max moment, $M_z = 0.125wL^2 = 17 \text{ kNm}$
 Max shear, $F_y = 0.5wL = 17 \text{ kN}$

Design of Steel Beam with no or little axial force --- Predominantly Flexural Member**1. Design Forces**

ULS Bending Moment $M_{max} = 16.6 \text{ kN-m}$
 ULS Shear Force $V_{max} = 16.6 \text{ kN}$

Axial Force $P_{max} = 0 \text{ kN}$
 Load factor, $\gamma_f = 1.5$

**2. Individual Member Properties**

Section size	200x100	21 kg/m	$E = 2.05 \times 10^5 \text{ N/mm}^2$
Beam depth, D	200	mm	Depth between fillets, $d = 184 \text{ mm}$
Beam width, B	100	mm	* Plastic modulus, $S_x = 2.69 \times 10^6 \text{ mm}^3$
Flange thickness, T	8	mm	
Web thickness, t	6	mm	[*calculated based on contribution from flanges & web]
Sectional area, A_g	7.36×10^3	mm^2	Radius of gyration, $r_x = 191.2 \text{ mm}$
Moment of inertia, I_x	2.69×10^8	mm^4	Radius of gyration, $r_y = 46.649 \text{ mm}$
Moment of inertia, I_y	1.60×10^7	mm^4	$A_{eff} = 7356 \text{ mm}^2$
Section Modulus, Z_x	1.20×10^6	mm^3	$Z_{x,eff} = 1.48 \times 10^5 \text{ mm}^3$
Section Modulus, Z_y	1.60×10^5	mm^3	$Z_{y,eff} = 2.67 \times 10^4 \text{ mm}^3$

3. Material Properties

Steel grade = 50
 Design strength, $p_y = 355 \text{ N/mm}^2$
 Elastic modulus = 205 kN/mm^2

4. Section Classification

Epsilon $= [275/p_y]^{0.5} = 0.880$
 Outstand element of compression flange, b/T
 Plastic - 7.9
 Compact - 8.8
 Semi-compact - 13.2
 Actual $b/T = 5.875$ **Flange is plastic.**
 Section classified as **Plastic**
 Web subject to compression throughout, d/t
 Plastic - 70.4 $r_1 = 0.00$
 Compact - 88.0
 Semi-compact - 105.62 $r_2 = 0.00$
 Actual $d/t = 30.67$ **Web is plastic.**

5. Design Capacity Check**5.1 Local/Section Capacity Check**

Shear capacity of beam, $P_v = 0.6p_yA_v = 256 \text{ kN}$ **> Design Value, OK**
 Since d/t is not more than 63 $\epsilon \rightarrow$ Shear buckling resistance check is not required *Clause 4.2.3*
 Moment Capacity of beam, $\min(P_y S_x, 1.2P_y Z_x)$ (about major axis) *Clause 4.2.5*
 M_{cx} beam section = 509.2 kN-m **> Design Value, OK**

5.2 Lateral Torsional Buckling Check*Clause 4.3*

$(A_{eff}/A_g)^{0.5} = 1.00$
 Effective length, $L_{xx} = 4 \text{ m}$
 Slenderness Ratio: $\lambda = 20.9$
 Design Lambda, $\lambda = 85.7$
 Effective length, $L_{yy} = 4.00 \text{ m}$
 Slenderness Ratio: $\lambda_y = 85.75$

Equivalent Slenderness Ratio, $\lambda_{LT} = u \lambda \sqrt{\beta_w} = 68.7$
 $\beta_w = 1.00$
 Buckling parameter, $u = 0.9$
 Slenderness factor, $v = 0.89$ } For Equal Flanges
 Torsional index, $x = 25.0$
 $\lambda/x = 3.4$
 Slenderness factor $v = 1 / ((1 + 0.05(\lambda/x)^2)^{0.25})$

Bending strength, $p_b = \frac{P_E P_r}{\phi_{LT} \sqrt{(\phi_{LT}^2 - P_E P_r)}} = 225.9 \text{ N/mm}^2$
 Euler strength P_E
 $P_E = (\pi^2 E / \lambda_{LT}^2) = 428.1$
 $\phi_{LT} = \frac{p_y + (\eta_{LT} + 1) p_E}{2} = 449$
 Limiting Slenderness Ratio $\lambda_{L0} = 30.20$
 $\alpha_{LT} = 7.00$
 $\eta_{LT} = \alpha_{LT} (\lambda_{LT} - \lambda_{L0}) / 1000 = 0.27$
 $\lambda_{L0} = 0.4(\pi^2 E / p_y)$
 It shall be ≥ 0.0
 $= 0.27$

Buckling Resis. Momt. M_b = 324.0 kNm **> Design Value, OK**
 Ratio of ULS BM / Capacity = 0.05

Use Simpli Supported Beam to calculate UDL & Deflection for the applied bending moment

Equivalent SLS UDL (SS) = 5.5 kN/m
 Span/deflection Ratio = 11926.9 **<250 OK**
 Deflection at Mid-Span (SS) $\delta = \frac{5wL^4}{384EI_{xx}} = 0.3 \text{ mm}$

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site Dubai Depot
Fuel Station (Rashidiya)

Date May '07

Job No. 068010

Designed by Rudy

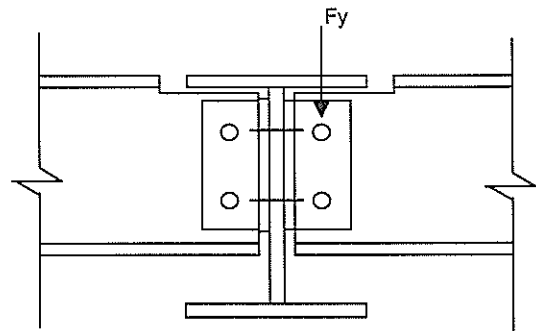
Sheet No.

Secondary Beam Connection Design

From the analysis for the perimeter beam

$F_y = 17 \text{ kN}$

Eccentricity of load, $e = 60 \text{ mm}$



- Assume shear due to bending is resisted by bolt on the flange
- Assume vertical and horizontal shear is resisted by bolt on the web

Check shear on the web

- for the secondary beam web

No of bolt provided	=	2
Shear per bolt	=	8.5 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, V_b	=	58.9 kN

- for the main beam web

No of bolt provided	=	2
Distance between bolt	=	50 mm
Tensile load per bolt	=	6.6 kN
Bolt diameter, ϕ	=	16 mm
Tensile capacity, V_b	=	70.7 kN

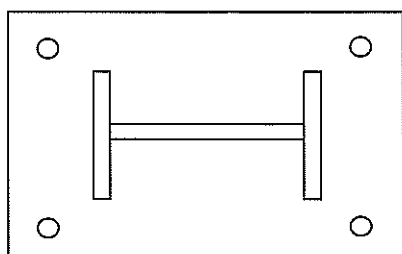
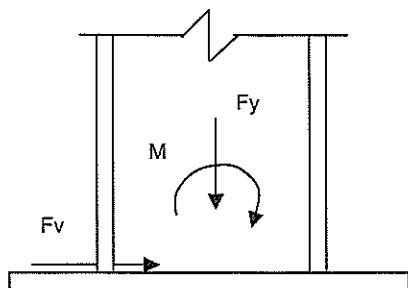
Calculation/Sketch

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Site	Dubai Depot Fuel Station (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Base Plate Design (Case 1 - Max Moment with corresponding axial load)



Applied Load

F_y	=	103.4 kN
F_v	=	4.31 kN
M	=	243.08 kNm

Column section

Depth, D	=	400 mm
Width, B	=	250 mm
Flange thickness, t_f	=	12 mm
Web thickness, t_w	=	8 mm

Base plate

Length, L_p	=	600 mm
Width, B_p	=	450 mm
Thickness, t_p	=	20 mm
$p_y p$	=	345 N/mm ²
f_{cu}	=	30 N/mm ²
$0.4 f_{cu}$	=	12 N/mm ²

Holding down bolt design

Distance between the outer bolts, db	=	500 mm
No of bolt provided at the outer line	=	3
Total no of bolt	=	6
Maximum tension per bolt due to bending and axial load, T_1	=	265.5 kN
Shear per bolt, V_1	=	0.7 kN

Using grade 8.8 bolt,

Bolt size	=	27 mm
Tension capacity, T_b	=	207 kN
Shear capacity, V_b	=	172 kN

$$T_1 / T_b + V_1 / V_b = 1.29 \leq 1.4, \text{ Ok}$$

Column to base plate weld design

Assuming moment is resisted by weld at the column flange

Force on the weld, T_2	=	$M / (D_c - t_f) - T_f$
	=	523.1 kN

Effective length of the weld resisting T_2	=	$2 \cdot B - t_w$
	=	492 mm

Weld size, s =	8 mm	(up to 15mm thick)
Weld strength, p_w =	1.4 kN/mm	(based on longitudinal capacity)

Total weld resistance	=	688.8 kN	Ok
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Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai Depot Fuel Station (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Assuming shear is resisted by weld at the web

Effective length of the weld resisting F_v = 340 mm

Weld size, s = 6 mm (up to 15mm thick)

Weld strength, p_w = 1.05 kN/mm (based on longitudinal capacity)

Total weld resistance = 357 kN Ok

Design plate thickness

Distance between the outer bolt to the column edge, x_1 = 50 mm

Maximum moment at the face of column, M_1
= force on the bolt * distance = 39.8 kNm

Minimum thickness of plate required, $t_{p \min}$ = $[6 * M_1 / 1.2 p_{yp}]^{0.5}$
= 24.0 mm > than provided

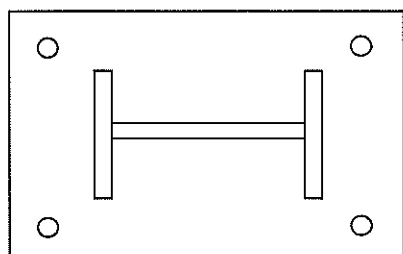
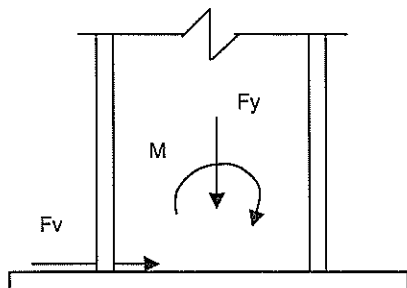
Calculation/Sketch

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Site	Dubai Depot Fuel Station (Rashidiya)	Date	May '07	Job No.	068010
Designed by Rudy				Sheet No.	

Base Plate Design (Case 1 - Min axial load with corresponding moment)



Applied Load

Fy	=	-26.12 kN
Fv	=	28.72 kN
M	=	188.86 kNm

Column section

Depth, D	=	400 mm
Width, B	=	250 mm
Flange thickness, tf	=	12 mm
Web thickness, tw	=	8 mm

Base plate

Length, Lp	=	600 mm
Width, Bp	=	450 mm
Thickness, tp	=	20 mm
pyp	=	345 N/mm ²
fcu	=	30 N/mm ²
0.4 fcu	=	12 N/mm ²

Holding down bolt design

Distance between the outer bolts, db	=	500 mm
No of bolt provided at the outer line	=	3
Total no of bolt	=	6
Maximum tension per bolt due to bending and axial load, T1	=	152.0 kN
Shear per bolt, V1	=	4.8 kN

Using grade 8.8 bolt,

Bolt size	=	27 mm
Tension capacity, Tb	=	207 kN
Shear capacity, Vb	=	172 kN

T1 / Tb + V1 / Vb	=	0.76 <= 1.4, Ok
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Column to base plate weld design

Assuming moment is resisted by weld at the column flange

Force on the weld, T2	=	M / (Dc - tf) - Tf
	=	512.9 kN

Effective length of the weld resisting T2	=	2*B-tw
	=	492 mm

Weld size, s =	8 mm	(up to 15mm thick)
Weld strength, pw =	1.4 kN/mm	(based on longitudinal capacity)

Total weld resistance	=	688.8 kN	Ok
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Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai Depot Fuel Station (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Assuming shear is resisted by weld at the web

Effective length of the weld resisting F_v = 340 mm

Weld size, s = 6 mm (up to 15mm thick)

Weld strength, p_w = 1.05 kN/mm (based on longitudinal capacity)

Total weld resistance = 357 kN Ok

Design plate thickness

Distance between the outer bolt to the column edge, x_1 = 50 mm

Maximum moment at the face of column, M_1
= force on the bolt * distance = 22.8 kNm

Minimum thickness of plate required, $t_{p \min}$ = $[6 * M_1 / 1.2 p_y]^{0.5}$
= 18.2 mm Ok

4 UNDERGROUND FUEL TANK DESIGN

Document No.: **DM001-E-MAV-CDP-DR-DCC-371953**

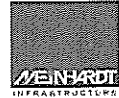
Rev A1

Date: 29 May 2007

Page

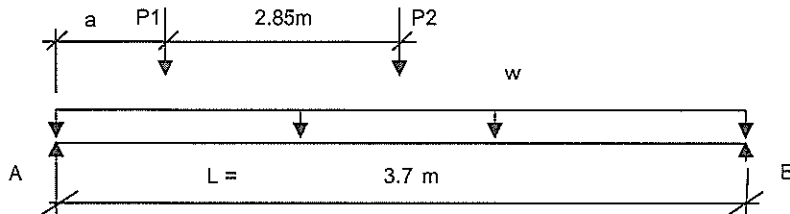
Calculation/Sketch

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Site	Dubai Depot Fuel Station - Fuel Tank (Rashidiya)	Date	May '07	Job No.	068010
			Designed by	Rudy	Sheet No.

Top Slab Design



w , slab self weight = 24×0.3 = 7.2 kN/m²
 $P1 = P2$ = medium fire vehicle load = 50 kN

Case 1 - One wheel of the vehicle at the centre of the span

$a = 1.85$ m & $P2 = 0$ kN

$M_{span} = 1.4 \times w \times L^2 / 8 + (1.6 \times P / L) \times [(a) \times (L - a)]$
 $= 91.2$ kNm

$V_{max} = 1.4 \times w \times L / 2 + 1.6 \times P / 2$
 $= 64.9$ kN

Case 1 - One wheel of the vehicle at the end of the span

$a = 0$ m & $P1 = P2 = 50$ kN

$M_{span} = 1.4 \times w \times L^2 / 8 + (1.6 \times P / L) \times [(a) \times (L - a) + (a + 2.85) \times (L - a - 2.85)]$
 $= 69.6$ kNm

$V_{max} = 1.4 \times w \times L / 2 + 1.6 \times P \times [1 + a / L]$
 $= 160.3$ kN

f_{cu}	=	40 N/mm ²
f_y	=	460 N/mm ²
slab thickness	=	300 mm
cover to the main reinforcement	=	58 mm
effective depth of the footing, d	=	242 mm
d'	=	58 mm

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
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Site	Dubai Depot Fuel Station - Fuel Tank (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Design bending reinforcement

For the support

$$\begin{aligned}K' &= 0.156 \\K = M/fcubd^2 &= 0.02213 \\z &= 143.6 \text{ mm} \\x &= 218.8 \text{ mm} \\As' \text{ req} &= 0.0 \text{ mm}^2 \\As \text{ req} &= 902.5 \text{ mm}^2 \\Provide &T16 - 150 \text{ c/c} \\As \text{ provided} &= 1340.4 \text{ mm}^2 \text{ Ok}\end{aligned}$$

For the span

$$\begin{aligned}K' &= 0.156 \\K = M/fcubd^2 &= 0.03895 \\z &= 131.2 \text{ mm} \\x &= 246.3 \text{ mm} \\As' \text{ req} &= 0.0 \text{ mm}^2 \\As \text{ req} &= 1738.5 \text{ mm}^2 \\Provide &T20 - 150 \text{ c/c} \\As \text{ provided} &= 2094.4 \text{ mm}^2 \text{ Ok}\end{aligned}$$

Design shear reinforcement

$$\begin{aligned}V_{max}/bd &= 0.66 \text{ N/mm}^2 & 100As/Bd &= 0.55389 \\& & 400/d &= 1.652893 \\& & f_{cu}/30 &= 1.333333 \\vc &= 0.69 \text{ N/mm}^2 \\vc + 0.4 &= 1.09 \text{ N/mm}^2 \\0.8f_{cu}^{0.5} &= 5.0 \text{ N/mm}^2 \leq 5 \text{ N/mm}^2 \\Asv/sv \text{ req} &= 0 \text{ mm}^2 / \text{mm} \\Provide &T @200 \text{ c/c} \quad (2 \text{ legs}) \\Asv/sv \text{ prov} &= 0 \text{ mm}^2 / \text{mm}\end{aligned}$$

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd

Site Dubai Depot

Date: Apr '07

Job No. 068010

Fuel Station - Fuel Tank (Rashidiya)

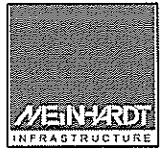
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Pudiy

Checked by

Ravi

Sheet No.



Bottom Slab Design

1. Material Properties

Concrete grade	f_{cu}	=	40	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Thickness of slab	h	=	300	mm
Width of slab	b	=	1000	mm
Short span	L_x	=	3740	mm
Long span	L_y	=	5400	mm
Concrete Ten. cover	c_1	=	75	mm
Bar dia. in short direc.	D_x	=	12	mm
Bar dia. in long direc.	D_y	=	12	mm
Effect.depth in short	d_x	=	219.0	mm
Effect.depth in long	d_y	=	213.0	mm
Concrete comp. cover	c_2	=	75	mm

3. Loadings

Self-weight of slab		=	7.20	KN/m ²
Floor finishes		=	0.00	KN/m ²
Partition		=	0.00	KN/m ²
M & E services		=	0.00	KN/m ²
Dead load		=	7.20	KN/m ²
Live load (Uplift)		=	53.40	KN/m ²
Design load (1.4Uplift - DL)		=	67.56	KN/m ²

4. Analysis & Reinforcement Design

Short Span

Neg. m'nt at suppt.	M_3	=	48.28	KNm
Pos. m'nt at mid-span	M_x	=	36.21	KNm
Neg. m'nt at suppt.	M_4	=	48.28	KNm

Steel req'd

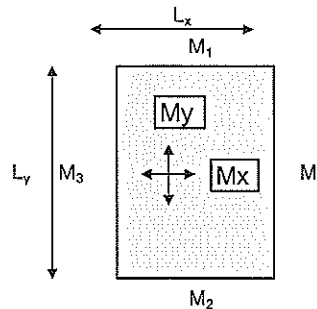
Neg. m'nt at suppt.	A_{3req}	=	580	mm ²
Pos. m'nt at mid-span	A_{xreq}	=	435	mm ²
Neg. m'nt at suppt.	A_{4req}	=	580	mm ²

Long Span

Neg. m'nt at suppt.	M_1	=	30.24	KNm
Pos. m'nt at mid-span	M_y	=	22.68	KNm
Neg. m'nt at suppt.	M_2	=	30.24	KNm

Steel req'd

Neg. m'nt at suppt.	A_{1req}	=	373	mm ²
Pos. m'nt at mid-span	A_{yreq}	=	280	mm ²
Neg. m'nt at suppt.	A_{2req}	=	373	mm ²



Type of Slab Panel :

1 - Interior panel

Span ratio	L_y/L_x	=	1.444	Two way slab design
No. of discont. edges		=	0.000	
Beta-y	β_y	=	0.024	
Gamma	γ	=	0.358	
End Conditions :			4/3 for continuous	
			0 for discontinuous	

Beta-1 (short)	β_1	=	1.333
Beta-2 (short)	β_2	=	1.333
Beta-3 (long)	β_3	=	1.333
Beta-4 (long)	β_4	=	1.333
Beta-x	β_x	=	0.038

Loads on Beams

	DL (KN/m)	LL (KN/m)
R1	=	
R2	=	
R3	=	
R4	=	

Steel prov'd

A_{3prov}	754	mm ²	T12 - 150 c/c	O.K
A_{xprov}	754	mm ²	T12 - 150 c/c	O.K
A_{4prov}	754	mm ²	T12 - 150 c/c	O.K
% of steel	0.34	%	within limit !	O.K

Steel prov'd

A_{1prov}	754	mm ²	T12 - 150 c/c	O.K
A_{yprov}	754	mm ²	T12 - 150 c/c	O.K
A_{2prov}	754	mm ²	T12 - 150 c/c	O.K
% of steel	0.35	%	within limit !	O.K

Site Dubai Depot
Fuel Station - Fuel Tank Wall

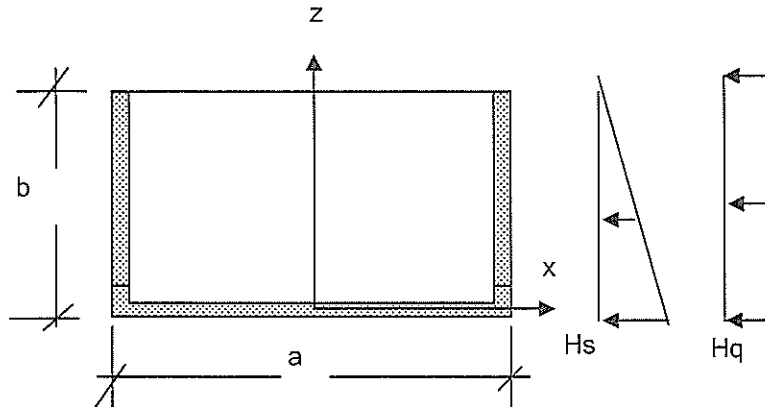
Date: May '07

Job No. 068010

Designed by Rudy

Checked by Ravi

Sheet No.



Assume

$$\gamma_{\text{soil}} = 20 \text{ kN/m}^3$$

$$\gamma_{\text{water}} = 10 \text{ kN/m}^3$$

$$q_{\text{surcharge}} = 50 \text{ kN/m}^2$$

$$K_o = 0.5$$

$$a = 5.4 \text{ m}$$

$$b = 4.6 \text{ m}$$

$$Z = \frac{t^2}{6} \text{ m}^3$$

$$= 0.015 \text{ m}^3$$

$$\frac{a}{b} = 1.17$$

$$\frac{t}{b} = 0.3 \text{ m}$$

$$H_s = [\gamma_{\text{water}} + K_o \gamma_{\text{soil}}] \cdot b \cdot \text{Load Factor (LF)}$$

$$= 92 \text{ kN/m}^2$$

$$= 129 \text{ kN/m}^2$$

$$\text{LF (SLS)} = 1.0$$

$$\text{LS (ULS)} = 1.4$$

$$H_s = K_o \cdot q_{\text{surcharge}}$$

$$= 25 \text{ kN/m}^2$$

$$= 40 \text{ kN/m}^2$$

$$\text{LF (SLS)} = 1.0$$

$$\text{LS (ULS)} = 1.6$$

Calculation/Sketch



Site	Dubai Depot Fuel Station - Fuel Tank Wall	Date:	May '07	Job No.	068010
Designed by	<i>Rudy</i>	Checked by	<i>Ravi</i>	Sheet No.	

Based on Roark's Formula for Stress and Strain for rectangular plate with 3 edges fixed and 1 edge free

For uniform load over the entire plate

For a/b	=	1.17			
β_1	=	0.462	γ_1	=	0.532
β_2	=	0.337	γ_2	=	0.631
β_3	=	0.322			

At x = 0, z = 0 (midpoint of bottom support - vertical reinforcement)

Max σ_b	=	$-\beta_1 q b^2 / t^2$	R	=	$\gamma_1 q b$		
	=	-2717 kN/m ²		=	98	kN/m	(ULS)
M	=	σZ					
	=	-65 kNm	(ULS)				

At x = 0, z = b (midpoint of vertical span - vertical reinforcement)

σ_b	=	$-\beta_2 q b^2 / t^2$			
	=	1982 kN/m ²			
M	=	48 kNm	(ULS)		

At x = a/2, z = b (top of side support - horizontal reinforcement)

σ_b	=	$-\beta_3 q b^2 / t^2$	R	=	$\gamma_2 q b$		
	=	-1890 kN/m ²		=	116	kN/m	(ULS)
M	=	-45 kNm	(ULS)				

For triangular load

For a/b	=	1.17			
β_1	=	0.249	γ_1	=	0.353
β_2	=	0.193	γ_2	=	0.195

At x = 0, z = 0 (midpoint of bottom support - vertical reinforcement)

Max σ_b	=	$-\beta_1 q b^2 / t^2$	R	=	$\gamma_1 q b$		
	=	-5392 kN/m ²		=	209	kN/m	(ULS)
M	=	σZ					
	=	-113 kNm	(ULS)				

At x = a/2, z = b (top of side support - horizontal reinforcement)

σ_b	=	$-\beta_2 q b^2 / t^2$	R	=	$\gamma_2 q b$		
	=	-4177 kN/m ²		=	115	kN/m	(ULS)
M	=	-88 kNm	(ULS)				

Summary of design forces

For vertical reinforcement

Msupport (bottom)	=	-178 kNm
Mspan	=	130 kNm
Rmax	=	307 kN

For horizontal reinforcement

Mmax	=	-133 kNm
Rmax	=	232 kN

Site Dubai Depot

Date: May '07

Job No. 068010

Fuel Station - Fuel Tank Wall 1 (Vertical)

Designed by RudyChecked by Ravi

Sheet No.

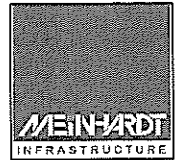
Slab design for ordinary load conditions:**Support**

Design Data	
Design moment M	178.4 kNm
Width of section b	1000 mm
Effective depth d	242.0 mm
$f'_{cu} = DIF \cdot f_{cu}$	40.00 N/mm ²
$f'_y = DIF \cdot f_y$	460.00 N/mm ²
Cover to Comp. Reinf.	35.00 mm
S.F for concrete r_{mc}	1.50
S.F for steel r_{ms}	1.15
$f_c = 0.67 f'_{cu} / r_{mc}$	17.87 N/mm ²
$f_t = f'_y / r_{ms}$	400.00 N/mm ²
$k = M / bd^2 f'_{cu}$	0.08
$0.5 + (0.25 - k/R)^{0.5}$	0.91
$z = d \dots$	219.41 mm
ULS $A_s = M / ft z$	2033 mm ²
$A_s' = M/ft.z$ (Req'd)	0.0
A_s' (Comp.) Provided	1608 mm²
No. of Bars/m Wdth	8 0
Dia. Of Bars	T 16 T 25
$A_s = M/ft.z$ (Req'd)	2033 mm ²
A_s (Tens.) Provided	2513 mm²
No. of Bars/m Wdth	8 0
Dia. Of Bars	T 20 T 32
%p	1.04% OK

Span

130.2 kNm
1000 mm
242.0 mm
40.00 N/mm ²
460.00 N/mm ²
35.00 mm
1.50
1.15
17.87 N/mm ²
400.00 N/mm ²
0.06
0.93
226.00 mm
1440 mm ²
0.0
2513 mm²
8 0
T 20 T 25
1440 mm ²
1608 mm²
8 0
T 16 T 32
0.66% OK

Shear V at Face	306.8 kN	ok
v	1.27 N/mm ²	
Shear V at d	260.8 kN	(85%)
A_s at d	2513	
v at d = V / bd	1.08 N/mm ²	ok
$2 \cdot v_c$ at d	1.70	
A_{sv} / S_v for d	0.00 mm ² /mm	ok
A_{sv} / S_v	0.00 mm²/mm	
A_{sv}/S_v Provided	0.00 mm²/mm	ok
Link Dia.	No. of Links	
0	200	200



Site	Dubai Depot	Date:	May '07	Job No.	068010
	Fuel Station - Fuel Tank Wall 1 (Horizontal)				
	Designed by <u>Pachy</u>	Checked by	<u>Ravi</u>	Sheet No.	

Slab design for ordinary load conditions:

Design Data	
Design moment M	133.1 kNm
Width of section b	1000 mm
Effective depth d	242.0 mm
$f'_{cu} = DIF \cdot f_{cu}$	40.00 N/mm ²
$f'_y = DIF \cdot f_y$	460.00 N/mm ²
Cover to Comp. Reinf.	35.00 mm
S.F for concrete r_{mc}	1.50
S.F for steel r_{ms}	1.15
$f_c = 0.67 f'_{cu} / r_{mc}$	17.87 N/mm ²
$f_t = f'_y / r_{ms}$	400.00 N/mm ²
$k = M / bd^2 f'_{cu}$	0.06
$0.5 + (0.25 - k/R)^{0.5}$	0.93
$z = d (...)$	225.61 mm
ULS $A_s = M / ft z$	1475 mm ²
$A_s' = M / ft z$ (Req'd)	0.0
A_s' (Comp.) Provided	1608 mm ²
No. of Bars/m Widt	8 0
Dia. Of Bars	T 16 T 25
$A_s = M / ft z$ (Req'd)	1475 mm ²
A_s (Tens.) Provided	1608 mm ²
No. of Bars/m Widt	8 0
Dia. Of Bars	T 16 T 32
%p	0.66% OK

Shear V at Face	231.5 kN	
v	0.96 N/mm ²	ok
Shear V at d	196.8 kN	(85%)
A_s at d	1608	
v at d = V / bd	0.81 N/mm ²	
$2 \cdot v_c$ at d	1.46	
A_{sv} / S_v for d	0.00 mm ² /mm	
A_{sv} / S_v	0.00 mm ² /mm	
A_{sv} / S_v Provided	0.00 mm ² /mm	ok
Link Dia.	No. of Links	Spacing - D
0	200	200

Site Dubai Depot
Fuel Station - Fuel Tank Wall

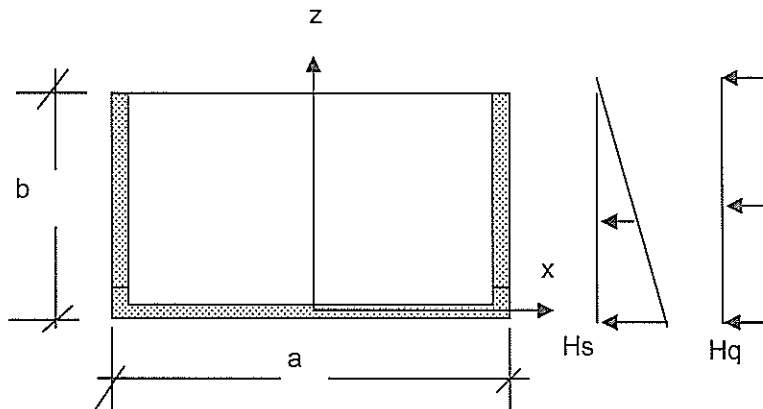
Date: May '07

Job No. 068010

Designed by Rudy

Checked by Ravi

Sheet No.



Assume

γ_{soil}	=	20 kN/m ³
γ_{water}	=	10 kN/m ³
$q_{\text{surcharge}}$	=	50 kN/m ²
K_o	=	0.5

$$\begin{aligned} a &= 4.04 \text{ m} \\ b &= 4.94 \text{ m} \\ Z &= t^2/6 \text{ m}^3 \\ &= 0.015 \text{ m}^3 \end{aligned}$$

$$\begin{aligned} a/b &= 0.82 \\ t &= 0.3 \text{ m} \end{aligned}$$

$$\begin{aligned} H_s &= [\gamma_{\text{water}} + K_o \gamma_{\text{soil}}] * b * \text{Load Factor (LF)} \\ &= 99 \text{ kN/m}^2 \\ &= 138 \text{ kN/m}^2 \end{aligned}$$

$$\begin{aligned} \text{LF (SLS)} &= 1.0 \\ \text{LS (ULS)} &= 1.4 \end{aligned}$$

$$\begin{aligned} H_s &= K_o * q_{\text{surcharge}} \\ &= 25 \text{ kN/m}^2 \\ &= 40 \text{ kN/m}^2 \end{aligned}$$

$$\begin{aligned} \text{LF (SLS)} &= 1.0 \\ \text{LS (ULS)} &= 1.6 \end{aligned}$$

Site	Dubai Depot Fuel Station - Fuel Tank Wall	Date:	May '07	Job No.	068010
Designed by	<u>Rudy</u>	Checked by	<u>Ravi</u>	Sheet No.	

Based on Roark's Formula for Stress and Strain for rectangular plate with 3 edges fixed and 1 edge free

For uniform load over the entire plate

For a/b	=	0.82			
β_1	=	0.213	γ_1	=	0.372
β_2	=	0.178	γ_2	=	0.409
β_3	=	0.255			

At x = 0, z = 0 (midpoint of bottom support - vertical reinforcement)

Max σ_b	=	$-\beta_1 q b^2 / t^2$	R	=	$\gamma_1 q b$		
	=	-1445 kN/m ²		=	74	kN/m	(ULS)
M	=	σ_z					
	=	-35 kNm	(ULS)				

At x = 0, z = b (midpoint of vertical span - vertical reinforcement)

σ_b	=	$-\beta_2 q b^2 / t^2$			
	=	1207 kN/m ²			
M	=	29 kNm	(ULS)		

At x = a/2, z = b (top of side support - horizontal reinforcement)

σ_b	=	$-\beta_3 q b^2 / t^2$	R	=	$\gamma_2 q b$		
	=	-1728 kN/m ²		=	81	kN/m	(ULS)
M	=	-41 kNm	(ULS)				

For triangular load

For a/b	=	0.82			
β_1	=	0.140	γ_1	=	0.281
β_2	=	0.136	γ_2	=	0.219

At x = 0, z = 0 (midpoint of bottom support - vertical reinforcement)

Max σ_b	=	$-\beta_1 q b^2 / t^2$	R	=	$\gamma_1 q b$		
	=	-3760 kN/m ²		=	192	kN/m	(ULS)
M	=	σ_z					
	=	-79 kNm	(ULS)				

At x = a/2, z = b (top of side support - horizontal reinforcement)

σ_b	=	$-\beta_2 q b^2 / t^2$	R	=	$\gamma_2 q b$		
	=	-3647 kN/m ²		=	150	kN/m	(ULS)
M	=	-77 kNm	(ULS)				

Summary of design forces

For vertical reinforcement

M _{support (bottom)}	=	-114 kNm
M _{span}	=	95 kNm
R _{max}	=	266 kN

For horizontal reinforcement

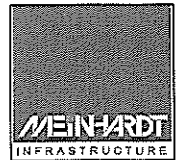
M _{max}	=	-118 kNm
R _{max}	=	231 kN

Site	Dubai Depot	Date:	May '07	Job No.	068010
	Fuel Station - Fuel Tank Wall 1 (Vertical)				
	Designed by <u>Rudy</u>	Checked by <u>Ravi</u>		Sheet No.	

Slab design for ordinary load conditions:

Support		Span	
Design Data			
Design moment M	113.6 kNm	95.0 kNm	
Width of section b	1000 mm	1000 mm	
Effective depth d	242.0 mm	242.0 mm	
$f'_{cu} = DIF \cdot f_{cu}$	40.00 N/mm ²	40.00 N/mm ²	
$f'_y = DIF \cdot f_y$	460.00 N/mm ²	460.00 N/mm ²	
Cover to Comp. Reinf.	35.00 mm	35.00 mm	
S.F for concrete r_{mc}	1.50	1.50	
S.F for steel r_{ms}	1.15	1.15	
$f_c = 0.67 f'_{cu} / r_{mc}$	17.87 N/mm ²	17.87 N/mm ²	
$f_t = f'_y / r_{ms}$	400.00 N/mm ²	400.00 N/mm ²	
$k = M / bd^2 f'_{cu}$	0.05	0.04	
$0.5 + (0.25 - k/R)^{0.5}$	0.94	0.95	
$z = d (...)$	228.17 mm	229.90 mm	
ULS $A_s = M / ft_z$	1245 mm ²	1033 mm ²	
$A_s' = M / ft_z$ (Req'd)	0.0	0.0	
A_s' (Comp.) Provided	1608 mm ²	1608 mm ²	
No. of Bars/m Width	8	0	
Dia. Of Bars	T 16	T 25	
$A_s = M / ft_z$ (Req'd)	1245 mm ²	1033 mm ²	
A_s (Tens.) Provided	1608 mm ²	1608 mm ²	
No. of Bars/m Width	8	0	
Dia. Of Bars	T 16	T 32	
%p	0.66% OK	0.66% OK	

Shear V at Face	265.6 kN	ok
v	1.10 N/mm ²	
Shear V at d	225.8 kN	(85%)
A_s at d	1608	
v at d = V / bd	0.93 N/mm ²	
$2 \cdot v_c$ at d	1.46	
A_{sv} / S_v for d	0.00 mm ² /mm	
A_{sv} / S_v	0.00 mm ² /mm	ok
A_{sv} / S_v Provided	0.00 mm ² /mm	
Link Dia.	No. of Links	Spacing - D
0	200	200



Site	Dubai Depot	Date:	May '07	Job No.	068010
	Fuel Station - Fuel Tank Wall 1 (Vertical)	Designed by	<i>Rudi</i>	Checked by	<i>Ravi</i>
				Sheet No.	

Slab design for ordinary load conditions:

Design Data	
Design moment M	118.1 kNm
Width of section b	1000 mm
Effective depth d	242.0 mm
$f'_{cu} = DIF \cdot f_{cu}$	40.00 N/mm ²
$f'_y = DIF \cdot f_y$	460.00 N/mm ²
Cover to Comp. Reinf.	35.00 mm
S.F for concrete r_{mc}	1.50
S.F for steel r_{ms}	1.15
$f_c = 0.67 f'_{cu} / r_{mc}$	17.87 N/mm ²
$f_t = f'_y / r_{ms}$	400.00 N/mm ²
$k = M / bd^2 f'_{cu}$	0.05
$0.5 + (0.25 - k/R)^{0.5}$	0.94
$z = d \dots$	227.59 mm
ULS $A_s = M / f_t z$	1297 mm ²
$A_s' = M / f_t z$ (Req'd)	0.0
A_s' (Comp.) Provided	1608 mm ²
No. of Bars/m Width	8 0
Dia. Of Bars	T 16 T 25
$A_s = M / f_t z$ (Req'd)	1297 mm ²
A_s (Tens.) Provided	1608 mm ²
No. of Bars/m Width	8 0
Dia. Of Bars	T 16 T 32
%p	0.66% OK

Shear V at Face	230.7 kN	ok
v	0.95 N/mm ²	
Shear V at d	196.1 kN	(85%)
A_s at d	1608	
v at d = V / bd	0.81 N/mm ²	
$2 \cdot v_c$ at d	1.46	
A_{sv} / S_v for d	0.00 mm ² /mm	
A_{sv} / S_v	0.00 mm ² /mm	ok
A_{sv}/S_v Provided	0.00 mm ² /mm	
Link Dia.	No. of Links	Spacing - D
0	200	200

Appendix 'B'
Red Line Rashidiya Depot
Diesel Fuel Station Building-21
Reply to Engineer Comments

**SYSTRA/PARSONS (The Engineer)**

The Engineer's Office, Union Square, P.O. Box 111081, Dubai, UAE
Tel.: +971 (0) 4 234 0202, Fax: +971 (0) 4 234 0203

**DUBAI METRO PROJECT**

Document title	Rashidiya Main Depot Building 21 – Diesel Fuel Station Enhanced Design Drawings and Report
Design Stage	DCP3
Discipline	Civil phase (1) – Red Line TL Design
Reviewer	Hazem Sharaf
Document Project Number	
Revision n°/Date	Rev. A1/June 27, 2007
Transmittal ID	TRANSMIT-007875

N°	Reference of the document / section	ENGINEER COMMENTS	CONTRACTOR ANSWERS	Clearance Date
1		The submission status is NOWC and contractor must incorporate all our comments in the next DCP submission.		
2	General	1. Correlate between drawings and computer model (i.e. vertical and horizontal bracing secondary beams). 2. The submitted calculation must include all straining action outputs for each case of loading separately (not the combinations only). 3. The submitted summarized design of sections	Please refer to attached revised calculation report for query items listed.	

N°	Reference of the document / section	ENGINEER COMMENTS	CONTRACTOR ANSWERS	Clearance Date
		is not sufficient; provide more detailed design for the frame, secondary beams showing the design parameters (i.e. buckling length of members, buckling factors, knee bracing locations)		
3	Calculations DCC - 371953 – A1	It has been noticed that the deflection at the tip of the cantilever frame due to dead & live load in the submitted calculation page (28) are (13.8 cm) and the sway due to dead & live & wind load is (5.1 cm). Check deflection & sway for the correct case of loading and verify.	Please refer to attached revised calculation report for query items listed.	
4		Provide the design of the embedded length of the anchor bolts inside the concrete footings.	Please refer to attached calculation for query item listed.	
5		Provide the design of missing connection.	Secondary beams have been removed in the revised design; hence connection design is not required (no changes to base plate design from previously).	
6		Mention the reference in analysis for the straining actions used in the design of the footings, and consider the critical direction of the moment with vertical force. Also the design due to applied loads must be checked, the calculation of stresses on the	Calculations have been revised and checked, please refer to attached calculation for query item listed.	

**SYSTRA/PARSONS (The Engineer)**

The Engineer's Office, Union Square, P.O. Box 111081, Dubai, UAE

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N°	Reference of the document / section	ENGINEER COMMENTS	CONTRACTOR ANSWERS	Clearance Date
		excel sheet is not correct, check and verify.		
7		Provide the check of uplift for the underground tank.	Please refer to attached calculation for query item listed.	
8	Drawings	1. Provide the design of purlins & side grits. 2. Correlate between drawings (371835 A1 and 371840 A1), two types of section for the purlins is used.	1. Please refer to attached calculation for design of 200C25 purlins. 2. Noted. DCP4 drawings will only show one type of purlin to be used (200C25).	

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site:	Dubai Metro Project - Rashidiya Fuel Station Loadings	Date:	19-Sep-07	Job No.:	68010.01
		Designed by:	CJA	Sheet No.:	

Column spacing = 4 m

Roof Loadings :**DL**

W_{DL} :

Sheeting	0.1 kPa		
Purlins	0.1 kPa		
Insulation	0.3 kPa		
	<u>0.5 kPa</u>	x 4 =	2 kN/m

LL

0.6 kPa x 4 = 2.4 kN/m

Notional horizontal load, Ph

(Load Case of 1.4DL + 1.6LL)

Tributary load area for the roof

= 6*4
= 24 m²

Total Dead Load
12 kN

Total Live Load
14 kN
(Including A/C equipment)

Tributary load area for the wall

= 4*4
= 16 m²

Total Dead Load
8 kN
(excluding the M&E Finishes)

Total Live Load

Therefore, total ULS load, W_u = 34 kN

Ph = 1.5 % of W_u
= 0.5 kN

Calculation/Sketch

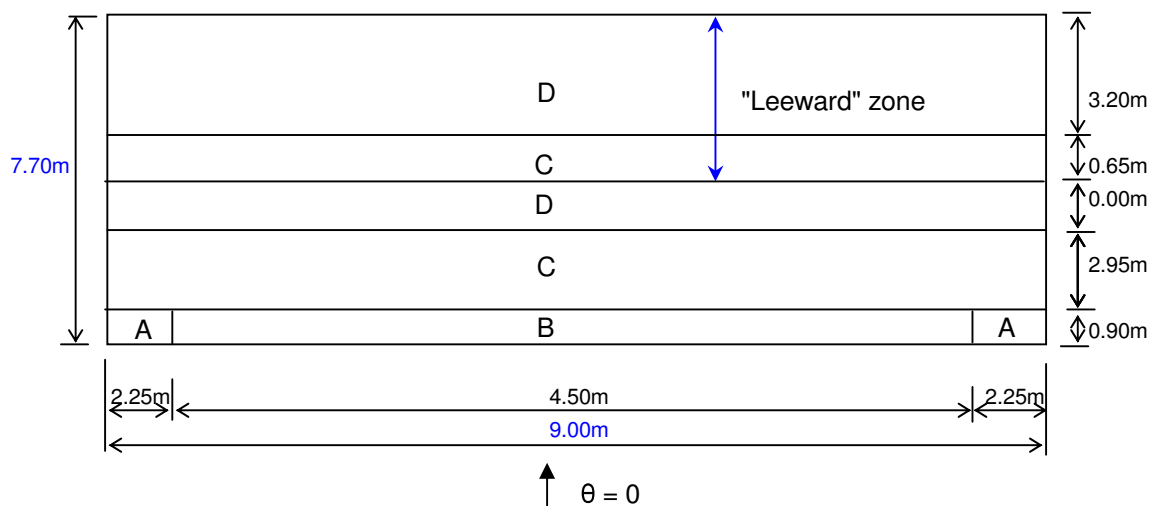
Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site:	Dubai Metro Project - Rashidiya Fuel Station Wind Load Computation	Date:	19-Sep-07	Job No.:	068010.01
		Designed by:	CJA	Sheet No.:	

Average Cpe for roof (Wind direction normal to eave)

PLAN



Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 9 \text{ m} \\
 H &= 5.2 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 9 \\
 &= 9 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	4.1	-2			
B	4.1	-1.4			
C	26.6	-0.7	5.9	-0.7	-0.7
D	0.0	-0.2	28.8	-0.2	0.2
Total	34.7		34.7		
Average		-0.93		-0.28	0.05

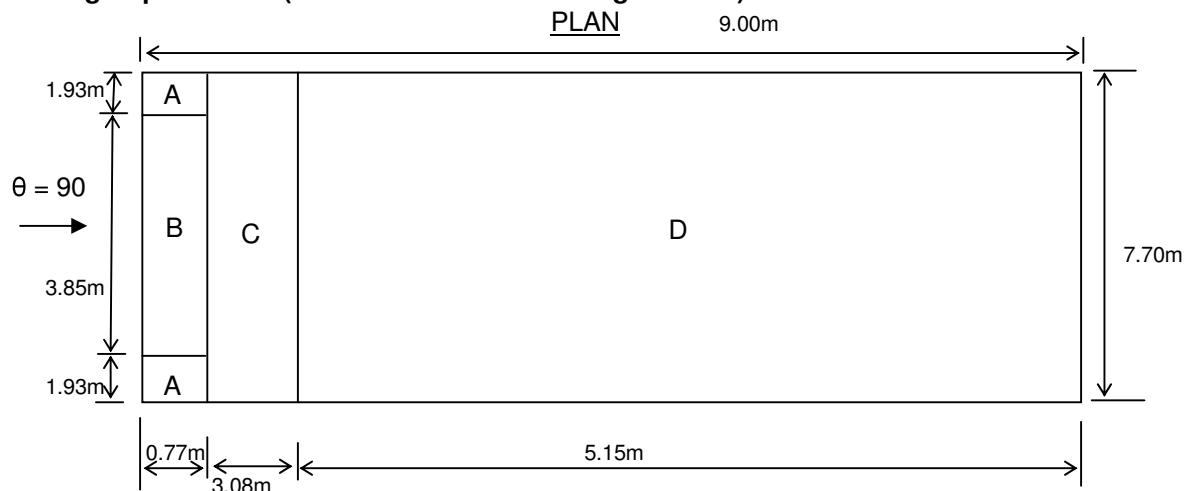
Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site:	Dubai Metro Project - Rashidiya Fuel Station Wind Load Computation	Date:	19-Sep-07	Job No.:	068010.01
		Designed by:	CJA	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)



Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 7.7 \text{ m} \\
 H &= 5.2 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 7.7 \\
 &= 7.7 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

Wind Load Computation

$$\begin{aligned}
 D &= 7.70 \text{ m} \\
 H &= 5.2 \text{ m} \\
 D/H &= 1.48
 \end{aligned}$$

Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.93	-0.28 0.05	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.81	-0.5 -0.5	-0.8 (Zone B)	-0.8 (Zone B)

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

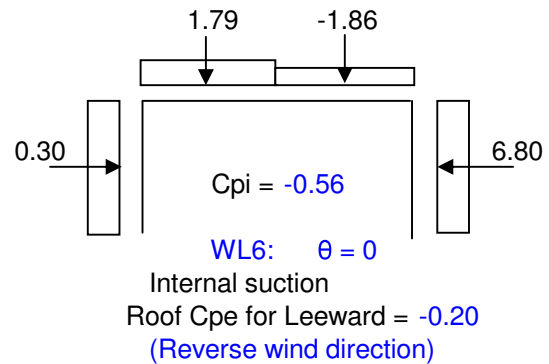
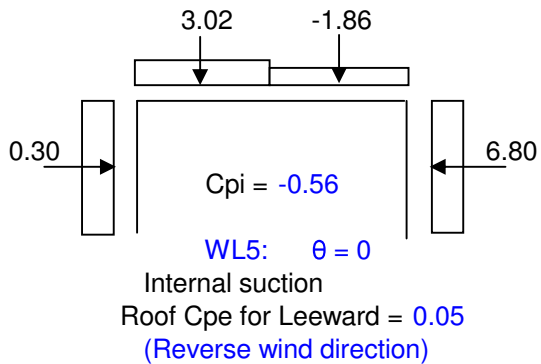
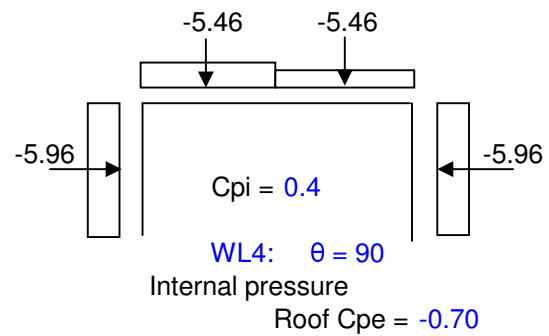
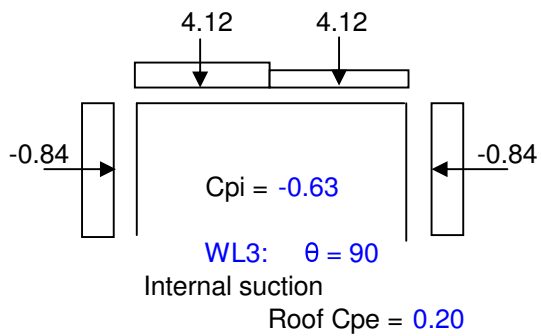
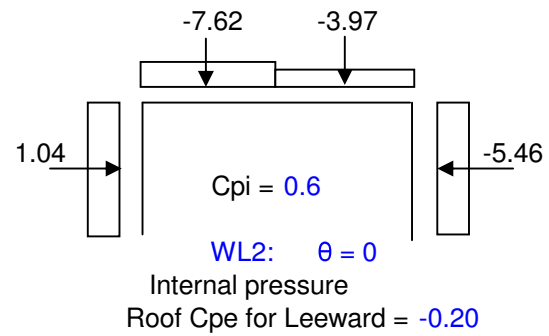
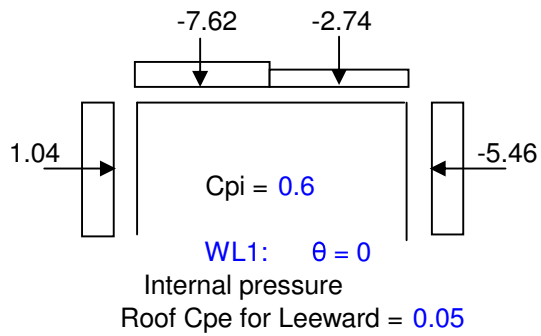


Site: **Dubai Metro Project - Rashidiya**
Fuel Station
Wind Load Computation

Date: 19-Sep-07 Job No.: 068010.01

Designed by: **CJA** Sheet No.:

$$\text{Wind Load} = 4 \times q \times (C_{pe} - C_{pi}) \quad \text{kN/m} \quad (\text{For unit width})$$



Site: **Dubai Metro Project - Rashidiya**
Fuel Station
Earthquake Loading

Date: 19-Sep-07

Job No.: 68010.01

Designed by: **CJA**

Sheet No.:

Estimation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
 Height of the Column h_n = Given below

Design Information:

- 1) Seismic Zone = 2A (from Design Brief)
 2) Occupancy Category = 1 (Tanle- 16-K, Conservatively adopt 1)
 3) Lateral Force procedure = **Static Procedure** (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

- 4) **Base Shear** $V_{base-S} = (C_v \times I \times W) / (R \times T)$ (Section 1630.2.1)
 Seismic Coefficient $C_v = 0.32$ (Table 16-R assuming soil profile type S_D)
 Importance Factor $I = 1.00$ (Table 16-K)
 Numeric. Coef: Strength & Ductility $R = 4.50$ (Table 16-N for OMRT steel only)
 Elastic Fundamental Period of Vibration $T = C_t (h_n)^{3/4}$ Sec.
 Numerical Coefficient $C_t = 0.0853$ (Steel frame t.b.c)

- 5) **Limiting Values of Base shear** $V_{base-S-Max} = (2.5 \times C_a \times I \times W) / R = 4.20$
 $V_{base-S-Min} = 0.11 \times C_a \times I \times W = 0.83$

Design Earthquake Loads

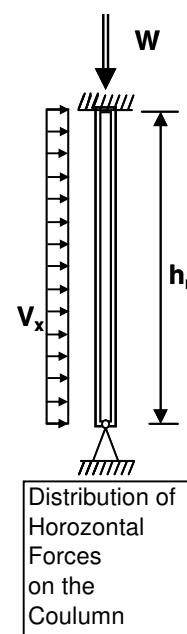
(Section 1630.6)

- $E_x = \rho \times \text{Min.} (E_h \text{ or } V_{base-S-Max}) + E_v$
 $E_{x-max} = \Omega \times \text{Min.} (E_h \text{ or } V_{base-S-Max})$
 $V_{base-S} = E_h =$ Earthquake load in the element due to the base shear, V , as set out above (for non-structural components refer to section 1632)
 Vert. component of EQ ground motion $E_v = 0.5 C_a I D$ (Table 16-K)
 Seismic Coefficient $C_a = 0.22$ (Table 16-Q, for soil type S_D)
 Seismic Force Amplification Factor $\Omega = 2.80$ (Table 16-N)
 Dead Load on the element $W = D =$ Given below
 $\rho = 2 - 6.1 / (r_{max} \sqrt{A_B}) = 1.25$ (conservative) ($1.0 \leq \rho \leq 1.5$)
 r_{max} = The maximum element-storey shear ratio
 A_B = Ground floor area of the structure

Distribution of Horizontal Forces:

Distribution of Horizontal Force on the Column $V_x = E_x / h_n$

Col	Gr. Load, W , kN	Ht h_n , m	Time, T sec.	V_{base-S} kN	V_{base-S} kN	E_v kN	E_x kN	E_{x-max} kN	V_x kN/m	V_{x-max} kN/m
	34	4.00	0.24	10.1	4.20	3.784	9.0	11.8	2.3	2.9





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Job No
068010.01

Sheet No

1

Rev

Part

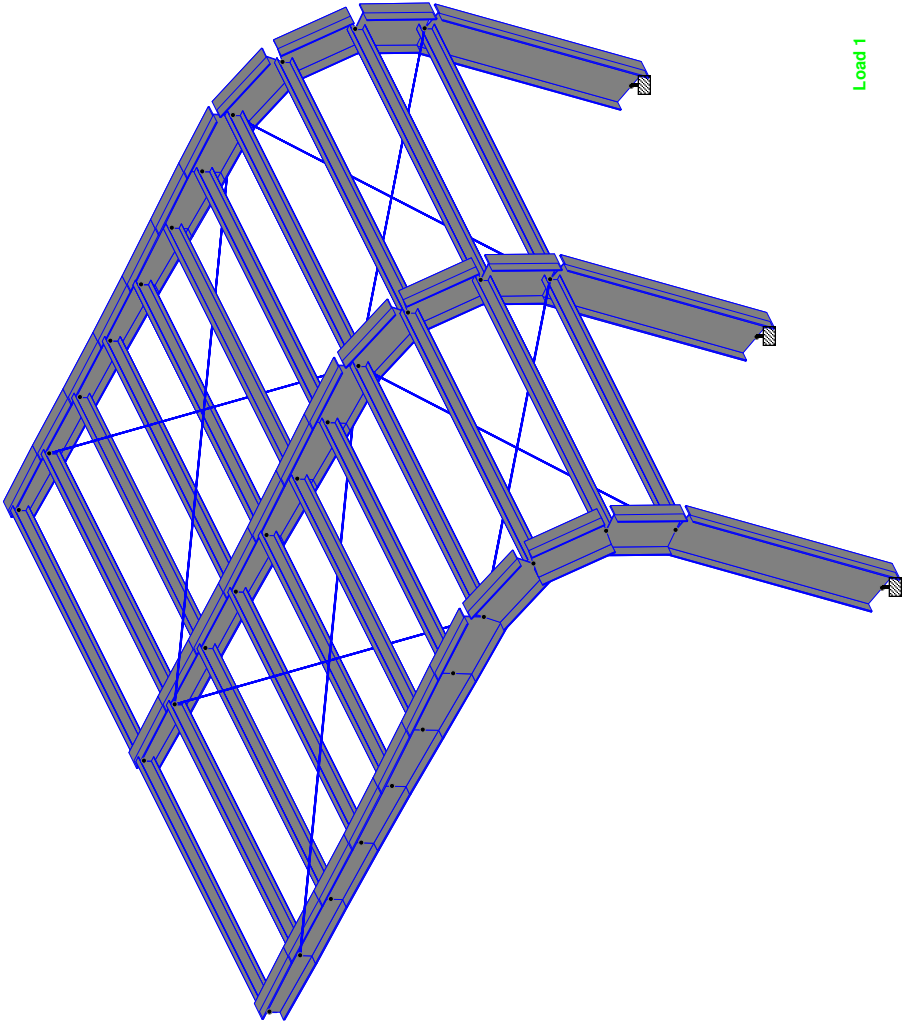
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By CJA

DateSep 2007

Chd

File Fuel Station-P3 Col550 T Date/Time 19-Sep-2007 12:02



Load 1

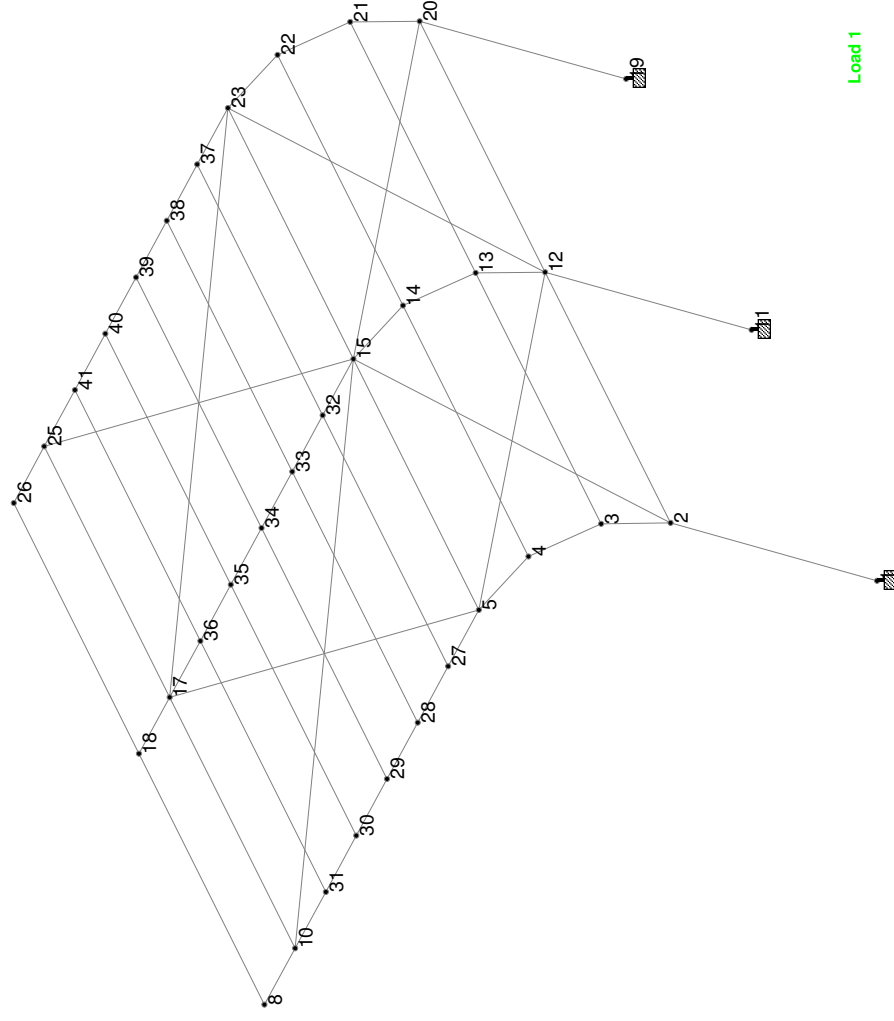
RASHIDIYA B21: STRUCTURE LAYOUT



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	Job No 068010.01		Sheet No 1	Rev
	Part			
	Ref			
	By CJA	Date Sep 2007	Chd	
Client Dubai RTA	File Fuel Station-R3 Col550 T			



RASHIDIYA B21: NODE NUMBERING LAYOUT

Load 1



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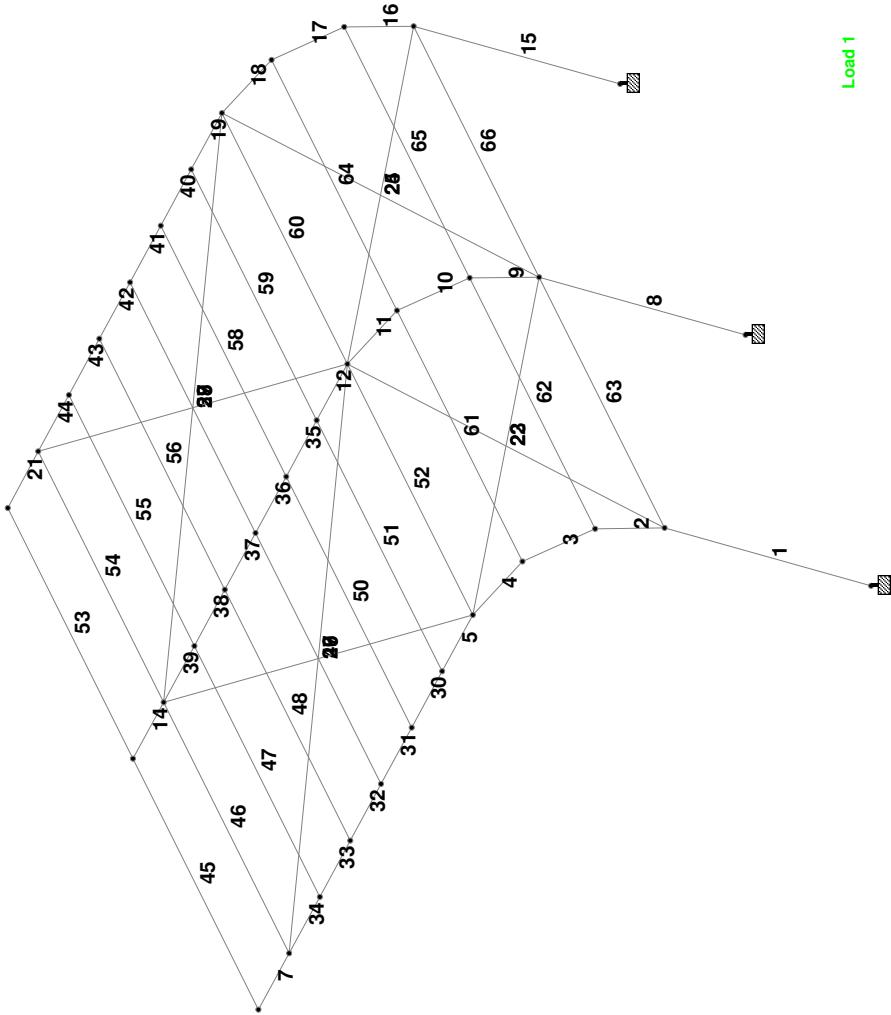
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By CJA

DateSep 2007

Ctd

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RASHIDIYA B21: BEAM NUMBERING LAYOUT



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Sheet No

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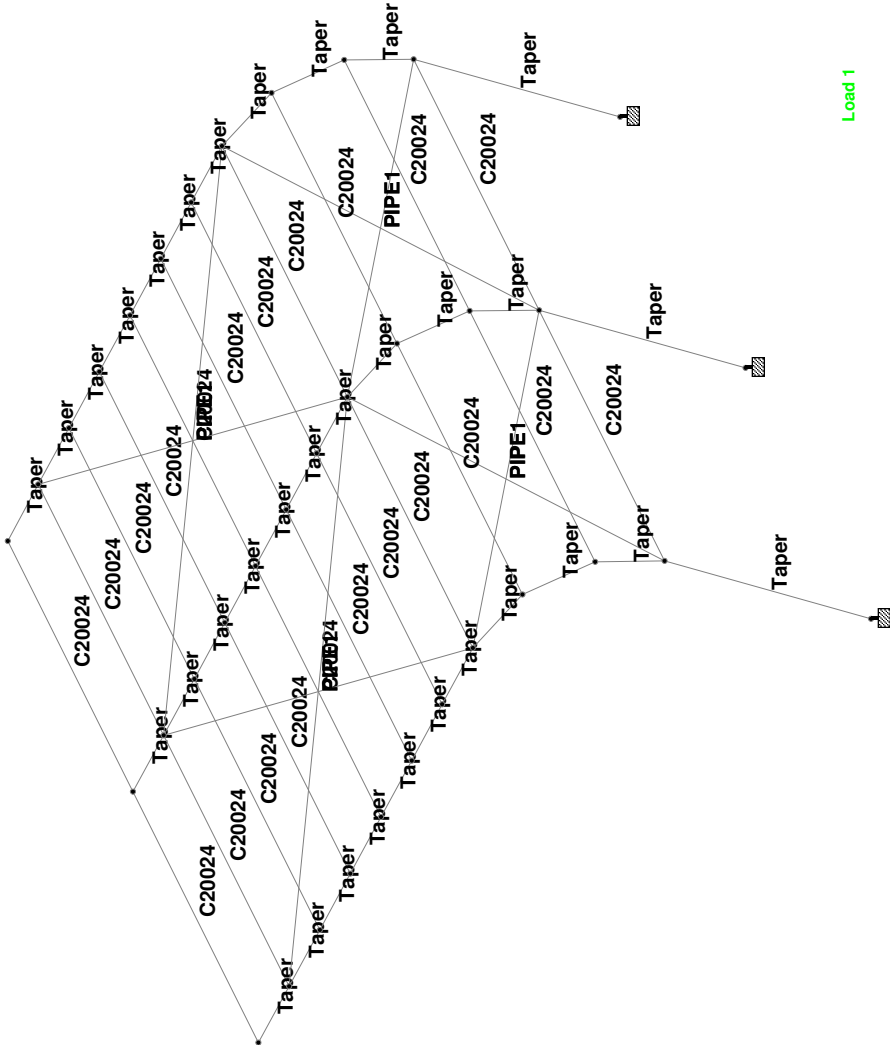
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RASHIDIYA B21: SECTION PROPERTY LAYOUT



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Sheet No

1

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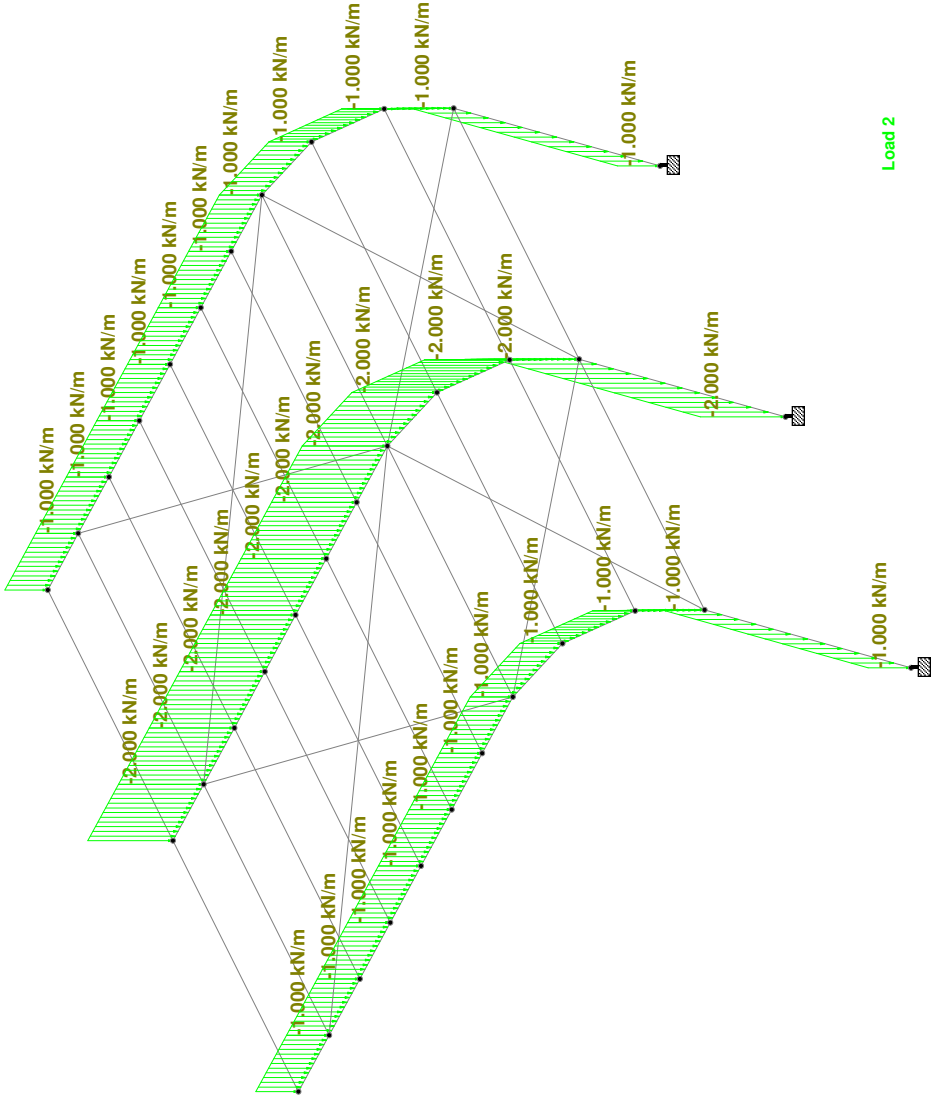
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RASHIDIYA B21: SDL

Load 2



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1

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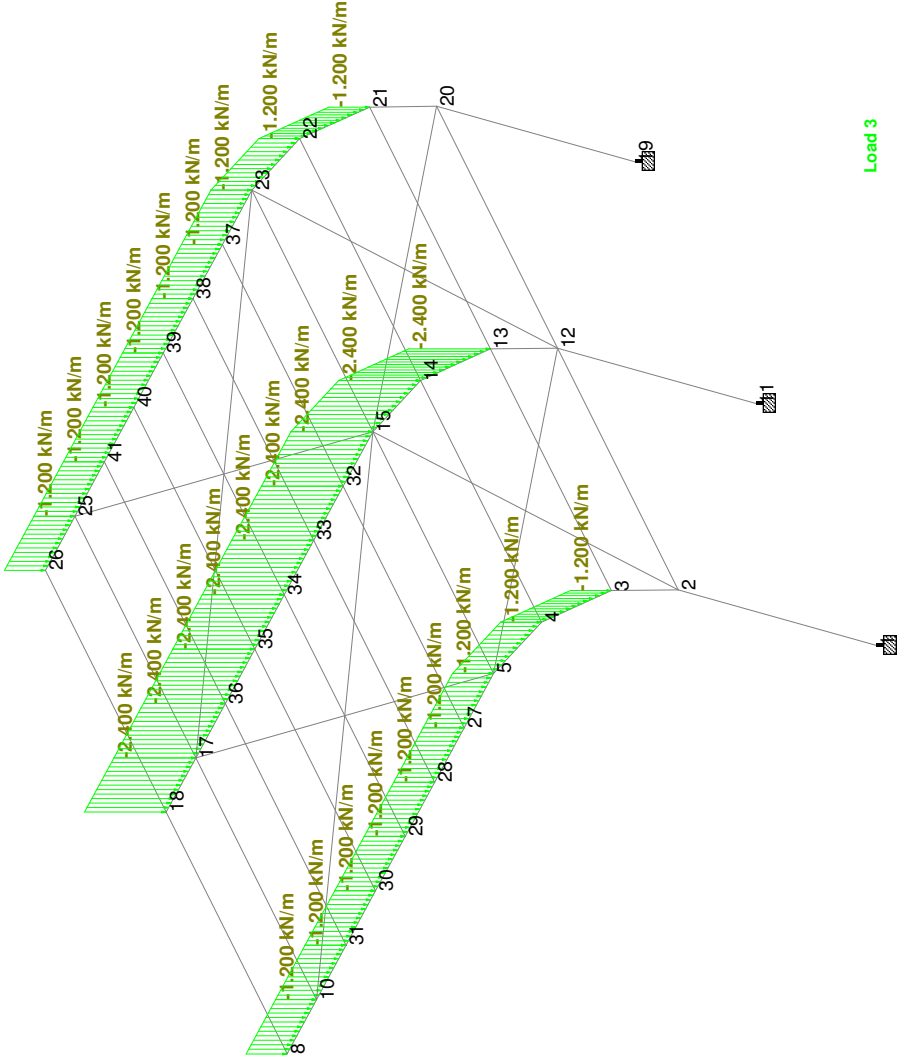
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068010.01

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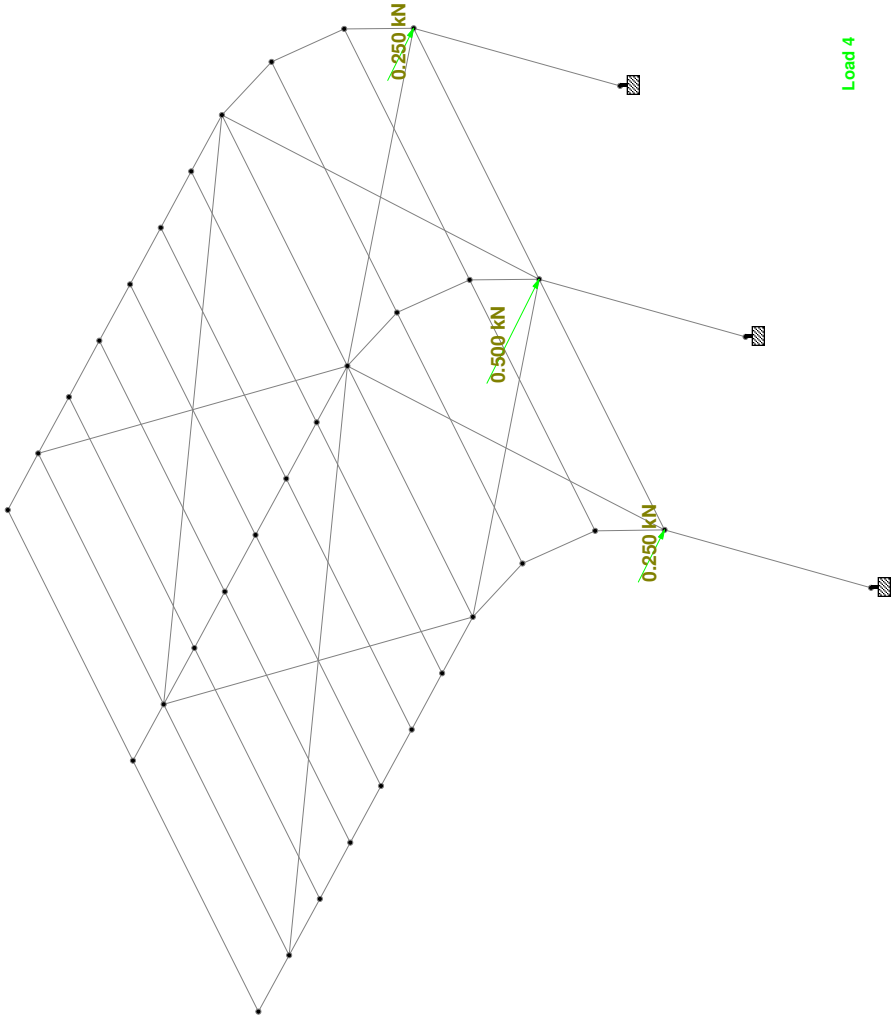
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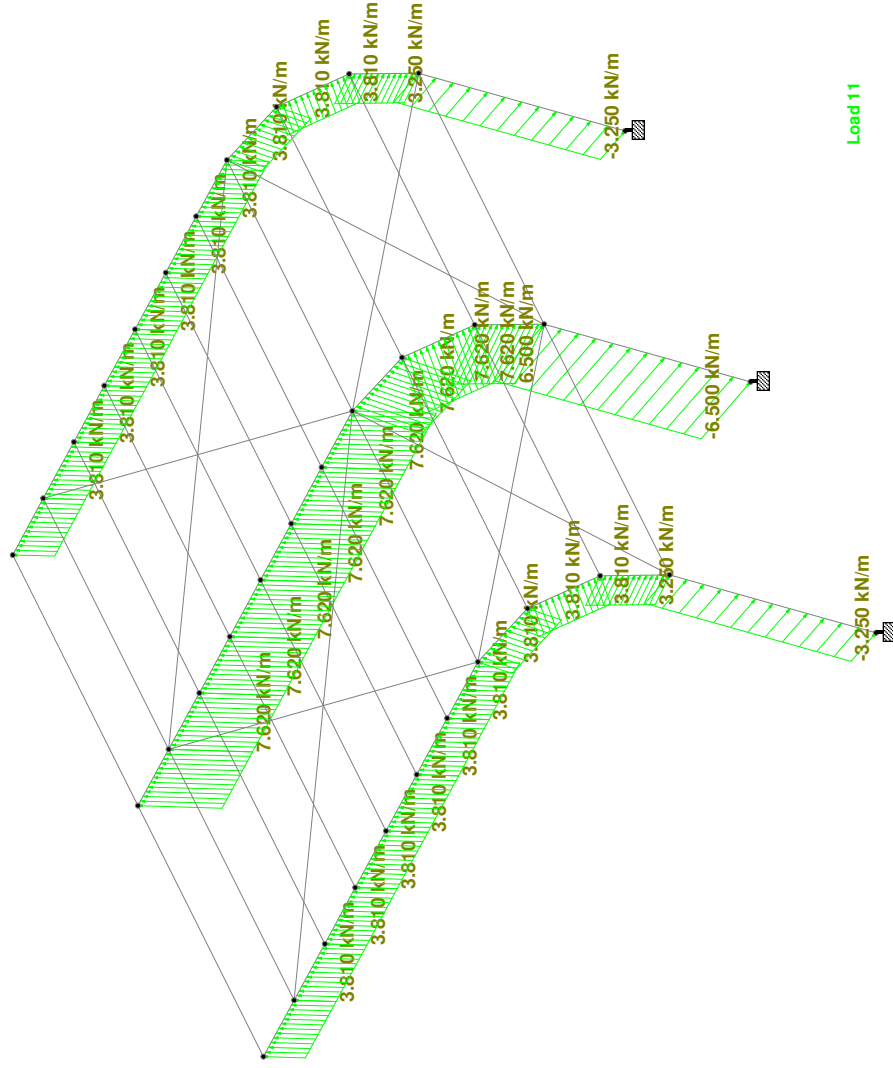
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RASHIDIYA B21: NOTIONAL LOAD (PH)

Load 4



RASHIDIYA B21: WL1



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Sheet No

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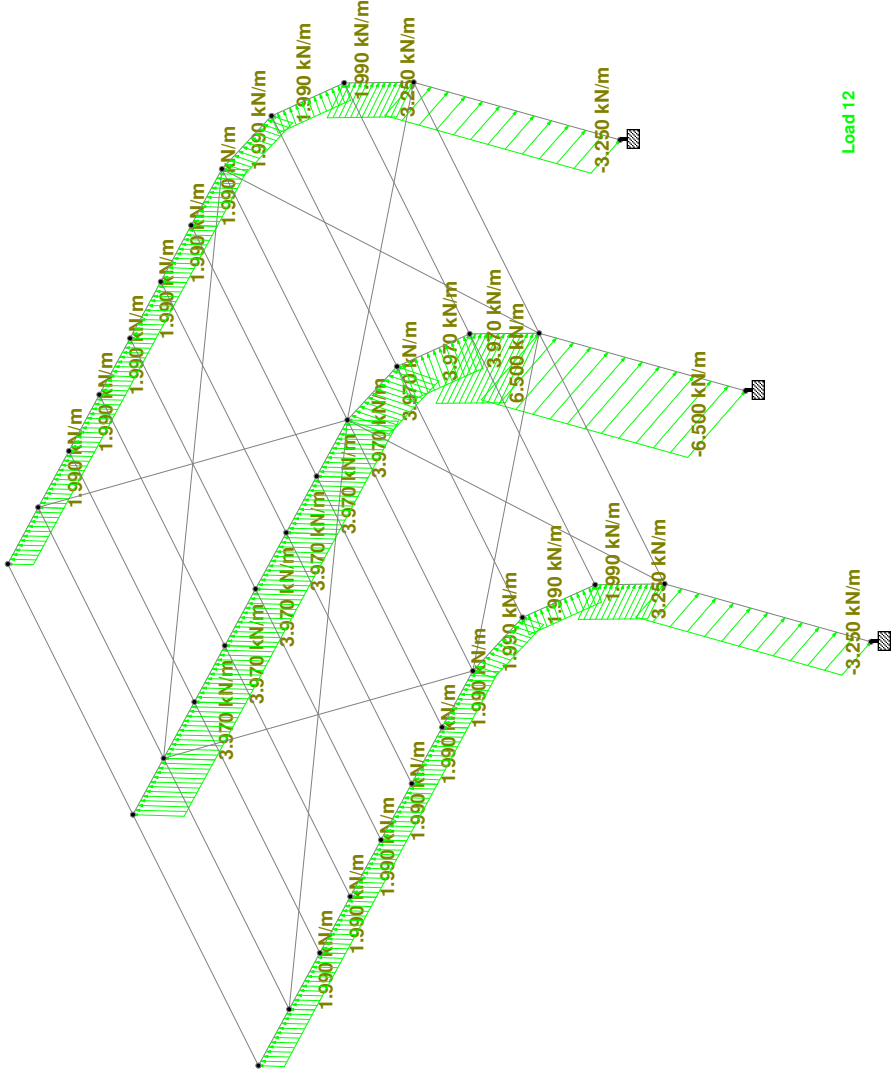
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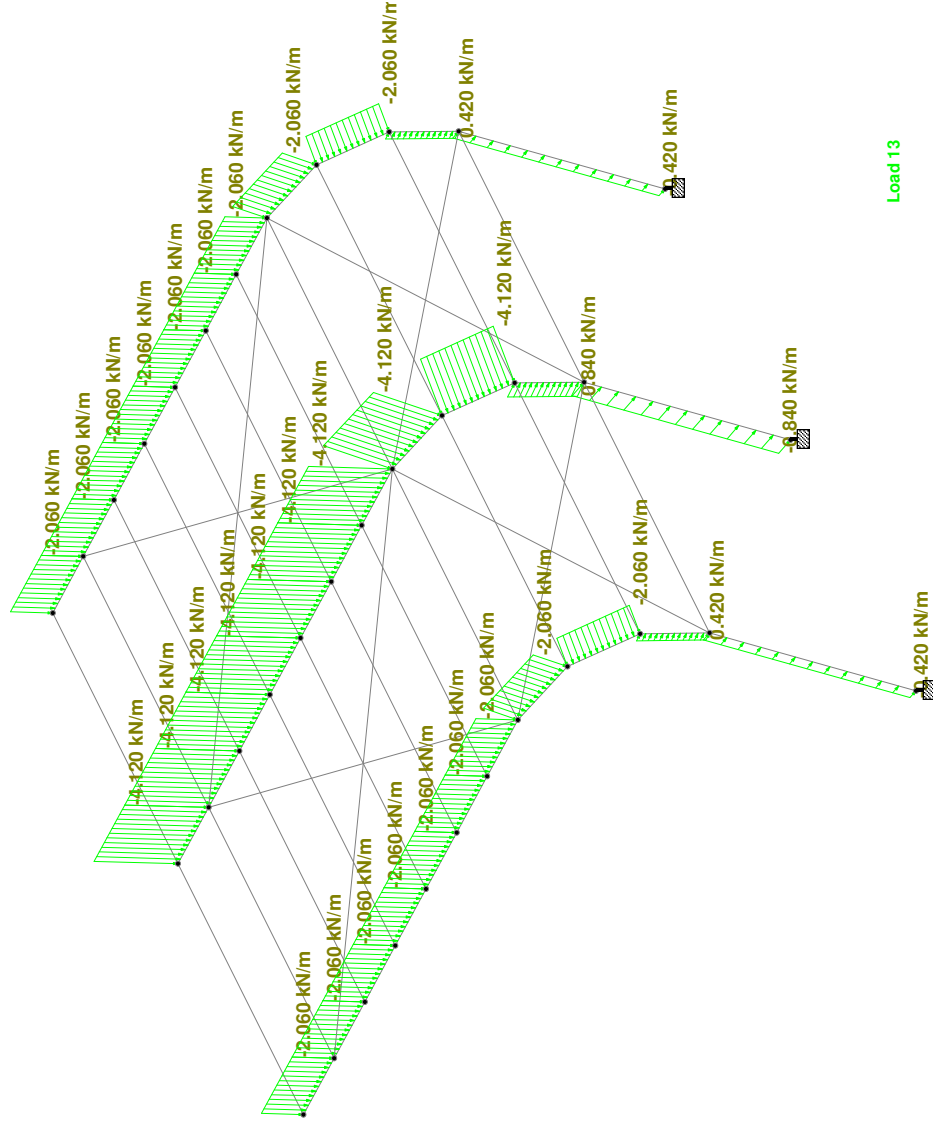
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RASHIDIYA B21: WL2



RASHIDIYA B21: WL3



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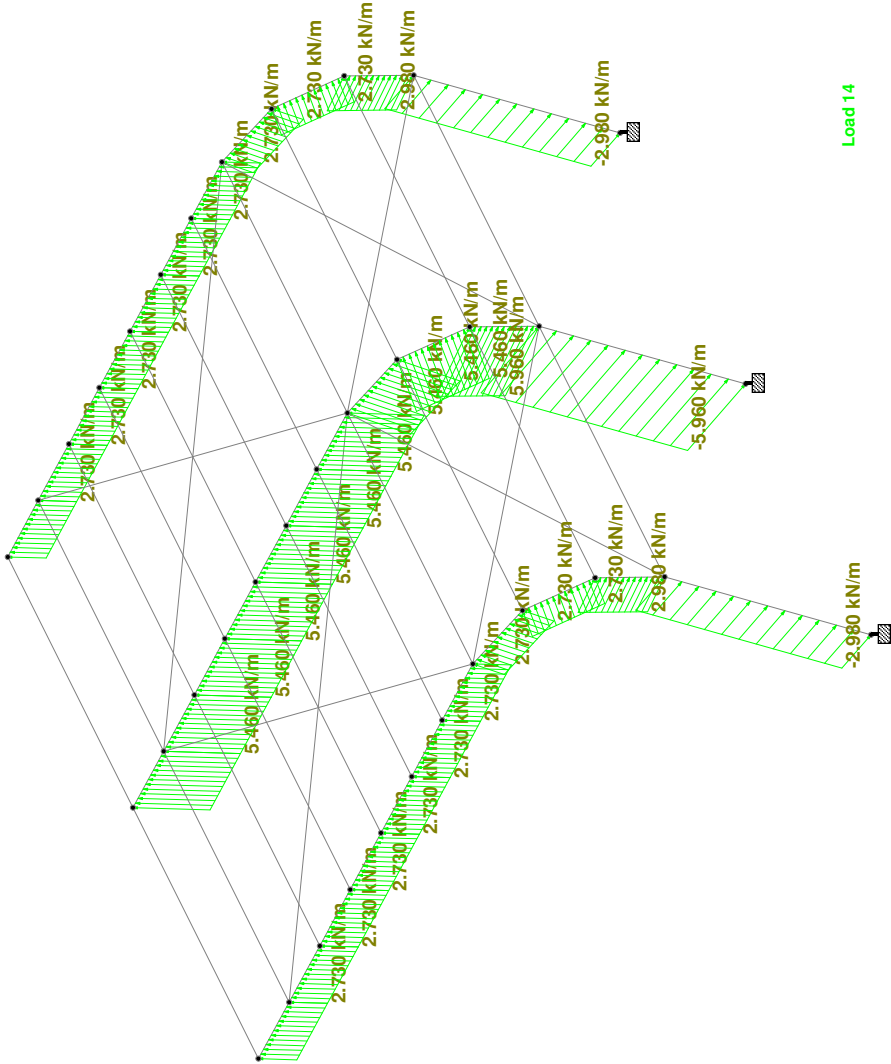
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Load 14

RASHIDIYA B21: WL4



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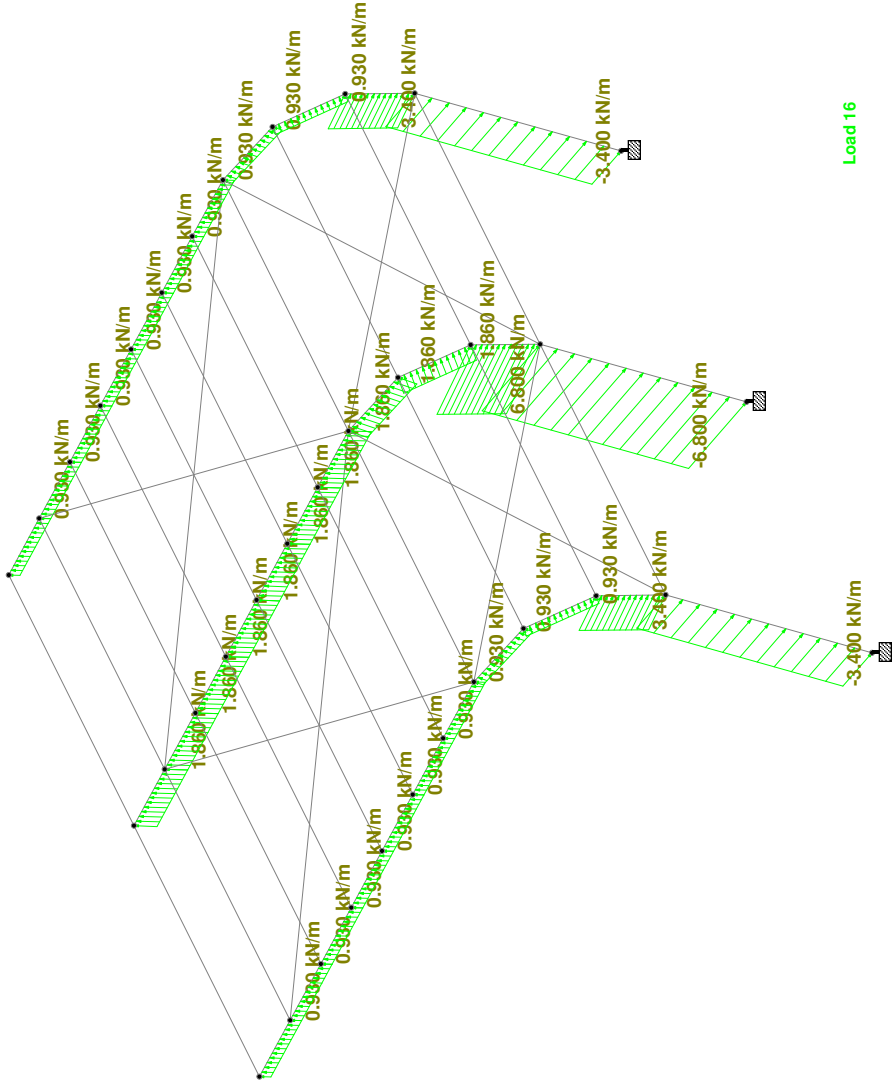
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RASHIDIYA B21: WL6



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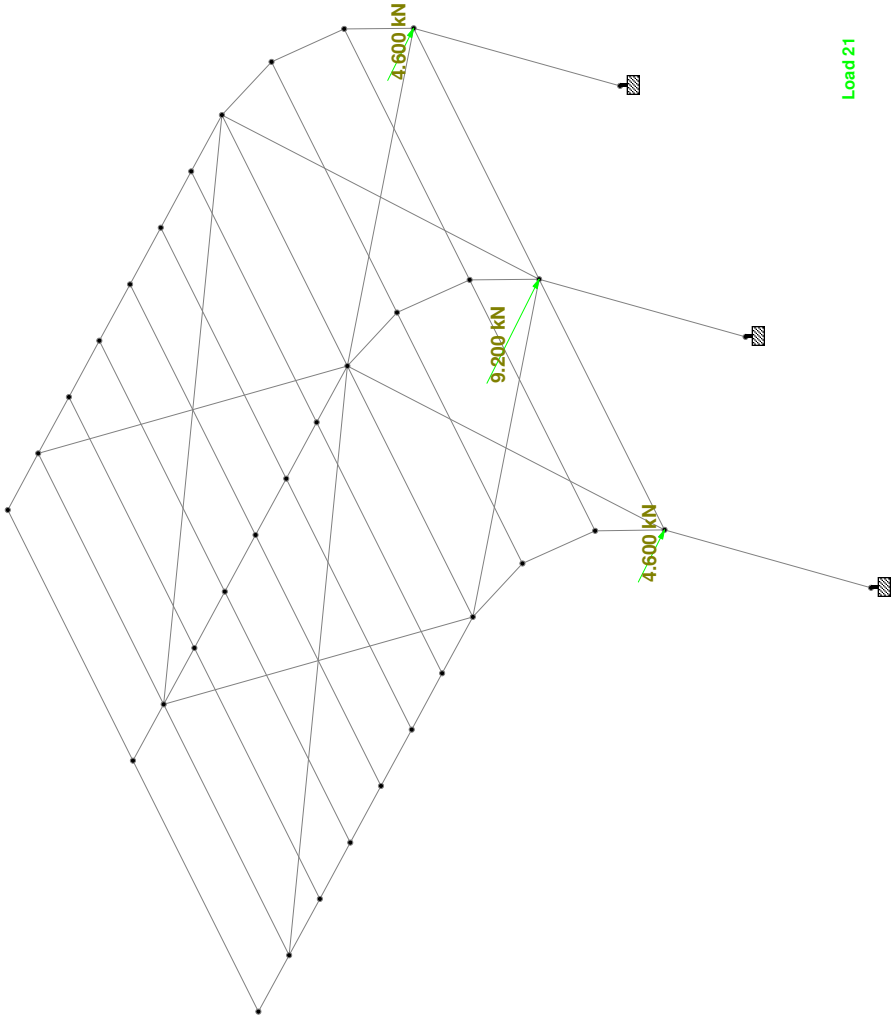
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RASHIDIYA B21: EQ1



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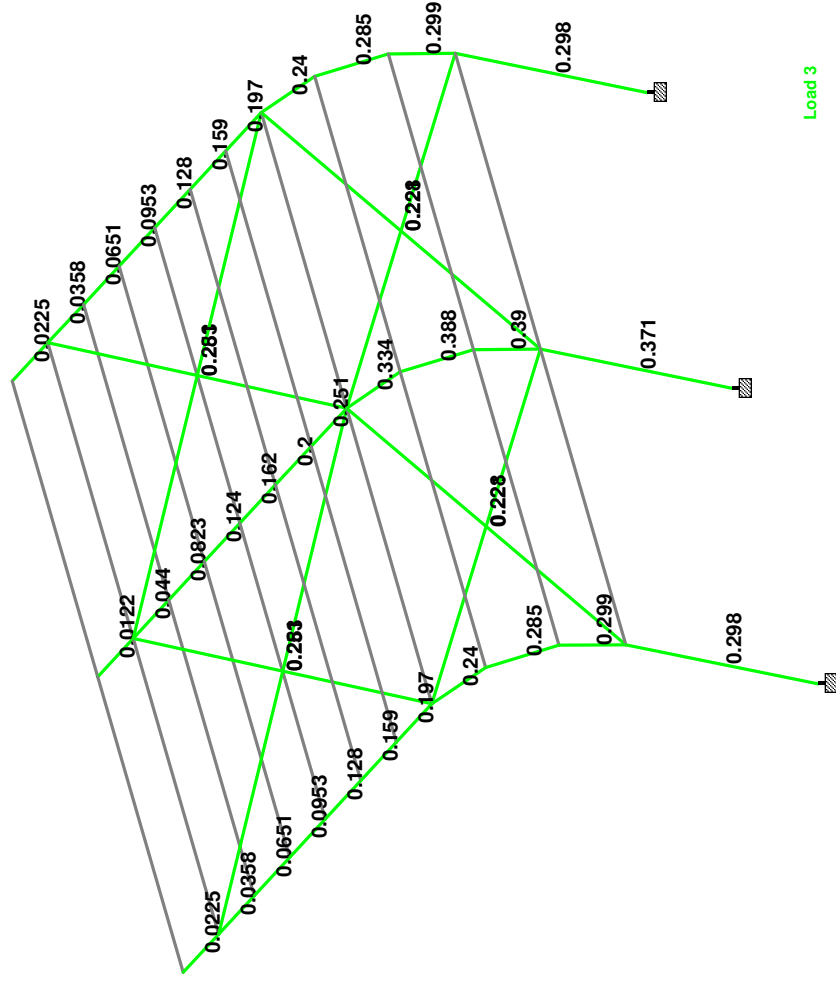
Client Dubai RTA

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File	Fuel Station-R3 Col550 T	Date/Time	21-Sep-2007 11:57
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File	Fuel Station-R3 Col550 T	Date/Time	21-Sep-2007 11:57
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Load 3

RASHIDIYA B21: DESIGN ANALYSIS RESULTS
***COLD-FORMED PURLINS UNCHECKED BY PROGRAM**

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*****
*
*          STAAD.Pro          *
*      Version  2006    Bld 1005.US      *
*      Proprietary Program of          *
*      Research Engineers, Intl.        *
*      Date=    SEP 21, 2007          *
*      Time=    11:58:54              *
*
*
*      USER ID: Meinhardt Infrastructure Pte Ltd  *
*****

```

```

1. STAAD SPACE
INPUT FILE: Fuel Station-R3 Col550 T1.STD
2. START JOB INFORMATION
3. ENGINEER DATE SEP 2007
4. JOB NAME DUBAI METRO PROJECT - RASHIDIYA B21
5. JOB CLIENT DUBAI RTA
6. JOB NO 068010.01
7. ENGINEER NAME CJA
8. JOB COMMENT *COLUMN SIZE INCREASED TO 550MM AS PER JV REQUEST
9. END JOB INFORMATION
10. INPUT WIDTH 79
11. UNIT METER KN
12. JOINT COORDINATES
13. 1 0 0 4; 2 0.921 3.062 4; 3 0.91 3.962 4; 4 0.386 4.693 4; 5 -0.463 4.993 4
14. 8 -6.759 5.213 4; 10 -5.85971 5.18158 4; 11 0 0 0; 12 0.921 3.062 0
15. 13 0.91 3.962 0; 14 0.386 4.693 0; 15 -0.463 4.993 0; 17 -5.85971 5.18158 0
16. 18 -6.759 5.213 0; 19 0 0 -4; 20 0.921 3.062 -4; 21 0.91 3.962 -4
17. 22 0.386 4.693 -4; 23 -0.463 4.993 -4; 25 -5.85971 5.18158 -4
18. 26 -6.759 5.213 -4; 27 -1.36245 5.02443 4; 28 -2.2619 5.05586 4
19. 29 -3.16136 5.08729 4; 30 -4.06081 5.11872 4; 31 -4.96026 5.15015 4
20. 32 -1.36245 5.02443 0; 33 -2.2619 5.05586 0; 34 -3.16136 5.08729 0
21. 35 -4.06081 5.11872 0; 36 -4.96026 5.15015 0; 37 -1.36245 5.02443 -4
22. 38 -2.2619 5.05586 -4; 39 -3.16136 5.08729 -4; 40 -4.06081 5.11872 -4
23. 41 -4.96026 5.15015 -4
24. MEMBER INCIDENCES
25. 1 1 2; 2 2 3; 3 3 4; 4 4 5; 5 5 27; 7 10 8; 8 11 12; 9 12 13; 10 13 14
26. 11 14 15; 12 15 32; 14 17 18; 15 19 20; 16 20 21; 17 21 22; 18 22 23; 19 23 37
27. 21 25 26; 22 5 12; 23 15 2; 24 15 20; 25 23 12; 26 5 17; 27 15 10; 28 15 25
28. 29 23 17; 30 27 28; 31 28 29; 32 29 30; 33 30 31; 34 31 10; 35 32 33; 36 33 34
29. 37 34 35; 38 35 36; 39 36 17; 40 37 38; 41 38 39; 42 39 40; 43 40 41; 44 41 25
30. 45 8 18; 46 10 17; 47 31 36; 48 30 35; 49 29 34; 50 28 33; 51 27 32; 52 5 15
31. 53 18 26; 54 17 25; 55 36 41; 56 35 40; 57 34 39; 58 33 38; 59 32 37; 60 15 23
32. 61 4 14; 62 3 13; 63 2 12; 64 14 22; 65 13 21; 66 12 20
33. START USER TABLE
34. TABLE 1
35. UNIT METER KN
36. PIPE
37. PIPE1
38. 0.012 0 0 0
39. END
40. DEFINE MATERIAL START
41. ISOTROPIC STEEL
42. E 2.05E+008
43. POISSON 0.3
44. DENSITY 76.8195
45. ALPHA 1.2E-005
46. DAMP 0.03
47. END DEFINE MATERIAL
48. MEMBER PROPERTY BRITISH
49. 1 TO 4 8 TO 11 15 TO 18 TAPERED 0.55 0.01 0.55 0.25 0.015 0.25 0.015
50. 19 TAPERED 0.55 0.01 0.514285 0.25 0.015 0.25 0.015
51. 7 14 21 TAPERED 0.33571 0.01 0.3 0.25 0.015 0.25 0.015
52. 22 TO 29 UPTABLE 1 PIPE1

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Friday, September 21, 2007, 11:59 AM

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54. 5 TAPERED 0.55 0.01 0.514285 0.25 0.015 0.25 0.015
55. 31 TAPERED 0.45476 0.01 0.395235 0.25 0.015 0.25 0.015
56. 32 TAPERED 0.395235 0.01 0.355552 0.25 0.015 0.25 0.015
57. 33 TAPERED 0.355552 0.01 0.339017 0.25 0.015 0.25 0.015
58. 34 TAPERED 0.339017 0.01 0.33571 0.25 0.015 0.25 0.015
59. 35 TAPERED 0.514285 0.01 0.45476 0.25 0.015 0.25 0.015
60. 12 TAPERED 0.55 0.01 0.514285 0.25 0.015 0.25 0.015
61. 36 TAPERED 0.45476 0.01 0.395235 0.25 0.015 0.25 0.015
62. 37 TAPERED 0.395235 0.01 0.355552 0.25 0.015 0.25 0.015
63. 38 TAPERED 0.355552 0.01 0.339017 0.25 0.015 0.25 0.015
64. 39 TAPERED 0.339017 0.01 0.33571 0.25 0.015 0.25 0.015
65. 40 TAPERED 0.514285 0.01 0.45476 0.25 0.015 0.25 0.015
66. 41 TAPERED 0.45476 0.01 0.395235 0.25 0.015 0.25 0.015
67. 42 TAPERED 0.395235 0.01 0.355552 0.25 0.015 0.25 0.015
68. 43 TAPERED 0.355552 0.01 0.339017 0.25 0.015 0.25 0.015
69. 44 TAPERED 0.339017 0.01 0.33571 0.25 0.015 0.25 0.015
70. MEMBER PROPERTY LYSAGHT
71. 45 TO 66 TABLE ST C20024
72. CONSTANTS
73. MATERIAL STEEL ALL
74. SUPPORTS
75. 1 11 19 FIXED
76. MEMBER CABLE
77. 22 TO 29 TENSION 5
78. MEMBER TENSION
79. 22 TO 29
80. LOAD 1 SELFWEIGHT
81. SELFWEIGHT Y -1
82. LOAD 2 SDL
83. MEMBER LOAD
84. 1 TO 5 7 15 TO 19 21 30 TO 34 40 TO 44 UNI GY -1.
85. 8 TO 12 14 35 TO 39 UNI GY -2
86. LOAD 3 LL
87. MEMBER LOAD
88. 3 TO 5 7 17 TO 19 21 30 TO 34 40 TO 44 UNI GY -1.2
89. 10 TO 12 14 35 TO 39 UNI GY -2.4
90. LOAD 4 PH (NOTIONAL HORIZONTAL LOAD)
91. JOINT LOAD
92. 2 20 FX 0.25
93. 12 FX 0.5
94. LOAD 11 WL1 (INTERNAL SUCTION, WIND ANGLE 0)
95. MEMBER LOAD
96. 2 16 UNI Y 3.25
97. 9 UNI Y 6.5
98. 3 TO 5 7 17 TO 19 21 30 TO 34 40 TO 44 UNI Y 3.81
99. 10 TO 12 14 35 TO 39 UNI Y 7.62
100. 1 15 UNI Y -3.25
101. 8 UNI Y -6.5
102. LOAD 12 WL2 (INTERNAL PRESSURE, WIND ANGLE 0)
103. MEMBER LOAD
104. 1 15 UNI Y -3.25
105. 8 UNI Y -6.5
106. 3 TO 5 7 17 TO 19 21 30 TO 34 40 TO 44 UNI Y 1.99
107. 10 TO 12 14 35 TO 39 UNI Y 3.97
108. 2 16 UNI Y 3.25
109. 9 UNI Y 6.5
110. LOAD 13 WL3 (INTERNAL SUCTION, WIND ANGLE 90)
111. MEMBER LOAD
112. 2 16 UNI Y 0.42
113. 9 UNI Y 0.84
114. 3 TO 5 7 17 TO 19 21 30 TO 34 40 TO 44 UNI Y -2.06
115. 10 TO 12 14 35 TO 39 UNI Y -4.12
116. 1 15 UNI Y -0.42
117. 8 UNI Y -0.84
118. LOAD 14 WL4 (INTERNAL PRESSURE, WIND ANGLE 90)
119. MEMBER LOAD
120. 1 15 UNI Y -2.98
121. 8 UNI Y -5.96
122. 2 16 UNI Y 2.98

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Friday, September 21, 2007, 11:59 AM

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124. 3 TO 5 7 17 TO 19 21 30 TO 34 40 TO 44 UNI Y 2.73
125. 10 TO 12 14 35 TO 39 UNI Y 5.46
126. LOAD 15 WL5 (INTERNAL SUCTION, WIND ANGLE 0, REVERSE WIND)
127. MEMBER LOAD
128. 1 15 UNI Y 3.2
129. 8 UNI Y 6.8
130. 2 16 UNI Y -3.2
131. 9 UNI Y -6.8
132. 3 TO 5 7 17 TO 19 21 30 TO 34 40 TO 44 UNI Y 1.51
133. 10 TO 12 14 35 TO 39 UNI Y 3.02
134. LOAD 16 WL6 (INTERNAL PRESSURE, WIND ANGLE 0, REVERSE WIND)
135. MEMBER LOAD
136. 1 15 UNI Y -3.4
137. 8 UNI Y -6.8
138. 2 16 UNI Y 3.4
139. 9 UNI Y 6.8
140. 3 TO 5 7 17 TO 19 21 30 TO 34 40 TO 44 UNI Y 0.93
141. 10 TO 12 14 35 TO 39 UNI Y 1.86
142. LOAD 21 EQ1 (EARTHQUAKE LOAD)
143. JOINT LOAD
144. 2 20 FX 4.6
145. 12 FX 9.2
146. LOAD COMB 1001 1.4DL + 1.6LL
147. 1 1.4 2 1.4 3 1.6
148. LOAD COMB 1002 1.4DL + 1.6LL + 1.0PH
149. 1 1.4 2 1.4 3 1.6 4 1.0
150. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
151. 1 1.2 2 1.2 3 1.2 11 1.2
152. LOAD COMB 1111 1.4DL + 1.4WL1
153. 1 1.4 2 1.4 11 1.4
154. LOAD COMB 1211 1.0DL + 1.4WL1
155. 1 1.0 2 1.0 11 1.4
156. LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
157. 1 1.2 2 1.2 3 1.2 12 1.2
158. LOAD COMB 1112 1.4DL + 1.4WL2
159. 1 1.4 2 1.4 12 1.4
160. LOAD COMB 1212 1.0DL + 1.4WL2
161. 1 1.0 2 1.0 12 1.4
162. LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3
163. 1 1.2 2 1.2 3 1.2 13 1.2
164. LOAD COMB 1113 1.4DL + 1.4WL3
165. 1 1.4 2 1.4 13 1.4
166. LOAD COMB 1213 1.0DL + 1.4WL3
167. 1 1.0 2 1.0 13 1.4
168. LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
169. 1 1.2 2 1.2 3 1.2 14 1.2
170. LOAD COMB 1114 1.4DL + 1.4WL4
171. 1 1.4 2 1.4 14 1.4
172. LOAD COMB 1214 1.0DL + 1.4WL4
173. 1 1.0 2 1.0 14 1.4
174. LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ1
175. 1 1.2 2 1.2 3 0.5 21 1.0
176. LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ1
177. 1 1.2 2 1.2 3 0.5 21 -1.0
178. LOAD COMB 1221 0.9DL + 1.0EQ1
179. 1 0.9 2 0.9 21 1.0
180. LOAD COMB 1321 0.9DL - 1.0EQ1
181. 1 0.9 2 0.9 21 -1.0
182. *SLS
183. LOAD COMB 2001 1.0DL + 1.0LL
184. 1 1.0 2 1.0 3 1.0
185. LOAD COMB 2002 1.0DL + 0.5LL
186. 1 1.0 2 1.0 3 0.5
187. LOAD COMB 2011 DL + LL + WL1
188. 1 1.0 2 1.0 3 1.0 11 1.0
189. LOAD COMB 2012 DL + LL + WL2
190. 1 1.0 2 1.0 3 1.0 12 1.0
191. LOAD COMB 2013 DL + LL + WL3
192. 1 1.0 2 1.0 3 1.0 13 1.0

```

Friday, September 21, 2007, 11:59 AM

194. 1 1.0 2 1.0 3 1.0 14 1.0
 195. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

 NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 36/ 63/ 3
 ORIGINAL/FINAL BAND-WIDTH= 19/ 10/ 60 DOF
 TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 198
 SIZE OF STIFFNESS MATRIX = 12 DOUBLE KILO-WORDS
 REQRD/AVAIL. DISK SPACE = 12.3/ 340868.4 MB

**NOTE-Tension/Compression converged after 1 iterations, Case= 1

**NOTE-Tension/Compression converged after 1 iterations, Case= 2

**NOTE-Tension/Compression converged after 1 iterations, Case= 3

**NOTE-Tension/Compression converged after 1 iterations, Case= 4

**NOTE-Tension/Compression converged after 1 iterations, Case= 11

**NOTE-Tension/Compression converged after 1 iterations, Case= 12

**NOTE-Tension/Compression converged after 1 iterations, Case= 13

**NOTE-Tension/Compression converged after 1 iterations, Case= 14

**NOTE-Tension/Compression converged after 1 iterations, Case= 15

**NOTE-Tension/Compression converged after 1 iterations, Case= 16

**NOTE-Tension/Compression converged after 1 iterations, Case= 21

Friday, September 21, 2007, 11:59 AM

196. LOAD LIST 1 TO 4 11 TO 16 21 1001 1002 1011 TO 1014 1021 1111 TO 1114 1121 -
 197. 1211 TO 1214 1221 1321 2001 2002 2011 TO 2014
 198. PRINT SUPPORT REACTION LIST 1 TO 5 8 10 TO 15 17 TO 23 25 TO 41

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	1	0.32	12.77	0.12	0.21	-0.07	-21.82
	2	0.99	13.07	0.12	0.21	-0.07	-28.44
	3	1.30	10.72	0.12	0.20	-0.07	-37.40
	4	-0.30	-0.01	0.15	0.26	-0.08	0.98
	11	-24.06	-28.95	0.24	0.41	-0.13	179.64
	12	-19.22	-13.54	0.19	0.33	-0.10	109.02
	13	3.65	17.84	0.10	0.17	-0.05	-75.64
	14	-20.04	-20.08	0.21	0.36	-0.12	135.14
	15	9.83	-16.10	0.19	0.34	-0.11	26.46
	16	-17.06	-4.42	0.17	0.28	-0.09	69.53
	21	-5.25	0.18	0.15	0.25	-0.08	17.22
	1001	3.91	53.33	0.53	0.92	-0.29	-130.21
	1002	3.61	53.33	0.68	1.18	-0.38	-129.22
	1011	-25.74	9.14	0.72	1.24	-0.40	110.38
	1111	-31.85	-4.35	0.67	1.16	-0.37	181.13
	1211	-32.37	-14.68	0.58	0.99	-0.32	201.24
	1012	-19.93	27.63	0.67	1.15	-0.37	25.63
	1112	-25.07	17.23	0.61	1.06	-0.34	82.26
	1212	-25.59	6.89	0.51	0.89	-0.28	102.36
	1013	7.51	65.29	0.56	0.96	-0.31	-195.96
	1113	6.94	61.16	0.48	0.83	-0.27	-176.26
	1213	6.42	50.82	0.38	0.66	-0.21	-156.15
	1014	-20.91	19.78	0.69	1.19	-0.38	56.97
	1114	-26.22	8.06	0.64	1.10	-0.35	118.83
	1214	-26.74	-2.27	0.54	0.93	-0.30	138.94
	1021	-3.03	36.56	0.50	0.86	-0.28	-61.80
	1121	7.47	36.19	0.21	0.36	-0.12	-96.23
	1221	-4.07	23.44	0.37	0.63	-0.20	-28.02
	1321	6.43	23.07	0.07	0.13	-0.04	-62.45
	2001	2.61	36.56	0.36	0.63	-0.20	-87.66
	2002	1.96	31.20	0.30	0.53	-0.17	-68.96
	2011	-21.45	7.62	0.60	1.04	-0.33	91.98
	2012	-16.61	23.03	0.56	0.96	-0.31	21.35
	2013	6.26	54.41	0.46	0.80	-0.25	-163.30
	2014	-17.43	16.48	0.57	0.99	-0.32	47.48
11	1	-0.64	14.52	0.00	0.00	0.00	-20.49
	2	-1.98	22.65	0.00	0.00	0.00	-22.73
	3	-2.60	17.43	0.00	0.00	0.00	-31.76
	4	-0.40	0.01	0.00	0.00	0.00	1.09
	11	-22.45	-47.15	0.00	0.00	0.00	177.39
	12	-23.02	-22.07	0.00	0.00	0.00	113.31
	13	-3.65	29.04	0.00	0.00	0.00	-66.94
	14	-20.82	-32.73	0.00	0.00	0.00	135.79
	15	25.08	-26.13	0.00	0.00	0.00	8.14
	16	-24.41	-7.31	0.00	0.00	0.00	78.10
	21	-7.90	-0.37	0.00	0.00	0.00	21.91
	1001	-7.82	79.94	0.00	0.00	0.00	-111.33
	1002	-8.22	79.95	0.00	0.00	0.00	-110.23

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
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Friday, September 21, 2007, 11:59 AM

1111	-35.10	-13.97	0.00	0.00	0.00	187.84
1211	-34.05	-28.84	0.00	0.00	0.00	205.13
1012	-33.88	39.05	0.00	0.00	0.00	45.99
1112	-35.89	21.15	0.00	0.00	0.00	98.12
1212	-34.84	6.28	0.00	0.00	0.00	115.41
1013	-10.64	100.38	0.00	0.00	0.00	-170.31
1113	-8.78	92.70	0.00	0.00	0.00	-154.22
1213	-7.73	77.83	0.00	0.00	0.00	-136.94
1014	-31.24	26.25	0.00	0.00	0.00	72.97
1114	-32.81	6.22	0.00	0.00	0.00	129.60
1214	-31.76	-8.65	0.00	0.00	0.00	146.89
1021	-12.34	52.96	0.00	0.00	0.00	-45.83
1121	3.46	53.70	0.00	0.00	0.00	-89.66
1221	-10.26	33.09	0.00	0.00	0.00	-16.99
1321	5.55	33.83	0.00	0.00	0.00	-60.81
2001	-5.22	54.61	0.00	0.00	0.00	-74.98
2002	-3.92	45.89	0.00	0.00	0.00	-59.10
2011	-27.67	7.45	0.00	0.00	0.00	102.41
2012	-28.23	32.54	0.00	0.00	0.00	38.33
2013	-8.87	83.65	0.00	0.00	0.00	-141.92
2014	-26.03	21.87	0.00	0.00	0.00	60.81
19 1	0.32	12.77	-0.12	-0.21	0.07	-21.82
2	0.99	13.07	-0.12	-0.21	0.07	-28.44
3	1.30	10.72	-0.12	-0.20	0.07	-37.40
4	-0.30	-0.01	-0.15	-0.26	0.08	0.98
11	-24.06	-28.95	-0.24	-0.41	0.13	179.64
12	-19.22	-13.54	-0.19	-0.33	0.10	109.02
13	3.65	17.84	-0.10	-0.17	0.05	-75.64
14	-20.04	-20.08	-0.21	-0.36	0.12	135.14
15	9.83	-16.10	-0.19	-0.34	0.11	26.46
16	-17.06	-4.42	-0.17	-0.28	0.09	69.53
21	-5.25	0.18	-0.15	-0.25	0.08	17.22
1001	3.91	53.33	-0.53	-0.92	0.29	-130.21
1002	3.61	53.33	-0.68	-1.18	0.38	-129.22
1011	-25.74	9.14	-0.72	-1.24	0.40	110.38
1111	-31.85	-4.35	-0.67	-1.16	0.37	181.13
1211	-32.37	-14.68	-0.58	-0.99	0.32	201.24
1012	-19.93	27.63	-0.67	-1.15	0.37	25.63
1112	-25.07	17.23	-0.61	-1.06	0.34	82.26
1212	-25.59	6.89	-0.51	-0.89	0.28	102.36
1013	7.51	65.29	-0.56	-0.96	0.31	-195.96
1113	6.94	61.16	-0.48	-0.83	0.27	-176.26
1213	6.42	50.82	-0.38	-0.66	0.21	-156.15
1014	-20.91	19.78	-0.69	-1.19	0.38	56.97
1114	-26.22	8.06	-0.64	-1.10	0.35	118.83
1214	-26.74	-2.27	-0.54	-0.93	0.30	138.94
1021	-3.03	36.56	-0.50	-0.86	0.28	-61.80

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1121	7.47	36.19	-0.21	-0.36	0.12	-96.23	
1221	-4.07	23.44	-0.37	-0.63	0.20	-28.02	
1321	6.43	23.07	-0.07	-0.13	0.04	-62.45	
2001	2.61	36.56	-0.36	-0.63	0.20	-87.66	
2002	1.96	31.20	-0.30	-0.53	0.17	-68.96	
2011	-21.45	7.62	-0.60	-1.04	0.33	91.98	
2012	-16.61	23.03	-0.56	-0.96	0.31	21.35	
2013	6.26	54.41	-0.46	-0.80	0.25	-163.30	
2014	-17.43	16.48	-0.57	-0.99	0.32	47.48	

***** END OF LATEST ANALYSIS RESULT *****

Friday, September 21, 2007, 11:59 AM

199. PRINT FORCE ENVELOPE NSECTION 5 LIST 1 TO 5 7 TO 12 14 TO 19 21 TO 66

MEMBER FORCE ENVELOPE

ALL UNITS ARE KN METE

MEMB	DISTANCE		FY	LD	MZ	LD	FZ	LD	MY	LD
1	0.00	MAX	29.25	1111	201.24	1211	0.72	1011	-0.14	1321
		MIN	-14.05	15	-195.96	1013	0.07	1321	-1.30	1011
	0.64	MAX	25.83	1111	185.16	1211	0.72	1011	-0.09	1321
		MIN	-12.01	15	-203.14	1013	0.07	1321	-0.84	1011
	1.28	MAX	22.41	1111	171.18	1211	0.72	1011	-0.04	1321
		MIN	-9.96	15	-209.84	1013	0.07	1321	-0.38	1011
	1.92	MAX	18.99	1111	159.30	1211	0.72	1011	0.08	1011
		MIN	-7.91	15	-216.06	1013	0.07	1321	0.00	1321
	2.56	MAX	15.57	1111	152.67	11	0.72	1011	0.54	1011
		MIN	-5.87	15	-221.78	1013	0.07	1321	0.05	1321
	3.20	MAX	12.63	1011	149.25	11	0.72	1011	1.00	1011
		MIN	-3.82	15	-227.03	1013	0.07	1321	0.10	1321

MAX/MIN FORCE VALUES FOR MEMB 1, AMONGST ALL SECT LOCATIONS										
	FY/	DIST	LD	MZ/	DIST	LD				
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD	
MAX.	29.25	0.00	1111	201.24	0.00	1211				
	0.72	0.00	1011	1.00	3.20	1011	64.68 C	0.00	1013	
MIN.	-14.05	0.00	15	-227.03	3.20	1013				
	0.07	3.20	1321	-1.30	0.00	1011	34.65 T	3.20	11	

2	0.00	MAX	5.24	1213	227.03	1013	2.47	1011	-0.19	13
		MIN	-24.24	1111	-149.25	11	0.35	13	-1.43	1011
	0.18	MAX	5.35	1213	226.17	1013	2.47	1011	-0.12	13
		MIN	-23.43	1111	-146.31	11	0.35	13	-0.98	1011
	0.36	MAX	5.45	1213	225.31	1013	2.47	1011	-0.06	1321
		MIN	-22.61	1111	-143.49	11	0.35	13	-0.54	1011
	0.54	MAX	5.55	1213	224.42	1013	2.47	1011	0.02	1321
		MIN	-21.80	1111	-140.77	11	0.35	13	-0.10	1111
	0.72	MAX	5.65	1213	223.53	1013	2.47	1011	0.40	1002
		MIN	-20.99	1111	-138.15	11	0.35	13	0.06	16
	0.90	MAX	5.75	1213	222.61	1013	2.47	1011	0.85	1002
		MIN	-20.18	1111	-135.64	11	0.35	13	0.12	13

MAX/MIN FORCE VALUES FOR MEMB 2, AMONGST ALL SECT LOCATIONS										
	FY/	DIST	LD	MZ/	DIST	LD				
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD	
MAX.	5.75	0.90	1213	227.03	0.00	1013				
	2.47	0.00	1011	0.85	0.90	1002	64.14 C	0.00	1013	
MIN.	-24.24	0.00	1111	-149.25	0.00	11				
	0.35	0.90	13	-1.43	0.00	1011	28.80 T	0.90	11	

3	0.00	MAX	39.50	1013	222.61	1013	0.94	1013	0.38	1002
		MIN	-27.71	11	-135.64	11	-0.24	11	0.05	16
	0.18	MAX	38.66	1013	215.58	1013	0.94	1013	0.53	1002
		MIN	-27.02	11	-130.72	11	-0.24	11	0.01	11

Friday, September 21, 2007, 11:59 AM

0.54	MAX	36.97	1013	201.98	1013	0.94	1013	0.85	1013
	MIN	-25.65	11	-121.24	11	-0.24	11	-0.08	11
0.72	MAX	36.12	1013	195.40	1013	0.94	1013	1.02	1013
	MIN	-24.97	11	-116.69	11	-0.24	11	-0.12	11
0.90	MAX	35.28	1013	188.98	1013	0.94	1013	1.18	1013
	MIN	-24.28	11	-112.26	11	-0.24	11	-0.16	11

MAX/MIN FORCE VALUES FOR MEMB 3, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	39.50	0.00	1013	222.61	0.00	1013			
	0.94	0.00	1013	1.18	0.90	1013	47.69 C	0.00	1013
MIN.	-27.71	0.00	11	-135.64	0.00	11			
	-0.24	0.90	11	-0.16	0.90	11	15.83 T	0.90	11

4	0.00	MAX	54.26	1013	188.98	1013	-0.70	1321	0.91	1013
		MIN	-28.92	11	-112.26	11	-5.60	1002	-0.15	11
	0.18	MAX	53.17	1013	179.31	1013	-0.70	1321	0.09	1321
		MIN	-28.23	11	-107.12	11	-5.60	1002	-0.63	1111
	0.36	MAX	52.08	1013	169.83	1013	-0.70	1321	-0.04	1321
		MIN	-27.55	11	-102.09	11	-5.60	1002	-1.55	1011
	0.54	MAX	50.99	1013	160.55	1013	-0.70	1321	-0.16	1321
		MIN	-26.86	11	-97.19	11	-5.60	1002	-2.54	1011
	0.72	MAX	49.89	1013	151.47	1013	-0.70	1321	-0.29	1321
		MIN	-26.17	11	-92.42	11	-5.60	1002	-3.54	1011
	0.90	MAX	48.80	1013	142.58	1013	-0.70	1321	-0.42	1321
		MIN	-25.49	11	-87.77	11	-5.60	1002	-4.53	1011

MAX/MIN FORCE VALUES FOR MEMB 4, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	54.26	0.00	1013	188.98	0.00	1013			
	-0.70	0.00	1321	0.91	0.00	1013	17.62 C	0.00	1002
MIN.	-28.92	0.00	11	-112.26	0.00	11			
	-5.60	0.90	1002	-4.53	0.90	1011	0.08 C	0.90	15

5	0.00	MAX	43.70	1013	142.58	1013	7.30	1002	-0.39	1321
		MIN	-26.75	11	-87.77	11	0.89	1321	-4.38	1011
	0.18	MAX	42.57	1013	134.82	1013	7.30	1002	-0.22	1321
		MIN	-26.07	11	-83.01	11	0.89	1321	-3.09	1011
	0.36	MAX	41.44	1013	127.25	1013	7.30	1002	-0.06	1321
		MIN	-25.38	11	-78.38	11	0.89	1321	-1.80	1111
	0.54	MAX	40.32	1013	119.90	1013	7.30	1002	0.19	1013
		MIN	-24.70	11	-73.87	11	0.89	1321	-0.62	1211
	0.72	MAX	39.19	1013	112.74	1013	7.30	1002	1.30	1013
		MIN	-24.01	11	-69.49	11	0.89	1321	-0.09	11
	0.90	MAX	38.07	1013	105.79	1013	7.30	1002	2.60	1002
		MIN	-23.32	11	-65.23	11	0.89	1321	0.27	11

MAX/MIN FORCE VALUES FOR MEMB 5, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	43.70	0.00	1013	142.58	0.00	1013			
	7.30	0.00	1002	2.60	0.90	1002	21.71 C	0.00	1011
MIN.	-26.75	0.00	11	-87.77	0.00	11			
	0.89	0.90	1321	-4.38	0.00	1011	2.48 C	0.90	13

Friday, September 21, 2007, 11:59 AM

7	0.00	MAX	6.59	1013	3.47	1013	3.37	1011	-0.32	1321
		MIN	-4.10	11	-2.15	11	0.33	1321	-2.97	1011
	0.18	MAX	5.49	1013	2.38	1013	3.37	1011	-0.26	1321
		MIN	-3.42	11	-1.47	11	0.33	1321	-2.36	1011
	0.36	MAX	4.40	1013	1.49	1013	3.37	1011	-0.20	1321
		MIN	-2.73	11	-0.92	11	0.33	1321	-1.75	1011
	0.54	MAX	3.31	1013	0.80	1013	3.37	1011	-0.14	1321
		MIN	-2.05	11	-0.49	11	0.33	1321	-1.14	1011
	0.72	MAX	2.22	1013	0.30	1013	3.37	1011	-0.08	1321
		MIN	-1.36	11	-0.18	11	0.33	1321	-0.56	1002
	0.90	MAX	1.13	1013	0.00	1013	3.37	1011	0.10	1211
		MIN	-0.68	11	0.00	1211	0.33	1321	-0.06	1013

MAX/MIN FORCE VALUES FOR MEMB					7, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD		MZ/	DIST	LD			
	FZ	DIST	LD		MY	DIST	LD	FX	DIST	LD

MAX.	6.59	0.00	1013		3.47	0.00	1013			
	3.37	0.00	1011		0.10	0.90	1211	0.15 C	0.00	1002
MIN.	-4.10	0.00	11		-2.15	0.00	11			
	0.33	0.90	1321		-2.97	0.00	1011	0.00 T	0.90	13

8	0.00	MAX	43.69	1012	205.13	1211	0.00	2014	0.00	2014
		MIN	-31.55	15	-170.31	1013	0.00	2014	0.00	2014
	0.64	MAX	38.05	1012	191.62	1211	0.00	2014	0.00	2014
		MIN	-27.20	15	-194.89	1013	0.00	2014	0.00	2014
	1.28	MAX	36.50	1013	182.19	1211	0.00	2014	0.00	2014
		MIN	-22.85	15	-218.65	1013	0.00	2014	0.00	2014
	1.92	MAX	35.19	1013	176.83	1211	0.00	2014	0.00	2014
		MIN	-18.50	15	-241.57	1013	0.00	2014	0.00	2014
	2.56	MAX	33.89	1013	178.40	11	0.00	2014	0.00	2014
		MIN	-14.15	15	-263.66	1013	0.00	2014	0.00	2014
	3.20	MAX	32.59	1013	185.30	11	0.00	2014	0.00	2014
		MIN	-12.87	11	-284.92	1013	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB					8, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD		MZ/	DIST	LD			
	FZ	DIST	LD		MY	DIST	LD	FX	DIST	LD

MAX.	43.69	0.00	1012		205.13	0.00	1211			
	0.00	0.00	1		0.00	0.00	1	93.06 C	0.00	1013
MIN.	-31.55	0.00	15		-284.92	3.20	1013			
	0.00	3.20	2014		0.00	3.20	2014	51.62 T	3.20	11

9	0.00	MAX	1.00	15	284.92	1013	0.00	2014	0.00	2014
		MIN	-23.17	1011	-185.30	11	0.00	2014	0.00	2014
	0.18	MAX	-0.22	15	288.04	1013	0.00	2014	0.00	2014
		MIN	-21.77	1011	-184.06	11	0.00	2014	0.00	2014
	0.36	MAX	-1.45	15	291.13	1013	0.00	2014	0.00	2014
		MIN	-20.37	1011	-183.03	11	0.00	2014	0.00	2014
	0.54	MAX	-1.53	21	294.19	1013	0.00	2014	0.00	2014
		MIN	-20.12	1002	-182.20	11	0.00	2014	0.00	2014
	0.72	MAX	-1.53	21	297.22	1013	0.00	2014	0.00	2014
		MIN	-20.13	1002	-181.59	11	0.00	2014	0.00	2014
	0.90	MAX	-0.90	16	300.22	1013	0.00	2014	0.00	2014
		MIN	-20.14	1002	-181.19	11	0.00	2014	0.00	2014

Friday, September 21, 2007, 11:59 AM

	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	1.00	0.00	15	300.22	0.90	1013			
	0.00	0.00	1	0.00	0.00	1	103.50 C	0.00	1013
MIN.	-23.17	0.00	1011	-185.30	0.00	11			
	0.00	0.90	2014	0.00	0.90	2014	46.93 T	0.90	11

10	0.00	MAX	43.86	1013	300.22	1013	0.00	2014	0.00	2014
		MIN	-28.17	11	-181.19	11	0.00	2014	0.00	2014
	0.18	MAX	42.30	1013	292.47	1013	0.00	2014	0.00	2014
		MIN	-26.80	11	-176.25	11	0.00	2014	0.00	2014
	0.36	MAX	40.73	1013	285.00	1013	0.00	2014	0.00	2014
		MIN	-25.43	11	-171.55	11	0.00	2014	0.00	2014
	0.54	MAX	39.17	1013	277.82	1013	0.00	2014	0.00	2014
		MIN	-24.06	11	-167.10	11	0.00	2014	0.00	2014
	0.72	MAX	37.60	1013	270.91	1013	0.00	2014	0.00	2014
		MIN	-22.69	11	-162.89	11	0.00	2014	0.00	2014
	0.90	MAX	36.04	1013	264.29	1013	0.00	2014	0.00	2014
		MIN	-21.32	11	-158.94	11	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB 10, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	43.86	0.00	1013	300.22	0.00	1013			
	0.00	0.00	1	0.00	0.00	1	91.60 C	0.00	1013
MIN.	-28.17	0.00	11	-181.19	0.00	11			
	0.00	0.90	2014	0.00	0.90	2014	37.44 T	0.90	11

11	0.00	MAX	79.37	1013	264.29	1013	0.00	2014	0.00	2014
		MIN	-39.02	11	-158.94	11	0.00	2014	0.00	2014
	0.18	MAX	77.38	1013	250.17	1013	0.00	2014	0.00	2014
		MIN	-37.65	11	-152.03	11	0.00	2014	0.00	2014
	0.36	MAX	75.40	1013	236.42	1013	0.00	2014	0.00	2014
		MIN	-36.27	11	-145.38	11	0.00	2014	0.00	2014
	0.54	MAX	73.41	1013	223.02	1013	0.00	2014	0.00	2014
		MIN	-34.90	11	-138.97	11	0.00	2014	0.00	2014
	0.72	MAX	71.42	1013	209.98	1013	0.00	2014	0.00	2014
		MIN	-33.53	11	-132.80	11	0.00	2014	0.00	2014
	0.90	MAX	69.44	1013	197.29	1013	0.00	2014	0.00	2014
		MIN	-32.16	11	-126.89	11	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB 11, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	79.37	0.00	1013	264.29	0.00	1013			
	0.00	0.00	1	0.00	0.00	1	50.55 C	0.00	1013
MIN.	-39.02	0.00	11	-158.94	0.00	11			
	0.00	0.90	2014	0.00	0.90	2014	18.50 T	0.90	11

12	0.00	MAX	65.73	1013	197.29	1013	0.00	2014	0.00	2014
		MIN	-42.50	11	-126.89	11	0.00	2014	0.00	2014
	0.18	MAX	63.68	1013	185.65	1013	0.00	2014	0.00	2014
		MIN	-41.13	11	-119.36	11	0.00	2014	0.00	2014
	0.36	MAX	61.63	1013	174.37	1013	0.00	2014	0.00	2014
		MIN	-39.76	11	-112.08	11	0.00	2014	0.00	2014
	0.54	MAX	59.58	1013	163.46	1013	0.00	2014	0.00	2014

Friday, September 21, 2007, 11:59 AM

0.72	MAX	57.53	1013	152.92	1013	0.00	2014	0.00	2014
	MIN	-37.02	11	-98.26	11	0.00	2014	0.00	2014
0.90	MAX	55.49	1013	142.75	1013	0.00	2014	0.00	2014
	MIN	-35.64	11	-91.72	11	0.00	2014	0.00	2014

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| MAX/MIN FORCE VALUES FOR MEMB      12, AMONGST ALL SECT LOCATIONS
|      FY/   DIST   LD      MZ/   DIST   LD
|      FZ    DIST   LD      MY    DIST   LD      FX      DIST   LD
|
| MAX.      65.73   0.00 1013      197.29   0.00 1013
|           0.00   0.00   1       0.00   0.00   1       47.93 C   0.00 1002
| MIN.     -42.50   0.00  11      -126.89   0.00  11
|           0.00   0.90 2014       0.00   0.90 2014       1.40 C   0.90  11
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14	0.00	MAX	8.47	1013	3.09	1013	0.00	2014	0.00	2014
		MIN	-5.50	11	-1.87	11	0.00	2014	0.00	2014
	0.18	MAX	6.46	1013	1.75	1013	0.00	2014	0.00	2014
		MIN	-4.13	11	-1.00	11	0.00	2014	0.00	2014
	0.36	MAX	4.44	1013	0.77	1013	0.00	2014	0.00	2014
		MIN	-2.76	11	-0.38	11	0.00	2014	0.00	2014
	0.54	MAX	2.43	1013	0.18	1001	0.00	2014	0.00	2014
		MIN	-1.39	11	-0.01	11	0.00	2014	0.00	2014
	0.72	MAX	0.50	1001	0.20	1111	0.00	2014	0.00	2014
		MIN	-0.02	11	-0.11	1013	0.00	2014	0.00	2014
	0.90	MAX	1.75	1211	0.00	1211	0.00	2014	0.00	2014
		MIN	-1.59	1013	0.00	1013	0.00	2014	0.00	2014

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|-----|
| MAX/MIN FORCE VALUES FOR MEMB      14, AMONGST ALL SECT LOCATIONS
|      FY/   DIST   LD      MZ/   DIST   LD
|      FZ    DIST   LD      MY    DIST   LD      FX      DIST   LD
|
| MAX.      8.47   0.00 1013      3.09   0.00 1013
|           0.00   0.00   1       0.00   0.00   1       0.26 C   0.00 1001
| MIN.     -5.50   0.00  11      -1.87   0.00  11
|           0.00   0.90 2014       0.00   0.90 2014       0.04 T   0.90 1211
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15	0.00	MAX	29.25	1111	201.24	1211	-0.07	1321	1.30	1011
		MIN	-14.05	15	-195.96	1013	-0.72	1011	0.14	1321
	0.64	MAX	25.83	1111	185.16	1211	-0.07	1321	0.84	1011
		MIN	-12.01	15	-203.14	1013	-0.72	1011	0.09	1321
	1.28	MAX	22.41	1111	171.18	1211	-0.07	1321	0.38	1011
		MIN	-9.96	15	-209.84	1013	-0.72	1011	0.04	1321
	1.92	MAX	18.99	1111	159.30	1211	-0.07	1321	0.00	1321
		MIN	-7.91	15	-216.06	1013	-0.72	1011	-0.08	1011
	2.56	MAX	15.57	1111	152.67	11	-0.07	1321	-0.05	1321
		MIN	-5.87	15	-221.78	1013	-0.72	1011	-0.54	1011
	3.20	MAX	12.63	1011	149.25	11	-0.07	1321	-0.10	1321
		MIN	-3.82	15	-227.03	1013	-0.72	1011	-1.00	1011

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| MAX/MIN FORCE VALUES FOR MEMB      15, AMONGST ALL SECT LOCATIONS
|      FY/   DIST   LD      MZ/   DIST   LD
|      FZ    DIST   LD      MY    DIST   LD      FX      DIST   LD
|
| MAX.      29.25   0.00 1111      201.24   0.00 1211
|           -0.07   0.00 1321       1.30   0.00 1011       64.68 C   0.00 1013
| MIN.     -14.05   0.00  15      -227.03   3.20 1013
|           -0.72   3.20 1011       -1.00   3.20 1011       34.65 T   3.20  11
|-----|

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Friday, September 21, 2007, 11:59 AM

16	0.00	MAX	5.24	1213	227.03	1013	-0.35	13	1.43	1011
		MIN	-24.24	1111	-149.25	11	-2.47	1011	0.19	13
	0.18	MAX	5.35	1213	226.17	1013	-0.35	13	0.98	1011
		MIN	-23.43	1111	-146.31	11	-2.47	1011	0.12	13
	0.36	MAX	5.45	1213	225.31	1013	-0.35	13	0.54	1011
		MIN	-22.61	1111	-143.49	11	-2.47	1011	0.06	1321
	0.54	MAX	5.55	1213	224.42	1013	-0.35	13	0.10	1111
		MIN	-21.80	1111	-140.77	11	-2.47	1011	-0.02	1321
	0.72	MAX	5.65	1213	223.53	1013	-0.35	13	-0.06	16
		MIN	-20.99	1111	-138.15	11	-2.47	1011	-0.40	1002
	0.90	MAX	5.75	1213	222.61	1013	-0.35	13	-0.12	13
		MIN	-20.18	1111	-135.64	11	-2.47	1011	-0.85	1002

	MAX/MIN FORCE VALUES FOR MEMB				16, AMONGST ALL SECT LOCATIONS					
		FY/	DIST	LD	MZ/	DIST	LD			
		FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
	MAX.	5.75	0.90	1213	227.03	0.00	1013			
		-0.35	0.00	13	1.43	0.00	1011	64.14 C	0.00	1013
	MIN.	-24.24	0.00	1111	-149.25	0.00	11			
		-2.47	0.90	1011	-0.85	0.90	1002	28.80 T	0.90	11

17	0.00	MAX	39.50	1013	222.61	1013	0.24	11	-0.05	16
		MIN	-27.71	11	-135.64	11	-0.94	1013	-0.38	1002
	0.18	MAX	38.66	1013	215.58	1013	0.24	11	-0.01	11
		MIN	-27.02	11	-130.72	11	-0.94	1013	-0.53	1002
	0.36	MAX	37.81	1013	208.70	1013	0.24	11	0.03	11
		MIN	-26.34	11	-125.92	11	-0.94	1013	-0.69	1002
	0.54	MAX	36.97	1013	201.98	1013	0.24	11	0.08	11
		MIN	-25.65	11	-121.24	11	-0.94	1013	-0.85	1013
	0.72	MAX	36.12	1013	195.40	1013	0.24	11	0.12	11
		MIN	-24.97	11	-116.69	11	-0.94	1013	-1.02	1013
	0.90	MAX	35.28	1013	188.98	1013	0.24	11	0.16	11
		MIN	-24.28	11	-112.26	11	-0.94	1013	-1.18	1013

	MAX/MIN FORCE VALUES FOR MEMB				17, AMONGST ALL SECT LOCATIONS					
		FY/	DIST	LD	MZ/	DIST	LD			
		FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
	MAX.	39.50	0.00	1013	222.61	0.00	1013			
		0.24	0.00	11	0.16	0.90	11	47.69 C	0.00	1013
	MIN.	-27.71	0.00	11	-135.64	0.00	11			
		-0.94	0.90	1013	-1.18	0.90	1013	15.83 T	0.90	11

18	0.00	MAX	54.26	1013	188.98	1013	5.60	1002	0.15	11
		MIN	-28.92	11	-112.26	11	0.70	1321	-0.91	1013
	0.18	MAX	53.17	1013	179.31	1013	5.60	1002	0.63	1111
		MIN	-28.23	11	-107.12	11	0.70	1321	-0.09	1321
	0.36	MAX	52.08	1013	169.83	1013	5.60	1002	1.55	1011
		MIN	-27.55	11	-102.09	11	0.70	1321	0.04	1321
	0.54	MAX	50.99	1013	160.55	1013	5.60	1002	2.54	1011
		MIN	-26.86	11	-97.19	11	0.70	1321	0.16	1321
	0.72	MAX	49.89	1013	151.47	1013	5.60	1002	3.54	1011
		MIN	-26.17	11	-92.42	11	0.70	1321	0.29	1321
	0.90	MAX	48.80	1013	142.58	1013	5.60	1002	4.53	1011
		MIN	-25.49	11	-87.77	11	0.70	1321	0.42	1321

	MAX/MIN FORCE VALUES FOR MEMB				18, AMONGST ALL SECT LOCATIONS					

Friday, September 21, 2007, 11:59 AM

	MAX.	54.26	0.00	1013	188.98	0.00	1013		
		5.60	0.00	1002	4.53	0.90	1011	17.62 C	0.00 1002
	MIN.	-28.92	0.00	11	-112.26	0.00	11		
		0.70	0.90	1321	-0.91	0.00	1013	0.08 C	0.90 15

19	0.00	MAX	43.70	1013	142.58	1013	-0.89	1321	4.38	1011
		MIN	-26.75	11	-87.77	11	-7.30	1002	0.39	1321
	0.18	MAX	42.57	1013	134.82	1013	-0.89	1321	3.09	1011
		MIN	-26.07	11	-83.01	11	-7.30	1002	0.22	1321
	0.36	MAX	41.44	1013	127.25	1013	-0.89	1321	1.80	1111
		MIN	-25.38	11	-78.38	11	-7.30	1002	0.06	1321
	0.54	MAX	40.32	1013	119.90	1013	-0.89	1321	0.62	1211
		MIN	-24.70	11	-73.87	11	-7.30	1002	-0.19	1013
	0.72	MAX	39.19	1013	112.74	1013	-0.89	1321	0.09	11
		MIN	-24.01	11	-69.49	11	-7.30	1002	-1.30	1013
	0.90	MAX	38.07	1013	105.79	1013	-0.89	1321	-0.27	11
		MIN	-23.32	11	-65.23	11	-7.30	1002	-2.60	1002

	MAX/MIN FORCE VALUES FOR MEMB				19, AMONGST ALL SECT LOCATIONS					
		FY/	DIST	LD	MZ/	DIST	LD			
		FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
	MAX.	43.70	0.00	1013	142.58	0.00	1013			
		-0.89	0.00	1321	4.38	0.00	1011	21.71 C	0.00	1011
	MIN.	-26.75	0.00	11	-87.77	0.00	11			
		-7.30	0.90	1002	-2.60	0.90	1002	2.48 C	0.90	13

21	0.00	MAX	6.59	1013	3.47	1013	-0.33	1321	2.97	1011
		MIN	-4.10	11	-2.15	11	-3.37	1011	0.32	1321
	0.18	MAX	5.49	1013	2.38	1013	-0.33	1321	2.36	1011
		MIN	-3.42	11	-1.47	11	-3.37	1011	0.26	1321
	0.36	MAX	4.40	1013	1.49	1013	-0.33	1321	1.75	1011
		MIN	-2.73	11	-0.92	11	-3.37	1011	0.20	1321
	0.54	MAX	3.31	1013	0.80	1013	-0.33	1321	1.14	1011
		MIN	-2.05	11	-0.49	11	-3.37	1011	0.14	1321
	0.72	MAX	2.22	1013	0.30	1013	-0.33	1321	0.56	1002
		MIN	-1.36	11	-0.18	11	-3.37	1011	0.08	1321
	0.90	MAX	1.13	1013	0.00	1013	-0.33	1321	0.06	1013
		MIN	-0.68	11	0.00	1211	-3.37	1011	-0.10	1211

	MAX/MIN FORCE VALUES FOR MEMB				21, AMONGST ALL SECT LOCATIONS					
		FY/	DIST	LD	MZ/	DIST	LD			
		FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
	MAX.	6.59	0.00	1013	3.47	0.00	1013			
		-0.33	0.00	1321	2.97	0.00	1011	0.15 C	0.00	1002
	MIN.	-4.10	0.00	11	-2.15	0.00	11			
		-3.37	0.90	1011	-0.10	0.90	1211	0.00 T	0.90	13

22	0.00	MAX	0.03	1114	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	21	0.00	2014	0.00	2014	0.00	2014
	0.93	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	1.86	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	2.79	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	3.72	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014

Friday, September 21, 2007, 11:59 AM

4.65	MAX	0.00	21	0.00	2014	0.00	2014	0.00	2014
	MIN	-0.03	1114	0.00	2014	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB 22, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.03	0.00	1001	0.00	0.00	1			
	0.00	0.00	1	0.00	0.00	1	2.56 T	4.65	1321
MIN.	-0.03	4.65	1114	0.00	4.65	2014			
	0.00	4.65	2014	0.00	4.65	2014	23.26 T	0.00	1011

23	0.00	MAX	0.03	1114	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	21	0.00	2014	0.00	2014	0.00	2014
	0.93	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	1.86	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	2.79	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	3.72	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	4.65	MAX	0.00	21	0.00	2014	0.00	2014	0.00	2014
		MIN	-0.03	1114	0.00	2014	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB 23, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.03	0.00	1001	0.00	0.00	1			
	0.00	0.00	1	0.00	0.00	1	2.63 T	4.65	1321
MIN.	-0.03	4.65	1114	0.00	4.65	2014			
	0.00	4.65	2014	0.00	4.65	2014	22.48 T	0.00	1011

24	0.00	MAX	0.03	1114	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	21	0.00	2014	0.00	2014	0.00	2014
	0.93	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	1.86	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	2.79	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	3.72	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	4.65	MAX	0.00	21	0.00	2014	0.00	2014	0.00	2014
		MIN	-0.03	1114	0.00	2014	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB 24, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.03	0.00	1001	0.00	0.00	1			
	0.00	0.00	1	0.00	0.00	1	2.63 T	4.65	1321
MIN.	-0.03	4.65	1114	0.00	4.65	2014			
	0.00	4.65	2014	0.00	4.65	2014	22.48 T	0.00	1011

25	0.00	MAX	0.03	1114	0.00	2014	0.00	2014	0.00	2014
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Friday, September 21, 2007, 11:59 AM

0.93	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
1.86	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
2.79	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
3.72	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
4.65	MAX	0.00	21	0.00	2014	0.00	2014	0.00	2014
	MIN	-0.03	1114	0.00	2014	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB 25, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.03	0.00	1001	0.00	0.00	1			
	0.00	0.00	1	0.00	0.00	1	2.56 T	4.65	1321
MIN.	-0.03	4.65	1114	0.00	4.65	2014			
	0.00	4.65	2014	0.00	4.65	2014	23.26 T	0.00	1011

26	0.00	MAX	0.04	1114	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	21	0.00	2014	0.00	2014	0.00	2014
	1.34	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	2.69	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	4.03	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	5.38	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	6.72	MAX	0.00	21	0.00	2014	0.00	2014	0.00	2014
		MIN	-0.04	1114	0.00	2014	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB 26, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.04	0.00	1001	0.00	0.00	1			
	0.00	0.00	1	0.00	0.00	1	1.09 T	0.00	11
MIN.	-0.04	6.72	1114	0.00	6.72	2014			
	0.00	6.72	2014	0.00	6.72	2014	28.57 T	6.72	1002

27	0.00	MAX	0.04	1114	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	21	0.00	2014	0.00	2014	0.00	2014
	1.34	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	2.69	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	4.03	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	5.38	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	6.72	MAX	0.00	21	0.00	2014	0.00	2014	0.00	2014
		MIN	-0.04	1114	0.00	2014	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB 27, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD

Friday, September 21, 2007, 11:59 AM

		0.00	0.00	1	0.00	0.00	1	2.81 T	0.00	1321
	MIN.	-0.04	6.72	1114	0.00	6.72	2014			
		0.00	6.72	2014	0.00	6.72	2014	25.79 T	6.72	1011

28	0.00	MAX	0.04	1114	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	21	0.00	2014	0.00	2014	0.00	2014
	1.34	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	2.69	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	4.03	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	5.38	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	6.72	MAX	0.00	21	0.00	2014	0.00	2014	0.00	2014
		MIN	-0.04	1114	0.00	2014	0.00	2014	0.00	2014

	MAX/MIN FORCE VALUES FOR MEMB 28, AMONGST ALL SECT LOCATIONS									
		FY/	DIST	LD	MZ/	DIST	LD			
		FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
	MAX.	0.04	0.00	1001	0.00	0.00	1			
		0.00	0.00	1	0.00	0.00	1	2.81 T	0.00	1321
	MIN.	-0.04	6.72	1114	0.00	6.72	2014			
		0.00	6.72	2014	0.00	6.72	2014	25.79 T	6.72	1011

29	0.00	MAX	0.04	1114	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	21	0.00	2014	0.00	2014	0.00	2014
	1.34	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	2.69	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	4.03	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	5.38	MAX	0.00	2014	0.00	2014	0.00	2014	0.00	2014
		MIN	0.00	2014	0.00	2014	0.00	2014	0.00	2014
	6.72	MAX	0.00	21	0.00	2014	0.00	2014	0.00	2014
		MIN	-0.04	1114	0.00	2014	0.00	2014	0.00	2014

	MAX/MIN FORCE VALUES FOR MEMB 29, AMONGST ALL SECT LOCATIONS									
		FY/	DIST	LD	MZ/	DIST	LD			
		FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
	MAX.	0.04	0.00	1001	0.00	0.00	1			
		0.00	0.00	1	0.00	0.00	1	1.09 T	0.00	11
	MIN.	-0.04	6.72	1114	0.00	6.72	2014			
		0.00	6.72	2014	0.00	6.72	2014	28.57 T	6.72	1002

30	0.00	MAX	37.75	1013	105.79	1013	0.15	11	2.61	1002
		MIN	-23.18	11	-65.23	11	-1.28	1013	0.19	11
	0.18	MAX	36.63	1013	99.09	1013	0.15	11	2.39	1002
		MIN	-22.50	11	-61.12	11	-1.28	1013	0.22	11
	0.36	MAX	35.50	1013	92.60	1013	0.15	11	2.17	1002
		MIN	-21.81	11	-57.13	11	-1.28	1013	0.24	11
	0.54	MAX	34.38	1013	86.31	1013	0.15	11	1.94	1002
		MIN	-21.13	11	-53.27	11	-1.28	1013	0.27	11
	0.72	MAX	33.27	1013	80.22	1013	0.15	11	1.72	1002
		MIN	-20.44	11	-49.53	11	-1.28	1013	0.25	1321
	0.90	MAX	32.15	1013	74.33	1013	0.15	11	1.50	1002

Friday, September 21, 2007, 11:59 AM

MAX/MIN FORCE VALUES FOR MEMB				30, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD

MAX.	37.75	0.00	1013	105.79	0.00	1013			
	0.15	0.00	11	2.61	0.00	1002	21.55 C	0.00	1011
MIN.	-23.18	0.00	11	-65.23	0.00	11			
	-1.28	0.90	1013	0.19	0.00	11	2.49 C	0.90	13

31	0.00	MAX	31.71	1013	74.33	1013	-0.08	11	1.54	1002
		MIN	-19.54	11	-45.91	11	-1.30	1002	0.23	1321
	0.18	MAX	30.59	1013	68.73	1013	-0.08	11	1.31	1002
		MIN	-18.86	11	-42.45	11	-1.30	1002	0.19	1321
	0.36	MAX	29.48	1013	63.32	1013	-0.08	11	1.07	1002
		MIN	-18.17	11	-39.12	11	-1.30	1002	0.15	1321
	0.54	MAX	28.37	1013	58.11	1013	-0.08	11	0.84	1002
		MIN	-17.49	11	-35.91	11	-1.30	1002	0.11	1321
	0.72	MAX	27.27	1013	53.11	1013	-0.08	11	0.62	1011
		MIN	-16.80	11	-32.83	11	-1.30	1002	0.07	1321
	0.90	MAX	26.16	1013	48.30	1013	-0.08	11	0.44	1011
		MIN	-16.11	11	-29.86	11	-1.30	1002	0.03	1321

MAX/MIN FORCE VALUES FOR MEMB				31, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD

MAX.	31.71	0.00	1013	74.33	0.00	1013			
	-0.08	0.00	11	1.54	0.00	1002	21.40 C	0.00	1011
MIN.	-19.54	0.00	11	-45.91	0.00	11			
	-1.30	0.90	1002	0.03	0.90	1321	2.45 C	0.90	1321

32	0.00	MAX	25.59	1013	48.30	1013	0.26	1111	0.43	1002
		MIN	-15.82	11	-29.86	11	-0.04	13	0.06	1321
	0.18	MAX	24.49	1013	43.79	1013	0.26	1111	0.43	1011
		MIN	-15.14	11	-27.08	11	-0.04	13	0.05	1321
	0.36	MAX	23.39	1013	39.48	1013	0.26	1111	0.47	1011
		MIN	-14.45	11	-24.41	11	-0.04	13	0.05	1321
	0.54	MAX	22.28	1013	35.37	1013	0.26	1111	0.51	1011
		MIN	-13.76	11	-21.88	11	-0.04	13	0.05	1321
	0.72	MAX	21.18	1013	31.46	1013	0.26	1111	0.55	1011
		MIN	-13.08	11	-19.46	11	-0.04	13	0.05	1321
	0.90	MAX	20.09	1013	27.75	1013	0.26	1111	0.59	1011
		MIN	-12.39	11	-17.17	11	-0.04	13	0.04	13

MAX/MIN FORCE VALUES FOR MEMB				32, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD

MAX.	25.59	0.00	1013	48.30	0.00	1013			
	0.26	0.00	1111	0.59	0.90	1011	21.26 C	0.00	1011
MIN.	-15.82	0.00	11	-29.86	0.00	11			
	-0.04	0.90	13	0.04	0.90	13	2.40 C	0.90	1321

33	0.00	MAX	19.39	1013	27.75	1013	0.13	1111	0.55	1011
		MIN	-12.01	11	-17.17	11	-0.04	1013	0.06	1321
	0.18	MAX	18.29	1013	24.35	1013	0.13	1111	0.57	1011

Friday, September 21, 2007, 11:59 AM

0.36	MAX	17.19	1013	21.16	1013	0.13	1111	0.59	1011
	MIN	-10.64	11	-13.09	11	-0.04	1013	0.06	1321
0.54	MAX	16.09	1013	18.17	1013	0.13	1111	0.60	1011
	MIN	-9.95	11	-11.24	11	-0.04	1013	0.06	1321
0.72	MAX	15.00	1013	15.37	1013	0.13	1111	0.62	1011
	MIN	-9.27	11	-9.51	11	-0.04	1013	0.05	1321
0.90	MAX	13.90	1013	12.77	1013	0.13	1111	0.64	1011
	MIN	-8.58	11	-7.90	11	-0.04	1013	0.05	13

MAX/MIN FORCE VALUES FOR MEMB 33, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	19.39	0.00	1013	27.75	0.00	1013			
	0.13	0.00	1111	0.64	0.90	1011	21.13 C	0.00	1011
MIN.	-12.01	0.00	11	-17.17	0.00	11			
	-0.04	0.90	1013	0.05	0.90	13	2.36 C	0.90	1321

34	0.00	MAX	13.07	1013	12.77	1013	-0.46	1321	0.61	1011
		MIN	-8.10	11	-7.90	11	-3.92	1011	0.08	1321
	0.18	MAX	11.97	1013	10.51	1013	-0.46	1321	-0.01	1321
		MIN	-7.42	11	-6.51	11	-3.92	1011	-0.10	1011
	0.36	MAX	10.88	1013	8.46	1013	-0.46	1321	-0.09	1321
		MIN	-6.73	11	-5.23	11	-3.92	1011	-0.80	1011
	0.54	MAX	9.78	1013	6.60	1013	-0.46	1321	-0.17	1321
		MIN	-6.05	11	-4.08	11	-3.92	1011	-1.51	1011
	0.72	MAX	8.69	1013	4.93	1013	-0.46	1321	-0.26	1321
		MIN	-5.36	11	-3.05	11	-3.92	1011	-2.21	1011
	0.90	MAX	7.59	1013	3.47	1013	-0.46	1321	-0.34	1321
		MIN	-4.68	11	-2.15	11	-3.92	1011	-2.92	1011

MAX/MIN FORCE VALUES FOR MEMB 34, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	13.07	0.00	1013	12.77	0.00	1013			
	-0.46	0.00	1321	0.61	0.00	1011	21.00 C	0.00	1011
MIN.	-8.10	0.00	11	-7.90	0.00	11			
	-3.92	0.90	1011	-2.92	0.90	1011	2.31 C	0.90	1321

35	0.00	MAX	55.46	1013	142.75	1013	0.00	2014	0.00	2014
		MIN	-35.93	11	-91.72	11	0.00	2014	0.00	2014
	0.18	MAX	53.41	1013	132.95	1013	0.00	2014	0.00	2014
		MIN	-34.56	11	-85.38	11	0.00	2014	0.00	2014
	0.36	MAX	51.37	1013	123.52	1013	0.00	2014	0.00	2014
		MIN	-33.19	11	-79.28	11	0.00	2014	0.00	2014
	0.54	MAX	49.33	1013	114.46	1013	0.00	2014	0.00	2014
		MIN	-31.81	11	-73.43	11	0.00	2014	0.00	2014
	0.72	MAX	47.29	1013	105.76	1013	0.00	2014	0.00	2014
		MIN	-30.44	11	-67.83	11	0.00	2014	0.00	2014
	0.90	MAX	45.26	1013	97.43	1013	0.00	2014	0.00	2014
		MIN	-29.07	11	-62.47	11	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB 35, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	55.46	0.00	1013	142.75	0.00	1013			

Friday, September 21, 2007, 11:59 AM

			0.00	0.90	2014		0.00	0.90	2014		1.47	C	0.90	11

36	0.00	MAX	45.48	1013		97.43	1013		0.00	2014		0.00	2014	
		MIN	-29.49	11		-62.47	11		0.00	2014		0.00	2014	
	0.18	MAX	43.45	1013		89.43	1013		0.00	2014		0.00	2014	
		MIN	-28.12	11		-57.29	11		0.00	2014		0.00	2014	
	0.36	MAX	41.41	1013		81.79	1013		0.00	2014		0.00	2014	
		MIN	-26.75	11		-52.35	11		0.00	2014		0.00	2014	
	0.54	MAX	39.38	1013		74.52	1013		0.00	2014		0.00	2014	
		MIN	-25.38	11		-47.66	11		0.00	2014		0.00	2014	
	0.72	MAX	37.36	1013		67.61	1013		0.00	2014		0.00	2014	
		MIN	-24.00	11		-43.22	11		0.00	2014		0.00	2014	
	0.90	MAX	35.33	1013		61.07	1013		0.00	2014		0.00	2014	
		MIN	-22.63	11		-39.02	11		0.00	2014		0.00	2014	

MAX/MIN FORCE VALUES FOR MEMB					36, AMONGST ALL SECT LOCATIONS									
		FY/	DIST	LD		MZ/	DIST	LD						
		FZ	DIST	LD		MY	DIST	LD		FX		DIST	LD	
	MAX.	45.48	0.00	1013		97.43	0.00	1013						
		0.00	0.00	1		0.00	0.00	1		47.35	C	0.00	1002	
	MIN.	-29.49	0.00	11		-62.47	0.00	11						
		0.00	0.90	2014		0.00	0.90	2014		1.53	C	0.90	11	

37	0.00	MAX	35.80	1013		61.07	1013		0.00	2014		0.00	2014	
		MIN	-23.22	11		-39.02	11		0.00	2014		0.00	2014	
	0.18	MAX	33.77	1013		54.81	1013		0.00	2014		0.00	2014	
		MIN	-21.85	11		-34.96	11		0.00	2014		0.00	2014	
	0.36	MAX	31.75	1013		48.91	1013		0.00	2014		0.00	2014	
		MIN	-20.48	11		-31.15	11		0.00	2014		0.00	2014	
	0.54	MAX	29.73	1013		43.38	1013		0.00	2014		0.00	2014	
		MIN	-19.10	11		-27.59	11		0.00	2014		0.00	2014	
	0.72	MAX	27.71	1013		38.21	1013		0.00	2014		0.00	2014	
		MIN	-17.73	11		-24.28	11		0.00	2014		0.00	2014	
	0.90	MAX	25.69	1013		33.40	1013		0.00	2014		0.00	2014	
		MIN	-16.36	11		-21.21	11		0.00	2014		0.00	2014	

MAX/MIN FORCE VALUES FOR MEMB					37, AMONGST ALL SECT LOCATIONS									
		FY/	DIST	LD		MZ/	DIST	LD						
		FZ	DIST	LD		MY	DIST	LD		FX		DIST	LD	
	MAX.	35.80	0.00	1013		61.07	0.00	1013						
		0.00	0.00	1		0.00	0.00	1		47.05	C	0.00	1002	
	MIN.	-23.22	0.00	11		-39.02	0.00	11						
		0.00	0.90	2014		0.00	0.90	2014		1.59	C	0.90	11	

38	0.00	MAX	26.42	1013		33.40	1013		0.00	2014		0.00	2014	
		MIN	-17.13	11		-21.21	11		0.00	2014		0.00	2014	
	0.18	MAX	24.40	1013		28.83	1013		0.00	2014		0.00	2014	
		MIN	-15.76	11		-18.25	11		0.00	2014		0.00	2014	
	0.36	MAX	22.39	1013		24.62	1013		0.00	2014		0.00	2014	
		MIN	-14.38	11		-15.54	11		0.00	2014		0.00	2014	
	0.54	MAX	20.37	1013		20.77	1013		0.00	2014		0.00	2014	
		MIN	-13.01	11		-13.07	11		0.00	2014		0.00	2014	
	0.72	MAX	18.35	1013		17.29	1013		0.00	2014		0.00	2014	
		MIN	-11.64	11		-10.85	11		0.00	2014		0.00	2014	
	0.90	MAX	16.34	1013		14.16	1013		0.00	2014		0.00	2014	
		MIN	-10.27	11		-8.88	11		0.00	2014		0.00	2014	

Friday, September 21, 2007, 11:59 AM

MAX/MIN FORCE VALUES FOR MEMB				38, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD

MAX.	26.42	0.00	1013	33.40	0.00	1013			
	0.00	0.00	1	0.00	0.00	1	46.75 C	0.00	1002
MIN.	-17.13	0.00	11	-21.21	0.00	11			
	0.00	0.90	2014	0.00	0.90	2014	1.64 C	0.90	11

39	0.00	MAX	17.34	1013	14.16	1013	0.00	2014	0.00	2014
		MIN	-11.22	11	-8.88	11	0.00	2014	0.00	2014
	0.18	MAX	15.32	1013	11.22	1013	0.00	2014	0.00	2014
		MIN	-9.85	11	-6.98	11	0.00	2014	0.00	2014
	0.36	MAX	13.31	1013	8.65	1013	0.00	2014	0.00	2014
		MIN	-8.48	11	-5.33	11	0.00	2014	0.00	2014
	0.54	MAX	11.29	1013	6.43	1013	0.00	2014	0.00	2014
		MIN	-7.11	11	-3.93	11	0.00	2014	0.00	2014
	0.72	MAX	9.28	1013	4.58	1013	0.00	2014	0.00	2014
		MIN	-5.73	11	-2.78	11	0.00	2014	0.00	2014
	0.90	MAX	7.26	1013	3.09	1013	0.00	2014	0.00	2014
		MIN	-4.36	11	-1.87	11	0.00	2014	0.00	2014

MAX/MIN FORCE VALUES FOR MEMB				39, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD

MAX.	17.34	0.00	1013	14.16	0.00	1013			
	0.00	0.00	1	0.00	0.00	1	46.45 C	0.00	1002
MIN.	-11.22	0.00	11	-8.88	0.00	11			
	0.00	0.90	2014	0.00	0.90	2014	1.68 C	0.90	11

40	0.00	MAX	37.75	1013	105.79	1013	1.28	1013	-0.19	11
		MIN	-23.18	11	-65.23	11	-0.15	11	-2.61	1002
	0.18	MAX	36.63	1013	99.09	1013	1.28	1013	-0.22	11
		MIN	-22.50	11	-61.12	11	-0.15	11	-2.39	1002
	0.36	MAX	35.50	1013	92.60	1013	1.28	1013	-0.24	11
		MIN	-21.81	11	-57.13	11	-0.15	11	-2.17	1002
	0.54	MAX	34.38	1013	86.31	1013	1.28	1013	-0.27	11
		MIN	-21.13	11	-53.27	11	-0.15	11	-1.94	1002
	0.72	MAX	33.27	1013	80.22	1013	1.28	1013	-0.25	1321
		MIN	-20.44	11	-49.53	11	-0.15	11	-1.72	1002
	0.90	MAX	32.15	1013	74.33	1013	1.28	1013	-0.21	1321
		MIN	-19.75	11	-45.91	11	-0.15	11	-1.50	1002

MAX/MIN FORCE VALUES FOR MEMB				40, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD

MAX.	37.75	0.00	1013	105.79	0.00	1013			
	1.28	0.00	1013	-0.19	0.00	11	21.55 C	0.00	1011
MIN.	-23.18	0.00	11	-65.23	0.00	11			
	-0.15	0.90	11	-2.61	0.00	1002	2.49 C	0.90	13

41	0.00	MAX	31.71	1013	74.33	1013	1.30	1002	-0.23	1321
		MIN	-19.54	11	-45.91	11	0.08	11	-1.54	1002
	0.18	MAX	30.59	1013	68.73	1013	1.30	1002	-0.19	1321
		MIN	-18.86	11	-42.45	11	0.08	11	-1.31	1002
	0.36	MAX	29.48	1013	63.32	1013	1.30	1002	-0.15	1321

Friday, September 21, 2007, 11:59 AM

0.54	MAX	28.37	1013	58.11	1013	1.30	1002	-0.11	1321
	MIN	-17.49	11	-35.91	11	0.08	11	-0.84	1002
0.72	MAX	27.27	1013	53.11	1013	1.30	1002	-0.07	1321
	MIN	-16.80	11	-32.83	11	0.08	11	-0.62	1011
0.90	MAX	26.16	1013	48.30	1013	1.30	1002	-0.03	1321
	MIN	-16.11	11	-29.86	11	0.08	11	-0.44	1011

MAX/MIN FORCE VALUES FOR MEMB 41, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	31.71	0.00	1013	74.33	0.00	1013			
	1.30	0.00	1002	-0.03	0.90	1321	21.40 C	0.00	1011
MIN.	-19.54	0.00	11	-45.91	0.00	11			
	0.08	0.90	11	-1.54	0.00	1002	2.45 C	0.90	1321

42	0.00	MAX	25.59	1013	48.30	1013	0.04	13	-0.06	1321
		MIN	-15.82	11	-29.86	11	-0.26	1111	-0.43	1002
	0.18	MAX	24.49	1013	43.79	1013	0.04	13	-0.05	1321
		MIN	-15.14	11	-27.08	11	-0.26	1111	-0.43	1011
	0.36	MAX	23.39	1013	39.48	1013	0.04	13	-0.05	1321
		MIN	-14.45	11	-24.41	11	-0.26	1111	-0.47	1011
	0.54	MAX	22.28	1013	35.37	1013	0.04	13	-0.05	1321
		MIN	-13.76	11	-21.88	11	-0.26	1111	-0.51	1011
	0.72	MAX	21.18	1013	31.46	1013	0.04	13	-0.05	1321
		MIN	-13.08	11	-19.46	11	-0.26	1111	-0.55	1011
	0.90	MAX	20.09	1013	27.75	1013	0.04	13	-0.04	13
		MIN	-12.39	11	-17.17	11	-0.26	1111	-0.59	1011

MAX/MIN FORCE VALUES FOR MEMB 42, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	25.59	0.00	1013	48.30	0.00	1013			
	0.04	0.00	13	-0.04	0.90	13	21.26 C	0.00	1011
MIN.	-15.82	0.00	11	-29.86	0.00	11			
	-0.26	0.90	1111	-0.59	0.90	1011	2.40 C	0.90	1321

43	0.00	MAX	19.39	1013	27.75	1013	0.04	1013	-0.06	1321
		MIN	-12.01	11	-17.17	11	-0.13	1111	-0.55	1011
	0.18	MAX	18.29	1013	24.35	1013	0.04	1013	-0.06	1321
		MIN	-11.32	11	-15.07	11	-0.13	1111	-0.57	1011
	0.36	MAX	17.19	1013	21.16	1013	0.04	1013	-0.06	1321
		MIN	-10.64	11	-13.09	11	-0.13	1111	-0.59	1011
	0.54	MAX	16.09	1013	18.17	1013	0.04	1013	-0.06	1321
		MIN	-9.95	11	-11.24	11	-0.13	1111	-0.60	1011
	0.72	MAX	15.00	1013	15.37	1013	0.04	1013	-0.05	1321
		MIN	-9.27	11	-9.51	11	-0.13	1111	-0.62	1011
	0.90	MAX	13.90	1013	12.77	1013	0.04	1013	-0.05	13
		MIN	-8.58	11	-7.90	11	-0.13	1111	-0.64	1011

MAX/MIN FORCE VALUES FOR MEMB 43, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	19.39	0.00	1013	27.75	0.00	1013			
	0.04	0.00	1013	-0.05	0.90	13	21.13 C	0.00	1011
MIN.	-12.01	0.00	11	-17.17	0.00	11			

Friday, September 21, 2007, 11:59 AM

44	0.00	MAX	13.07	1013	12.77	1013	3.92	1011	-0.08	1321
		MIN	-8.10	11	-7.90	11	0.46	1321	-0.61	1011
	0.18	MAX	11.97	1013	10.51	1013	3.92	1011	0.10	1011
		MIN	-7.42	11	-6.51	11	0.46	1321	0.01	1321
	0.36	MAX	10.88	1013	8.46	1013	3.92	1011	0.80	1011
		MIN	-6.73	11	-5.23	11	0.46	1321	0.09	1321
	0.54	MAX	9.78	1013	6.60	1013	3.92	1011	1.51	1011
		MIN	-6.05	11	-4.08	11	0.46	1321	0.17	1321
	0.72	MAX	8.69	1013	4.93	1013	3.92	1011	2.21	1011
		MIN	-5.36	11	-3.05	11	0.46	1321	0.26	1321
	0.90	MAX	7.59	1013	3.47	1013	3.92	1011	2.92	1011
		MIN	-4.68	11	-2.15	11	0.46	1321	0.34	1321

MAX/MIN FORCE VALUES FOR MEMB					44, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD		MZ/	DIST	LD			
	FZ	DIST	LD		MY	DIST	LD	FX	DIST	LD
	MAX.	13.07	0.00 1013		12.77	0.00 1013				
		3.92	0.00 1011		2.92	0.90 1011		21.00 C	0.00 1011	
	MIN.	-8.10	0.00 11		-7.90	0.00 11				
		0.46	0.90 1321		-0.61	0.00 1011		2.31 C	0.90 1321	

45	0.00	MAX	1.13	1013	0.05	1013	0.04	1013	0.10	1211
		MIN	-0.68	11	-0.03	1211	-0.05	1211	-0.06	1013
	0.80	MAX	1.06	1013	0.51	11	0.04	1013	0.06	1211
		MIN	-0.68	11	-0.82	1013	-0.05	1211	-0.03	1013
	1.60	MAX	0.99	1013	1.05	11	0.04	1013	0.02	1111
		MIN	-0.71	1211	-1.65	1013	-0.05	1211	0.00	13
	2.40	MAX	0.93	1013	1.60	1211	0.04	1013	0.03	1013
		MIN	-0.76	1211	-2.41	1013	-0.05	1211	-0.02	1211
	3.20	MAX	0.86	1013	2.23	1211	0.04	1013	0.06	1013
		MIN	-0.82	1211	-3.13	1013	-0.05	1211	-0.06	1211
	4.00	MAX	0.79	1013	2.91	1211	0.04	1013	0.09	1013
		MIN	-0.88	1211	-3.79	1013	-0.05	1211	-0.11	1211

MAX/MIN FORCE VALUES FOR MEMB					45, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD		MZ/	DIST	LD			
	FZ	DIST	LD		MY	DIST	LD	FX	DIST	LD
	MAX.	1.13	0.00 1013		2.91	4.00 1211				
		0.04	0.00 1013		0.10	0.00 1211		3.37 C	0.00 1011	
	MIN.	-0.88	4.00 1211		-3.79	4.00 1013				
		-0.05	4.00 1211		-0.11	4.00 1211		0.33 C	4.00 1321	

46	0.00	MAX	0.97	1013	0.00	1111	0.04	1013	0.09	1211
		MIN	-0.57	11	0.00	15	-0.05	1211	-0.08	1013
	0.80	MAX	0.90	1013	0.46	11	0.04	1013	0.05	1211
		MIN	-0.57	11	-0.75	1013	-0.05	1211	-0.05	1013
	1.60	MAX	0.84	1013	0.91	11	0.04	1013	0.01	1211
		MIN	-0.60	1211	-1.45	1013	-0.05	1211	-0.01	1013
	2.40	MAX	0.77	1013	1.37	1211	0.04	1013	0.03	1013
		MIN	-0.65	1211	-2.09	1013	-0.05	1211	-0.02	1211
	3.20	MAX	0.70	1013	1.92	1211	0.04	1013	0.06	1013
		MIN	-0.71	1211	-2.68	1013	-0.05	1211	-0.06	1211
	4.00	MAX	0.64	1013	2.51	1211	0.04	1013	0.10	1013
		MIN	-0.77	1211	-3.22	1013	-0.05	1211	-0.10	1211

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Friday, September 21, 2007, 11:59 AM

	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.97	0.00	1013	2.51	4.00	1211			
	0.04	0.00	1013	0.10	4.00	1013	8.05 C	0.00	1011
MIN.	-0.77	4.00	1211	-3.22	4.00	1013			
	-0.05	4.00	1211	-0.10	4.00	1211	0.88 C	4.00	1321

47	0.00	MAX	0.83	1013	0.00	11	0.05	1013	0.07	1211
		MIN	-0.48	11	0.00	1013	-0.04	1211	-0.10	1013
	0.80	MAX	0.77	1013	0.38	11	0.05	1013	0.04	1211
		MIN	-0.48	11	-0.64	1013	-0.04	1211	-0.06	1013
	1.60	MAX	0.70	1013	0.76	11	0.05	1013	0.01	11
		MIN	-0.50	1211	-1.23	1013	-0.04	1211	-0.02	1013
	2.40	MAX	0.63	1013	1.14	11	0.05	1013	0.02	1013
		MIN	-0.56	1211	-1.76	1013	-0.04	1211	-0.02	1211
	3.20	MAX	0.57	1013	1.60	1211	0.05	1013	0.06	1013
		MIN	-0.61	1211	-2.24	1013	-0.04	1211	-0.05	1211
	4.00	MAX	0.50	1013	2.11	1211	0.05	1013	0.10	1013
		MIN	-0.67	1111	-2.67	1013	-0.04	1211	-0.09	1211

MAX/MIN FORCE VALUES FOR MEMB 47, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.83	0.00	1013	2.11	4.00	1211			
	0.05	0.00	1013	0.10	4.00	1013	4.02 C	0.00	1011
MIN.	-0.67	4.00	1111	-2.67	4.00	1013			
	-0.04	4.00	1211	-0.10	0.00	1013	0.45 C	4.00	1321

48	0.00	MAX	0.70	1013	0.00	1211	0.05	1013	0.08	1211
		MIN	-0.38	11	0.00	1013	-0.04	1211	-0.09	1013
	0.80	MAX	0.63	1013	0.31	11	0.05	1013	0.05	1211
		MIN	-0.38	11	-0.54	1013	-0.04	1211	-0.05	1013
	1.60	MAX	0.56	1013	0.61	11	0.05	1013	0.01	1211
		MIN	-0.40	1211	-1.01	1013	-0.04	1211	-0.02	1013
	2.40	MAX	0.50	1013	0.92	11	0.05	1013	0.02	1013
		MIN	-0.46	1211	-1.44	1013	-0.04	1211	-0.02	1211
	3.20	MAX	0.43	1013	1.29	1211	0.05	1013	0.06	1013
		MIN	-0.51	1211	-1.81	1013	-0.04	1211	-0.06	1211
	4.00	MAX	0.36	1013	1.72	1211	0.05	1013	0.10	1013
		MIN	-0.58	1111	-2.12	1013	-0.04	1211	-0.09	1211

MAX/MIN FORCE VALUES FOR MEMB 48, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.70	0.00	1013	1.72	4.00	1211			
	0.05	0.00	1013	0.10	4.00	1013	0.14 C	0.00	1111
MIN.	-0.58	4.00	1111	-2.12	4.00	1013			
	-0.04	4.00	1211	-0.09	0.00	1013	0.01 T	4.00	13

49	0.00	MAX	0.57	1013	0.00	1211	0.04	1013	0.08	1211
		MIN	-0.29	11	-0.01	1013	-0.04	1211	-0.09	1013
	0.80	MAX	0.50	1013	0.24	11	0.04	1013	0.05	1211
		MIN	-0.29	11	-0.43	1013	-0.04	1211	-0.05	1013
	1.60	MAX	0.43	1013	0.47	11	0.04	1013	0.01	1211
		MIN	-0.31	1211	-0.80	1013	-0.04	1211	-0.01	1013
	2.40	MAX	0.37	1013	0.71	11	0.04	1013	0.02	1013

Friday, September 21, 2007, 11:59 AM

3.20	MAX	0.30	1013	0.99	1211	0.04	1013	0.06	1013
	MIN	-0.42	1111	-1.39	1013	-0.04	1211	-0.05	1211
4.00	MAX	0.23	1013	1.35	1211	0.04	1013	0.09	1013
	MIN	-0.50	1111	-1.60	1013	-0.04	1211	-0.09	1211

MAX/MIN FORCE VALUES FOR MEMB				49, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD

MAX.	0.57	0.00	1013	1.35	4.00	1211			
	0.04	0.00	1013	0.09	4.00	1013	0.20 T	0.00	1321
MIN.	-0.50	4.00	1111	-1.60	4.00	1013			
	-0.04	4.00	1211	-0.09	4.00	1211	1.34 T	4.00	1002

50	0.00	MAX	0.44	1013	0.00	11	0.04	1013	0.09	1211
		MIN	-0.21	11	-0.01	1013	-0.05	1211	-0.07	1013
	0.80	MAX	0.38	1013	0.17	11	0.04	1013	0.06	1211
		MIN	-0.21	11	-0.34	1013	-0.05	1211	-0.04	1013
	1.60	MAX	0.31	1013	0.34	11	0.04	1013	0.02	1211
		MIN	-0.22	1211	-0.61	1013	-0.05	1211	-0.01	1013
	2.40	MAX	0.24	1013	0.51	11	0.04	1013	0.02	1013
		MIN	-0.28	1211	-0.83	1013	-0.05	1211	-0.02	1211
	3.20	MAX	0.18	1013	0.71	1211	0.04	1013	0.05	1013
		MIN	-0.35	1111	-1.00	1013	-0.05	1211	-0.06	1211
	4.00	MAX	0.12	13	1.00	1211	0.04	1013	0.08	1013
		MIN	-0.43	1111	-1.11	1013	-0.05	1211	-0.09	1211

MAX/MIN FORCE VALUES FOR MEMB				50, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD

MAX.	0.44	0.00	1013	1.00	4.00	1211			
	0.04	0.00	1013	0.09	0.00	1211	0.33 C	0.00	1111
MIN.	-0.43	4.00	1111	-1.11	4.00	1013			
	-0.05	4.00	1211	-0.09	4.00	1211	0.05 T	4.00	1213

51	0.00	MAX	0.32	1013	-0.01	1321	0.03	1013	0.11	1211
		MIN	-0.14	11	-0.06	1002	-0.05	1211	-0.04	1013
	0.80	MAX	0.25	1013	0.10	11	0.03	1013	0.07	1111
		MIN	-0.14	11	-0.28	1013	-0.05	1211	-0.02	1013
	1.60	MAX	0.18	1013	0.21	11	0.03	1013	0.03	1111
		MIN	-0.16	1211	-0.45	1013	-0.05	1211	0.00	13
	2.40	MAX	0.12	1013	0.32	11	0.03	1013	0.02	1013
		MIN	-0.22	1111	-0.57	1013	-0.05	1211	-0.02	1211
	3.20	MAX	0.07	13	0.47	1211	0.03	1013	0.05	1013
		MIN	-0.30	1111	-0.63	1013	-0.05	1211	-0.06	1211
	4.00	MAX	0.07	13	0.71	1211	0.03	1013	0.07	1013
		MIN	-0.38	1111	-0.65	1013	-0.05	1211	-0.10	1211

MAX/MIN FORCE VALUES FOR MEMB				51, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD

MAX.	0.32	0.00	1013	0.71	4.00	1211			
	0.03	0.00	1013	0.11	0.00	1211	8.54 C	0.00	1002
MIN.	-0.38	4.00	1111	-0.65	4.00	1013			
	-0.05	4.00	1211	-0.10	4.00	1211	1.15 C	4.00	1321

Friday, September 21, 2007, 11:59 AM

52	0.00	MAX	0.08	1321	-0.11	1321	0.03	1013	0.10	1211
		MIN	-0.34	1011	-1.20	1011	-0.05	1211	-0.05	1013
	0.80	MAX	0.03	1321	-0.14	13	0.03	1013	0.06	1211
		MIN	-0.41	1111	-0.93	1002	-0.05	1211	-0.02	1013
	1.60	MAX	-0.02	13	-0.06	11	0.03	1013	0.02	1111
		MIN	-0.49	1111	-0.68	1002	-0.05	1211	0.00	13
	2.40	MAX	-0.02	13	0.13	11	0.03	1013	0.02	1013
		MIN	-0.57	1111	-0.42	1013	-0.05	1211	-0.02	1211
	3.20	MAX	-0.02	13	0.44	1111	0.03	1013	0.05	1013
		MIN	-0.65	1111	-0.16	1013	-0.05	1211	-0.05	1211
	4.00	MAX	-0.02	13	0.99	1111	0.03	1013	0.07	1013
		MIN	-0.72	1111	-0.09	13	-0.05	1211	-0.09	1211

MAX/MIN FORCE VALUES FOR MEMB					52, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD		MZ/	DIST	LD			
	FZ	DIST	LD		MY	DIST	LD	FX	DIST	LD

MAX.	0.08	0.00	1321		0.99	4.00	1111			
	0.03	0.00	1013		0.10	0.00	1211	23.30 C	0.00	1002
MIN.	-0.72	4.00	1111		-1.20	0.00	1011			
	-0.05	4.00	1211		-0.09	4.00	1211	3.30 C	4.00	11

53	0.00	MAX	0.88	1211	2.91	1211	0.05	1211	0.09	1013
		MIN	-0.79	1013	-3.79	1013	-0.04	1013	-0.11	1211
	0.80	MAX	0.82	1211	2.23	1211	0.05	1211	0.06	1013
		MIN	-0.86	1013	-3.13	1013	-0.04	1013	-0.06	1211
	1.60	MAX	0.76	1211	1.60	1211	0.05	1211	0.03	1013
		MIN	-0.93	1013	-2.41	1013	-0.04	1013	-0.02	1211
	2.40	MAX	0.71	1211	1.05	11	0.05	1211	0.02	1111
		MIN	-0.99	1013	-1.65	1013	-0.04	1013	0.00	13
	3.20	MAX	0.68	11	0.51	11	0.05	1211	0.06	1211
		MIN	-1.06	1013	-0.82	1013	-0.04	1013	-0.03	1013
	4.00	MAX	0.68	11	0.05	1013	0.05	1211	0.10	1211
		MIN	-1.13	1013	-0.03	1211	-0.04	1013	-0.06	1013

MAX/MIN FORCE VALUES FOR MEMB					53, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD		MZ/	DIST	LD			
	FZ	DIST	LD		MY	DIST	LD	FX	DIST	LD

MAX.	0.88	0.00	1211		2.91	0.00	1211			
	0.05	0.00	1211		0.10	4.00	1211	3.37 C	0.00	1011
MIN.	-1.13	4.00	1013		-3.79	0.00	1013			
	-0.04	4.00	1013		-0.11	0.00	1211	0.33 C	4.00	1321

54	0.00	MAX	0.77	1211	2.51	1211	0.05	1211	0.10	1013
		MIN	-0.64	1013	-3.22	1013	-0.04	1013	-0.10	1211
	0.80	MAX	0.71	1211	1.92	1211	0.05	1211	0.06	1013
		MIN	-0.70	1013	-2.68	1013	-0.04	1013	-0.06	1211
	1.60	MAX	0.65	1211	1.37	1211	0.05	1211	0.03	1013
		MIN	-0.77	1013	-2.09	1013	-0.04	1013	-0.02	1211
	2.40	MAX	0.60	1211	0.91	11	0.05	1211	0.01	1211
		MIN	-0.84	1013	-1.45	1013	-0.04	1013	-0.01	1013
	3.20	MAX	0.57	11	0.46	11	0.05	1211	0.05	1211
		MIN	-0.90	1013	-0.75	1013	-0.04	1013	-0.05	1013
	4.00	MAX	0.57	11	0.00	1111	0.05	1211	0.09	1211
		MIN	-0.97	1013	0.00	15	-0.04	1013	-0.08	1013

MAX/MIN FORCE VALUES FOR MEMB					54, AMONGST ALL SECT LOCATIONS					
	FY/	DIST	LD		MZ/	DIST	LD			

Friday, September 21, 2007, 11:59 AM

	MAX.	0.77	0.00	1211	2.51	0.00	1211		
		0.05	0.00	1211	0.10	0.00	1013	8.05 C	0.00 1011
	MIN.	-0.97	4.00	1013	-3.22	0.00	1013		
		-0.04	4.00	1013	-0.10	0.00	1211	0.88 C	4.00 1321

55	0.00	MAX	0.67	1111	2.11	1211	0.04	1211	0.10	1013
		MIN	-0.50	1013	-2.67	1013	-0.05	1013	-0.09	1211
	0.80	MAX	0.61	1211	1.60	1211	0.04	1211	0.06	1013
		MIN	-0.57	1013	-2.24	1013	-0.05	1013	-0.05	1211
	1.60	MAX	0.56	1211	1.14	11	0.04	1211	0.02	1013
		MIN	-0.63	1013	-1.76	1013	-0.05	1013	-0.02	1211
	2.40	MAX	0.50	1211	0.76	11	0.04	1211	0.01	11
		MIN	-0.70	1013	-1.23	1013	-0.05	1013	-0.02	1013
	3.20	MAX	0.48	11	0.38	11	0.04	1211	0.04	1211
		MIN	-0.77	1013	-0.64	1013	-0.05	1013	-0.06	1013
	4.00	MAX	0.48	11	0.00	11	0.04	1211	0.07	1211
		MIN	-0.83	1013	0.00	1013	-0.05	1013	-0.10	1013

	MAX/MIN FORCE VALUES FOR MEMB				55, AMONGST ALL SECT LOCATIONS					
		FY/	DIST	LD	MZ/	DIST	LD			
		FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
	MAX.	0.67	0.00	1111	2.11	0.00	1211			
		0.04	0.00	1211	0.10	0.00	1013	4.02 C	0.00	1011
	MIN.	-0.83	4.00	1013	-2.67	0.00	1013			
		-0.05	4.00	1013	-0.10	4.00	1013	0.45 C	4.00	1321

56	0.00	MAX	0.58	1111	1.72	1211	0.04	1211	0.10	1013
		MIN	-0.36	1013	-2.12	1013	-0.05	1013	-0.09	1211
	0.80	MAX	0.51	1211	1.29	1211	0.04	1211	0.06	1013
		MIN	-0.43	1013	-1.81	1013	-0.05	1013	-0.06	1211
	1.60	MAX	0.46	1211	0.92	11	0.04	1211	0.02	1013
		MIN	-0.50	1013	-1.44	1013	-0.05	1013	-0.02	1211
	2.40	MAX	0.40	1211	0.61	11	0.04	1211	0.01	1211
		MIN	-0.56	1013	-1.01	1013	-0.05	1013	-0.02	1013
	3.20	MAX	0.38	11	0.31	11	0.04	1211	0.05	1211
		MIN	-0.63	1013	-0.54	1013	-0.05	1013	-0.05	1013
	4.00	MAX	0.38	11	0.00	1211	0.04	1211	0.08	1211
		MIN	-0.70	1013	0.00	1013	-0.05	1013	-0.09	1013

	MAX/MIN FORCE VALUES FOR MEMB				56, AMONGST ALL SECT LOCATIONS					
		FY/	DIST	LD	MZ/	DIST	LD			
		FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
	MAX.	0.58	0.00	1111	1.72	0.00	1211			
		0.04	0.00	1211	0.10	0.00	1013	0.14 C	0.00	1111
	MIN.	-0.70	4.00	1013	-2.12	0.00	1013			
		-0.05	4.00	1013	-0.09	4.00	1013	0.01 T	4.00	13

57	0.00	MAX	0.50	1111	1.35	1211	0.04	1211	0.09	1013
		MIN	-0.23	1013	-1.60	1013	-0.04	1013	-0.09	1211
	0.80	MAX	0.42	1111	0.99	1211	0.04	1211	0.06	1013
		MIN	-0.30	1013	-1.39	1013	-0.04	1013	-0.05	1211
	1.60	MAX	0.36	1211	0.71	11	0.04	1211	0.02	1013
		MIN	-0.37	1013	-1.12	1013	-0.04	1013	-0.02	1211
	2.40	MAX	0.31	1211	0.47	11	0.04	1211	0.01	1211
		MIN	-0.43	1013	-0.80	1013	-0.04	1013	-0.01	1013
	3.20	MAX	0.29	11	0.24	11	0.04	1211	0.05	1211

Friday, September 21, 2007, 11:59 AM

4.00	MAX	0.29	11	0.00	1211	0.04	1211	0.08	1211
	MIN	-0.57	1013	-0.01	1013	-0.04	1013	-0.09	1013

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| MAX/MIN FORCE VALUES FOR MEMB      57, AMONGST ALL SECT LOCATIONS
|      FY/   DIST  LD      MZ/   DIST  LD
|      FZ    DIST  LD      MY    DIST  LD      FX      DIST  LD
|
| MAX.      0.50   0.00 1111      1.35   0.00 1211
|            0.04   0.00 1211      0.09   0.00 1013      0.20 T   0.00 1321
| MIN.     -0.57   4.00 1013     -1.60   0.00 1013
|            -0.04   4.00 1013     -0.09   0.00 1211      1.34 T   4.00 1002
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58	0.00	MAX	0.43	1111	1.00	1211	0.05	1211	0.08	1013
		MIN	-0.12	13	-1.11	1013	-0.04	1013	-0.09	1211
	0.80	MAX	0.35	1111	0.71	1211	0.05	1211	0.05	1013
		MIN	-0.18	1013	-1.00	1013	-0.04	1013	-0.06	1211
	1.60	MAX	0.28	1211	0.51	11	0.05	1211	0.02	1013
		MIN	-0.24	1013	-0.83	1013	-0.04	1013	-0.02	1211
	2.40	MAX	0.22	1211	0.34	11	0.05	1211	0.02	1211
		MIN	-0.31	1013	-0.61	1013	-0.04	1013	-0.01	1013
	3.20	MAX	0.21	11	0.17	11	0.05	1211	0.06	1211
		MIN	-0.38	1013	-0.34	1013	-0.04	1013	-0.04	1013
	4.00	MAX	0.21	11	0.00	11	0.05	1211	0.09	1211
		MIN	-0.44	1013	-0.01	1013	-0.04	1013	-0.07	1013

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| MAX/MIN FORCE VALUES FOR MEMB      58, AMONGST ALL SECT LOCATIONS
|      FY/   DIST  LD      MZ/   DIST  LD
|      FZ    DIST  LD      MY    DIST  LD      FX      DIST  LD
|
| MAX.      0.43   0.00 1111      1.00   0.00 1211
|            0.05   0.00 1211      0.09   4.00 1211      0.33 C   0.00 1111
| MIN.     -0.44   4.00 1013     -1.11   0.00 1013
|            -0.04   4.00 1013     -0.09   0.00 1211      0.05 T   4.00 1213
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59	0.00	MAX	0.38	1111	0.71	1211	0.05	1211	0.07	1013
		MIN	-0.07	13	-0.65	1013	-0.03	1013	-0.10	1211
	0.80	MAX	0.30	1111	0.47	1211	0.05	1211	0.05	1013
		MIN	-0.07	13	-0.63	1013	-0.03	1013	-0.06	1211
	1.60	MAX	0.22	1111	0.32	11	0.05	1211	0.02	1013
		MIN	-0.12	1013	-0.57	1013	-0.03	1013	-0.02	1211
	2.40	MAX	0.16	1211	0.21	11	0.05	1211	0.03	1111
		MIN	-0.18	1013	-0.45	1013	-0.03	1013	0.00	13
	3.20	MAX	0.14	11	0.10	11	0.05	1211	0.07	1111
		MIN	-0.25	1013	-0.28	1013	-0.03	1013	-0.02	1013
	4.00	MAX	0.14	11	-0.01	1321	0.05	1211	0.11	1211
		MIN	-0.32	1013	-0.06	1002	-0.03	1013	-0.04	1013

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| MAX/MIN FORCE VALUES FOR MEMB      59, AMONGST ALL SECT LOCATIONS
|      FY/   DIST  LD      MZ/   DIST  LD
|      FZ    DIST  LD      MY    DIST  LD      FX      DIST  LD
|
| MAX.      0.38   0.00 1111      0.71   0.00 1211
|            0.05   0.00 1211      0.11   4.00 1211      8.54 C   0.00 1002
| MIN.     -0.32   4.00 1013     -0.65   0.00 1013
|            -0.03   4.00 1013     -0.10   0.00 1211      1.15 C   4.00 1321
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Friday, September 21, 2007, 11:59 AM

0.80	MAX	0.65	1111	0.44	1111	0.05	1211	0.05	1013
	MIN	0.02	13	-0.16	1013	-0.03	1013	-0.05	1211
1.60	MAX	0.57	1111	0.13	11	0.05	1211	0.02	1013
	MIN	0.02	13	-0.42	1013	-0.03	1013	-0.02	1211
2.40	MAX	0.49	1111	-0.06	11	0.05	1211	0.02	1111
	MIN	0.02	13	-0.68	1002	-0.03	1013	0.00	13
3.20	MAX	0.41	1111	-0.14	13	0.05	1211	0.06	1211
	MIN	-0.03	1321	-0.93	1002	-0.03	1013	-0.02	1013
4.00	MAX	0.34	1011	-0.11	1321	0.05	1211	0.10	1211
	MIN	-0.08	1321	-1.20	1011	-0.03	1013	-0.05	1013

MAX/MIN FORCE VALUES FOR MEMB 60, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.72	0.00	1111	0.99	0.00	1111			
	0.05	0.00	1211	0.10	4.00	1211	23.30 C	0.00	1002
MIN.	-0.08	4.00	1321	-1.20	4.00	1011			
	-0.03	4.00	1013	-0.09	0.00	1211	3.30 C	4.00	11

61	0.00	MAX	0.37	1002	0.55	1013	0.06	1013	0.04	11
		MIN	-0.05	11	-0.08	11	-0.02	11	-0.13	1013
	0.80	MAX	0.30	1013	0.28	1013	0.06	1013	0.03	11
		MIN	-0.05	11	-0.04	11	-0.02	11	-0.08	1013
	1.60	MAX	0.23	1013	0.07	1013	0.06	1013	0.01	11
		MIN	-0.05	11	-0.03	1	-0.02	11	-0.04	1013
	2.40	MAX	0.17	1013	0.04	11	0.06	1013	0.01	1013
		MIN	-0.05	11	-0.09	1113	-0.02	11	-0.02	1211
	3.20	MAX	0.10	1013	0.08	11	0.06	1013	0.05	1013
		MIN	-0.10	1111	-0.20	1013	-0.02	11	-0.03	11
	4.00	MAX	0.06	13	0.16	1211	0.06	1013	0.10	1013
		MIN	-0.18	1111	-0.25	1013	-0.02	11	-0.05	11

MAX/MIN FORCE VALUES FOR MEMB 61, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.37	0.00	1002	0.55	0.00	1013			
	0.06	0.00	1013	0.10	4.00	1013	6.44 C	0.00	1002
MIN.	-0.18	4.00	1111	-0.25	4.00	1013			
	-0.02	4.00	11	-0.13	0.00	1013	0.96 C	4.00	1321

62	0.00	MAX	0.35	1002	0.53	1011	0.04	1013	0.01	11
		MIN	0.02	13	0.06	13	-0.01	11	-0.09	1002
	0.80	MAX	0.27	1002	0.30	1011	0.04	1013	0.00	11
		MIN	0.02	13	0.03	1321	-0.01	11	-0.07	1002
	1.60	MAX	0.19	1011	0.11	1011	0.04	1013	0.00	11
		MIN	0.02	13	-0.02	1321	-0.01	11	-0.04	1002
	2.40	MAX	0.13	1011	0.02	15	0.04	1013	0.00	1321
		MIN	-0.01	1321	-0.04	1121	-0.01	11	-0.02	1111
	3.20	MAX	0.06	1011	0.00	1321	0.04	1013	0.03	1013
		MIN	-0.06	1321	-0.09	1011	-0.01	11	-0.01	11
	4.00	MAX	0.05	11	0.06	1321	0.04	1013	0.06	1013
		MIN	-0.12	1121	-0.11	1011	-0.01	11	-0.02	11

MAX/MIN FORCE VALUES FOR MEMB 62, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD

Friday, September 21, 2007, 11:59 AM

			0.04	0.00	1013		0.06	4.00	1013		2.14	C	0.00	1111
	MIN.		-0.12	4.00	1121		-0.11	4.00	1011					
			-0.01	4.00	11		-0.09	0.00	1002		0.10	C	4.00	13

63	0.00	MAX	0.32	1002		0.47	1011		-0.01	1321		0.17	1011	
		MIN	0.02	13		0.05	13		-0.07	1011		0.01	1321	
	0.80	MAX	0.24	1002		0.25	1011		-0.01	1321		0.12	1011	
		MIN	0.02	13		0.03	1321		-0.07	1011		0.01	1321	
	1.60	MAX	0.17	1011		0.09	1011		-0.01	1321		0.07	1011	
		MIN	0.02	13		-0.02	1321		-0.07	1011		0.01	1321	
	2.40	MAX	0.10	1011		0.01	15		-0.01	1321		0.01	1002	
		MIN	-0.01	1321		-0.04	1121		-0.07	1011		0.00	16	
	3.20	MAX	0.05	11		0.00	1321		-0.01	1321		0.00	1321	
		MIN	-0.06	1321		-0.08	1011		-0.07	1011		-0.04	1011	
	4.00	MAX	0.05	11		0.07	1321		-0.01	1321		-0.01	1321	
		MIN	-0.13	1121		-0.08	1011		-0.07	1011		-0.10	1011	

MAX/MIN FORCE VALUES FOR MEMB 63, AMONGST ALL SECT LOCATIONS														
			FY/	DIST	LD		MZ/	DIST	LD					
			FZ	DIST	LD		MY	DIST	LD		FX		DIST	LD
	MAX.		0.32	0.00	1002		0.47	0.00	1011					
			-0.01	0.00	1321		0.17	0.00	1011		16.13	C	0.00	1011
	MIN.		-0.13	4.00	1121		-0.08	4.00	1011					
			-0.07	4.00	1011		-0.10	4.00	1011		1.77	C	4.00	1321

64	0.00	MAX	0.18	1111		0.16	1211		0.02	11		0.10	1013	
		MIN	-0.06	13		-0.25	1013		-0.06	1013		-0.05	11	
	0.80	MAX	0.10	1111		0.08	11		0.02	11		0.05	1013	
		MIN	-0.10	1013		-0.20	1013		-0.06	1013		-0.03	11	
	1.60	MAX	0.05	11		0.04	11		0.02	11		0.01	1013	
		MIN	-0.17	1013		-0.09	1113		-0.06	1013		-0.02	1211	
	2.40	MAX	0.05	11		0.07	1013		0.02	11		0.01	11	
		MIN	-0.23	1013		-0.03	1		-0.06	1013		-0.04	1013	
	3.20	MAX	0.05	11		0.28	1013		0.02	11		0.03	11	
		MIN	-0.30	1013		-0.04	11		-0.06	1013		-0.08	1013	
	4.00	MAX	0.05	11		0.55	1013		0.02	11		0.04	11	
		MIN	-0.37	1002		-0.08	11		-0.06	1013		-0.13	1013	

MAX/MIN FORCE VALUES FOR MEMB 64, AMONGST ALL SECT LOCATIONS														
			FY/	DIST	LD		MZ/	DIST	LD					
			FZ	DIST	LD		MY	DIST	LD		FX		DIST	LD
	MAX.		0.18	0.00	1111		0.55	4.00	1013					
			0.02	0.00	11		0.10	0.00	1013		6.44	C	0.00	1002
	MIN.		-0.37	4.00	1002		-0.25	0.00	1013					
			-0.06	4.00	1013		-0.13	4.00	1013		0.96	C	4.00	1321

65	0.00	MAX	0.12	1121		0.06	1321		0.01	11		0.06	1013	
		MIN	-0.05	11		-0.11	1011		-0.04	1013		-0.02	11	
	0.80	MAX	0.06	1321		0.00	1321		0.01	11		0.03	1013	
		MIN	-0.06	1011		-0.09	1011		-0.04	1013		-0.01	11	
	1.60	MAX	0.01	1321		0.02	15		0.01	11		0.00	1321	
		MIN	-0.13	1011		-0.04	1121		-0.04	1013		-0.02	1111	
	2.40	MAX	-0.02	13		0.11	1011		0.01	11		0.00	11	
		MIN	-0.19	1011		-0.02	1321		-0.04	1013		-0.04	1002	
	3.20	MAX	-0.02	13		0.30	1011		0.01	11		0.00	11	
		MIN	-0.27	1002		0.03	1321		-0.04	1013		-0.07	1002	
	4.00	MAX	-0.02	13		0.53	1011		0.01	11		0.01	11	

Friday, September 21, 2007, 11:59 AM

MAX/MIN FORCE VALUES FOR MEMB 65, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.12	0.00	1121	0.53	4.00	1011			
	0.01	0.00	11	0.06	0.00	1013	2.14 C	0.00	1111
MIN.	-0.35	4.00	1002	-0.11	0.00	1011			
	-0.04	4.00	1013	-0.09	4.00	1002	0.10 C	4.00	13

66	0.00	MAX	0.13	1121	0.07	1321	0.07	1011	-0.01	1321
		MIN	-0.05	11	-0.08	1011	0.01	1321	-0.10	1011
	0.80	MAX	0.06	1321	0.00	1321	0.07	1011	0.00	1321
		MIN	-0.05	11	-0.08	1011	0.01	1321	-0.04	1011
	1.60	MAX	0.01	1321	0.01	15	0.07	1011	0.01	1002
		MIN	-0.10	1011	-0.04	1121	0.01	1321	0.00	16
	2.40	MAX	-0.02	13	0.09	1011	0.07	1011	0.07	1011
		MIN	-0.17	1011	-0.02	1321	0.01	1321	0.01	1321
	3.20	MAX	-0.02	13	0.25	1011	0.07	1011	0.12	1011
		MIN	-0.24	1002	0.03	1321	0.01	1321	0.01	1321
	4.00	MAX	-0.02	13	0.47	1011	0.07	1011	0.17	1011
		MIN	-0.32	1002	0.05	13	0.01	1321	0.01	1321

MAX/MIN FORCE VALUES FOR MEMB 66, AMONGST ALL SECT LOCATIONS									
	FY/	DIST	LD	MZ/	DIST	LD			
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	0.13	0.00	1121	0.47	4.00	1011			
	0.07	0.00	1011	0.17	4.00	1011	16.13 C	0.00	1011
MIN.	-0.32	4.00	1002	-0.08	0.00	1011			
	0.01	4.00	1321	-0.10	0.00	1011	1.77 C	4.00	1321

***** END OF FORCE ENVELOPE FROM INTERNAL STORAGE *****

200. PRINT JOINT DISPLACEMENTS LIST 2 TO 5 8 10 12 TO 15 17 18 20 TO 23 25 TO 41

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
2	1	-0.0903	0.0258	-0.0061	0.0000	0.0002	0.0006
	2	-0.1126	0.0324	-0.0059	0.0000	0.0002	0.0008
	3	-0.1428	0.0416	-0.0057	0.0000	0.0002	0.0010
	4	0.0027	-0.0008	-0.0073	0.0001	0.0002	0.0000
	11	0.6168	-0.1811	-0.0114	0.0001	0.0004	-0.0039
	12	0.3590	-0.1056	-0.0091	0.0001	0.0003	-0.0022
	13	-0.2812	0.0823	-0.0047	0.0000	0.0002	0.0018
	14	0.4576	-0.1344	-0.0101	0.0001	0.0003	-0.0028
	15	0.1355	-0.0391	-0.0099	0.0001	0.0003	-0.0010
	16	0.2129	-0.0629	-0.0078	0.0001	0.0002	-0.0012
	21	0.0467	-0.0139	-0.0069	0.0001	0.0002	-0.0002
	1001	-0.5126	0.1480	-0.0258	0.0002	0.0009	0.0035
	1002	-0.5099	0.1472	-0.0331	0.0003	0.0011	0.0035
	1011	0.3253	-0.0975	-0.0348	0.0003	0.0011	-0.0018
	1111	0.5794	-0.1720	-0.0327	0.0003	0.0011	-0.0034
	1211	0.6606	-0.1953	-0.0279	0.0002	0.0009	-0.0040

Friday, September 21, 2007, 11:59 AM

1112	0.2185	-0.0664	-0.0295	0.0002	0.0010	-0.0011
1212	0.2997	-0.0896	-0.0247	0.0002	0.0008	-0.0016
1013	-0.7523	0.2185	-0.0267	0.0002	0.0009	0.0051
1113	-0.6778	0.1966	-0.0233	0.0002	0.0008	0.0046
1213	-0.5966	0.1734	-0.0185	0.0002	0.0006	0.0040
1014	0.1342	-0.0416	-0.0333	0.0003	0.0011	-0.0006
1114	0.3565	-0.1067	-0.0309	0.0002	0.0010	-0.0020
1214	0.4377	-0.1300	-0.0261	0.0002	0.0008	-0.0026
1021	-0.2682	0.0767	-0.0241	0.0002	0.0008	0.0020
1121	-0.3616	0.1045	-0.0102	0.0001	0.0003	0.0024
1221	-0.1359	0.0385	-0.0177	0.0001	0.0006	0.0011
1321	-0.2293	0.0663	-0.0038	0.0000	0.0001	0.0015
2001	-0.3457	0.0998	-0.0176	0.0001	0.0006	0.0024
2002	-0.2743	0.0790	-0.0148	0.0001	0.0005	0.0019
2011	0.2711	-0.0813	-0.0290	0.0002	0.0009	-0.0015
2012	0.0132	-0.0058	-0.0267	0.0002	0.0009	0.0002
2013	-0.6270	0.1821	-0.0223	0.0002	0.0008	0.0042
2014	0.1118	-0.0346	-0.0277	0.0002	0.0009	-0.0005
3	1	-0.1570	0.0246	-0.0008	0.0001	-0.0001
	2	-0.1932	0.0311	-0.0005	0.0001	-0.0001
	3	-0.2426	0.0400	-0.0005	0.0001	-0.0001
	4	0.0040	-0.0008	-0.0009	0.0001	-0.0001
	11	1.0106	-0.1753	-0.0020	0.0001	-0.0002
	12	0.5801	-0.1024	-0.0014	0.0001	-0.0001
	13	-0.4720	0.0793	-0.0002	0.0000	-0.0001
	14	0.7463	-0.1302	-0.0016	0.0001	-0.0002
	15	0.2459	-0.0374	-0.0014	0.0001	-0.0001
	16	0.3353	-0.0612	-0.0011	0.0001	-0.0001
	21	0.0668	-0.0137	-0.0008	0.0001	-0.0001
1001	-0.8783	0.1418	-0.0026	0.0003	-0.0005	0.0046
1002	-0.8743	0.1410	-0.0035	0.0003	-0.0006	0.0046

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	1011	0.5015	-0.0956	-0.0045	0.0003	-0.0006	-0.0020
	1111	0.9247	-0.1675	-0.0046	0.0003	-0.0005	-0.0041
	1211	1.0647	-0.1897	-0.0041	0.0003	-0.0004	-0.0049
	1012	-0.0151	-0.0082	-0.0039	0.0003	-0.0006	0.0005
	1112	0.3219	-0.0655	-0.0038	0.0003	-0.0005	-0.0011
	1212	0.4620	-0.0877	-0.0033	0.0002	-0.0004	-0.0019
	1013	-1.2776	0.2099	-0.0024	0.0003	-0.0005	0.0066
	1113	-1.1510	0.1889	-0.0022	0.0002	-0.0004	0.0059
	1213	-1.0109	0.1666	-0.0016	0.0002	-0.0003	0.0052
	1014	0.1844	-0.0415	-0.0042	0.0003	-0.0006	-0.0005
	1114	0.5547	-0.1044	-0.0042	0.0003	-0.0005	-0.0023
	1214	0.6947	-0.1266	-0.0036	0.0002	-0.0004	-0.0031
	1021	-0.4747	0.0730	-0.0027	0.0002	-0.0004	0.0026
	1121	-0.6082	0.1005	-0.0010	0.0001	-0.0002	0.0031
	1221	-0.2484	0.0364	-0.0020	0.0002	-0.0003	0.0015
	1321	-0.3819	0.0638	-0.0004	0.0000	-0.0001	0.0019
	2001	-0.5927	0.0956	-0.0018	0.0002	-0.0003	0.0031
	2002	-0.4714	0.0756	-0.0016	0.0001	-0.0003	0.0025
	2011	0.4179	-0.0797	-0.0038	0.0003	-0.0005	-0.0017
	2012	-0.0126	-0.0068	-0.0032	0.0003	-0.0005	0.0004
	2013	-1.0647	0.1749	-0.0020	0.0002	-0.0004	0.0055
	2014	0.1536	-0.0346	-0.0035	0.0003	-0.0005	-0.0004
4	1	-0.2272	-0.0261	-0.0027	0.0000	-0.0001	0.0010
	2	-0.2766	-0.0291	-0.0025	0.0000	-0.0001	0.0012
	3	-0.3451	-0.0339	-0.0025	0.0000	-0.0001	0.0015
	4	0.0044	-0.0006	-0.0026	0.0000	-0.0001	0.0000
	11	1.3976	0.1028	-0.0029	0.0001	-0.0001	-0.0056
	12	0.7945	0.0516	-0.0028	0.0001	-0.0001	-0.0031
	13	-0.6642	-0.0590	-0.0025	0.0000	-0.0001	0.0028

Friday, September 21, 2007, 11:59 AM

15	0.3615	0.0457	-0.0026	0.0001	-0.0001	-0.0017
16	0.4507	0.0216	-0.0027	0.0000	-0.0001	-0.0017
21	0.0826	-0.0025	-0.0026	0.0000	-0.0001	-0.0002
1001	-1.2573	-0.1315	-0.0113	0.0001	-0.0004	0.0056
1002	-1.2529	-0.1322	-0.0139	0.0002	-0.0005	0.0056
1011	0.6586	0.0165	-0.0127	0.0002	-0.0005	-0.0022
1111	1.2514	0.0667	-0.0113	0.0002	-0.0004	-0.0047
1211	1.4529	0.0888	-0.0092	0.0002	-0.0003	-0.0056
1012	-0.0652	-0.0450	-0.0126	0.0002	-0.0005	0.0008
1112	0.4071	-0.0050	-0.0112	0.0002	-0.0004	-0.0012
1212	0.6086	0.0170	-0.0091	0.0001	-0.0003	-0.0021
1013	-1.8156	-0.1777	-0.0122	0.0001	-0.0005	0.0080
1113	-1.6351	-0.1599	-0.0108	0.0001	-0.0004	0.0072
1213	-1.4336	-0.1378	-0.0087	0.0001	-0.0003	0.0062
1014	0.2160	-0.0196	-0.0126	0.0002	-0.0005	-0.0004
1114	0.7351	0.0246	-0.0112	0.0002	-0.0004	-0.0026
1214	0.9366	0.0467	-0.0091	0.0001	-0.0003	-0.0035
1021	-0.6944	-0.0856	-0.0101	0.0001	-0.0004	0.0033

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
1121		-0.8596	-0.0807	-0.0049	0.0000	-0.0002	0.0037
1221		-0.3708	-0.0521	-0.0073	0.0001	-0.0003	0.0018
1321		-0.5359	-0.0472	-0.0021	0.0000	-0.0001	0.0023
2001		-0.8488	-0.0891	-0.0077	0.0001	-0.0003	0.0038
2002		-0.6763	-0.0721	-0.0065	0.0001	-0.0002	0.0030
2011		0.5488	0.0137	-0.0106	0.0002	-0.0004	-0.0018
2012		-0.0543	-0.0375	-0.0105	0.0001	-0.0004	0.0007
2013		-1.5130	-0.1481	-0.0102	0.0001	-0.0004	0.0066
2014		0.1800	-0.0163	-0.0105	0.0001	-0.0004	-0.0003
5	1	-0.2615	-0.1238	-0.0090	-0.0003	0.0001	0.0012
	2	-0.3171	-0.1440	-0.0095	-0.0003	0.0001	0.0014
	3	-0.3946	-0.1745	-0.0096	-0.0003	0.0001	0.0017
	4	0.0044	-0.0011	-0.0090	-0.0002	0.0000	0.0000
	11	1.5792	0.6166	-0.0071	0.0001	0.0000	-0.0063
	12	0.8942	0.3337	-0.0081	-0.0001	0.0000	-0.0035
	13	-0.7558	-0.3185	-0.0101	-0.0004	0.0001	0.0032
	14	1.1610	0.4467	-0.0077	0.0000	0.0000	-0.0046
	15	0.4178	0.2050	-0.0082	-0.0001	0.0000	-0.0020
	16	0.5035	0.1706	-0.0086	-0.0001	0.0000	-0.0018
	21	0.0888	0.0149	-0.0090	-0.0002	0.0000	-0.0002
1001		-1.4414	-0.6540	-0.0413	-0.0014	0.0003	0.0065
1002		-1.4371	-0.6551	-0.0503	-0.0016	0.0003	0.0065
1011		0.7272	0.2093	-0.0423	-0.0010	0.0002	-0.0023
1111		1.4009	0.4884	-0.0359	-0.0007	0.0001	-0.0052
1211		1.6323	0.5955	-0.0285	-0.0004	0.0001	-0.0062
1012		-0.0948	-0.1303	-0.0434	-0.0012	0.0002	0.0011
1112		0.4419	0.0923	-0.0373	-0.0009	0.0002	-0.0012
1212		0.6733	0.1994	-0.0298	-0.0007	0.0001	-0.0022
1013		-2.0748	-0.9128	-0.0458	-0.0016	0.0003	0.0091
1113		-1.8681	-0.8207	-0.0401	-0.0014	0.0003	0.0081
1213		-1.6367	-0.7136	-0.0326	-0.0012	0.0002	0.0071
1014		0.2253	0.0053	-0.0429	-0.0011	0.0002	-0.0003
1114		0.8153	0.2505	-0.0367	-0.0008	0.0002	-0.0027
1214		1.0468	0.3576	-0.0293	-0.0006	0.0001	-0.0038
1021		-0.8028	-0.3936	-0.0361	-0.0011	0.0002	0.0038
1121		-0.9805	-0.4235	-0.0180	-0.0007	0.0001	0.0042
1221		-0.4319	-0.2261	-0.0257	-0.0008	0.0001	0.0022
1321		-0.6096	-0.2559	-0.0077	-0.0003	0.0001	0.0026
2001		-0.9732	-0.4422	-0.0281	-0.0009	0.0002	0.0044
2002		-0.7759	-0.3550	-0.0233	-0.0008	0.0001	0.0035
2011		0.6060	0.1744	-0.0352	-0.0008	0.0002	-0.0019
2012		-0.0790	-0.1086	-0.0362	-0.0010	0.0002	0.0009

Friday, September 21, 2007, 11:59 AM

2014	0.1878	0.0044	-0.0358	-0.0009	0.0002	-0.0002
8 1	-0.2946	-1.0965	-0.0013	-0.0005	0.0000	0.0017
2	-0.3561	-1.2832	-0.0011	-0.0012	0.0001	0.0020
3	-0.4423	-1.5607	-0.0011	-0.0014	0.0001	0.0024
4	0.0051	-0.0047	-0.0014	0.0000	0.0000	0.0000

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	11	1.7522	5.5207	-0.0026	0.0045	-0.0001	-0.0085
	12	0.9886	2.9963	-0.0021	0.0023	0.0000	-0.0046
	13	-0.8431	-2.8347	-0.0008	-0.0025	0.0001	0.0044
	14	1.2867	4.0031	-0.0023	0.0032	-0.0001	-0.0062
	15	0.4753	1.8202	-0.0016	0.0018	0.0000	-0.0028
	16	0.5526	1.5419	-0.0018	0.0011	0.0000	-0.0024
	21	0.0943	0.1438	-0.0014	0.0000	0.0000	-0.0002
	1001	-1.6186	-5.8286	-0.0051	-0.0047	0.0003	0.0091
	1002	-1.6134	-5.8333	-0.0064	-0.0047	0.0003	0.0091
	1011	0.7911	1.8965	-0.0073	0.0016	0.0001	-0.0028
	1111	1.5421	4.3975	-0.0070	0.0039	0.0000	-0.0067
	1211	1.8024	5.3494	-0.0060	0.0046	0.0000	-0.0082
	1012	-0.1252	-1.1328	-0.0067	-0.0010	0.0002	0.0018
	1112	0.4731	0.8633	-0.0063	0.0009	0.0001	-0.0013
	1212	0.7334	1.8151	-0.0053	0.0016	0.0001	-0.0027
	1013	-2.3233	-8.1300	-0.0051	-0.0068	0.0004	0.0126
	1113	-2.0913	-7.3001	-0.0045	-0.0059	0.0003	0.0113
	1213	-1.8310	-6.3482	-0.0035	-0.0052	0.0003	0.0098
	1014	0.2325	0.0754	-0.0069	0.0001	0.0002	0.0000
	1114	0.8904	2.2728	-0.0065	0.0021	0.0001	-0.0034
	1214	1.1507	3.2247	-0.0056	0.0028	0.0000	-0.0049
	1021	-0.9077	-3.4921	-0.0049	-0.0028	0.0002	0.0055
	1121	-1.0963	-3.7797	-0.0020	-0.0027	0.0001	0.0059
	1221	-0.4913	-1.9979	-0.0036	-0.0016	0.0001	0.0031
	1321	-0.6799	-2.2854	-0.0007	-0.0015	0.0001	0.0035
	2001	-1.0929	-3.9403	-0.0035	-0.0032	0.0002	0.0061
	2002	-0.8718	-3.1600	-0.0029	-0.0024	0.0002	0.0049
	2011	0.6593	1.5804	-0.0061	0.0014	0.0001	-0.0024
	2012	-0.1043	-0.9440	-0.0055	-0.0008	0.0002	0.0015
	2013	-1.9360	-6.7750	-0.0042	-0.0057	0.0003	0.0105
	2014	0.1938	0.0628	-0.0057	0.0001	0.0001	0.0000
10	1	-0.2893	-0.9431	-0.0031	-0.0004	0.0000	0.0017
	2	-0.3498	-1.1035	-0.0028	-0.0011	0.0000	0.0020
	3	-0.4346	-1.3424	-0.0027	-0.0013	0.0000	0.0024
	4	0.0052	-0.0041	-0.0033	0.0000	0.0000	0.0000
	11	1.7255	4.7561	-0.0058	0.0039	-0.0001	-0.0085
	12	0.9742	2.5825	-0.0047	0.0020	-0.0001	-0.0046
	13	-0.8293	-2.4404	-0.0021	-0.0022	0.0001	0.0044
	14	1.2674	3.4492	-0.0051	0.0028	-0.0001	-0.0062
	15	0.4663	1.5641	-0.0038	0.0015	-0.0001	-0.0028
	16	0.5452	1.3305	-0.0042	0.0009	0.0000	-0.0023
	21	0.0937	0.1255	-0.0035	0.0000	0.0000	-0.0002
	1001	-1.5901	-5.0131	-0.0127	-0.0041	0.0001	0.0091
	1002	-1.5849	-5.0172	-0.0160	-0.0041	0.0001	0.0091
	1011	0.7822	1.6405	-0.0174	0.0014	0.0000	-0.0028
	1111	1.5210	3.7932	-0.0165	0.0034	-0.0001	-0.0067
	1211	1.7766	4.6119	-0.0141	0.0040	-0.0001	-0.0082
	1012	-0.1194	-0.9678	-0.0161	-0.0009	0.0000	0.0018

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

Friday, September 21, 2007, 11:59 AM

1112	0.4691	0.7503	-0.0150	0.0007	0.0000	-0.0013
1212	0.7248	1.5689	-0.0126	0.0013	0.0000	-0.0027
1013	-2.2836	-6.9954	-0.0130	-0.0059	0.0002	0.0126
1113	-2.0557	-6.2819	-0.0114	-0.0051	0.0002	0.0113
1213	-1.8001	-5.4632	-0.0090	-0.0045	0.0002	0.0098
1014	0.2324	0.0722	-0.0166	0.0001	0.0000	0.0000
1114	0.8796	1.9636	-0.0156	0.0018	-0.0001	-0.0034
1214	1.1352	2.7822	-0.0132	0.0024	-0.0001	-0.0049
1021	-0.8905	-3.0017	-0.0120	-0.0025	0.0001	0.0054
1121	-1.0778	-3.2526	-0.0051	-0.0024	0.0001	0.0059
1221	-0.4815	-1.7165	-0.0089	-0.0014	0.0000	0.0031
1321	-0.6688	-1.9674	-0.0019	-0.0013	0.0000	0.0035
2001	-1.0737	-3.3890	-0.0087	-0.0027	0.0001	0.0061
2002	-0.8564	-2.7178	-0.0073	-0.0021	0.0001	0.0049
2011	0.6518	1.3670	-0.0145	0.0012	0.0000	-0.0024
2012	-0.0995	-0.8065	-0.0134	-0.0007	0.0000	0.0015
2013	-1.9030	-5.8295	-0.0109	-0.0049	0.0002	0.0105
2014	0.1937	0.0601	-0.0138	0.0000	0.0000	0.0000
12	1	-0.0900	0.0255	0.0000	0.0000	0.0007
	2	-0.1093	0.0306	0.0000	0.0000	0.0008
	3	-0.1405	0.0402	0.0000	0.0000	0.0010
	4	0.0028	-0.0008	0.0000	0.0000	0.0000
	11	0.6372	-0.1850	0.0000	0.0000	-0.0042
	12	0.3776	-0.1100	0.0000	0.0000	-0.0024
	13	-0.2815	0.0812	0.0000	0.0000	0.0020
	14	0.4755	-0.1382	0.0000	0.0000	-0.0031
	15	0.1174	-0.0330	0.0000	0.0000	-0.0011
	16	0.2314	-0.0678	0.0000	0.0000	-0.0013
	21	0.0563	-0.0166	0.0000	0.0000	-0.0002
1001	-0.5037	0.1428	0.0000	0.0000	0.0000	0.0038
1002	-0.5010	0.1420	0.0000	0.0000	0.0000	0.0038
1011	0.3570	-0.1065	0.0000	0.0000	0.0000	-0.0020
1111	0.6131	-0.1806	0.0000	0.0000	0.0000	-0.0038
1211	0.6928	-0.2030	0.0000	0.0000	0.0000	-0.0044
1012	0.0454	-0.0165	0.0000	0.0000	0.0000	0.0002
1112	0.2497	-0.0755	0.0000	0.0000	0.0000	-0.0012
1212	0.3294	-0.0980	0.0000	0.0000	0.0000	-0.0018
1013	-0.7455	0.2130	0.0000	0.0000	0.0000	0.0055
1113	-0.6731	0.1922	0.0000	0.0000	0.0000	0.0049
1213	-0.5934	0.1698	0.0000	0.0000	0.0000	0.0043
1014	0.1629	-0.0504	0.0000	0.0000	0.0000	-0.0006
1114	0.3867	-0.1150	0.0000	0.0000	0.0000	-0.0022
1214	0.4664	-0.1375	0.0000	0.0000	0.0000	-0.0028
1021	-0.2530	0.0708	0.0000	0.0000	0.0000	0.0021
1121	-0.3657	0.1040	0.0000	0.0000	0.0000	0.0026
1221	-0.1230	0.0338	0.0000	0.0000	0.0000	0.0011
1321	-0.2357	0.0671	0.0000	0.0000	0.0000	0.0016
2001	-0.3397	0.0963	0.0000	0.0000	0.0000	0.0025

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
2002	-0.2695	0.0762	0.0000	0.0000	0.0000	0.0000	0.0020
2011	0.2975	-0.0888	0.0000	0.0000	0.0000	0.0000	-0.0016
2012	0.0379	-0.0138	0.0000	0.0000	0.0000	0.0000	0.0002
2013	-0.6212	0.1775	0.0000	0.0000	0.0000	0.0000	0.0046
2014	0.1358	-0.0420	0.0000	0.0000	0.0000	0.0000	-0.0005
13	1	-0.1594	0.0242	0.0000	0.0000	0.0000	0.0009
	2	-0.1985	0.0288	0.0000	0.0000	0.0000	0.0012
	3	-0.2507	0.0381	0.0000	0.0000	0.0000	0.0014
	4	0.0040	-0.0009	0.0000	0.0000	0.0000	0.0000
	11	1.0710	-0.1781	0.0000	0.0000	0.0000	-0.0054
	12	0.6202	-0.1063	0.0000	0.0000	0.0000	-0.0030

Friday, September 21, 2007, 11:59 AM

	14	0.7932	-0.1332	0.0000	0.0000	0.0000	-0.0040
	15	0.2410	-0.0310	0.0000	0.0000	0.0000	-0.0016
	16	0.3647	-0.0659	0.0000	0.0000	0.0000	-0.0016
	21	0.0774	-0.0165	0.0000	0.0000	0.0000	-0.0002
1001	-0.9022	0.1352	0.0000	0.0000	0.0000	0.0000	0.0052
1002	-0.8982	0.1343	0.0000	0.0000	0.0000	0.0000	0.0052
1011	0.5548	-0.1044	0.0000	0.0000	0.0000	0.0000	-0.0023
1111	0.9982	-0.1752	0.0000	0.0000	0.0000	0.0000	-0.0047
1211	1.1414	-0.1964	0.0000	0.0000	0.0000	0.0000	-0.0055
1012	0.0139	-0.0182	0.0000	0.0000	0.0000	0.0000	0.0006
1112	0.3672	-0.0746	0.0000	0.0000	0.0000	0.0000	-0.0013
1212	0.5104	-0.0958	0.0000	0.0000	0.0000	0.0000	-0.0021
1013	-1.3218	0.2024	0.0000	0.0000	0.0000	0.0000	0.0074
1113	-1.1912	0.1828	0.0000	0.0000	0.0000	0.0000	0.0067
1213	-1.0480	0.1616	0.0000	0.0000	0.0000	0.0000	0.0058
1014	0.2215	-0.0505	0.0000	0.0000	0.0000	0.0000	-0.0005
1114	0.6093	-0.1123	0.0000	0.0000	0.0000	0.0000	-0.0026
1214	0.7525	-0.1335	0.0000	0.0000	0.0000	0.0000	-0.0035
1021	-0.4775	0.0662	0.0000	0.0000	0.0000	0.0000	0.0030
1121	-0.6322	0.0991	0.0000	0.0000	0.0000	0.0000	0.0034
1221	-0.2448	0.0312	0.0000	0.0000	0.0000	0.0000	0.0016
1321	-0.3995	0.0642	0.0000	0.0000	0.0000	0.0000	0.0021
2001	-0.6086	0.0911	0.0000	0.0000	0.0000	0.0000	0.0035
2002	-0.4833	0.0720	0.0000	0.0000	0.0000	0.0000	0.0028
2011	0.4623	-0.0870	0.0000	0.0000	0.0000	0.0000	-0.0019
2012	0.0116	-0.0152	0.0000	0.0000	0.0000	0.0000	0.0005
2013	-1.1015	0.1687	0.0000	0.0000	0.0000	0.0000	0.0062
2014	0.1846	-0.0421	0.0000	0.0000	0.0000	0.0000	-0.0005
14	1	-0.2344	-0.0302	0.0000	0.0000	0.0000	0.0011
	2	-0.2964	-0.0420	0.0000	0.0000	0.0000	0.0015
	3	-0.3704	-0.0485	0.0000	0.0000	0.0000	0.0018
	4	0.0037	-0.0014	0.0000	0.0000	0.0000	0.0000
	11	1.5131	0.1404	0.0000	0.0000	0.0000	-0.0066
	12	0.8630	0.0685	0.0000	0.0000	0.0000	-0.0036
	13	-0.7160	-0.0835	0.0000	0.0000	0.0000	0.0033
	14	1.1150	0.0985	0.0000	0.0000	0.0000	-0.0048

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	15	0.3778	0.0675	0.0000	0.0000	0.0000	-0.0021
	16	0.4931	0.0265	0.0000	0.0000	0.0000	-0.0019
	21	0.0926	-0.0057	0.0000	0.0000	0.0000	-0.0002
1001	-1.3359	-0.1787	0.0000	0.0000	0.0000	0.0000	0.0066
1002	-1.3322	-0.1801	0.0000	0.0000	0.0000	0.0000	0.0066
1011	0.7341	0.0236	0.0000	0.0000	0.0000	0.0000	-0.0026
1111	1.3751	0.0955	0.0000	0.0000	0.0000	0.0000	-0.0055
1211	1.5874	0.1243	0.0000	0.0000	0.0000	0.0000	-0.0066
1012	-0.0460	-0.0627	0.0000	0.0000	0.0000	0.0000	0.0010
1112	0.4649	-0.0052	0.0000	0.0000	0.0000	0.0000	-0.0014
1212	0.6773	0.0237	0.0000	0.0000	0.0000	0.0000	-0.0024
1013	-1.9408	-0.2450	0.0000	0.0000	0.0000	0.0000	0.0093
1113	-1.7457	-0.2180	0.0000	0.0000	0.0000	0.0000	0.0083
1213	-1.5333	-0.1891	0.0000	0.0000	0.0000	0.0000	0.0073
1014	0.2564	-0.0266	0.0000	0.0000	0.0000	0.0000	-0.0004
1114	0.8177	0.0369	0.0000	0.0000	0.0000	0.0000	-0.0030
1214	1.0301	0.0657	0.0000	0.0000	0.0000	0.0000	-0.0041
1021	-0.7296	-0.1166	0.0000	0.0000	0.0000	0.0000	0.0038
1121	-0.9149	-0.1052	0.0000	0.0000	0.0000	0.0000	0.0042
1221	-0.3852	-0.0707	0.0000	0.0000	0.0000	0.0000	0.0022
1321	-0.5704	-0.0593	0.0000	0.0000	0.0000	0.0000	0.0026
2001	-0.9013	-0.1207	0.0000	0.0000	0.0000	0.0000	0.0044
2002	-0.7161	-0.0965	0.0000	0.0000	0.0000	0.0000	0.0035
2011	0.6118	0.0197	0.0000	0.0000	0.0000	0.0000	-0.0021

Friday, September 21, 2007, 11:59 AM

2013	-1.6173	-0.2042	0.0000	0.0000	0.0000	0.0078
2014	0.2137	-0.0222	0.0000	0.0000	0.0000	-0.0003
15	1	-0.2718	-0.1367	0.0000	0.0000	0.0013
	2	-0.3457	-0.1824	0.0000	0.0000	0.0017
	3	-0.4304	-0.2192	0.0000	0.0000	0.0021
	4	0.0030	-0.0036	0.0000	0.0000	0.0000
	11	1.7266	0.7465	0.0000	0.0000	-0.0075
	12	0.9790	0.3978	0.0000	0.0000	-0.0041
	13	-0.8257	-0.3954	0.0000	0.0000	0.0039
	14	1.2698	0.5381	0.0000	0.0000	-0.0054
	15	0.4464	0.2618	0.0000	0.0000	-0.0024
	16	0.5532	0.1968	0.0000	0.0000	-0.0021
	21	0.0984	0.0103	0.0000	0.0000	-0.0002
1001	-1.5532	-0.7976	0.0000	0.0000	0.0000	0.0077
1002	-1.5501	-0.8012	0.0000	0.0000	0.0000	0.0077
1011	0.8144	0.2497	0.0000	0.0000	0.0000	-0.0028
1111	1.5526	0.5982	0.0000	0.0000	0.0000	-0.0062
1211	1.7996	0.7259	0.0000	0.0000	0.0000	-0.0075
1012	-0.0827	-0.1687	0.0000	0.0000	0.0000	0.0013
1112	0.5061	0.1101	0.0000	0.0000	0.0000	-0.0014
1212	0.7531	0.2378	0.0000	0.0000	0.0000	-0.0026
1013	-2.2483	-1.1206	0.0000	0.0000	0.0000	0.0109
1113	-2.0205	-1.0004	0.0000	0.0000	0.0000	0.0097

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
1213	-1.7735	-0.8727	0.0000	0.0000	0.0000	0.0000	0.0085
1014	0.2663	-0.0004	0.0000	0.0000	0.0000	0.0000	-0.0003
1114	0.9132	0.3065	0.0000	0.0000	0.0000	0.0000	-0.0033
1214	1.1602	0.4342	0.0000	0.0000	0.0000	0.0000	-0.0046
1021	-0.8579	-0.4823	0.0000	0.0000	0.0000	0.0000	0.0045
1121	-1.0546	-0.5029	0.0000	0.0000	0.0000	0.0000	0.0049
1221	-0.4574	-0.2770	0.0000	0.0000	0.0000	0.0000	0.0026
1321	-0.6542	-0.2975	0.0000	0.0000	0.0000	0.0000	0.0030
2001	-1.0479	-0.5384	0.0000	0.0000	0.0000	0.0000	0.0052
2002	-0.8327	-0.4288	0.0000	0.0000	0.0000	0.0000	0.0041
2011	0.6786	0.2080	0.0000	0.0000	0.0000	0.0000	-0.0023
2012	-0.0689	-0.1406	0.0000	0.0000	0.0000	0.0000	0.0011
2013	-1.8736	-0.9338	0.0000	0.0000	0.0000	0.0000	0.0091
2014	0.2219	-0.0003	0.0000	0.0000	0.0000	0.0000	-0.0003
17	1	-0.3014	-1.0386	0.0000	0.0000	0.0000	0.0019
	2	-0.3857	-1.3864	0.0000	0.0000	0.0000	0.0025
	3	-0.4792	-1.6795	0.0000	0.0000	0.0000	0.0030
	4	0.0045	-0.0132	0.0000	0.0000	0.0000	0.0000
	11	1.9037	5.8050	0.0000	0.0000	0.0000	-0.0104
	12	1.0752	3.1225	0.0000	0.0000	0.0000	-0.0056
	13	-0.9150	-3.0226	0.0000	0.0000	0.0000	0.0054
	14	1.3985	4.1978	0.0000	0.0000	0.0000	-0.0075
	15	0.5078	1.9769	0.0000	0.0000	0.0000	-0.0036
	16	0.6028	1.5783	0.0000	0.0000	0.0000	-0.0028
	21	0.1038	0.1165	0.0000	0.0000	0.0000	-0.0002
1001	-1.7287	-6.0822	0.0000	0.0000	0.0000	0.0000	0.0110
1002	-1.7242	-6.0953	0.0000	0.0000	0.0000	0.0000	0.0110
1011	0.8849	2.0406	0.0000	0.0000	0.0000	0.0000	-0.0036
1111	1.7032	4.7320	0.0000	0.0000	0.0000	0.0000	-0.0084
1211	1.9781	5.7020	0.0000	0.0000	0.0000	0.0000	-0.0102
1012	-0.1093	-1.1784	0.0000	0.0000	0.0000	0.0000	0.0022
1112	0.5433	0.9765	0.0000	0.0000	0.0000	0.0000	-0.0017
1212	0.8182	1.9465	0.0000	0.0000	0.0000	0.0000	-0.0034
1013	-2.4976	-8.5525	0.0000	0.0000	0.0000	0.0000	0.0154
1113	-2.2430	-7.6267	0.0000	0.0000	0.0000	0.0000	0.0137
1213	-1.9682	-6.6567	0.0000	0.0000	0.0000	0.0000	0.0120
1014	0.2786	0.1120	0.0000	0.0000	0.0000	0.0000	-0.0001

1214	1.2707	3.4520	0.0000	0.0000	0.0000	-0.0062
1021	-0.9604	-3.6333	0.0000	0.0000	0.0000	0.0066
1121	-1.1679	-3.8662	0.0000	0.0000	0.0000	0.0070
1221	-0.5146	-2.0660	0.0000	0.0000	0.0000	0.0037
1321	-0.7222	-2.2990	0.0000	0.0000	0.0000	0.0041
2001	-1.1663	-4.1045	0.0000	0.0000	0.0000	0.0074
2002	-0.9267	-3.2647	0.0000	0.0000	0.0000	0.0059
2011	0.7374	1.7005	0.0000	0.0000	0.0000	-0.0030
2012	-0.0911	-0.9820	0.0000	0.0000	0.0000	0.0018
2013	-2.0814	-7.1271	0.0000	0.0000	0.0000	0.0128

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
18	2014	0.2321	0.0934	0.0000	0.0000	0.0000	-0.0001
	1	-0.3073	-1.2079	0.0000	0.0000	0.0000	0.0019
	2	-0.3935	-1.6114	0.0000	0.0000	0.0000	0.0025
	3	-0.4887	-1.9522	0.0000	0.0000	0.0000	0.0030
	4	0.0044	-0.0145	0.0000	0.0000	0.0000	0.0000
	11	1.9364	6.7417	0.0000	0.0000	0.0000	-0.0104
	12	1.0928	3.6255	0.0000	0.0000	0.0000	-0.0056
	13	-0.9321	-3.5107	0.0000	0.0000	0.0000	0.0054
	14	1.4221	4.8750	0.0000	0.0000	0.0000	-0.0075
	15	0.5191	2.3000	0.0000	0.0000	0.0000	-0.0036
	16	0.6116	1.8316	0.0000	0.0000	0.0000	-0.0028
	21	0.1044	0.1344	0.0000	0.0000	0.0000	-0.0002
	1001	-1.7632	-7.0705	0.0000	0.0000	0.0000	0.0110
	1002	-1.7588	-7.0850	0.0000	0.0000	0.0000	0.0110
	1011	0.8962	2.3643	0.0000	0.0000	0.0000	-0.0036
	1111	1.7298	5.4914	0.0000	0.0000	0.0000	-0.0084
	1211	2.0101	6.6191	0.0000	0.0000	0.0000	-0.0102
	1012	-0.1162	-1.3751	0.0000	0.0000	0.0000	0.0022
	1112	0.5486	1.1287	0.0000	0.0000	0.0000	-0.0017
	1212	0.8290	2.2565	0.0000	0.0000	0.0000	-0.0034
	1013	-2.5460	-9.9386	0.0000	0.0000	0.0000	0.0154
1113	-2.2862	-8.8620	0.0000	0.0000	0.0000	0.0137	
1213	-2.0058	-7.7343	0.0000	0.0000	0.0000	0.0120	
1014	0.2790	0.1242	0.0000	0.0000	0.0000	-0.0001	
1114	1.0097	2.8780	0.0000	0.0000	0.0000	-0.0044	
1214	1.2901	4.0057	0.0000	0.0000	0.0000	-0.0062	
1021	-0.9810	-4.2248	0.0000	0.0000	0.0000	0.0066	
1121	-1.1898	-4.4936	0.0000	0.0000	0.0000	0.0070	
1221	-0.5264	-2.4030	0.0000	0.0000	0.0000	0.0037	
1321	-0.7352	-2.6718	0.0000	0.0000	0.0000	0.0041	
2001	-1.1896	-4.7715	0.0000	0.0000	0.0000	0.0074	
2002	-0.9452	-3.7954	0.0000	0.0000	0.0000	0.0059	
2011	0.7468	1.9703	0.0000	0.0000	0.0000	-0.0030	
2012	-0.0968	-1.1459	0.0000	0.0000	0.0000	0.0018	
2013	-2.1217	-8.2822	0.0000	0.0000	0.0000	0.0128	
2014	0.2325	0.1035	0.0000	0.0000	0.0000	-0.0001	
20	1	-0.0903	0.0258	0.0061	0.0000	-0.0002	0.0006
	2	-0.1126	0.0324	0.0059	0.0000	-0.0002	0.0008
	3	-0.1428	0.0416	0.0057	0.0000	-0.0002	0.0010
	4	0.0027	-0.0008	0.0073	-0.0001	-0.0002	0.0000
	11	0.6168	-0.1811	0.0114	-0.0001	-0.0004	-0.0039
	12	0.3590	-0.1056	0.0091	-0.0001	-0.0003	-0.0022
	13	-0.2812	0.0823	0.0047	0.0000	-0.0002	0.0018
	14	0.4576	-0.1344	0.0101	-0.0001	-0.0003	-0.0028
	15	0.1355	-0.0391	0.0099	-0.0001	-0.0003	-0.0010
	16	0.2129	-0.0629	0.0078	-0.0001	-0.0002	-0.0012
	21	0.0467	-0.0139	0.0069	-0.0001	-0.0002	-0.0002
	1001	-0.5126	0.1480	0.0258	-0.0002	-0.0009	0.0035

Friday, September 21, 2007, 11:59 AM

JOINT DISPLACEMENT (CM		RADIANS)		STRUCTURE TYPE = SPACE			

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	1002	-0.5099	0.1472	0.0331	-0.0003	-0.0011	0.0035
	1011	0.3253	-0.0975	0.0348	-0.0003	-0.0011	-0.0018
	1111	0.5794	-0.1720	0.0327	-0.0003	-0.0011	-0.0034
	1211	0.6606	-0.1953	0.0279	-0.0002	-0.0009	-0.0040
	1012	0.0159	-0.0070	0.0321	-0.0003	-0.0011	0.0002
	1112	0.2185	-0.0664	0.0295	-0.0002	-0.0010	-0.0011
	1212	0.2997	-0.0896	0.0247	-0.0002	-0.0008	-0.0016
	1013	-0.7523	0.2185	0.0267	-0.0002	-0.0009	0.0051
	1113	-0.6778	0.1966	0.0233	-0.0002	-0.0008	0.0046
	1213	-0.5966	0.1734	0.0185	-0.0002	-0.0006	0.0040
	1014	0.1342	-0.0416	0.0333	-0.0003	-0.0011	-0.0006
	1114	0.3565	-0.1067	0.0309	-0.0002	-0.0010	-0.0020
	1214	0.4377	-0.1300	0.0261	-0.0002	-0.0008	-0.0026
	1021	-0.2682	0.0767	0.0241	-0.0002	-0.0008	0.0020
	1121	-0.3616	0.1045	0.0102	-0.0001	-0.0003	0.0024
	1221	-0.1359	0.0385	0.0177	-0.0001	-0.0006	0.0011
	1321	-0.2293	0.0663	0.0038	0.0000	-0.0001	0.0015
	2001	-0.3457	0.0998	0.0176	-0.0001	-0.0006	0.0024
	2002	-0.2743	0.0790	0.0148	-0.0001	-0.0005	0.0019
	2011	0.2711	-0.0813	0.0290	-0.0002	-0.0009	-0.0015
	2012	0.0132	-0.0058	0.0267	-0.0002	-0.0009	0.0002
	2013	-0.6270	0.1821	0.0223	-0.0002	-0.0008	0.0042
	2014	0.1118	-0.0346	0.0277	-0.0002	-0.0009	-0.0005
21	1	-0.1570	0.0246	0.0008	-0.0001	0.0001	0.0008
	2	-0.1932	0.0311	0.0005	-0.0001	0.0001	0.0010
	3	-0.2426	0.0400	0.0005	-0.0001	0.0001	0.0013
	4	0.0040	-0.0008	0.0009	-0.0001	0.0001	0.0000
	11	1.0106	-0.1753	0.0020	-0.0001	0.0002	-0.0048
	12	0.5801	-0.1024	0.0014	-0.0001	0.0001	-0.0027
	13	-0.4720	0.0793	0.0002	0.0000	0.0001	0.0024
	14	0.7463	-0.1302	0.0016	-0.0001	0.0002	-0.0035
	15	0.2459	-0.0374	0.0014	-0.0001	0.0001	-0.0014
	16	0.3353	-0.0612	0.0011	-0.0001	0.0001	-0.0015
	21	0.0668	-0.0137	0.0008	-0.0001	0.0001	-0.0002
	1001	-0.8783	0.1418	0.0026	-0.0003	0.0005	0.0046
	1002	-0.8743	0.1410	0.0035	-0.0003	0.0006	0.0046
	1011	0.5015	-0.0956	0.0045	-0.0003	0.0006	-0.0020
	1111	0.9247	-0.1675	0.0046	-0.0003	0.0005	-0.0041
	1211	1.0647	-0.1897	0.0041	-0.0003	0.0004	-0.0049
	1012	-0.0151	-0.0082	0.0039	-0.0003	0.0006	0.0005
	1112	0.3219	-0.0655	0.0038	-0.0003	0.0005	-0.0011
	1212	0.4620	-0.0877	0.0033	-0.0002	0.0004	-0.0019
	1013	-1.2776	0.2099	0.0024	-0.0003	0.0005	0.0066
	1113	-1.1510	0.1889	0.0022	-0.0002	0.0004	0.0059
	1213	-1.0109	0.1666	0.0016	-0.0002	0.0003	0.0052
	1014	0.1844	-0.0415	0.0042	-0.0003	0.0006	-0.0005
	1114	0.5547	-0.1044	0.0042	-0.0003	0.0005	-0.0023
	1214	0.6947	-0.1266	0.0036	-0.0002	0.0004	-0.0031

JOINT DISPLACEMENT (CM		RADIANS)		STRUCTURE TYPE = SPACE			

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	1021	-0.4747	0.0730	0.0027	-0.0002	0.0004	0.0026
	1121	-0.6082	0.1005	0.0010	-0.0001	0.0002	0.0031
	1221	-0.2484	0.0364	0.0020	-0.0002	0.0003	0.0015
	1321	-0.3819	0.0638	0.0004	0.0000	0.0001	0.0019
	2001	-0.5927	0.0956	0.0018	-0.0002	0.0003	0.0031
	2002	-0.4714	0.0756	0.0016	-0.0001	0.0003	0.0025

Friday, September 21, 2007, 11:59 AM

2012	-0.0126	-0.0068	0.0032	-0.0003	0.0005	0.0004
2013	-1.0647	0.1749	0.0020	-0.0002	0.0004	0.0055
2014	0.1536	-0.0346	0.0035	-0.0003	0.0005	-0.0004
22 1	-0.2272	-0.0261	0.0027	0.0000	0.0001	0.0010
2	-0.2766	-0.0291	0.0025	0.0000	0.0001	0.0012
3	-0.3451	-0.0339	0.0025	0.0000	0.0001	0.0015
4	0.0044	-0.0006	0.0026	0.0000	0.0001	0.0000
11	1.3976	0.1028	0.0029	-0.0001	0.0001	-0.0056
12	0.7945	0.0516	0.0028	-0.0001	0.0001	-0.0031
13	-0.6642	-0.0590	0.0025	0.0000	0.0001	0.0028
14	1.0288	0.0728	0.0028	-0.0001	0.0001	-0.0041
15	0.3615	0.0457	0.0026	-0.0001	0.0001	-0.0017
16	0.4507	0.0216	0.0027	0.0000	0.0001	-0.0017
21	0.0826	-0.0025	0.0026	0.0000	0.0001	-0.0002
1001	-1.2573	-0.1315	0.0113	-0.0001	0.0004	0.0056
1002	-1.2529	-0.1322	0.0139	-0.0002	0.0005	0.0056
1011	0.6586	0.0165	0.0127	-0.0002	0.0005	-0.0022
1111	1.2514	0.0667	0.0113	-0.0002	0.0004	-0.0047
1211	1.4529	0.0888	0.0092	-0.0002	0.0003	-0.0056
1012	-0.0652	-0.0450	0.0126	-0.0002	0.0005	0.0008
1112	0.4071	-0.0050	0.0112	-0.0002	0.0004	-0.0012
1212	0.6086	0.0170	0.0091	-0.0001	0.0003	-0.0021
1013	-1.8156	-0.1777	0.0122	-0.0001	0.0005	0.0080
1113	-1.6351	-0.1599	0.0108	-0.0001	0.0004	0.0072
1213	-1.4336	-0.1378	0.0087	-0.0001	0.0003	0.0062
1014	0.2160	-0.0196	0.0126	-0.0002	0.0005	-0.0004
1114	0.7351	0.0246	0.0112	-0.0002	0.0004	-0.0026
1214	0.9366	0.0467	0.0091	-0.0001	0.0003	-0.0035
1021	-0.6944	-0.0856	0.0101	-0.0001	0.0004	0.0033
1121	-0.8596	-0.0807	0.0049	0.0000	0.0002	0.0037
1221	-0.3708	-0.0521	0.0073	-0.0001	0.0003	0.0018
1321	-0.5359	-0.0472	0.0021	0.0000	0.0001	0.0023
2001	-0.8488	-0.0891	0.0077	-0.0001	0.0003	0.0038
2002	-0.6763	-0.0721	0.0065	-0.0001	0.0002	0.0030
2011	0.5488	0.0137	0.0106	-0.0002	0.0004	-0.0018
2012	-0.0543	-0.0375	0.0105	-0.0001	0.0004	0.0007
2013	-1.5130	-0.1481	0.0102	-0.0001	0.0004	0.0066
2014	0.1800	-0.0163	0.0105	-0.0001	0.0004	-0.0003
23 1	-0.2615	-0.1238	0.0090	0.0003	-0.0001	0.0012
2	-0.3171	-0.1440	0.0095	0.0003	-0.0001	0.0014
3	-0.3946	-0.1745	0.0096	0.0003	-0.0001	0.0017

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
4		0.0044	-0.0011	0.0090	0.0002	0.0000	0.0000
11		1.5792	0.6166	0.0071	-0.0001	0.0000	-0.0063
12		0.8942	0.3337	0.0081	0.0001	0.0000	-0.0035
13		-0.7558	-0.3185	0.0101	0.0004	-0.0001	0.0032
14		1.1610	0.4467	0.0077	0.0000	0.0000	-0.0046
15		0.4178	0.2050	0.0082	0.0001	0.0000	-0.0020
16		0.5035	0.1706	0.0086	0.0001	0.0000	-0.0018
21		0.0888	0.0149	0.0090	0.0002	0.0000	-0.0002
1001		-1.4414	-0.6540	0.0413	0.0014	-0.0003	0.0065
1002		-1.4371	-0.6551	0.0503	0.0016	-0.0003	0.0065
1011		0.7272	0.2093	0.0423	0.0010	-0.0002	-0.0023
1111		1.4009	0.4884	0.0359	0.0007	-0.0001	-0.0052
1211		1.6323	0.5955	0.0285	0.0004	-0.0001	-0.0062
1012		-0.0948	-0.1303	0.0434	0.0012	-0.0002	0.0011
1112		0.4419	0.0923	0.0373	0.0009	-0.0002	-0.0012
1212		0.6733	0.1994	0.0298	0.0007	-0.0001	-0.0022
1013		-2.0748	-0.9128	0.0458	0.0016	-0.0003	0.0091
1113		-1.8681	-0.8207	0.0401	0.0014	-0.0003	0.0081
1213		-1.6367	-0.7136	0.0326	0.0012	-0.0002	0.0071

Friday, September 21, 2007, 11:59 AM

1114	0.8153	0.2505	0.0367	0.0008	-0.0002	-0.0027
1214	1.0468	0.3576	0.0293	0.0006	-0.0001	-0.0038
1021	-0.8028	-0.3936	0.0361	0.0011	-0.0002	0.0038
1121	-0.9805	-0.4235	0.0180	0.0007	-0.0001	0.0042
1221	-0.4319	-0.2261	0.0257	0.0008	-0.0001	0.0022
1321	-0.6096	-0.2559	0.0077	0.0003	-0.0001	0.0026
2001	-0.9732	-0.4422	0.0281	0.0009	-0.0002	0.0044
2002	-0.7759	-0.3550	0.0233	0.0008	-0.0001	0.0035
2011	0.6060	0.1744	0.0352	0.0008	-0.0002	-0.0019
2012	-0.0790	-0.1086	0.0362	0.0010	-0.0002	0.0009
2013	-1.7290	-0.7607	0.0382	0.0014	-0.0003	0.0076
2014	0.1878	0.0044	0.0358	0.0009	-0.0002	-0.0002
25	1	-0.2893	-0.9431	0.0031	0.0004	0.0000
	2	-0.3498	-1.1035	0.0028	0.0011	0.0000
	3	-0.4346	-1.3424	0.0027	0.0013	0.0000
	4	0.0052	-0.0041	0.0033	0.0000	0.0000
	11	1.7255	4.7561	0.0058	-0.0039	0.0001
	12	0.9742	2.5825	0.0047	-0.0020	0.0001
	13	-0.8293	-2.4404	0.0021	0.0022	-0.0001
	14	1.2674	3.4492	0.0051	-0.0028	0.0001
	15	0.4663	1.5641	0.0038	-0.0015	0.0001
	16	0.5452	1.3305	0.0042	-0.0009	0.0000
	21	0.0937	0.1255	0.0035	0.0000	0.0000
1001	-1.5901	-5.0131	0.0127	0.0041	-0.0001	0.0091
1002	-1.5849	-5.0172	0.0160	0.0041	-0.0001	0.0091
1011	0.7822	1.6405	0.0174	-0.0014	0.0000	-0.0028
1111	1.5210	3.7932	0.0165	-0.0034	0.0001	-0.0067
1211	1.7766	4.6119	0.0141	-0.0040	0.0001	-0.0082

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	1012	-0.1194	-0.9678	0.0161	0.0009	0.0000	0.0018
	1112	0.4691	0.7503	0.0150	-0.0007	0.0000	-0.0013
	1212	0.7248	1.5689	0.0126	-0.0013	0.0000	-0.0027
	1013	-2.2836	-6.9954	0.0130	0.0059	-0.0002	0.0126
	1113	-2.0557	-6.2819	0.0114	0.0051	-0.0002	0.0113
	1213	-1.8001	-5.4632	0.0090	0.0045	-0.0002	0.0098
	1014	0.2324	0.0722	0.0166	-0.0001	0.0000	0.0000
	1114	0.8796	1.9636	0.0156	-0.0018	0.0001	-0.0034
	1214	1.1352	2.7822	0.0132	-0.0024	0.0001	-0.0049
	1021	-0.8905	-3.0017	0.0120	0.0025	-0.0001	0.0054
	1121	-1.0778	-3.2526	0.0051	0.0024	-0.0001	0.0059
	1221	-0.4815	-1.7165	0.0089	0.0014	0.0000	0.0031
	1321	-0.6688	-1.9674	0.0019	0.0013	0.0000	0.0035
	2001	-1.0737	-3.3890	0.0087	0.0027	-0.0001	0.0061
	2002	-0.8564	-2.7178	0.0073	0.0021	-0.0001	0.0049
	2011	0.6518	1.3670	0.0145	-0.0012	0.0000	-0.0024
	2012	-0.0995	-0.8065	0.0134	0.0007	0.0000	0.0015
	2013	-1.9030	-5.8295	0.0109	0.0049	-0.0002	0.0105
	2014	0.1937	0.0601	0.0138	0.0000	0.0000	0.0000
26	1	-0.2946	-1.0965	0.0013	0.0005	0.0000	0.0017
	2	-0.3561	-1.2832	0.0011	0.0012	-0.0001	0.0020
	3	-0.4423	-1.5607	0.0011	0.0014	-0.0001	0.0024
	4	0.0051	-0.0047	0.0014	0.0000	0.0000	0.0000
	11	1.7522	5.5207	0.0026	-0.0045	0.0001	-0.0085
	12	0.9886	2.9963	0.0021	-0.0023	0.0000	-0.0046
	13	-0.8431	-2.8347	0.0008	0.0025	-0.0001	0.0044
	14	1.2867	4.0031	0.0023	-0.0032	0.0001	-0.0062
	15	0.4753	1.8202	0.0016	-0.0018	0.0000	-0.0028
	16	0.5526	1.5419	0.0018	-0.0011	0.0000	-0.0024
	21	0.0943	0.1438	0.0014	0.0000	0.0000	-0.0002
	1001	-1.6186	-5.8286	0.0051	0.0047	-0.0003	0.0091
	1002	-1.6134	-5.8333	0.0064	0.0047	-0.0003	0.0091

Friday, September 21, 2007, 11:59 AM

1111	1.5421	4.3975	0.0070	-0.0039	0.0000	-0.0067
1211	1.8024	5.3494	0.0060	-0.0046	0.0000	-0.0082
1012	-0.1252	-1.1328	0.0067	0.0010	-0.0002	0.0018
1112	0.4731	0.8633	0.0063	-0.0009	-0.0001	-0.0013
1212	0.7334	1.8151	0.0053	-0.0016	-0.0001	-0.0027
1013	-2.3233	-8.1300	0.0051	0.0068	-0.0004	0.0126
1113	-2.0913	-7.3001	0.0045	0.0059	-0.0003	0.0113
1213	-1.8310	-6.3482	0.0035	0.0052	-0.0003	0.0098
1014	0.2325	0.0754	0.0069	-0.0001	-0.0002	0.0000
1114	0.8904	2.2728	0.0065	-0.0021	-0.0001	-0.0034
1214	1.1507	3.2247	0.0056	-0.0028	0.0000	-0.0049
1021	-0.9077	-3.4921	0.0049	0.0028	-0.0002	0.0055
1121	-1.0963	-3.7797	0.0020	0.0027	-0.0001	0.0059
1221	-0.4913	-1.9979	0.0036	0.0016	-0.0001	0.0031
1321	-0.6799	-2.2854	0.0007	0.0015	-0.0001	0.0035

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
2001		-1.0929	-3.9403	0.0035	0.0032	-0.0002	0.0061
2002		-0.8718	-3.1600	0.0029	0.0024	-0.0002	0.0049
2011		0.6593	1.5804	0.0061	-0.0014	-0.0001	-0.0024
2012		-0.1043	-0.9440	0.0055	0.0008	-0.0002	0.0015
2013		-1.9360	-6.7750	0.0042	0.0057	-0.0003	0.0105
2014		0.1938	0.0628	0.0057	-0.0001	-0.0001	0.0000
27	1	-0.2654	-0.2383	-0.0033	-0.0002	0.0001	0.0013
	2	-0.3216	-0.2784	-0.0034	-0.0003	0.0001	0.0016
	3	-0.4002	-0.3385	-0.0034	-0.0003	0.0001	0.0019
	4	0.0045	-0.0015	-0.0036	0.0000	0.0001	0.0000
	11	1.6002	1.2096	-0.0041	0.0009	0.0000	-0.0069
	12	0.9058	0.6580	-0.0038	0.0005	0.0000	-0.0037
	13	-0.7662	-0.6195	-0.0033	-0.0005	0.0001	0.0035
	14	1.1763	0.8777	-0.0039	0.0006	0.0000	-0.0050
	15	0.4245	0.3932	-0.0038	0.0004	0.0000	-0.0022
	16	0.5096	0.3404	-0.0037	0.0002	0.0001	-0.0020
	21	0.0896	0.0335	-0.0036	0.0000	0.0001	-0.0002
1001		-1.4622	-1.2649	-0.0149	-0.0011	0.0003	0.0071
1002		-1.4577	-1.2664	-0.0184	-0.0012	0.0004	0.0071
1011		0.7355	0.4253	-0.0170	0.0002	0.0003	-0.0025
1111		1.4184	0.9701	-0.0151	0.0007	0.0002	-0.0055
1211		1.6532	1.1768	-0.0124	0.0008	0.0002	-0.0067
1012		-0.0978	-0.2366	-0.0167	-0.0004	0.0003	0.0013
1112		0.4462	0.1978	-0.0148	0.0000	0.0002	-0.0012
1212		0.6811	0.4045	-0.0121	0.0002	0.0002	-0.0023
1013		-2.1041	-1.7696	-0.0161	-0.0016	0.0004	0.0100
1113		-1.8945	-1.5906	-0.0140	-0.0014	0.0003	0.0089
1213		-1.6597	-1.3839	-0.0114	-0.0012	0.0003	0.0078
1014		0.2268	0.0270	-0.0168	-0.0001	0.0003	-0.0002
1114		0.8249	0.5054	-0.0149	0.0003	0.0002	-0.0029
1214		1.0597	0.7121	-0.0122	0.0005	0.0002	-0.0041
1021		-0.8149	-0.7557	-0.0133	-0.0007	0.0003	0.0042
1121		-0.9942	-0.8227	-0.0062	-0.0007	0.0001	0.0046
1221		-0.4387	-0.4315	-0.0096	-0.0004	0.0002	0.0024
1321		-0.6180	-0.4985	-0.0025	-0.0004	0.0001	0.0028
2001		-0.9873	-0.8552	-0.0101	-0.0008	0.0002	0.0048
2002		-0.7871	-0.6859	-0.0084	-0.0006	0.0002	0.0039
2011		0.6129	0.3544	-0.0142	0.0001	0.0002	-0.0021
2012		-0.0815	-0.1972	-0.0139	-0.0003	0.0003	0.0011
2013		-1.7534	-1.4747	-0.0134	-0.0013	0.0003	0.0083
2014		0.1890	0.0225	-0.0140	-0.0001	0.0002	-0.0002
28	1	-0.2696	-0.3635	0.0000	-0.0002	0.0000	0.0014
	2	-0.3266	-0.4250	0.0000	-0.0004	0.0000	0.0017
	3	-0.4063	-0.5172	0.0000	-0.0005	0.0000	0.0021
	4	0.0046	-0.0019	-0.0001	0.0000	0.0000	0.0000

Friday, September 21, 2007, 11:59 AM

12	0.9182	1.0065	-0.0003	0.0007	0.0000	-0.0040
13	-0.7775	-0.9458	0.0001	-0.0008	0.0000	0.0038

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
14	1.1927	1.3420	-0.0004	0.0010	0.0000	-0.0053	
15	0.4319	0.5997	-0.0003	0.0006	0.0000	-0.0024	
16	0.5161	0.5215	-0.0002	0.0003	0.0000	-0.0021	
21	0.0904	0.0520	-0.0001	0.0000	0.0000	-0.0002	
1001	-1.4849	-1.9315	0.0000	-0.0016	0.0001	0.0077	
1002	-1.4803	-1.9334	-0.0001	-0.0016	0.0002	0.0077	
1011	0.7442	0.6521	-0.0006	0.0004	0.0001	-0.0026	
1111	1.4371	1.4849	-0.0007	0.0012	0.0000	-0.0059	
1211	1.6756	1.8003	-0.0007	0.0014	0.0000	-0.0072	
1012	-0.1013	-0.3590	-0.0004	-0.0004	0.0001	0.0014	
1112	0.4507	0.3052	-0.0005	0.0002	0.0001	-0.0012	
1212	0.6892	0.6206	-0.0004	0.0004	0.0001	-0.0025	
1013	-2.1361	-2.7018	0.0001	-0.0023	0.0002	0.0108	
1113	-1.9233	-2.4280	0.0001	-0.0020	0.0001	0.0097	
1213	-1.6848	-2.1126	0.0001	-0.0018	0.0001	0.0084	
1014	0.2282	0.0435	-0.0005	-0.0001	0.0001	-0.0002	
1114	0.8350	0.7749	-0.0006	0.0006	0.0001	-0.0031	
1214	1.0736	1.0903	-0.0006	0.0008	0.0000	-0.0043	
1021	-0.8283	-1.1528	-0.0001	-0.0010	0.0001	0.0046	
1121	-1.0091	-1.2568	0.0001	-0.0010	0.0001	0.0050	
1221	-0.4462	-0.6577	-0.0001	-0.0006	0.0001	0.0026	
1321	-0.6271	-0.7616	0.0001	-0.0005	0.0000	0.0030	
2001	-1.0026	-1.3057	0.0000	-0.0011	0.0001	0.0052	
2002	-0.7994	-1.0471	0.0000	-0.0009	0.0001	0.0042	
2011	0.6202	0.5434	-0.0005	0.0004	0.0001	-0.0021	
2012	-0.0844	-0.2992	-0.0003	-0.0004	0.0001	0.0012	
2013	-1.7801	-2.2515	0.0001	-0.0019	0.0001	0.0090	
2014	0.1901	0.0363	-0.0004	-0.0001	0.0001	-0.0001	
29	1	-0.2742	-0.4983	0.0005	-0.0003	0.0000	0.0015
	2	-0.3320	-0.5828	0.0005	-0.0006	0.0000	0.0018
	3	-0.4129	-0.7094	0.0005	-0.0007	0.0000	0.0022
	4	0.0047	-0.0024	0.0005	0.0000	0.0000	0.0000
	11	1.6469	2.5310	0.0005	0.0020	-0.0001	-0.0078
	12	0.9313	1.3772	0.0005	0.0010	0.0000	-0.0042
	13	-0.7896	-1.2951	0.0005	-0.0011	0.0000	0.0040
	14	1.2103	1.8367	0.0005	0.0014	-0.0001	-0.0057
	15	0.4399	0.8231	0.0005	0.0008	0.0000	-0.0026
	16	0.5230	0.7128	0.0005	0.0005	0.0000	-0.0022
	21	0.0912	0.0704	0.0005	0.0000	0.0000	-0.0002
1001	-1.5094	-2.6486	0.0024	-0.0022	0.0001	0.0082	
1002	-1.5046	-2.6510	0.0029	-0.0022	0.0001	0.0082	
1011	0.7533	0.8886	0.0026	0.0006	0.0000	-0.0027	
1111	1.4569	2.0299	0.0022	0.0017	-0.0001	-0.0062	
1211	1.6994	2.4623	0.0018	0.0020	-0.0001	-0.0076	
1012	-0.1053	-0.4960	0.0026	-0.0005	0.0000	0.0016	
1112	0.4552	0.4145	0.0023	0.0003	0.0000	-0.0012	
1212	0.6977	0.8469	0.0018	0.0006	0.0000	-0.0026	
1013	-2.1705	-3.7027	0.0026	-0.0031	0.0001	0.0115	

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
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Friday, September 21, 2007, 11:59 AM

	1014	0.2294	0.0554	0.0026	0.0000	0.0000	-0.0001
	1114	0.8457	1.0578	0.0023	0.0009	0.0000	-0.0032
	1214	1.0881	1.4902	0.0018	0.0012	0.0000	-0.0046
	1021	-0.8427	-1.5816	0.0021	-0.0013	0.0000	0.0049
	1121	-1.0251	-1.7225	0.0010	-0.0013	0.0000	0.0053
	1221	-0.4544	-0.9026	0.0015	-0.0008	0.0000	0.0028
	1321	-0.6368	-1.0435	0.0004	-0.0007	0.0000	0.0032
	2001	-1.0191	-1.7905	0.0016	-0.0015	0.0000	0.0056
	2002	-0.8127	-1.4358	0.0013	-0.0011	0.0000	0.0045
	2011	0.6278	0.7405	0.0021	0.0005	0.0000	-0.0022
	2012	-0.0878	-0.4133	0.0022	-0.0004	0.0000	0.0013
	2013	-1.8087	-3.0856	0.0022	-0.0026	0.0001	0.0096
	2014	0.1911	0.0462	0.0021	0.0000	0.0000	-0.0001
30	1	-0.2790	-0.6415	0.0000	-0.0003	0.0000	0.0016
	2	-0.3377	-0.7504	0.0000	-0.0007	0.0000	0.0019
	3	-0.4199	-0.9132	0.0000	-0.0009	0.0000	0.0023
	4	0.0049	-0.0029	0.0000	0.0000	0.0000	0.0000
	11	1.6723	3.2500	-0.0002	0.0026	-0.0001	-0.0082
	12	0.9452	1.7671	-0.0001	0.0013	-0.0001	-0.0044
	13	-0.8024	-1.6645	0.0000	-0.0015	0.0000	0.0042
	14	1.2287	2.3579	-0.0001	0.0019	-0.0001	-0.0059
	15	0.4484	1.0611	-0.0001	0.0010	0.0000	-0.0027
	16	0.5302	0.9132	-0.0001	0.0006	0.0000	-0.0023
	21	0.0920	0.0888	0.0000	0.0000	0.0000	-0.0002
	1001	-1.5353	-3.4097	-0.0001	-0.0028	0.0001	0.0087
	1002	-1.5305	-3.4127	-0.0001	-0.0028	0.0000	0.0087
	1011	0.7627	1.1339	-0.0003	0.0009	-0.0001	-0.0028
	1111	1.4777	2.6013	-0.0003	0.0022	-0.0001	-0.0065
	1211	1.7244	3.1581	-0.0003	0.0026	-0.0001	-0.0079
	1012	-0.1098	-0.6455	-0.0002	-0.0006	0.0000	0.0017
	1112	0.4598	0.5253	-0.0002	0.0004	-0.0001	-0.0012
	1212	0.7065	1.0821	-0.0002	0.0009	-0.0001	-0.0027
	1013	-2.2069	-4.7635	0.0000	-0.0040	0.0001	0.0121
	1113	-1.9869	-4.2789	0.0000	-0.0035	0.0001	0.0108
	1213	-1.7402	-3.7222	0.0000	-0.0031	0.0001	0.0094
	1014	0.2304	0.0634	-0.0002	0.0000	-0.0001	-0.0001
	1114	0.8567	1.3524	-0.0002	0.0012	-0.0001	-0.0033
	1214	1.1034	1.9092	-0.0002	0.0016	-0.0001	-0.0047
	1021	-0.8580	-2.0380	-0.0001	-0.0017	0.0000	0.0052
	1121	-1.0421	-2.2157	0.0000	-0.0016	0.0000	0.0056
	1221	-0.4631	-1.1639	-0.0001	-0.0010	0.0000	0.0030
	1321	-0.6471	-1.3415	0.0000	-0.0009	0.0000	0.0034
	2001	-1.0367	-2.3051	0.0000	-0.0019	0.0000	0.0059
	2002	-0.8267	-1.8485	0.0000	-0.0015	0.0000	0.0047
	2011	0.6356	0.9449	-0.0002	0.0007	-0.0001	-0.0023
	2012	-0.0915	-0.5380	-0.0002	-0.0005	0.0000	0.0014

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	2013	-1.8391	-3.9696	0.0000	-0.0034	0.0001	0.0101
	2014	0.1920	0.0528	-0.0002	0.0000	0.0000	-0.0001
31	1	-0.2841	-0.7907	-0.0016	-0.0004	0.0000	0.0017
	2	-0.3437	-0.9251	-0.0014	-0.0009	0.0000	0.0020
	3	-0.4272	-1.1255	-0.0014	-0.0011	0.0000	0.0024
	4	0.0050	-0.0035	-0.0016	0.0000	0.0000	0.0000
	11	1.6986	3.9959	-0.0029	0.0033	-0.0002	-0.0084
	12	0.9596	2.1711	-0.0024	0.0017	-0.0001	-0.0045
	13	-0.8157	-2.0486	-0.0011	-0.0018	0.0000	0.0043
	14	1.2479	2.8984	-0.0025	0.0023	-0.0001	-0.0061
	15	0.4573	1.3098	-0.0019	0.0013	-0.0001	-0.0028
	16	0.5376	1.1201	-0.0021	0.0008	-0.0001	-0.0023
	21	0.0928	0.1072	-0.0017	0.0000	0.0000	-0.0002
	1001	-1.5624	-4.2029	-0.0064	-0.0035	0.0000	0.0089

Friday, September 21, 2007, 11:59 AM

1011	0.7724	1.3855	-0.0087	0.0011	-0.0002	-0.0028
1111	1.4992	3.1921	-0.0082	0.0028	-0.0002	-0.0066
1211	1.7503	3.8785	-0.0070	0.0033	-0.0002	-0.0081
1012	-0.1145	-0.8042	-0.0081	-0.0008	-0.0001	0.0018
1112	0.4645	0.6375	-0.0075	0.0006	-0.0001	-0.0013
1212	0.7156	1.3238	-0.0063	0.0011	-0.0001	-0.0027
1013	-2.2449	-5.8679	-0.0065	-0.0050	0.0001	0.0124
1113	-2.0209	-5.2701	-0.0057	-0.0043	0.0001	0.0112
1213	-1.7698	-4.5838	-0.0045	-0.0038	0.0001	0.0097
1014	0.2314	0.0685	-0.0083	0.0000	-0.0001	0.0000
1114	0.8681	1.6557	-0.0078	0.0015	-0.0002	-0.0034
1214	1.1192	2.3420	-0.0066	0.0020	-0.0002	-0.0049
1021	-0.8741	-2.5145	-0.0060	-0.0021	0.0000	0.0054
1121	-1.0598	-2.7288	-0.0025	-0.0020	0.0000	0.0058
1221	-0.4722	-1.4370	-0.0044	-0.0012	0.0000	0.0031
1321	-0.6579	-1.6513	-0.0010	-0.0011	0.0000	0.0035
2001	-1.0550	-2.8413	-0.0044	-0.0023	0.0000	0.0060
2002	-0.8414	-2.2785	-0.0037	-0.0018	0.0000	0.0048
2011	0.6437	1.1546	-0.0072	0.0010	-0.0001	-0.0024
2012	-0.0954	-0.6702	-0.0067	-0.0006	-0.0001	0.0015
2013	-1.8707	-4.8899	-0.0055	-0.0041	0.0001	0.0104
2014	0.1929	0.0571	-0.0069	0.0000	-0.0001	0.0000
32 1	-0.2759	-0.2621	0.0000	0.0000	0.0000	0.0015
2	-0.3512	-0.3490	0.0000	0.0000	0.0000	0.0020
3	-0.4371	-0.4216	0.0000	0.0000	0.0000	0.0024
4	0.0032	-0.0055	0.0000	0.0000	0.0000	0.0000
11	1.7515	1.4587	0.0000	0.0000	0.0000	-0.0083
12	0.9926	0.7835	0.0000	0.0000	0.0000	-0.0045
13	-0.8382	-0.7630	0.0000	0.0000	0.0000	0.0043
14	1.2880	1.0542	0.0000	0.0000	0.0000	-0.0060
15	0.4548	0.4957	0.0000	0.0000	0.0000	-0.0028
16	0.5603	0.3949	0.0000	0.0000	0.0000	-0.0023
21	0.0992	0.0277	0.0000	0.0000	0.0000	-0.0002

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
1001	-1.5774	-1.5301	0.0000	0.0000	0.0000	0.0000	0.0086
1002	-1.5741	-1.5356	0.0000	0.0000	0.0000	0.0000	0.0086
1011	0.8247	0.5112	0.0000	0.0000	0.0000	0.0000	-0.0030
1111	1.5741	1.1866	0.0000	0.0000	0.0000	0.0000	-0.0068
1211	1.8250	1.4311	0.0000	0.0000	0.0000	0.0000	-0.0082
1012	-0.0859	-0.2990	0.0000	0.0000	0.0000	0.0000	0.0016
1112	0.5117	0.2415	0.0000	0.0000	0.0000	0.0000	-0.0015
1212	0.7626	0.4859	0.0000	0.0000	0.0000	0.0000	-0.0029
1013	-2.2829	-2.1548	0.0000	0.0000	0.0000	0.0000	0.0121
1113	-2.0514	-1.9236	0.0000	0.0000	0.0000	0.0000	0.0108
1213	-1.8006	-1.6792	0.0000	0.0000	0.0000	0.0000	0.0094
1014	0.2685	0.0258	0.0000	0.0000	0.0000	0.0000	-0.0003
1114	0.9252	0.6204	0.0000	0.0000	0.0000	0.0000	-0.0036
1214	1.1760	0.8648	0.0000	0.0000	0.0000	0.0000	-0.0050
1021	-0.8719	-0.9164	0.0000	0.0000	0.0000	0.0000	0.0051
1121	-1.0704	-0.9718	0.0000	0.0000	0.0000	0.0000	0.0055
1221	-0.4652	-0.5223	0.0000	0.0000	0.0000	0.0000	0.0029
1321	-0.6637	-0.5777	0.0000	0.0000	0.0000	0.0000	0.0033
2001	-1.0642	-1.0327	0.0000	0.0000	0.0000	0.0000	0.0058
2002	-0.8457	-0.8219	0.0000	0.0000	0.0000	0.0000	0.0046
2011	0.6873	0.4260	0.0000	0.0000	0.0000	0.0000	-0.0025
2012	-0.0716	-0.2492	0.0000	0.0000	0.0000	0.0000	0.0013
2013	-1.9024	-1.7957	0.0000	0.0000	0.0000	0.0000	0.0101
2014	0.2237	0.0215	0.0000	0.0000	0.0000	0.0000	-0.0002
33 1	-0.2804	-0.3996	0.0000	0.0000	0.0000	0.0000	0.0016
2	-0.3573	-0.5326	0.0000	0.0000	0.0000	0.0000	0.0021
3	-0.4445	-0.6445	0.0000	0.0000	0.0000	0.0000	0.0026

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

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Friday, September 21, 2007, 11:59 AM

2002	-0.8755	-1.7220	0.0000	0.0000	0.0000	0.0054
2011	0.7062	0.9069	0.0000	0.0000	0.0000	-0.0028
2012	-0.0784	-0.5112	0.0000	0.0000	0.0000	0.0016
2013	-1.9684	-3.7641	0.0000	0.0000	0.0000	0.0117
2014	0.2272	0.0569	0.0000	0.0000	0.0000	-0.0002
35	1	-0.2905	-0.7058	0.0000	0.0000	0.0018
	2	-0.3709	-0.9425	0.0000	0.0000	0.0024

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	3	-0.4612	-1.1415	0.0000	0.0000	0.0000	0.0029
	4	0.0039	-0.0104	0.0000	0.0000	0.0000	0.0000
	11	1.8389	3.9552	0.0000	0.0000	0.0000	-0.0100
	12	1.0401	2.1287	0.0000	0.0000	0.0000	-0.0054
	13	-0.8822	-2.0591	0.0000	0.0000	0.0000	0.0052
	14	1.3515	2.8606	0.0000	0.0000	0.0000	-0.0073
	15	0.4851	1.3403	0.0000	0.0000	0.0000	-0.0034
	16	0.5848	1.0776	0.0000	0.0000	0.0000	-0.0027
	21	0.1019	0.0807	0.0000	0.0000	0.0000	-0.0002
1001		-1.6638	-4.1341	0.0000	0.0000	0.0000	0.0105
1002		-1.6599	-4.1444	0.0000	0.0000	0.0000	0.0106
1011		0.8596	1.3985	0.0000	0.0000	0.0000	-0.0035
1111		1.6486	3.2297	0.0000	0.0000	0.0000	-0.0082
1211		1.9131	3.8890	0.0000	0.0000	0.0000	-0.0099
1012		-0.0989	-0.7933	0.0000	0.0000	0.0000	0.0021
1112		0.5302	0.6727	0.0000	0.0000	0.0000	-0.0017
1212		0.7948	1.3320	0.0000	0.0000	0.0000	-0.0034
1013		-2.4058	-5.8187	0.0000	0.0000	0.0000	0.0148
1113		-2.1611	-5.1903	0.0000	0.0000	0.0000	0.0132
1213		-1.8965	-4.5310	0.0000	0.0000	0.0000	0.0115
1014		0.2747	0.0849	0.0000	0.0000	0.0000	-0.0002
1114		0.9661	1.6972	0.0000	0.0000	0.0000	-0.0043
1214		1.2307	2.3565	0.0000	0.0000	0.0000	-0.0060
1021		-0.9223	-2.4680	0.0000	0.0000	0.0000	0.0063
1121		-1.1262	-2.6294	0.0000	0.0000	0.0000	0.0067
1221		-0.4933	-1.4028	0.0000	0.0000	0.0000	0.0036
1321		-0.6972	-1.5641	0.0000	0.0000	0.0000	0.0040
2001		-1.1226	-2.7898	0.0000	0.0000	0.0000	0.0071
2002		-0.8920	-2.2191	0.0000	0.0000	0.0000	0.0057
2011		0.7164	1.1654	0.0000	0.0000	0.0000	-0.0029
2012		-0.0824	-0.6611	0.0000	0.0000	0.0000	0.0017
2013		-2.0048	-4.8489	0.0000	0.0000	0.0000	0.0123
2014		0.2289	0.0708	0.0000	0.0000	0.0000	-0.0001
36	1	-0.2959	-0.8704	0.0000	0.0000	0.0000	0.0019
	2	-0.3782	-1.1623	0.0000	0.0000	0.0000	0.0025
	3	-0.4701	-1.4080	0.0000	0.0000	0.0000	0.0030
	4	0.0042	-0.0118	0.0000	0.0000	0.0000	0.0000
	11	1.8710	4.8721	0.0000	0.0000	0.0000	-0.0103
	12	1.0575	2.6214	0.0000	0.0000	0.0000	-0.0055
	13	-0.8985	-2.5365	0.0000	0.0000	0.0000	0.0054
	14	1.3748	3.5235	0.0000	0.0000	0.0000	-0.0075
	15	0.4963	1.6554	0.0000	0.0000	0.0000	-0.0036
	16	0.5937	1.3260	0.0000	0.0000	0.0000	-0.0028
	21	0.1028	0.0985	0.0000	0.0000	0.0000	-0.0002
1001		-1.6959	-5.0987	0.0000	0.0000	0.0000	0.0109
1002		-1.6917	-5.1105	0.0000	0.0000	0.0000	0.0109
1011		0.8722	1.7176	0.0000	0.0000	0.0000	-0.0036
1111		1.6757	3.9750	0.0000	0.0000	0.0000	-0.0084

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

Friday, September 21, 2007, 11:59 AM

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	1211	1.9453	4.7881	0.0000	0.0000	0.0000	-0.0101
	1012	-0.1040	-0.9832	0.0000	0.0000	0.0000	0.0021
	1112	0.5368	0.8241	0.0000	0.0000	0.0000	-0.0017
	1212	0.8064	1.6372	0.0000	0.0000	0.0000	-0.0034
	1013	-2.4512	-7.1728	0.0000	0.0000	0.0000	0.0152
	1113	-2.2016	-6.3970	0.0000	0.0000	0.0000	0.0136
	1213	-1.9320	-5.5839	0.0000	0.0000	0.0000	0.0119
	1014	0.2767	0.0992	0.0000	0.0000	0.0000	-0.0001
	1114	0.9809	2.0870	0.0000	0.0000	0.0000	-0.0044
	1214	1.2505	2.9001	0.0000	0.0000	0.0000	-0.0061
	1021	-0.9411	-3.0448	0.0000	0.0000	0.0000	0.0065
	1121	-1.1468	-3.2419	0.0000	0.0000	0.0000	0.0069
	1221	-0.5038	-1.7309	0.0000	0.0000	0.0000	0.0037
	1321	-0.7095	-1.9280	0.0000	0.0000	0.0000	0.0041
	2001	-1.1442	-3.4408	0.0000	0.0000	0.0000	0.0073
	2002	-0.9092	-2.7368	0.0000	0.0000	0.0000	0.0058
	2011	0.7268	1.4313	0.0000	0.0000	0.0000	-0.0030
	2012	-0.0867	-0.8194	0.0000	0.0000	0.0000	0.0018
	2013	-2.0427	-5.9773	0.0000	0.0000	0.0000	0.0127
	2014	0.2305	0.0827	0.0000	0.0000	0.0000	-0.0001
37	1	-0.2654	-0.2383	0.0033	0.0002	-0.0001	0.0013
	2	-0.3216	-0.2784	0.0034	0.0003	-0.0001	0.0016
	3	-0.4002	-0.3385	0.0034	0.0003	-0.0001	0.0019
	4	0.0045	-0.0015	0.0036	0.0000	-0.0001	0.0000
	11	1.6002	1.2096	0.0041	-0.0009	0.0000	-0.0069
	12	0.9058	0.6580	0.0038	-0.0005	0.0000	-0.0037
	13	-0.7662	-0.6195	0.0033	0.0005	-0.0001	0.0035
	14	1.1763	0.8777	0.0039	-0.0006	0.0000	-0.0050
	15	0.4245	0.3932	0.0038	-0.0004	0.0000	-0.0022
	16	0.5096	0.3404	0.0037	-0.0002	-0.0001	-0.0020
	21	0.0896	0.0335	0.0036	0.0000	-0.0001	-0.0002
1001		-1.4622	-1.2649	0.0149	0.0011	-0.0003	0.0071
1002		-1.4577	-1.2664	0.0184	0.0012	-0.0004	0.0071
1011		0.7355	0.4253	0.0170	-0.0002	-0.0003	-0.0025
1111		1.4184	0.9701	0.0151	-0.0007	-0.0002	-0.0055
1211		1.6532	1.1768	0.0124	-0.0008	-0.0002	-0.0067
1012		-0.0978	-0.2366	0.0167	0.0004	-0.0003	0.0013
1112		0.4462	0.1978	0.0148	0.0000	-0.0002	-0.0012
1212		0.6811	0.4045	0.0121	-0.0002	-0.0002	-0.0023
1013		-2.1041	-1.7696	0.0161	0.0016	-0.0004	0.0100
1113		-1.8945	-1.5906	0.0140	0.0014	-0.0003	0.0089
1213		-1.6597	-1.3839	0.0114	0.0012	-0.0003	0.0078
1014		0.2268	0.0270	0.0168	0.0001	-0.0003	-0.0002
1114		0.8249	0.5054	0.0149	-0.0003	-0.0002	-0.0029
1214		1.0597	0.7121	0.0122	-0.0005	-0.0002	-0.0041
1021		-0.8149	-0.7557	0.0133	0.0007	-0.0003	0.0042
1121		-0.9942	-0.8227	0.0062	0.0007	-0.0001	0.0046
1221		-0.4387	-0.4315	0.0096	0.0004	-0.0002	0.0024

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	1321	-0.6180	-0.4985	0.0025	0.0004	-0.0001	0.0028
	2001	-0.9873	-0.8552	0.0101	0.0008	-0.0002	0.0048
	2002	-0.7871	-0.6859	0.0084	0.0006	-0.0002	0.0039
	2011	0.6129	0.3544	0.0142	-0.0001	-0.0002	-0.0021
	2012	-0.0815	-0.1972	0.0139	0.0003	-0.0003	0.0011
	2013	-1.7534	-1.4747	0.0134	0.0013	-0.0003	0.0083
	2014	0.1890	0.0225	0.0140	0.0001	-0.0002	-0.0002
38	1	-0.2696	-0.3635	0.0000	0.0002	0.0000	0.0014
	2	-0.3266	-0.4250	0.0000	0.0004	0.0000	0.0017

Friday, September 21, 2007, 11:59 AM

	4	0.0046	-0.0019	0.0001	0.0000	0.0000	0.0000
	11	1.6228	1.8491	0.0005	-0.0014	0.0000	-0.0074
	12	0.9182	1.0065	0.0003	-0.0007	0.0000	-0.0040
	13	-0.7775	-0.9458	-0.0001	0.0008	0.0000	0.0038
	14	1.1927	1.3420	0.0004	-0.0010	0.0000	-0.0053
	15	0.4319	0.5997	0.0003	-0.0006	0.0000	-0.0024
	16	0.5161	0.5215	0.0002	-0.0003	0.0000	-0.0021
	21	0.0904	0.0520	0.0001	0.0000	0.0000	-0.0002
	1001	-1.4849	-1.9315	0.0000	0.0016	-0.0001	0.0077
	1002	-1.4803	-1.9334	0.0001	0.0016	-0.0002	0.0077
	1011	0.7442	0.6521	0.0006	-0.0004	-0.0001	-0.0026
	1111	1.4371	1.4849	0.0007	-0.0012	0.0000	-0.0059
	1211	1.6756	1.8003	0.0007	-0.0014	0.0000	-0.0072
	1012	-0.1013	-0.3590	0.0004	0.0004	-0.0001	0.0014
	1112	0.4507	0.3052	0.0005	-0.0002	-0.0001	-0.0012
	1212	0.6892	0.6206	0.0004	-0.0004	-0.0001	-0.0025
	1013	-2.1361	-2.7018	-0.0001	0.0023	-0.0002	0.0108
	1113	-1.9233	-2.4280	-0.0001	0.0020	-0.0001	0.0097
	1213	-1.6848	-2.1126	-0.0001	0.0018	-0.0001	0.0084
	1014	0.2282	0.0435	0.0005	0.0001	-0.0001	-0.0002
	1114	0.8350	0.7749	0.0006	-0.0006	-0.0001	-0.0031
	1214	1.0736	1.0903	0.0006	-0.0008	0.0000	-0.0043
	1021	-0.8283	-1.1528	0.0001	0.0010	-0.0001	0.0046
	1121	-1.0091	-1.2568	-0.0001	0.0010	-0.0001	0.0050
	1221	-0.4462	-0.6577	0.0001	0.0006	-0.0001	0.0026
	1321	-0.6271	-0.7616	-0.0001	0.0005	0.0000	0.0030
	2001	-1.0026	-1.3057	0.0000	0.0011	-0.0001	0.0052
	2002	-0.7994	-1.0471	0.0000	0.0009	-0.0001	0.0042
	2011	0.6202	0.5434	0.0005	-0.0004	-0.0001	-0.0021
	2012	-0.0844	-0.2992	0.0003	0.0004	-0.0001	0.0012
	2013	-1.7801	-2.2515	-0.0001	0.0019	-0.0001	0.0090
	2014	0.1901	0.0363	0.0004	0.0001	-0.0001	-0.0001
39	1	-0.2742	-0.4983	-0.0005	0.0003	0.0000	0.0015
	2	-0.3320	-0.5828	-0.0005	0.0006	0.0000	0.0018
	3	-0.4129	-0.7094	-0.0005	0.0007	0.0000	0.0022
	4	0.0047	-0.0024	-0.0005	0.0000	0.0000	0.0000
	11	1.6469	2.5310	-0.0005	-0.0020	0.0001	-0.0078
	12	0.9313	1.3772	-0.0005	-0.0010	0.0000	-0.0042

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	13	-0.7896	-1.2951	-0.0005	0.0011	0.0000	0.0040
	14	1.2103	1.8367	-0.0005	-0.0014	0.0001	-0.0057
	15	0.4399	0.8231	-0.0005	-0.0008	0.0000	-0.0026
	16	0.5230	0.7128	-0.0005	-0.0005	0.0000	-0.0022
	21	0.0912	0.0704	-0.0005	0.0000	0.0000	-0.0002
	1001	-1.5094	-2.6486	-0.0024	0.0022	-0.0001	0.0082
	1002	-1.5046	-2.6510	-0.0029	0.0022	-0.0001	0.0082
	1011	0.7533	0.8886	-0.0026	-0.0006	0.0000	-0.0027
	1111	1.4569	2.0299	-0.0022	-0.0017	0.0001	-0.0062
	1211	1.6994	2.4623	-0.0018	-0.0020	0.0001	-0.0076
	1012	-0.1053	-0.4960	-0.0026	0.0005	0.0000	0.0016
	1112	0.4552	0.4145	-0.0023	-0.0003	0.0000	-0.0012
	1212	0.6977	0.8469	-0.0018	-0.0006	0.0000	-0.0026
	1013	-2.1705	-3.7027	-0.0026	0.0031	-0.0001	0.0115
	1113	-1.9541	-3.3267	-0.0023	0.0027	-0.0001	0.0103
	1213	-1.7117	-2.8943	-0.0018	0.0024	-0.0001	0.0090
	1014	0.2294	0.0554	-0.0026	0.0000	0.0000	-0.0001
	1114	0.8457	1.0578	-0.0023	-0.0009	0.0000	-0.0032
	1214	1.0881	1.4902	-0.0018	-0.0012	0.0000	-0.0046
	1021	-0.8427	-1.5816	-0.0021	0.0013	0.0000	0.0049
	1121	-1.0251	-1.7225	-0.0010	0.0013	0.0000	0.0053
	1221	-0.4544	-0.9026	-0.0015	0.0008	0.0000	0.0028

Friday, September 21, 2007, 11:59 AM

	2001	-1.0191	-1.7905	-0.0016	0.0015	0.0000	0.0056
	2002	-0.8127	-1.4358	-0.0013	0.0011	0.0000	0.0045
	2011	0.6278	0.7405	-0.0021	-0.0005	0.0000	-0.0022
	2012	-0.0878	-0.4133	-0.0022	0.0004	0.0000	0.0013
	2013	-1.8087	-3.0856	-0.0022	0.0026	-0.0001	0.0096
	2014	0.1911	0.0462	-0.0021	0.0000	0.0000	-0.0001
40	1	-0.2790	-0.6415	0.0000	0.0003	0.0000	0.0016
	2	-0.3377	-0.7504	0.0000	0.0007	0.0000	0.0019
	3	-0.4199	-0.9132	0.0000	0.0009	0.0000	0.0023
	4	0.0049	-0.0029	0.0000	0.0000	0.0000	0.0000
	11	1.6723	3.2500	0.0002	-0.0026	0.0001	-0.0082
	12	0.9452	1.7671	0.0001	-0.0013	0.0001	-0.0044
	13	-0.8024	-1.6645	0.0000	0.0015	0.0000	0.0042
	14	1.2287	2.3579	0.0001	-0.0019	0.0001	-0.0059
	15	0.4484	1.0611	0.0001	-0.0010	0.0000	-0.0027
	16	0.5302	0.9132	0.0001	-0.0006	0.0000	-0.0023
	21	0.0920	0.0888	0.0000	0.0000	0.0000	-0.0002
1001		-1.5353	-3.4097	0.0001	0.0028	-0.0001	0.0087
1002		-1.5305	-3.4127	0.0001	0.0028	0.0000	0.0087
1011		0.7627	1.1339	0.0003	-0.0009	0.0001	-0.0028
1111		1.4777	2.6013	0.0003	-0.0022	0.0001	-0.0065
1211		1.7244	3.1581	0.0003	-0.0026	0.0001	-0.0079
1012		-0.1098	-0.6455	0.0002	0.0006	0.0000	0.0017
1112		0.4598	0.5253	0.0002	-0.0004	0.0001	-0.0012
1212		0.7065	1.0821	0.0002	-0.0009	0.0001	-0.0027

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	1013	-2.2069	-4.7635	0.0000	0.0040	-0.0001	0.0121
	1113	-1.9869	-4.2789	0.0000	0.0035	-0.0001	0.0108
	1213	-1.7402	-3.7222	0.0000	0.0031	-0.0001	0.0094
	1014	0.2304	0.0634	0.0002	0.0000	0.0001	-0.0001
	1114	0.8567	1.3524	0.0002	-0.0012	0.0001	-0.0033
	1214	1.1034	1.9092	0.0002	-0.0016	0.0001	-0.0047
	1021	-0.8580	-2.0380	0.0001	0.0017	0.0000	0.0052
	1121	-1.0421	-2.2157	0.0000	0.0016	0.0000	0.0056
	1221	-0.4631	-1.1639	0.0001	0.0010	0.0000	0.0030
	1321	-0.6471	-1.3415	0.0000	0.0009	0.0000	0.0034
	2001	-1.0367	-2.3051	0.0000	0.0019	0.0000	0.0059
	2002	-0.8267	-1.8485	0.0000	0.0015	0.0000	0.0047
	2011	0.6356	0.9449	0.0002	-0.0007	0.0001	-0.0023
	2012	-0.0915	-0.5380	0.0002	0.0005	0.0000	0.0014
	2013	-1.8391	-3.9696	0.0000	0.0034	-0.0001	0.0101
	2014	0.1920	0.0528	0.0002	0.0000	0.0000	-0.0001
41	1	-0.2841	-0.7907	0.0016	0.0004	0.0000	0.0017
	2	-0.3437	-0.9251	0.0014	0.0009	0.0000	0.0020
	3	-0.4272	-1.1255	0.0014	0.0011	0.0000	0.0024
	4	0.0050	-0.0035	0.0016	0.0000	0.0000	0.0000
	11	1.6986	3.9959	0.0029	-0.0033	0.0002	-0.0084
	12	0.9596	2.1711	0.0024	-0.0017	0.0001	-0.0045
	13	-0.8157	-2.0486	0.0011	0.0018	0.0000	0.0043
	14	1.2479	2.8984	0.0025	-0.0023	0.0001	-0.0061
	15	0.4573	1.3098	0.0019	-0.0013	0.0001	-0.0028
	16	0.5376	1.1201	0.0021	-0.0008	0.0001	-0.0023
	21	0.0928	0.1072	0.0017	0.0000	0.0000	-0.0002
1001		-1.5624	-4.2029	0.0064	0.0035	0.0000	0.0089
1002		-1.5574	-4.2064	0.0080	0.0035	0.0000	0.0089
1011		0.7724	1.3855	0.0087	-0.0011	0.0002	-0.0028
1111		1.4992	3.1921	0.0082	-0.0028	0.0002	-0.0066
1211		1.7503	3.8785	0.0070	-0.0033	0.0002	-0.0081
1012		-0.1145	-0.8042	0.0081	0.0008	0.0001	0.0018
1112		0.4645	0.6375	0.0075	-0.0006	0.0001	-0.0013
1212		0.7156	1.3238	0.0063	-0.0011	0.0001	-0.0027

Friday, September 21, 2007, 11:59 AM

1113	-2.0209	-5.2701	0.0057	0.0043	-0.0001	0.0112
1213	-1.7698	-4.5838	0.0045	0.0038	-0.0001	0.0097
1014	0.2314	0.0685	0.0083	0.0000	0.0001	0.0000
1114	0.8681	1.6557	0.0078	-0.0015	0.0002	-0.0034
1214	1.1192	2.3420	0.0066	-0.0020	0.0002	-0.0049
1021	-0.8741	-2.5145	0.0060	0.0021	0.0000	0.0054
1121	-1.0598	-2.7288	0.0025	0.0020	0.0000	0.0058
1221	-0.4722	-1.4370	0.0044	0.0012	0.0000	0.0031
1321	-0.6579	-1.6513	0.0010	0.0011	0.0000	0.0035
2001	-1.0550	-2.8413	0.0044	0.0023	0.0000	0.0060
2002	-0.8414	-2.2785	0.0037	0.0018	0.0000	0.0048
2011	0.6437	1.1546	0.0072	-0.0010	0.0001	-0.0024

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
2012		-0.0954	-0.6702	0.0067	0.0006	0.0001	0.0015
2013		-1.8707	-4.8899	0.0055	0.0041	-0.0001	0.0104
2014		0.1929	0.0571	0.0069	0.0000	0.0001	0.0000

***** END OF LATEST ANALYSIS RESULT *****

201. PARAMETER 1
 202. CODE BS5950
 203. PY 345000 MEMB 1 TO 5 7 TO 12 14 TO 19 21 30 TO 66
 204. PY 900000 MEMB 22 TO 29
 205. SBLT 1 MEMB 1 TO 5 7 TO 12 14 TO 19 21 30 TO 44
 206. SBLT 2 MEMB 45 TO 66
 207. SGR 1 MEMB 1 TO 5 7 TO 12 14 TO 19 21 30 TO 44
 208. *UNL 0.9 MEMB 1 TO 5 7 TO 12 14 TO 19 21 30 TO 44
 209. TRACK 2 MEMB 1 TO 5 7 TO 12 14 TO 19 21 TO 44
 210. CHECK CODE MEMB 1 TO 5 7 TO 12 14 TO 19 21 TO 44

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.9_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
1	TAPERED	PASS	BS-4.7 (C)	0.298	1013
		57.42 C	-0.77	227.03	3.20

CALCULATED CAPACITIES FOR MEMB 1 UNIT - kN,m SECTION CLASS 2					
MCZ= 820.5 MCY= 161.9 PC= 3015.1 PT= 4381.5 MB= 698.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 1076 kN : qw = 207 N/mm2					
d = 519 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

2	TAPERED	PASS	BS-4.7 (C)	0.299	1013
		64.14 C	-1.14	227.03	0.00

Friday, September 21, 2007, 11:59 AM

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| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw      = 207 N/mm2 |
| d   = 519 mm   :   t   =   9 mm           : a   =   0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2      status = PASS              : BS-4.4.3.3 status = PASS      |
|-----|
|      3      TAPERED      PASS      BS-4.7      (C)      0.285      1013
|              47.69 C              0.34              222.61      0.00
|-----|
| CALCULATED CAPACITIES FOR MEMB      3 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 820.5   MCY= 161.9   PC= 4117.5   PT= 4381.5   MB= 925.4 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw      = 207 N/mm2 |
| d   = 519 mm   :   t   =   9 mm           : a   =   0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2      status = PASS              : BS-4.4.3.3 status = PASS      |
|-----|
|      4      TAPERED      PASS      BS-4.7      (C)      0.240      1013
|              16.57 C              0.91              188.98      0.00
|-----|
| CALCULATED CAPACITIES FOR MEMB      4 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 820.5   MCY= 161.9   PC= 4117.0   PT= 0.0      MB= 925.4 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw      = 207 N/mm2 |
| d   = 519 mm   :   t   =   9 mm           : a   =   0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2      status = PASS              : BS-4.4.3.3 status = PASS      |
|-----|
|      5      TAPERED      PASS      BS-4.7      (C)      0.197      1013
|              16.18 C              -3.16              142.58      0.00

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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|-----|
| CALCULATED CAPACITIES FOR MEMB      5 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 820.5   MCY= 161.9   PC= 4120.0   PT= 0.0      MB= 925.4 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw      = 207 N/mm2 |
| d   = 519 mm   :   t   =   9 mm           : a   =   0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2      status = PASS              : BS-4.4.3.3 status = PASS      |
|-----|
|      7      TAPERED      PASS      BS-4.7      (C)      0.022      1002
|              0.15 C              -2.71              2.55      0.00
|-----|
| CALCULATED CAPACITIES FOR MEMB      7 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 445.6   MCY= 161.8   PC= 3431.1   PT= 3642.2   MB= 495.5 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 632 kN : qw      = 207 N/mm2 |
| d   = 305 mm   :   t   =   9 mm           : a   =   0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2      status = PASS              : BS-4.4.3.3 status = PASS      |
|-----|
|      8      TAPERED      PASS      BS-4.7      (C)      0.371      1013
|              82.13 C              0.00              284.92      3.20
|-----|
| CALCULATED CAPACITIES FOR MEMB      8 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 820.5   MCY= 161.9   PC= 3018.2   PT= 4381.5   MB= 677.3 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw      = 207 N/mm2 |
| d   = 519 mm   :   t   =   9 mm           : a   =   0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2      status = PASS              : BS-4.4.3.3 status = PASS      |
|-----|
|      9      TAPERED      PASS      BS-4.7      (C)      0.390      1013
|              100.29 C              0.00              -300.22      0.90
|-----|
| CALCULATED CAPACITIES FOR MEMB      9 UNIT - kN,m SECTION CLASS 2 |

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Friday, September 21, 2007, 11:59 AM

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| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw      = 207 N/mm2 |
| d   = 519 mm   :   t   =   9 mm           : a   =   0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2     status = PASS              : BS-4.4.3.3 status = PASS |
|-----|
| 10    TAPERED      PASS    BS-4.7 (C)      0.388      1013 |
|                91.60 C              0.00      300.22      0.00 |

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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|-----|
| CALCULATED CAPACITIES FOR MEMB 10 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 820.5 MCY= 161.9 PC= 4117.5 PT= 4381.5 MB= 925.4 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw      = 207 N/mm2 |
| d   = 519 mm   :   t   =   9 mm           : a   =   0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2     status = PASS              : BS-4.4.3.3 status = PASS |
|-----|
| 11    TAPERED      PASS    BS-4.7 (C)      0.334      1013 |
|                50.55 C              0.00      264.29      0.00 |

```

```

|-----|
| CALCULATED CAPACITIES FOR MEMB 11 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 820.5 MCY= 161.9 PC= 4117.0 PT= 4381.5 MB= 925.4 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw      = 207 N/mm2 |
| d   = 519 mm   :   t   =   9 mm           : a   =   0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2     status = PASS              : BS-4.4.3.3 status = PASS |
|-----|
| 12    TAPERED      PASS    BS-4.7 (C)      0.251      1013 |
|                45.52 C              0.00      197.29      0.00 |

```

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|-----|
| CALCULATED CAPACITIES FOR MEMB 12 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 820.5 MCY= 161.9 PC= 4120.0 PT= 0.0 MB= 925.4 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw      = 207 N/mm2 |
| d   = 519 mm   :   t   =   9 mm           : a   =   0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2     status = PASS              : BS-4.4.3.3 status = PASS |
|-----|
| 14    TAPERED      PASS    BS-4.2.3-(Y)    0.012      1013 |
|                0.22 C              0.00      3.09      0.00 |

```

```

|-----|
| CALCULATED CAPACITIES FOR MEMB 14 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 445.6 MCY= 161.8 PC= 3431.1 PT= 3642.2 MB= 468.6 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 632 kN : qw      = 207 N/mm2 |
| d   = 305 mm   :   t   =   9 mm           : a   =   0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2     status = PASS              : BS-4.4.3.3 status = PASS |
|-----|
| 15    TAPERED      PASS    BS-4.7 (C)      0.298      1013 |
|                57.42 C              0.77      227.03      3.20 |

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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Friday, September 21, 2007, 11:59 AM

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| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw = 207 N/mm2 |
| d = 519 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS |
|-----|
| 16 TAPERED PASS BS-4.7 (C) 0.299 1013 |
| 64.14 C 1.14 227.03 0.00 |
|-----|
| CALCULATED CAPACITIES FOR MEMB 16 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 820.5 MCY= 161.9 PC= 4117.2 PT= 4381.5 MB= 925.4 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw = 207 N/mm2 |
| d = 519 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS |
|-----|
| 17 TAPERED PASS BS-4.7 (C) 0.285 1013 |
| 47.69 C -0.34 222.61 0.00 |
|-----|
| CALCULATED CAPACITIES FOR MEMB 17 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 820.5 MCY= 161.9 PC= 4117.5 PT= 4381.5 MB= 925.4 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw = 207 N/mm2 |
| d = 519 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS |
|-----|
| 18 TAPERED PASS BS-4.7 (C) 0.240 1013 |
| 16.57 C -0.91 188.98 0.00 |
|-----|
| CALCULATED CAPACITIES FOR MEMB 18 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 820.5 MCY= 161.9 PC= 4117.0 PT= 0.0 MB= 925.4 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw = 207 N/mm2 |
| d = 519 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS |
|-----|
| 19 TAPERED PASS BS-4.7 (C) 0.197 1013 |
| 16.18 C 3.16 142.58 0.00 |

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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|-----|
| CALCULATED CAPACITIES FOR MEMB 19 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 820.5 MCY= 161.9 PC= 4120.0 PT= 0.0 MB= 925.4 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1076 kN : qw = 207 N/mm2 |
| d = 519 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS |
|-----|
| 21 TAPERED PASS BS-4.7 (C) 0.022 1002 |
| 0.15 C 2.71 2.55 0.00 |
|-----|
| CALCULATED CAPACITIES FOR MEMB 21 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 445.6 MCY= 161.8 PC= 3431.1 PT= 3642.2 MB= 495.5 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 632 kN : qw = 207 N/mm2 |
| d = 305 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS |
|-----|

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Friday, September 21, 2007, 11:59 AM

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
22 ST PIPE1	PASS	BS-4.6 (T)	0.228	1011	
	23.26 T	0.00	0.00	0.00	

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 899 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 465.23
Gross Area = 1.13 Net Area = 1.13 Eff. Area = 1.13

		z-z axis	y-y axis
Moment of inertia	:	0.102	0.102
Plastic modulus	:	0.288	0.288
Elastic modulus	:	0.170	0.170
Effective modulus	:	0.170	0.170
Shear Area	:	0.679	0.679

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : PLASTIC

		z-z axis	y-y axis
Tension Capacity	:	101.8	101.8
Shear Capacity	:	36.6	36.6

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.228	1011	23.3	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
23 ST PIPE1	PASS	BS-4.6 (T)	0.221	1011	
	22.48 T	0.00	0.00	0.00	

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 899 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 465.23
Gross Area = 1.13 Net Area = 1.13 Eff. Area = 1.13

		z-z axis	y-y axis
Moment of inertia	:	0.102	0.102
Plastic modulus	:	0.288	0.288
Elastic modulus	:	0.170	0.170
Effective modulus	:	0.170	0.170

Friday, September 21, 2007, 11:59 AM

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : PLASTIC

		z-z axis	y-y axis
Tension Capacity	:	101.8	101.8
Shear Capacity	:	36.6	36.6

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.221	1011	22.5	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

24 ST PIPE1	PASS	BS-4.6 (T)	0.221	1011
	22.48 T	0.00	0.00	0.00

MATERIAL DATA

Grade of steel	=	S 275
Modulus of elasticity	=	205 kN/mm2
Design Strength (py)	=	899 N/mm2

SECTION PROPERTIES (units - cm)

Member Length =	465.23
Gross Area =	1.13 Net Area = 1.13 Eff. Area = 1.13

		z-z axis	y-y axis
Moment of inertia	:	0.102	0.102
Plastic modulus	:	0.288	0.288
Elastic modulus	:	0.170	0.170
Effective modulus	:	0.170	0.170
Shear Area	:	0.679	0.679

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : PLASTIC

		z-z axis	y-y axis
Tension Capacity	:	101.8	101.8
Shear Capacity	:	36.6	36.6

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.221	1011	22.5	-	-	-	-

Torsion and deflections have not been considered in the design.

Friday, September 21, 2007, 11:59 AM

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
25 ST PIPE1	PASS	BS-4.6 (T)	0.228	1011	
	23.26 T	0.00	0.00	0.00	

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm²
Design Strength (py) = 899 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 465.23
Gross Area = 1.13 Net Area = 1.13 Eff. Area = 1.13

	z-z axis	y-y axis
Moment of inertia	: 0.102	0.102
Plastic modulus	: 0.288	0.288
Elastic modulus	: 0.170	0.170
Effective modulus	: 0.170	0.170
Shear Area	: 0.679	0.679

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : PLASTIC

	z-z axis	y-y axis
Tension Capacity	: 101.8	101.8
Shear Capacity	: 36.6	36.6

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.228	1011	23.3	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
26 ST PIPE1	PASS	BS-4.6 (T)	0.281	1002	
	28.57 T	0.00	0.00	6.72	

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm²
Design Strength (py) = 899 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 672.01
Gross Area = 1.13 Net Area = 1.13 Eff. Area = 1.13

	z-z axis	y-y axis
Moment of inertia	: 0.102	0.102
Plastic modulus	: 0.288	0.288
Elastic modulus	: 0.170	0.170

Friday, September 21, 2007, 11:59 AM

Shear Area : 0.679 0.679

DESIGN DATA (units - kN,m) BS5950-1/2000
 Section Class : PLASTIC

		z-z axis	y-y axis
Tension Capacity	:	101.8	101.8
Shear Capacity	:	36.6	36.6

BUCKLING CALCULATIONS (units - kN,m)
 (axis nomenclature as per design code)

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.281	1002	28.6	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
27 ST PIPE1		PASS	BS-4.6 (T)	0.253	1011
		25.79 T	0.00	0.00	6.72
=====					

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm2
 Design Strength (py) = 899 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 672.01
 Gross Area = 1.13 Net Area = 1.13 Eff. Area = 1.13

		z-z axis	y-y axis
Moment of inertia	:	0.102	0.102
Plastic modulus	:	0.288	0.288
Elastic modulus	:	0.170	0.170
Effective modulus	:	0.170	0.170
Shear Area	:	0.679	0.679

DESIGN DATA (units - kN,m) BS5950-1/2000
 Section Class : PLASTIC

		z-z axis	y-y axis
Tension Capacity	:	101.8	101.8
Shear Capacity	:	36.6	36.6

BUCKLING CALCULATIONS (units - kN,m)
 (axis nomenclature as per design code)

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.253	1011	25.8	-	-	-	-

Torsion and deflections have not been considered in the design.

Friday, September 21, 2007, 11:59 AM

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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=====

28 ST PIPE1	PASS	BS-4.6 (T)	0.253	1011
	25.79 T	0.00	0.00	6.72

=====

MATERIAL DATA

Grade of steel	=	S 275
Modulus of elasticity	=	205 kN/mm2
Design Strength (py)	=	899 N/mm2

SECTION PROPERTIES (units - cm)

Member Length =	672.01
Gross Area =	1.13
Net Area =	1.13
Eff. Area =	1.13

	z-z axis	y-y axis
Moment of inertia	: 0.102	0.102
Plastic modulus	: 0.288	0.288
Elastic modulus	: 0.170	0.170
Effective modulus	: 0.170	0.170
Shear Area	: 0.679	0.679

DESIGN DATA (units - kN,m)

Section Class	: PLASTIC
---------------	-----------

	z-z axis	y-y axis
Tension Capacity	: 101.8	101.8
Shear Capacity	: 36.6	36.6

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.253	1011	25.8	-	-	-	-

Torsion and deflections have not been considered in the design.

=====

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

=====

29 ST PIPE1	PASS	BS-4.6 (T)	0.281	1002
	28.57 T	0.00	0.00	6.72

=====

MATERIAL DATA

Grade of steel	=	S 275
Modulus of elasticity	=	205 kN/mm2
Design Strength (py)	=	899 N/mm2

SECTION PROPERTIES (units - cm)

Member Length =	672.01
Gross Area =	1.13
Net Area =	1.13
Eff. Area =	1.13

	z-z axis	y-y axis
Moment of inertia	: 0.102	0.102
Plastic modulus	: 0.288	0.288

Friday, September 21, 2007, 11:59 AM

Effective modulus	:	0.170	0.170
Shear Area	:	0.679	0.679

DESIGN DATA (units - kN,m) BS5950-1/2000
 Section Class : PLASTIC

		z-z axis	y-y axis
Tension Capacity	:	101.8	101.8
Shear Capacity	:	36.6	36.6

BUCKLING CALCULATIONS (units - kN,m)
 (axis nomenclature as per design code)

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.281	1002	28.6	-	-	-	-

Torsion and deflections have not been considered in the design.

30	TAPERED	PASS	BS-4.7 (C)	0.159	1013
		16.07 C	2.46	105.79	0.00

-----|
 | CALCULATED CAPACITIES FOR MEMB 30 UNIT - kN,m SECTION CLASS 2 |
 | MCZ= 754.3 MCY= 161.9 PC= 4011.4 PT= 0.0 MB= 848.2 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 | Shear Buckling check is required : Vb = 1002 kN : qw = 207 N/mm2 |
 | d = 484 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS

31	TAPERED	PASS	BS-4.7 (C)	0.128	1013
		15.98 C	1.39	74.33	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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=====

-----|
 | CALCULATED CAPACITIES FOR MEMB 31 UNIT - kN,m SECTION CLASS 2 |
 | MCZ= 647.3 MCY= 161.9 PC= 3818.0 PT= 0.0 MB= 724.6 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 | Shear Buckling check is required : Vb = 879 kN : qw = 207 N/mm2 |
 | d = 424 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS

32	TAPERED	PASS	BS-4.7 (C)	0.095	1013
		15.89 C	0.37	48.30	0.00

-----|
 | CALCULATED CAPACITIES FOR MEMB 32 UNIT - kN,m SECTION CLASS 2 |
 | MCZ= 544.4 MCY= 161.8 PC= 3624.5 PT= 0.0 MB= 607.0 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 | Shear Buckling check is required : Vb = 756 kN : qw = 207 N/mm2 |
 | d = 365 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS

33	TAPERED	PASS	BS-4.7 (C)	0.065	1013
		15.80 C	0.43	27.75	0.00

-----|
 | CALCULATED CAPACITIES FOR MEMB 33 UNIT - kN,m SECTION CLASS 2 |
 | MCZ= 478.1 MCY= 161.8 PC= 3495.5 PT= 0.0 MB= 532.0 |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
 | Shear Buckling check is required : Vb = 673 kN : qw = 207 N/mm2 |
d = 325 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2

Friday, September 21, 2007, 11:59 AM

34	TAPERED	PASS	BS-4.7	(C)	0.036	1013
		15.70 C		0.49	12.77	0.00

CALCULATED CAPACITIES FOR MEMB 34 UNIT - kN,m SECTION CLASS 2						
MCZ= 451.0 MCY= 161.8 PC= 3441.8 PT= 0.0 MB= 501.6						
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00						
Shear Buckling check is required : Vb = 639 kN : qw = 207 N/mm2						
d = 309 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2						
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS						

35	TAPERED	PASS	BS-4.7	(C)	0.200	1013
		45.26 C		0.00	142.75	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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=====

CALCULATED CAPACITIES FOR MEMB 35 UNIT - kN,m SECTION CLASS 2						
MCZ= 754.3 MCY= 161.9 PC= 4011.4 PT= 0.0 MB= 848.2						
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00						
Shear Buckling check is required : Vb = 1002 kN : qw = 207 N/mm2						
d = 484 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2						
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS						

36	TAPERED	PASS	BS-4.7	(C)	0.162	1013
		44.99 C		0.00	97.43	0.00

CALCULATED CAPACITIES FOR MEMB 36 UNIT - kN,m SECTION CLASS 2						
MCZ= 647.3 MCY= 161.9 PC= 3818.0 PT= 0.0 MB= 724.6						
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00						
Shear Buckling check is required : Vb = 879 kN : qw = 207 N/mm2						
d = 424 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2						
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS						

37	TAPERED	PASS	BS-4.7	(C)	0.124	1013
		44.72 C		0.00	61.07	0.00

CALCULATED CAPACITIES FOR MEMB 37 UNIT - kN,m SECTION CLASS 2						
MCZ= 544.4 MCY= 161.8 PC= 3624.5 PT= 0.0 MB= 607.0						
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00						
Shear Buckling check is required : Vb = 756 kN : qw = 207 N/mm2						
d = 365 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2						
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS						

38	TAPERED	PASS	BS-4.7	(C)	0.082	1013
		44.45 C		0.00	33.40	0.00

CALCULATED CAPACITIES FOR MEMB 38 UNIT - kN,m SECTION CLASS 2						
MCZ= 478.1 MCY= 161.8 PC= 3495.5 PT= 0.0 MB= 532.0						
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00						
Shear Buckling check is required : Vb = 673 kN : qw = 207 N/mm2						
d = 325 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2						
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS						

39	TAPERED	PASS	BS-4.7	(C)	0.044	1013
		44.19 C		0.00	14.16	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

Friday, September 21, 2007, 11:59 AM

```

|-----|
| CALCULATED CAPACITIES FOR MEMB    39 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 451.0  MCY= 161.8  PC= 3441.8  PT= 0.0  MB= 497.0 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 639 kN : qw = 207 N/mm2 |
| d = 309 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS |
|-----|
| 40  TAPERED  PASS  BS-4.7  (C)  0.159  1013 |
|      16.07 C      -2.46      105.79      0.00 |
|-----|
| CALCULATED CAPACITIES FOR MEMB    40 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 754.3  MCY= 161.9  PC= 4011.4  PT= 0.0  MB= 848.2 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 1002 kN : qw = 207 N/mm2 |
| d = 484 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS |
|-----|
| 41  TAPERED  PASS  BS-4.7  (C)  0.128  1013 |
|      15.98 C      -1.39      74.33      0.00 |
|-----|
| CALCULATED CAPACITIES FOR MEMB    41 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 647.3  MCY= 161.9  PC= 3818.0  PT= 0.0  MB= 724.6 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 879 kN : qw = 207 N/mm2 |
| d = 424 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS |
|-----|
| 42  TAPERED  PASS  BS-4.7  (C)  0.095  1013 |
|      15.89 C      -0.37      48.30      0.00 |
|-----|
| CALCULATED CAPACITIES FOR MEMB    42 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 544.4  MCY= 161.8  PC= 3624.5  PT= 0.0  MB= 607.0 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 756 kN : qw = 207 N/mm2 |
| d = 365 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS |
|-----|
| 43  TAPERED  PASS  BS-4.7  (C)  0.065  1013 |
|      15.80 C      -0.43      27.75      0.00 |

```

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

```

|-----|
| CALCULATED CAPACITIES FOR MEMB    43 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 478.1  MCY= 161.8  PC= 3495.5  PT= 0.0  MB= 532.0 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 673 kN : qw = 207 N/mm2 |
| d = 325 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |
| BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS |
|-----|
| 44  TAPERED  PASS  BS-4.7  (C)  0.036  1013 |
|      15.70 C      -0.49      12.77      0.00 |
|-----|
| CALCULATED CAPACITIES FOR MEMB    44 UNIT - kN,m SECTION CLASS 2 |
| MCZ= 451.0  MCY= 161.8  PC= 3441.8  PT= 0.0  MB= 501.6 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 |
| Shear Buckling check is required : Vb = 639 kN : qw = 207 N/mm2 |
| d = 309 mm : t = 9 mm : a = 0 mm : pyf = 344 N/mm2 |

```

Friday, September 21, 2007, 11:59 AM

***** END OF TABULATED RESULT OF DESIGN *****

211. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= SEP 21,2007 TIME= 11:58:58 ****

```
*****
*           For questions on STAAD.Pro, please contact           *
*   Research Engineers Offices at the following locations         *
*                                                                 *
*           Telephone                                           Email                       *
*   USA:      +1 (714)974-2500      support@reiusa.com          *
*   CANADA    +1 (905)632-4771      detech@odandetech.com       *
*   CANADA    +1 (604)629 6087      staad@dwco.com              *
*   UK         +44(1454)207-000      support@reel.co.uk          *
*   FRANCE    +33(0)1 64551084      support@reel.co.uk          *
*   GERMANY    +49/931/40468-71      info@reig.de                *
*   NORWAY     +47 67 57 21 30      staad@edr.no                *
*   SINGAPORE +65 6225-6158         support@reiusa.com          *
*   INDIA      +91(033)2357-3575     support@calcutta.reiusa.com *
*   JAPAN      +81(03)5952-6500      eng-eye@crc.co.jp          *
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*                                                                 *
*   North America      support@reiusa.com                      *
*   Europe              support@reel.co.uk                     *
*   Asia                support@reiasia.net                     *
*****
```

Information about the key files in the current distribution

Modification Date	CRC	Size (Bytes)	File Name
03/13/2007	0xb200	13737984	SProStaad.exe
03/12/2007	0xac00	06275072	SProStaadStl.exe
09/19/2003	0x2fc0	00081970	CMesh.dll
10/25/2006	0x4601	02486272	dbSectionInterface.dll
01/23/2001	0x9b40	00073728	LoadGen.dll
09/25/2003	0x6340	00704512	MeshEngine.dll
09/22/2003	0xce00	00069632	QuadPlateEngine.dll
12/22/2005	0x4181	00094208	SurfMesh.dll
02/14/2007	0x5140	00604160	AISCSections.mdb
06/13/2006	0xd4c1	01204224	AISCSectionsRCeco.mdb
01/05/2005	0x79c1	00319488	aiscsections_all_editions.mdb
01/05/2005	0x4b81	01810432	aiscsteeljoists.mdb
01/05/2005	0xcac1	03651584	aitctimbersections.mdb
01/27/2005	0xeb01	00552960	aluminumsections.mdb
01/05/2005	0xcd01	00163840	australiansections.mdb
01/05/2005	0x6a41	00229376	britishsections.mdb
09/12/2006	0xf4c0	00466944	bscoldformedsections.mdb
06/28/2005	0x8201	00327680	butlercoldformedsections.mdb
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10/12/2006	0x5fc1	00532480	ChineseSections.mdb
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Friday, September 21, 2007, 12:00 PM

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01/19/2006	0x8e40	00159744	venezuelansections.mdb



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Job No
068010.01Sheet No
1

Rev

Part

Job Title Dubai Metro Project - Rashidiya B21

Ref

By CJA

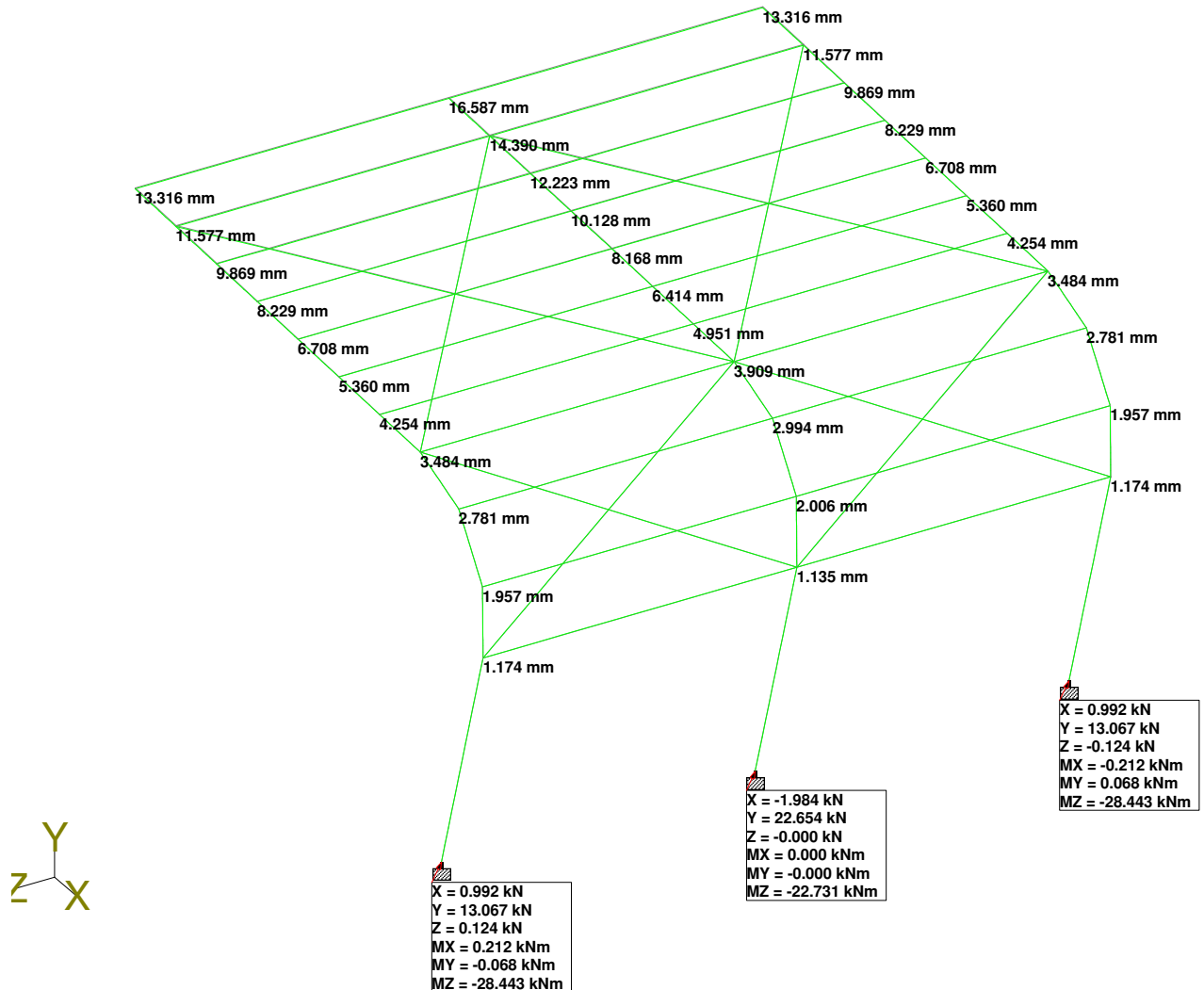
Date Sep 2007

Chd

Client Dubai RTA

File Fuel Station-R3 Col550 T

Date/Time 21-Sep-2007 11:57

DL OnlyLoad 2 : Displacement
Displacement - mm



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068010.01Sheet No
1

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Part

Job Title Dubai Metro Project - Rashidiya B21

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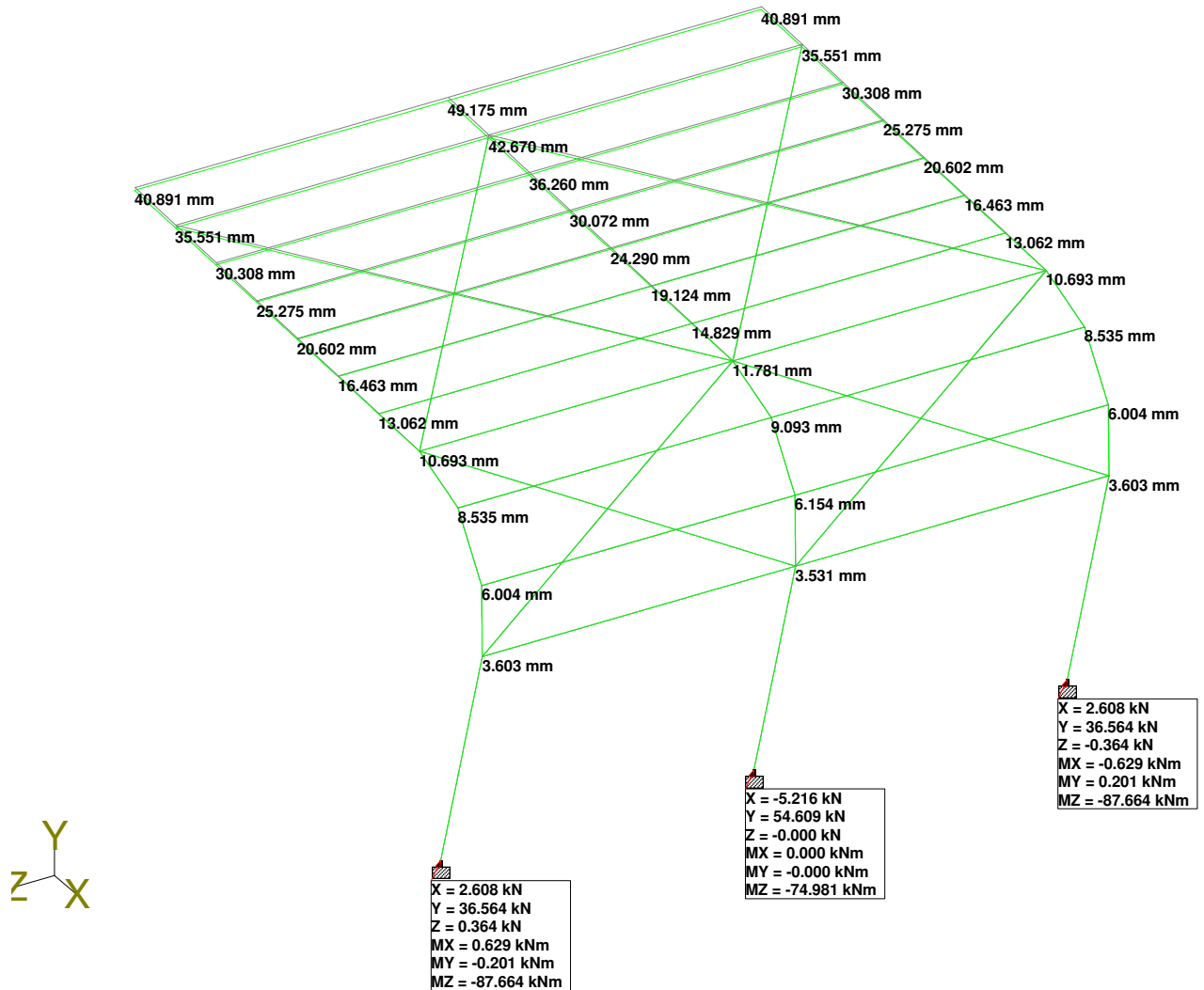
Date Sep 2007

Chd

Client Dubai RTA

File Fuel Station-R3 Col550 T

Date/Time 21-Sep-2007 11:57

DL + LLLoad 2001 : Displacement
Displacement - mm



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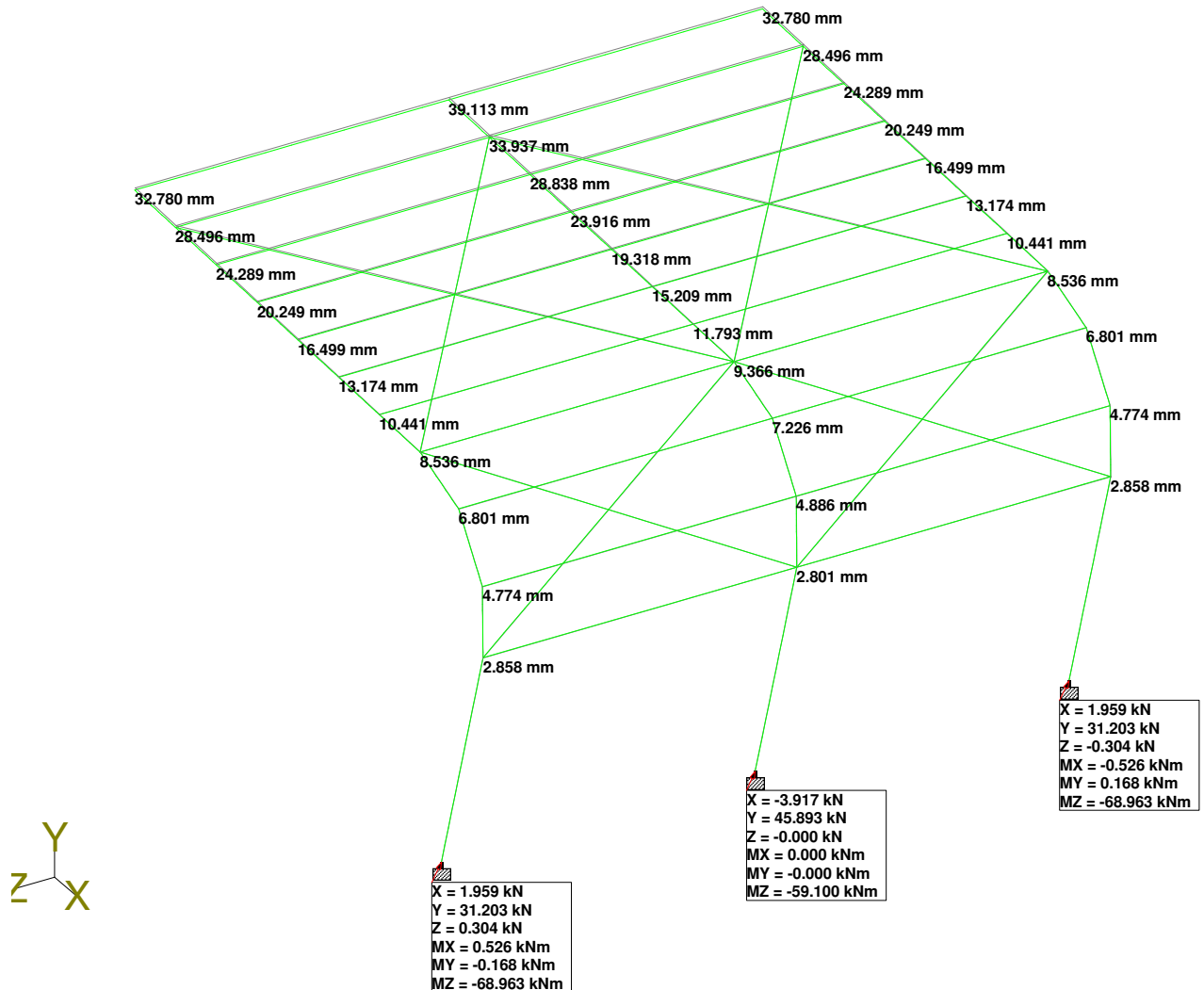
Date Sep 2007

Chd

Client Dubai RTA

File Fuel Station-R3 Col550 T

Date/Time 21-Sep-2007 11:57

DL + 0.5LLLoad 2002 : Displacement
Displacement - mm



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1

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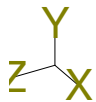
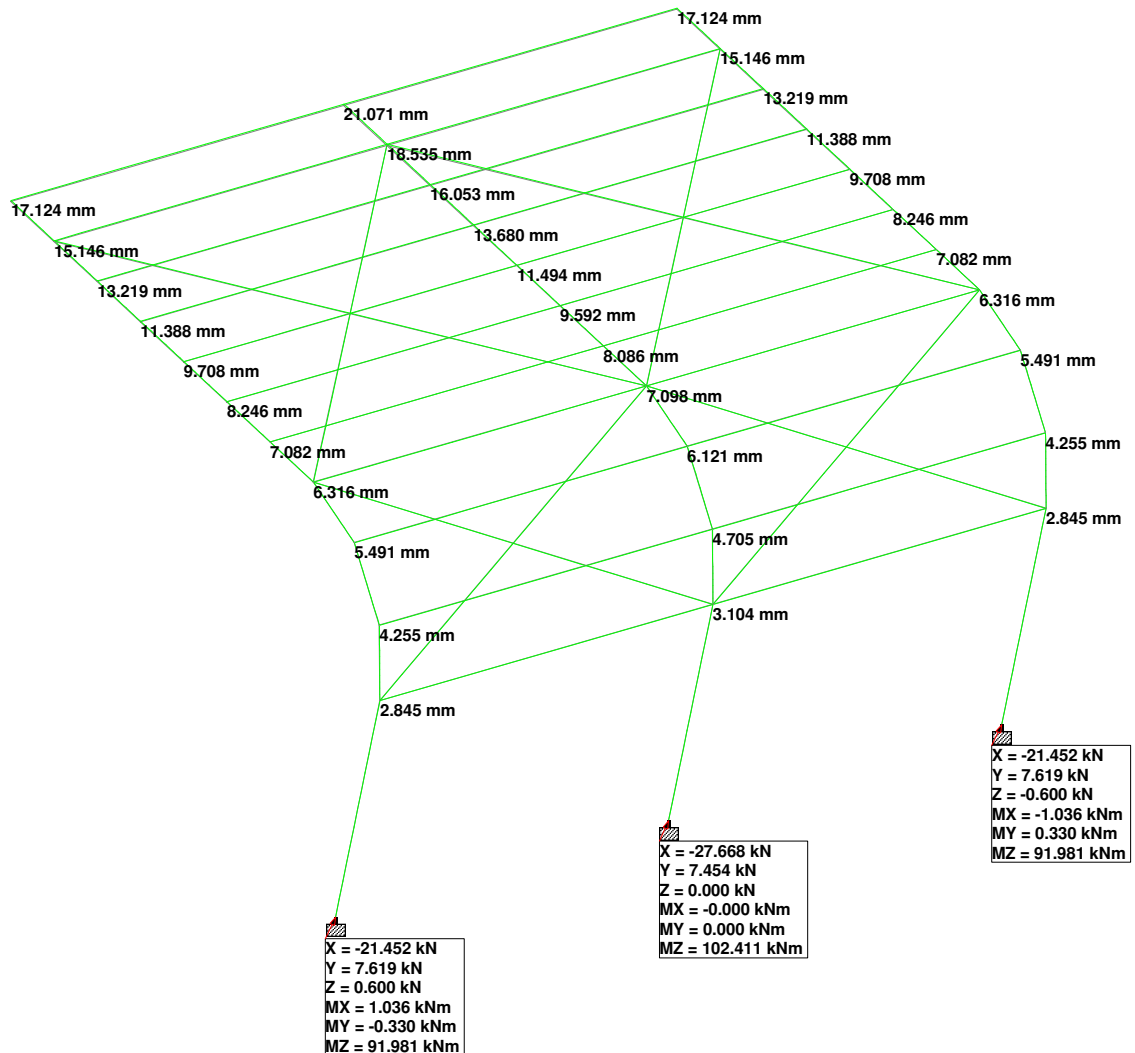
Date Sep 2007

Chd

Client Dubai RTA

File Fuel Station-R3 Col550 T

Date/Time 21-Sep-2007 11:57



DL + LL + WL1 Load 2011 : Displacement
Displacement - mm



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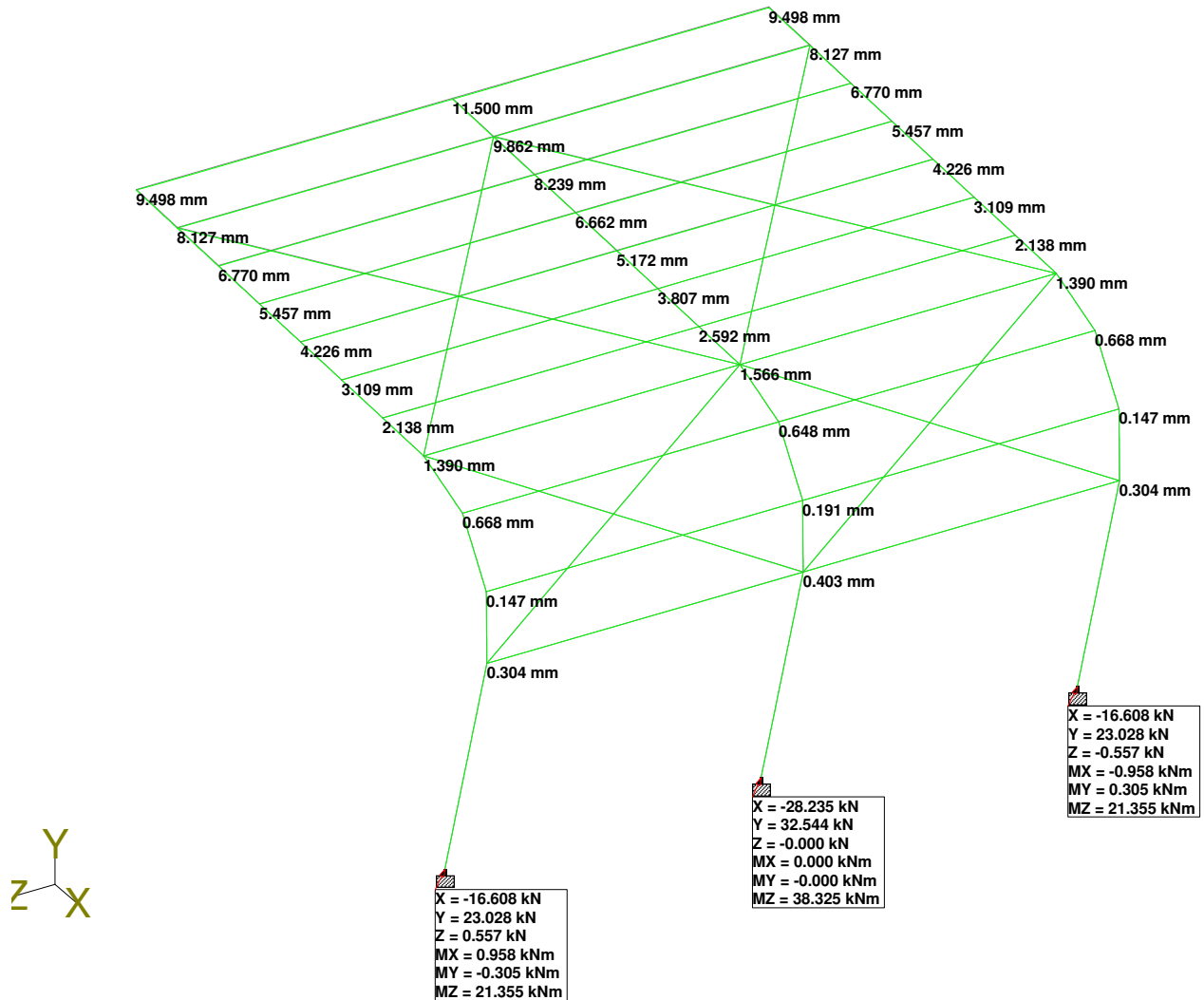
Date Sep 2007

Chd

Client Dubai RTA

File Fuel Station-R3 Col550 T

Date/Time 21-Sep-2007 11:57

DL + LL + WL2Load 2012 : Displacement
Displacement - mm



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1

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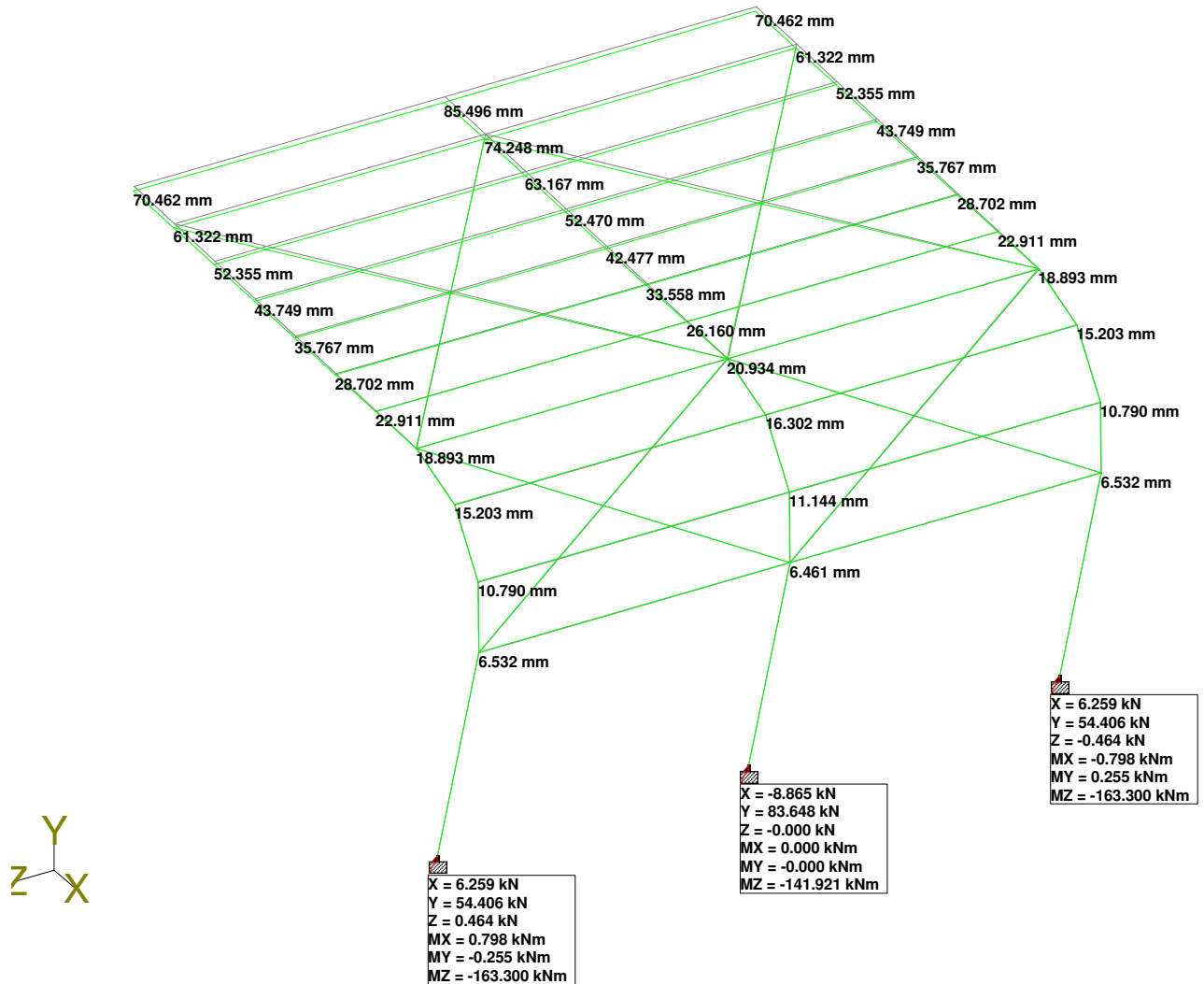
Date Sep 2007

Chd

Client Dubai RTA

File Fuel Station-R3 Col550 T

Date/Time 21-Sep-2007 11:57

**DL + LL + WL3**Load 2013 : Displacement
Displacement - mm



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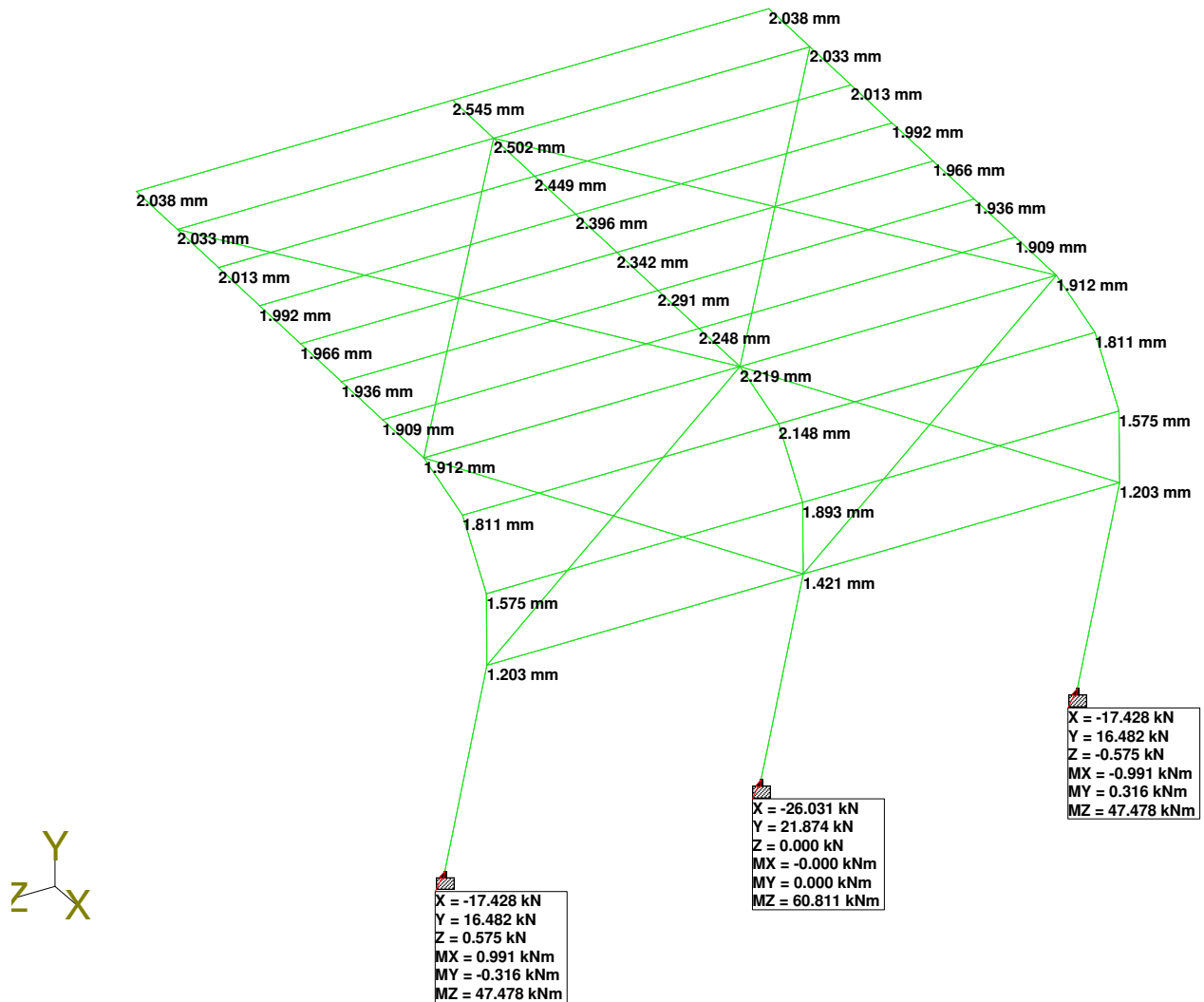
Date Sep 2007

Chd

Client Dubai RTA

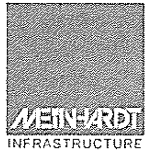
File Fuel Station-R3 Col550 T

Date/Time 21-Sep-2007 11:57

**DL + LL + WL4**Load 2014 : Displacement
Displacement - mm

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site DUBAI METRO PROJECT

Date SEP 2007

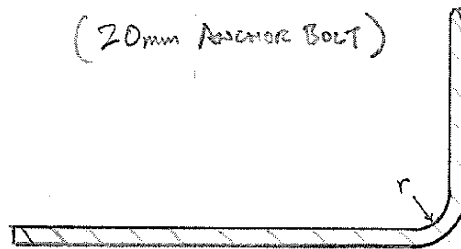
Job No.

RKSHIDIYA B21

Designed by CSA

Sheet No.

DESIGN OF ANCHORAGE LENGTH IN FOOTING



$\phi = 20 \text{ mm} \Rightarrow$ FROM TABLE 3 (ATTACHED), $r_{\min} = 60 \text{ mm}$

$f_y = 460 \text{ N/mm}^2$, FROM TABLE 3.2.9, TENSION ANCHORAGE = 32ϕ

FROM SHEET NO. 1.9, EQUIVALENT ANCHORAGE LENGTH = 12ϕ

$\Rightarrow$ FOR HOOK ANCHORAGE :

$$\begin{aligned} \text{ANCHORAGE LENGTH REQD.} &= 4\phi + r + (32 - 12)\phi \\ &= 4(20) + 60 + (20)20 \\ &= 540 \text{ mm} \end{aligned}$$

$\Rightarrow$ ANCHORAGE LENGTH PROVIDED = 600 mm (OK?)

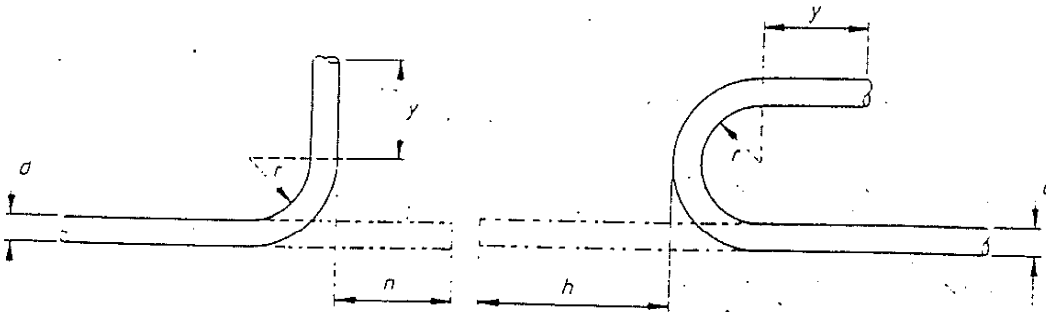


NATIONAL CERTIFICATE IN CONSTRUCTION SUPERVISION

Table 3. Minimum former radii, bend and hook allowances

Bend (bob)

Hook



Bar size	Type and grade R and type and grade S				Type and grade T and type and grade S			Fabric complying with BS 4483			
	d	r	n	h	r	n	h	d	r	n	h
6*		12	100	100	18	100	100	5	15	100	100
8		16	100	100	24	100	100	6	18	100	100
10		20	100	100	30	100	110	7	21	100	100
12		24	100	110	36	100	140	8	24	100	100
16		32	100	150	48	100	180	9	27	120	135
20		40	100	180	60	110	220	10	30	120	135
25		50	130	230	100	180	350	12	36	130	145
32		64	160	290	128	230	450	—	—	—	—
40		80	200	360	160	280	560	—	—	—	—
50*		100	250	450	200	350	700	—	—	—	—

NOTE 1. The use of hooks as end anchorages is not normally necessary with deformed bars.

NOTE 2. See clause 8 for definition of y .

*These are non-preferred sizes.

Table 4. Cutting and bending tolerances

	Tolerance
	mm
Cutting of straight lengths (including reinforcement for subsequent bending)	+25 -25
Bending ≤ 1000 mm	+5 -5
> 1000 mm to ≤ 2000 mm	+5 -10
> 2000 mm	+5 -25



CP 65 : Part 1 : 1999

**Table 3.29 – Ultimate anchorage bond lengths
and lap lengths as multiples of bar size**

Reinforcement type	Grade 250 plain	Grade 460		
		Plain	Deformed type 1	Deformed type 2
Concrete cube strength 30				
Tension anchorage and lap length	36	66	46	37
1.4 x tension lap	50	92	64	52
2.0 x tension lap	71	131	92	74
Compression anchorage length	29	53	37	29
Compression lap length	36	66	46	37
Concrete cube strength 35				
Tension anchorage and lap length	33	61	43	34
1.4 x tension lap	46	85	60	48
2.0 x tension lap	66	121	85	68
Compression anchorage length	27	49	34	27
Compression lap length	33	61	43	34
Concrete cube strength 40				
Tension anchorage and lap length	31	57	40	32
1.4 x tension lap	43	80	56	45
2.0 x tension lap	62	113	80	64
Compression anchorage length	25	46	32	26
Compression lap length	31	57	40	32

NOTE – The values are rounded up to the whole number and the lengths derived from these values may differ slightly from those calculated directly for each bar or wire size.

		Tension lap lengths		
		Bars in top of section as cast with cover < 2 ϕ	Corner bars not in top of section with cover < 2 ϕ	Otherwise
Clear distance between laps	≥ 75 mm and $\geq 6\phi$	$1.4l_b$	$1.4l_b$	$1.0l_b$
	Otherwise	$2.0l_b$	$1.4l_b$	$1.4l_b$

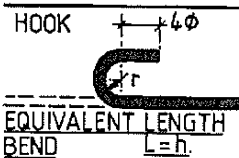
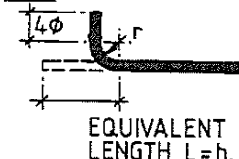
l_b = basic anchorage length, ϕ = bar size.

REINFORCEMENT, ANCHORAGE, LAP LENGTHS AND LINKS

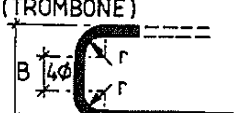
SHEET NO. 1.9

ANCHORAGE (BS8110 CLAUSE 3.12.8)

(a) Equivalent straight anchorage lengths
of standard hooks and bends (mm)

ANCHORAGE TYPE	VALUE	f_y N/mm ²	BAR SIZE, mm							
			8	10	12	16	20	25	32	40
 EQUIVALENT LENGTH BEND L=h.	16φ	250	128	160	192	256	320	400	512	640
	24φ	460	192	240	288	384	480	600	768	960
 EQUIVALENT LENGTH L=h.	8φ	250	64	80	96	128	160	200	256	320
	12φ	460	96	120	144	192	240	300	384	480

(b) Minimum overall depth of various U-bars (mm)

SHAPE CODE	B	r	f_y N/mm ²	8	10	12	16	20	25	32	40
 (HOOK) (HAIRPIN)	6φ	2φ	250	50	60	75	100	120	150	195	240
	8φ	3φ	460	65	80	100	130	160	-	-	-
	10φ	4φ	460	-	-	-	-	-	250	320	400
 (TROMBONE)	10φ	2φ	250	80	100	120	160	200	250	320	400
	12φ	3φ	460	100	120	145	195	240	-	-	-
	14φ	4φ	460	-	-	-	-	-	350	450	560

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site Dubai Depot - Rashidiya B21
Diesel Fuel Station

Date Sep '07 Job No. 068010

Designed by Rudy Sheet No.

Footing Design Check

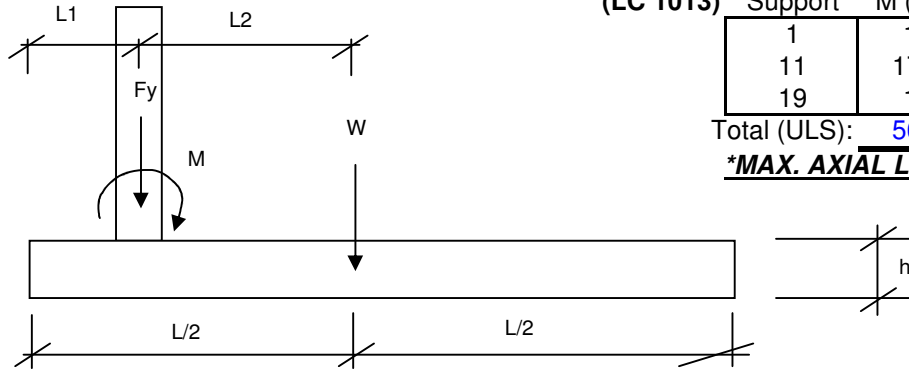
Max Column Reactions (From STAAD Pro Analysis)

(LC 1013) Support M (kNm) Fy (kN)

1	196	65.29
11	170.3	100.4
19	196	65.29

Total (ULS): 562.3 231.0

***MAX. AXIAL LOAD CONDITION**



Assume footing dimension

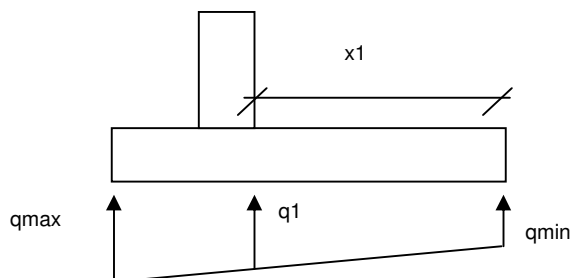
L	=	6.91 m	Depth of soil above the footing, Hs	=	0.0 m
B	=	9.0 m	Density of the backfill soil, γ_{soil}	=	20.0 kN/m ³
h	=	0.355 m	W (soil)	=	0.0 kN
L1	=	0.700 m			
L2	=	2.76 m	Safe bearing capacity	=	150.0 kN/m ²
W (selfwt)	=	529.9 kN	Column Width	=	0.550 m

Equivalent bending moment at the center of the footing, Mc

$$= (M - F_y \cdot L_2) / \gamma_L \quad \gamma_L = 1.4$$
$$= -52.9 \text{ kNm}$$

Total vertical load at the center of the footing, Ft

$$= F_y / \gamma_L + W(\text{soil}) + W(\text{selfwt})$$
$$= 694.8 \text{ kN}$$



Eccentricity due to the resultant force, e

$$= M_c / F_t$$
$$= -0.076 \text{ m} \quad \text{not greater than } L/6 = 1.152 \text{ m}$$

The whole base is under compression

Effective stress length, L'	=	6.91 m
Base pressure, qmin	=	10.43 kN/m ²
qmax	=	11.91 kN/m ²

Ok

$$q_{\max} = \frac{F_y}{B \times L} + \frac{6M}{B \times L^2}$$
$$q_{\min} = \frac{F_y}{B \times L} - \frac{6M}{B \times L^2}$$

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai Depot - Rashidiya B21 Diesel Fuel Station	Date	Sep '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Design of footing reinforcement

Distance between the footing edge to the column edge, x_1 = 5.935 m
Stress at the face of the column, q_1 = 11.70 kN/m²

cover to the main reinforcement = 83 mm
effective depth of the footing, d = 367 mm
 d' = 83 mm
 f_{cu} = 40 N/mm²
 f_y = 460 N/mm²

Design bending reinforcement

Maximum ULS moment at the face of column, M_1
= $B * \{ [q_{min} * x_1 * (x_1/2) + 0.5 * (q_1 - q_{min}) * x_1 * (x_1/3)] - [24 * x_1 * h * x_1/2] \} * \gamma_L$
= 518.7 kNm
 K' = 0.156
 $K = M_1 / f_{cu} B d^2$ = 0.0107
 z = 235.2 mm
 x = 292.8 mm

A_s' req = 0.0 mm²
 A_s req = 5509.5 mm²
 A_s req (min) = 585.0 mm² / m

Provide 60 T12
As provided 6785.8 mm² (spacing = 149.7 mm) Ok
= 755.35 mm² / m

Design shear reinforcement

Maximum ULS shear at the face of column, V_1
= $B * 0.5 * [q_1 + q_{min}] * x_1 * \gamma_L$ = 827.7 kN

$V_1 / b d$ = 0.25 N/mm² $100 A_s / B d$ = 0.205445
400/d = 1.089918

v_c = 0.41 N/mm²
 $v_c + 0.4$ = 0.81 N/mm²
 $0.8 f_{cu}^{0.5}$ = 5.0 N/mm² <= 5 N/mm²

A_{sv} / s_v req = 0 mm² / mm
Provide T @300 c/c (2 legs)
 A_{sv} / s_v prov = 0 mm² / mm

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



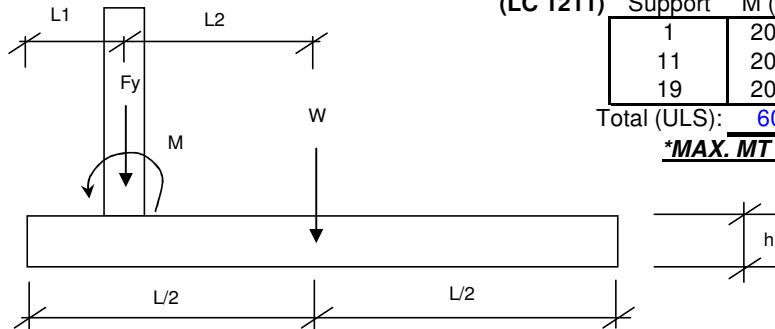
Site	Dubai Depot - Rashidiya B21 Diesel Fuel Station	Date	Sep '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Footing Design Check

Max Column Reactions (From STAAD Pro Analysis)

(LC 1211)	Support	M (kNm)	Fy (kN)
	1	201.24	-32.374
	11	205.13	-34.05
	19	201.24	-32.374
Total (ULS):		607.6	-98.8

***MAX. MT CONDITION**



Assume footing dimension

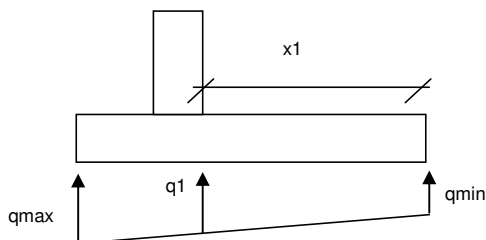
L	=	6.910 m	Depth of soil above the footing, Hs	=	0.0 m
B	=	9.00 m	Density of the backfill soil, γ_{soil}	=	20.0 kN/m ³
h	=	0.355 m	W (soil)	=	0.0 kN
L1	=	0.700 m			
L2	=	2.755 m	Safe bearing capacity	=	150.0 kN/m ²
W (selfwt)	=	529.9 kN	Column Width	=	0.550 m

Equivalent bending moment at the center of the footing, M_c

$$\begin{aligned} &= (M - F_y \cdot L_2) / \gamma_L & \gamma_L &= 1.4 \\ &= 239.6 \text{ kNm} \end{aligned}$$

Total vertical load at the center of the footing, F_t

$$\begin{aligned} &= F_y / \gamma_L + W(\text{soil}) + W(\text{selfwt}) \\ &= 459.3 \text{ kN} \end{aligned}$$



Eccentricity due to the resultant force, e

$$\begin{aligned} &= M_c / F_t \\ &= 0.522 \text{ m} \end{aligned}$$

not greater than $L/6 = 1.152 \text{ m}$

The whole base is under compression

Effective stress length, L'	=	6.91 m
Base pressure, q_{max}	=	10.73 kN/m ²
q_{min}	=	4.04 kN/m ²

Ok

$$\begin{aligned} q_{\text{max}} &= \frac{F_y}{B \times L} + \frac{6M}{B \times L^2} \\ q_{\text{min}} &= \frac{F_y}{B \times L} - \frac{6M}{B \times L^2} \end{aligned}$$

Calculation/Sketch

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Consulting Engineers, Planners & Managers



Site	Dubai Depot - Rashidiya B21 Diesel Fuel Station	Date	Sep '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Design of footing reinforcement

Distance between the footing edge to the column edge, x_1	=	5.935 m
Stress at the face of the column, q_1	=	9.79 kN/m ²
cover to the main reinforcement	=	83 mm
effective depth of the footing, d	=	367 mm
d'	=	83 mm
f_{cu}	=	40 N/mm ²
f_y	=	460 N/mm ²

Design bending reinforcementMaximum ULS moment at the face of column, M_1

$$= B \{ [q_{min} \cdot x_1 \cdot (x_1/2) + 0.5 \cdot (q_1 - q_{min}) \cdot x_1 \cdot (x_1/3)] - [24 \cdot x_1 \cdot h \cdot x_1/2] \} \cdot \gamma_L$$

$$= 569.1 \text{ kNm}$$

K'	=	0.156
$K = M_1 / f_{cu} B d^2$	=	0.01174
z	=	233.3 mm
x	=	297.0 mm

A_s' req	=	0.0 mm ²
A_s req	=	6094.1 mm ²
A_s req (min)	=	585.0 mm ² / m

Provide **60 T12**

A_s provided	=	6785.8 mm ² (spacing =	149.7 mm)	Ok
	=	755.35 mm ² / m		

Design shear reinforcementMaximum ULS shear at the face of column, V_1

$$= B \cdot 0.5 \cdot [q_1 + q_{min}] \cdot x_1 \cdot \gamma_L$$

$$= 517.0 \text{ kN}$$

$V_1 / b d$	=	0.16 N/mm ²	$100 A_s / B d$	=	0.205445
			$400 / d$	=	1.089918

v_c	=	0.41 N/mm ²
$v_c + 0.4$	=	0.81 N/mm ²
$0.8 f_{cu}^{0.5}$	=	5.0 N/mm ² <= 5 N/mm ²

A_{sv} / s_v req	=	0 mm ² / mm
Provide	=	T @300 c/c (2 legs)
A_{sv} / s_v prov	=	0 mm ² / mm

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



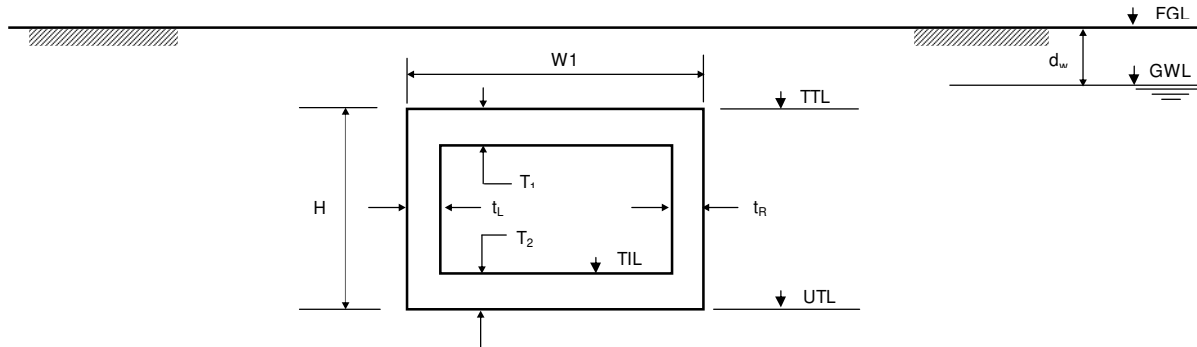
Site: **Dubai Metro Project - Rashidiya Depot**
Building 21
RC Tank Uplift - Simplified Check

Date: 19-Sep-07

Job No.: 68010.01

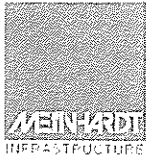
Designed by: **OEH**

Sheet No.:



Finished ground level, FGL	=	18.275	m	Soil bulk density, γ	=	20.0	kN/m ³
Ground water level, GWL	=	18.275	m	Soil submerged density, γ'	=	10.0	kN/m ³
RC Tank invert level, TIL	=	13.575	m	Water density, γ_w	=	10.0	kN/m ³
Top slab thickness, T_1	=	0.30	m	Underside of RC Tank level, UTL	=	13.275	m
Bottom slab thickness, T_2	=	0.30	m	Top of RC Tank level, TTL	=	18.275	m
RC Tank clear height	=	4.40	m	Height of soil above GWL, d_w	=	0.000	m
Overall RC Tank height, H	=	5.000	m	Height of water above TTL, h_w	=	NONE	m
Side wall thickness, t_L	=	0.30	m	Sand bulk density, γ_{sand}	=	18.0	kN/m ³
Overall RC Tank width (longer side), W1	=	5.70	m				
RC Tank clear width (longer side)	=	5.10	m				
Overall RC Tank width (shorter side), W2	=	4.04	m				
RC Tank clear width (shorter side)	=	3.44	m				
Total Slab Volume	=	13.82	m ³				
Total Wall Volume	=	24.13	m ³				
Total RC Tank Volume	=	37.95	m ³				
Weight of RC Tank	=	910.7	kN				
Volume of Fuel Tank	=	22.000	m ³				
Volume of void inside RC Tank	=	0.814	m ³				
Volume of sand inside RC Tank	=	54.38	m ³				
Weight of sand inside RC Tank	=	978.8	kN				
Weight of empty Fuel Tank	=	26.0	kN				
Restraining Force, R	=	1604.5	kN	(Weight of backfill, sand/1.30 + Weight of tank/1.10)			
Uplift Force, U	=	1151.4	kN				
Factor of Safety	=	R/U					
	=	1.39	> 1.15 OK.				

NOTE: Shear resistance between RC structure and the surrounding soil has been ignored



Site DUBAI METRO PROJECT

Date

Job No.

RASHIDIYA B21

Designed by CSA

Sheet No. 1/2

PURLIN ANALYSIS (TYPICAL):

DL:

SUSPENDING: 0.1 kPa

INSULATION: 0.3 kPa

0.4 kPa

 $\times 0.9 \text{m SPACING} = 0.36 \text{ kN/m}$

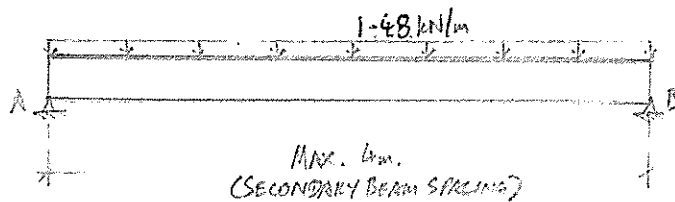
S/W:

7.73 kg/m = 0.08 kN/m

0.44 kN/m

LL:

0.6 kPa

 $\times 0.9 \text{m SPACING} = 0.54 \text{ kN/m}$ ULS: $1.4(0.44) + 1.6(0.54) = 1.48 \text{ kN/m}$ 

$$V_R = \frac{1.5 \times 4}{2} = 3 \text{ kN}$$

$$M_{\text{max}} = \frac{1.5 \times 4^2}{8} = 3 \text{ kNm}$$

PURLIN: 200C25 @ 0.9m SPACING

MANUT STEEL TABLES:

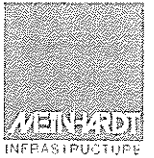
$$\left. \begin{array}{l} \text{ALLOWABLE SHEAR FORCE } V_{ax} = 43.95 \text{ kN} \\ V_{ay} = 45.55 \text{ kN} \end{array} \right\} > 3 \text{ kN}$$

$$\left. \begin{array}{l} \text{ALLOWABLE BENDING MT. } M_{ax0} = 11.69 \text{ kNm} \\ M_{ax01} = 7.02 \text{ kNm} \\ M_{ax02} = 7.6 \text{ kNm} \end{array} \right\} > 3 \text{ kNm}$$

$$\text{ASSUMING WL: } 0.6 \text{ kPa} \times 0.9 \text{m SPACING} \times 5.3 \text{m HT.} = 2.9 \text{ kN}$$

$$\left. \begin{array}{l} \text{ALLOWABLE AXIAL FORCE } P_{a0} = 82.02 \text{ kN} \\ T_a = 105.29 \text{ kN} \end{array} \right\} > 2.9 \text{ kN}$$

 $\Rightarrow$ USE OF 200C25 PURLINS OK!

Site DUBAI METRO PROJECT

Date

Job No.

Designed by CJA

Sheet No.

2/2CHECK DEFLECTION :

$$\begin{aligned} \text{Using SLS : } & 1.0 \text{ DL} + 1.0 \text{ LL} \\ & = 1(0.44) + 1(0.54) \\ & = 1 \text{ kN/m.} \end{aligned}$$

$$\begin{aligned} \Rightarrow \delta_{\max} &= \frac{5 w L^4}{384 E I} \\ &= \frac{5 \times 1 \text{ kN/m} \times 4^4 \text{ m}}{384 \times 205 \times 10^3 \text{ N/mm}^2 \times 6.17 \times 10^6 \text{ mm}^4} \\ &= 2.64 \text{ mm} < \frac{4000}{200} = 20 \text{ mm} \end{aligned}$$

$\Rightarrow$ DEFLECTION CHECK FOR 200C25 PURLINS OK!